2^{nd} INTERNATIONAL CONFERENCE ON BLOCKCHAIN & ARTIFICIAL INTELLIGENCE

September 17, 2025

ICBCAI 2025

Organized by

PG DEPARTMENT OF INFORMATION TECHNOLOGY AND BCA

Dwaraka Doss Goverdhan Doss Vaishnav College

(Autonomous)

Re-accredited with A++ by NAAC (3^{rd} Cycle)

College with Potential for Excellence

Linguistic Minority Institution

Affiliated to University of Madras

Arumbakkam, Chennai – 600 106.

Phone: 044 2363 5101

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PREFACE

Dwaraka Doss Goverdhan Doss Vaishnav College, a linguistic minority institution established in the year 1964 by the Rajasthanis and Gujaratis settled in Chennai for the cause of higher education. The college has seen a phenomenal growth in terms of its infrastructure, its constantly restructured and revamped curriculum to cater to the specific needs of the students community. Outstanding performance of the students in academics and extension activities has enabled the college to emerge one of the premier institutions of higher learning.

The Department of Computer Applications (BCA) was started in the year 1998 and the M.Sc. IT department was started in the year 2001 with an aim of providing conducive ambience for learning Management & Career oriented subjects, keeping in view the changing trends in education. Students every year take part in various cultural and extracurricular activities and win many prizes and bring laurels to the department and to the college.

The International Conference on Blockchain & Artificial Intelligence (ICBCAI 2025) was the Second International Conference organized by the PG Department of Information Technology and BCA on 17th September 2025. The main objective of this conference is to provide platform for researchers, industry experts, and practitioners to share their latest advancements and insights in Blockchain and Artificial Intelligence transformative technologies.

The book consists of seventy-seven extend research abstracts that have been through peer review process covering innovative research strategies and their applications with multi-disciplinary perspectives in the domains of Data Security and Privacy on Block Chain, DAOs and AI Governance, Supply Chain Transparency with AI, Ethical Consideration, Decentralised AI Marketplaces and beyond.

This compilation will definitely be a handy reference and an eye-opener for the academicians and young researchers pursuing their research in computer applications.

We sincerely thank our authorities for their unconditional support for the success of this venture.

September 2025 **Dr. K. ANGAYARKKANI**
Head i/c

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CHIEF EDITOR

Greetings!

The meaning of education has transformed greatly in today’s technology driven and digitally connected world that we live in. An educator in the present times has to adopt a multi-dimensional approach having knowledge creation, confidence building and honing leadership skills at its core. While many of our students have been greatly contributing to various renowned and reputed organisations as exemplar leaders, the institution also focuses on developing entrepreneurship skills among students so that they would have the courage and conviction to establish an enterprise and create a legacy. In the sphere of research, understanding the complexities of the various issues, exploring new avenues and contributing to development of knowledge has been our priority. Having carved a niche for itself over the year, through its myriad achievements, this institution today stands as a symbol of immense possibilities and innumerable opportunities. The task ahead is clearly defined-educate, enlighten and empower. As Benjamin Franklin said, “An investment in knowledge pays the best interest”.

Capt. Dr. S. Santhosh Baboo
Principal, Patron & Convenor

MESSAGE FROM PRINCIPAL

A heartfelt welcome to this global gathering of minds!

In today's dynamic landscape, the fusion of Blockchain and Artificial Intelligence (AI) stands as a transformative force reshaping industries and societies. This harmonious integration is driving innovation in finance, healthcare, supply chain, and beyond, promising enhanced efficiency, heightened security, and a paradigm shift in how we interact with and leverage data. As we navigate this technological frontier, the collaboration between Blockchain and AI continues to redefine the contemporary world, offering unprecedented solutions and opportunities for a smarter and more connected future.

Congratulations to the entire team for your outstanding efforts! Your hard work, dedication, and collaborative spirit have truly shone through.

I congratulate the department for taking initiative to organise the Second **International Conference on Blockchain and Artificial Intelligence** and would like to see them to soar to new heights and success in all their endeavours.

Capt. Dr. S. Santhosh Baboo
Principal, Patron & Convenor

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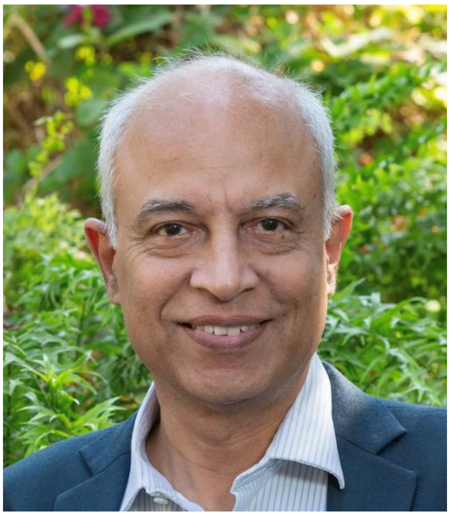
MESSAGE FROM SECRETARY

My heartily greetings. It gives me a great pleasure to release the proceedings of Second "**International conference on Blockchain and Artificial Intelligence**".

Blockchain and Artificial Intelligence (AI) represent two pillars of technological prowess that, when combined, create a synergy capable of transforming entire industries. Blockchain, a decentralized ledger, ensures a tamper-resistant and transparent record of transactions. Simultaneously, AI introduces the power of machine learning, enabling systems to adapt and learn from data patterns. The integration of these technologies enhances security, streamlines processes, and unlocks new potentials for innovation. Together, it redefine the landscape of our digital future, promising efficiency, security, and a paradigm shift in how we perceive and interact with information. I congratulate and appreciate the efforts taken by the whole team for organising this Second International conference on **Blockchain and Artificial Intelligence** on **17th Sep 2025.** Wishing the department continued success in all future endeavours.

Dr. Shri Ashok Kumar Mundhra
Secretary & Chief Patron

KEYNOTE ADDRESS FROM CHIEF GUEST



It is an honor and a privilege to address this gathering at the Second International Conference on Block Chain and Artificial Intelligence ICBCAI 2025. AI is no longer confined to laboratories or data centers; it permeates healthcare, education, agriculture, finance, logistics, and creative industries. This diffusion has given rise to a new class of professionals AI engineers, data scientists, machine learning specialists, AI ethicists, prompt engineers, and automation designers. Equally important are roles that blend AI with human insight educators integrating AI tools in pedagogy, entrepreneurs building AI-driven startups, and policymakers crafting ethical frameworks for responsible innovation.

The future of employment will be defined not by replacement but by *reinvention*. Routine, repetitive tasks will be automated, but this shift will free human potential to focus on higher-order functions such as decision-making, empathy-driven service, innovation, and interdisciplinary problem-solving. Thus, AI is not eliminating human work; it is transforming the very meaning of it. The true success of the AI era will not be measured merely by its algorithms, but by how it uplifts human capabilities and contributes to equitable growth.

I wish this conference grand success and hope that every participant leaves inspired, informed, and empowered to contribute meaningfully to the evolving digital ecosystem.

Dr. Kannan Moudgalya
Professor, IITB, Mumbai

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Prediction of Diabetes Mellitus through Patient Health Data using Machine Learning Techniques

Dr. S. Santhosh Baboo[1], Ms. B. Aarthi[2]
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Abstract: Diabetes mellitus is a chronic condition that requires early diagnosis and treatment since it poses serious health concerns to people worldwide. Machine learning approaches are now crucial tools for diagnosing and treating these diseases due to the exponential rise of healthcare data. Using patient health data, this study investigates the use of three supervised learning algorithms— Naïve Bayes, Decision Tree, and Random Forest—for the prediction of diabetes mellitus. The dataset used in this study comprises a number of medical parameters that are established risk factors for diabetes, including age, blood pressure, BMI, and glucose level. Preprocessing methods were used to improve model performance, handle missing values, and normalize data. Using common classification criteria, such as accuracy, precision, recall, and F1-score, the models were assessed. The Random Forest classifier outperformed the other two algorithms in terms of accuracy, demonstrating its resilience and potential for clinical decision assistance. According to the results, machine learning can help healthcare providers make well-informed decisions and play a major role in the early detection of diabetes

1. Introduction

The use of machine learning (ML) in healthcare is still a gamechanger, altering how illnesses are identified and treated. ML, or machine learning which basically uses models or historical data to program computers to perform better, has emerged as a crucial element in clinical development. One of its applications is the analysis of huge datasets, from which valuable insights are gleaned to aid clinical professionals in making decisions and enhancing patient care. Large-scale clinical datasets have been subjected to a variety of machine learning algorithms and approaches, including choice trees, irregular woods, and guiltless bayes, in order to find hidden patterns and associations. These findings have a significant favorable influence on patient outcomes and drug compliance in addition to helping with early disease detection. The importance of early disease detection emphasizes how urgently it is needed (Alanazi, 2022). Depending on how they are combined or interpreted, the 132 side effects in the viable dataset may result in around 41 distinct illnesses. The goal of this study is to create an expectation model that predicts the infirmity that patients are likely to experience based on their side effects, using data from 4920 patient records. Table 1 shows the symptoms of 41 diseases.

Table 1. Symptoms of the 41 Diseases

Back Pain	Bloody Stool	Scurrying
Constipation	Depression	Passage of Gases
Abdominal Pain	Irritation in Anus	Weakness in Limbs
Diarrhea	Neck Pain	Fast Heart Rate
Mild Fever	Dizziness	Internal Itching
Yellow Urine	Cramps	Toxic Look
Yellowing of Eyes	Bruising	Palpitations
Acute Liver Failure	Obesity	Painful Walking
Fluid Overload	Swollen Legs	Prominent Veinson Calf
Swelling of Stomach	Irritability	Fluid Overload
Swelled Lymph Nodes	Swollen Blood Vessels	Excessive Hunger
Sinus Pressure	Slurred Speech	Hip Joint Pain
Runny Nose	Knee Pain	Polyuria
Congestion	Skin Peeling	Family History
Chest Pain	Extra Marital Contacts	Swollen Extremities
Yellow Crust Ooze	Swelling Joints	Coma Unsteadiness
Lose of Smell	Stiff Neck	Unsteadiness
Movement Stiffness	Muscle Weakness	Drying and Tingling Lips

2.Literature Review

Machine learning techniques are increasingly used for early disease detection by identifying hidden patterns in medical data. Convolutional Neural Networks (CNN) are effective in automatic feature extraction for accurate disease classification. K-Nearest Neighbor (KNN) helps in predicting diseases by measuring symptom similarity with historical cases. Studies show that hybrid models often outperform traditional algorithms like Naïve Bayes, Decision Tree, and Logistic Regression. The medical field is rich in data but lacks effective tools to uncover hidden patterns and relationships within it. Data mining techniques such as classification, clustering, and association rule mining are used to analyze heart-related medical conditions. The MAFIA algorithm is applied for classification, with accuracy assessed using entropy-based cross- validation and partitioning methods. C4.5 decision tree algorithm is used for training, and K-means clustering helps in filtering relevant heart attack data from the database. Disease prediction involves analyzing patient symptoms to estimate the likelihood of specific illnesses for early and effective treatment. Previous models relied solely on symptoms without data transformation, resulting in poor accuracy. The proposed model improves accuracy by transforming the dataset using rarity-based weighting and combining Random Forest, LSTM, and SVM algorithms. This hybrid approach enhances prediction performance and supports automation in the healthcare industry.Integrating Ayurveda with machine learning addresses the rising demand for accessible and preventive healthcare solutions.The "Disease Prediction and Ayurvedic Guidance" system predicts common ailments like headaches and stomach issues based on user-reported symptoms.A Decision Tree algorithm is used for disease diagnosis, achieving 94.74% accuracy, 85.71% recall, and an F1 Score of 87.88%.This approach combines the predictive capabilities of ML with the holistic principles of Ayurveda to enhance modern healthcare. Diabetes is a chronic condition with serious complications, and its global prevalence is projected to rise from 422 million in 2014 to 642 million by 2040. Machine learning models are increasingly used to predict diabetes by identifying key risk factors using statistical methods like logistic regression. Classifiers such as Naïve Bayes, Decision Tree, AdaBoost, and Random Forest have been applied for effective diabetes prediction. Model performance is typically evaluated using metrics like accuracy and AUC, with various data partitioning protocols (K2, K5, K10) ensuring robustness through repeated trials. Diabetes is a chronic disease that can lead to severe complications if left undiagnosed and untreated.

This study aims to develop a machine learning model to predict the likelihood of diabetes with high accuracy. Decision Tree, SVM, and Naive Bayes algorithms are applied on the Pima Indians Diabetes Database for early detection. Naive Bayes achieved the highest accuracy of 76.30%, with performance evaluated using metrics like Precision, Recall, F-Measure, and ROC curves.

1. Materials and Methods

Materials and Methods is a section in research that describes the tools, data, and resources used for the study along with the detailed procedures followed to conduct the research. It explains what materials (such as datasets, software, or equipment) were utilized and how the experiments or analysis were carried out. This section ensures that the study can be replicated by others by providing a clear and systematic description of the methodology. Overall, it serves as a guide to understand the research process from data collection to analysis

3.1 Existing System

Several existing systems have applied machine learning techniques to predict diabetes using patient health data. Common algorithms like Naïve Bayes, Decision Tree, and Random Forest are widely used due to their effectiveness in classification tasks. These systems typically utilize datasets such as the Pima Indians Diabetes Database, which contains various health indicators like glucose level, BMI, age, and blood pressure. Naïve Bayes offers simplicity and fast computation but may sometimes lack accuracy due to its assumption of feature independence. Decision Trees provide interpretable models but can be prone to overfitting if not properly pruned. Random Forest, an ensemble of decision trees, generally delivers higher accuracy and robustness by reducing overfitting through averaging multiple trees. Existing research shows that combining these algorithms and comparing their performance helps in selecting the best predictive model for early diabetes detection.

3.1.1 Proprocessing

The preprocessing strategies utilized in Prediction Model 1 are:

Data Cleaning: Information is scrubbed through cycles, hence settling the irregularities in the information.

Data Decrease: The investigation turns out to be hard when managing enormous databases. Subsequently, those free variables (symptoms) are disposed of, which may not affect the objective variable (disease). In this paper, 95 of the 132 side effects firmly connected with illnesses are selected.

3.1.1 Decision Tree Classifier

A particular kind of machine learning model called the Decision Tree algorithm divides data into branches that resemble trees in order to make choices. Beginning at the very beginning, it assesses many qualities to determine which are most useful for decision- making. These choices result in more branches until they reach leaf nodes, when majority voting is used to make predictions. According to Grampurohit and Sagarnal (2020), the decision tree algorithm aids in disease prediction by examining symptoms. It separates the data by identifying the most crucial symptoms. This continues until it reaches a limit or discovers groups with comparable Symptoms. It assigns the most prevalent disease label to the last group after using the tree route based on symptoms to predict diseases for new instances.

4.Result and Discussion

The framework was prepared based on a clinical record of 4920 patients inclined to 41 infections, which were expected to have a blend of different side effects. 95 side effects out of 132 were considered to keep away from overfitting. The K overlay cross- validation procedure (K = 5) was utilized to really look at the exhibition of each of the three calculations on the dataset Disease Prediction Result Page expected illness, together with information on the illness, safety measures, and urgency—that is, whether or not the patient has to see a doctor right now. Additionally, it offers a link to an additional site with information regarding the anticipated illness. It has a text field where the patient's location can be entered. The patient is taken to the "PRACTO" website, which lists the local physicians who are capable of curing the illness, after clicking the submit button

5. Conclusion

From the authentic advancement of AI and its applications in clinical areas, it has been shown that frameworks and procedures have arisen that have empowered refined information examination through straightforward and direct utilization of AI calculations. This paper presents a far- reaching relative investigation of four calculations executed on a clinical record, each yielding a precision of up to 95 percent. The presentation is broken down into a confusion matrix and an accuracy score. Artificial knowledge will assume a significantly more significant role in information examination in the future because of the accessibility of enormous amounts of information created and put away by cutting-edge innovation.

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AI in Web Development: Using Prompt

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Abstract— AI is a fast-developing technology which has been usedby all the technologies to enhance performance across industries. One such area is web development, which has become essential for all types of businesses from large corporations to startups. Traditionally, building a website required significant time, financial investment, and human resources. But with the emergence of prompt-based AI systems allow users to describe what they want in plain language, and the system automatically generates the website code, layout, and even content. Tools like Lovable, Durable, and Frame AI can create responsive and SEO-friendly websites within seconds using AI models like LLMs and generative design. This paper explores how AI, especially prompt-based systems, is transforming the way websites are developed. A Comparison is done between traditional and AI-driven approaches by collecting data, identifying key problem areas, and evaluating performance through frameworks such as user experience and optimization. While AI helps save time and reduce dependency on manual work, challenges like creativity, customization, and scalability still exist. Based on the findings, it is likely that future web development workflows will benefit most from a hybrid approach that combines AI automation with human input.

I. Introduction

Traditional way of building a website is a hardest way. To build that is needs the knowledge of different programming languages like HTML, CSS, JavaScript, and backend technologies. And also, the better understand of the design principles, database management, and how to connect everything together to build a complete website. For this it needs a lot of time, money, and manpower. But, with the recent development of AI, a text-prompt is enough to build the website. By this websites building has become much simpler and faster. Now, users can just describe what kind of website they need using plain text, and the AI generates the full code, layout, and content automatically. Tools like Lovable and Framer AI allow prompts to build website within minutes and it was functional and editable. These platforms use large language models and generative design to build unique, responsive, and SEO-friendly websites. This approach is reducing the technical barrier and allowing even people with no coding background to turn their ideas into real, working websites.

II. Literature review

The use of AI in web development has brought about some significant changes in how it can design, build, and deploy applications. A number of researchers have looked into the many ways AI tools and prompt engineering are making things better for both developers and users.

For instance, Ranjan and Sanket [1] gave a good overview of how AI tools and their prompts work. They pointed out that prompts that can adapt and change based on user feedback and

preferences are very useful. Their research really highlighted the importance of having prompts that are both personalized and easy to understand, especially in fields like customer service, education, and healthcare. Building on this, Islam [2] focused on how prompt engineering helps developers, even those who might not have a strong technical background. He found that when you write prompts well, you can guide AI models to handle all sorts of repetitive tasks, such as generating code, running tests, and writing documentation. This frees up developers to spend more time on the creative and unique parts of a project.

Sharma [3] took a step back and looked at the bigger picture. He noted that AI's impact goes far beyond just coding and testing. He saw that AI is also making a difference in website design, personalization, and performance. By looking at a lot of different studies and examples, Sharma emphasized how AI is bringing in new ways of thinking about how to create websites that can adapt and change on their own. In a similar vein, Jadhav and Gholve [4] also argued that AI-powered tools are simplifying the development process by cutting down on manual work and making things more efficient. Their study backed up the idea that AI can speed up development timelines and make users happier with features like predictive analytics and personalized content.

Taken together, all these studies show that AI, especially when used with smart prompt engineering, has the potential to make web development accessible to more people, boost productivity, and create more user-friendly experiences. However, there's still more work to be done. It need to come up with some standard guidelines for designing prompts and also figure out how to handle the ethical issues that come with using AI in development.

III. Methodology

For the methodology, this paper has been using the comparative analysis approach, by using previous researchers' papers.

a. Research Design

The research will follow the qualitative descriptive design. And do not involve primary data collection. Instead of the primary data collection, this paper uses the secondary sources such as research articles, case studies. This comparative approach is used to analyse traditional website development methods alongside AI-drive prompt-based development. By this paer can identify the key difference, advantages, and challenges between the traditional way and AI-prompt.

b. Data Source

This paper used the data for this study was collected from secondary sources. This paper includes the works of Ranjan and Sanket, Islam, Sharma, and Jadhav and Gholve, who have examined different dimensions of AI in web development and prompt-based. Their studies provide insights into how AI tools and prompts are applied, the advantages of prompt-based for both technical and non-technical developers, and the larger impact of AI on design, personalization, and efficiency in web applications.

In addition to these, the paper use the online resources such as technology blogs, articles, and case studies that explain how AI-driven website builders' function in real-world scenarios. Official tools like Lovable, Durable, and Framer AI were particularly useful for

understanding how prompt-based systems generate websites, reduce development time, and simplify the overall process. These secondary sources together offered both theoretical knowledge and practical examples, making it possible to carry out a comparative analysis between traditional website development methods and AI-based approaches.

IV. Result and Discussion

This is the result obtained by the comparative analysis of traditional website building and AI prompt-based website building.

- **Development Time:** Traditional method consumes more time, required months to build a website, But AI tools can generate in minutes.
- **Required Skills:** Traditional method demands expertise in multiple programming languages, design and database management knowledge, but using AI tools easy to build a website just by the idea and with minimal training.
- **Cost of Development:** Traditional way cost high due to the professionals' skills required, but as AI tools there are many free tools and some cost much less compare to the traditional way.
- **Customization and Creativity:** The traditional way have great flexibility and originality and required more effort. In AI the system is improving and they still rely heavy on templates and pre-build designs.
- **Scalability And Maintenance:** Traditional way enable high scalability but required updates. With the advanced customization AI automates many updates and scale quickly.
- **User Experience:** AI personalized through predictive analytic make the website more user-focused which was compared with the traditional way.

By this the overall result this paper obtained that AI prompt system gives best performance compare with the traditional way in terms of speed, accessibility and in cost efficient also and have some drawbacks also but they can be overcome able.

b. Discussion

In the discussion this paper say that the prompt-based AI tools are really changing the way of building websites. AI tools making it so much easier and cheaper for people to build a website. This will be a huge help for startups, small businesses, or just individuals who don't have the money to hire a developer. They can see it with platforms like Lovable and Framer AI, where we can just type what we want, and it turns our words into a real website that good for SEO. It's shifted the idea of coding from to just have a good idea.

But this paper also found that it want the clear ideas about our website that how it can give it perfectly. It's super-fast at automating this paper says that the future of web development will be a mix of both:

- AI for the boring stuff like automation, making prototypes, and personalizing things.

- Humans for all the creative work, custom designs, and new ideas.

So, this paper thinks that while AI definitely makes things more efficient and accessible, this paper shows the best results will come from combining AI and human skills.

V. Limitations and Future Research

Even the AI prompt-based development has advantages over traditional method, there are still some limitations remain. Current AI tool still depends on pre-build templets and doesn’t have unique brand identities or customized features. While AI reduce the code errors and speed up the development time, still need human overview to check the quality, creativity, and alignment with the client requirements. The limitations are that this paper relies on secondary sources meaning the findings reflect on current state of the literature and available tools rather the first-hand empirical testing. The scope of this paper is limited because of the commonly used AI tools like Lovable, Durable, Framer AI. By these this paper shows you a meaningful overview of how AI is used to build a website.

From this the papers say that the future research should do some tests on how much time and money AI website builders actually save in real world situations, and how satisfied users are with the final websites. It will also help to test these tools by using them on large projects for the better understand of them problems like scalability and security. As it still improving, can expect for more creative designs, better personalization, and strong integrations with technologies. And the most effective approach to web development is hybrid model.

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A Comprehensive Review of Cyber Attack Prevention Techniques

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Abstarct

Cybersecurity has emerged as one of the most pressing global concerns in the digital era. The rapid rise in the frequency and sophistication of cyber-attacks, including ransomware, phishing, distributed denial-of- service (DDoS), and supply-chain exploits, underscores the need for robust prevention mechanisms. This paper reviews the current landscape of prevention techniques, analyzing their strengths, limitations, and applications. The study synthesizes recent literature to classify prevention techniques into network-level, system-level, data- centric, human-centric, and advanced intelligence-driven approaches. It further identifies critical challenges such as evolving zero-day vulnerabilities, cost barriers, and human negligence. The review concludes that no single defense mechanism is sufficient; instead, a multi-layered, hybrid security framework, combining technical controls, awareness programs, and regulatory compliance, is necessary for sustainable protection.

Keywords: Cybersecurity, Cyber- attacks, Prevention Techniques, Ransomware, Zero Trust, Network Security, Human-centric Security

2. INTRODUCTION:

The accelerated digitization of industries, services, and social interactions has led to an exponential rise in cyber risks. From financial institutions to healthcare and government, virtually every sector faces the threat of targeted cyberattacks. Cybersecurity is no longer a purely technical concern but a critical factor for national security, business continuity, and personal privacy. The fundamental aim of cybersecurity is to ensure confidentiality, integrity, and availability (CIA) of digital assets. While organizations invest heavily in detection and response mechanisms, the cost of breaches reveals that prevention is more cost-effective than recovery. For example, the global average cost of a data breach in 2024 reached $4.45 million, reflecting both direct losses and reputational damage. The primary objective of this paper is to review prevention techniques against modern cyber threats. Unlike traditional reviews that combine threats and solutions, this work focuses exclusively on defenses, with emphasis on their classification, effectiveness, and practical challenge.

3. LITERATURE REVIEW:

[1] Khurana S. (2017) This review highlights different cyber-attacks such as malware, phishing, and keylogging, and presents preventive measures like firewalls, antivirus software, and security tokens. The study emphasizes that although technical remedies exist, most attacks exploit user negligence, requiring both technical and human-centric defenses.

[2] Patil, A. A. (2024) The work examines cybersecurity challenges faced in 2023–24, including ransomware, IoT vulnerabilities, and mobile banking malware. It underlines the evolving nature of threats and stresses proactive defenses like strong authentication, encrypte d communication, and frequent patching.

[3] Hattimare, S. S., Pande, S. M., & Malekar, S. V. (2023) This study focuses on cyber-attack prevention, evaluating malware, phishing, ransomware, and insider threats. The authors recommend multi-layered prevention strategies including regular updates, employee training, strong authentication mechanisms, and real-time monitoring to counter emerging risks.

[4] Najem, D. F., & Kareem, S. M. (2025) The paper presents a broad review of cyber threats and classifies them into malware, DDoS, phishing, and botnets. It stresses the importance of the CIA triad (Confidentiality,Integrity, Availability) in prevention frameworks, while also noting that zero-day exploits and workforce skill shortages remain pressing challenges.

[5] IBM Security (2024) This industrial report provides global insights into the cost of a data breach, averaging $4.45 million in 2024. It emphasizes preventive approaches like encryption, backup systems, and Zero Trust architectures, showing how proactive investments reduce long-term losses.

[6] Verizon (2024) The Data Breach Investigations Report (DBIR) identifies phishing and stolen credentials as the top attack vectors. It highlights the role of employee training, phishing simulations, and MFA adoption in reducing successful attack rates.

[7] Jain, A. K., Deb, D., & Engelsma, J. J. (2021) Although primarily on biometric authentication, this review draws attention to issues of fairness, explainability, and privacy. Its relevance lies in stressing that trust and usability must be factored.

[8] Shukla, S., Varshney, G., Singh, S., & Goel, S. (2024) The study reviews evolving biometric technologies for authentication, noting gains in accuracy but also risks like spoofing. It suggests layered biometric systems integrated with MFA as an effective prevention strategy.

[9] MITRE ATT&CK Framework (2023) This framework provides a globally recognized taxonomy of attacker tactics and techniques. It is widely used to structure prevention strategies and align organizational defenses against adversarial behaviors.

3. CLASSIFICATION OF PREVENTION TECHNIQUES:

Cybersecurity prevention techniques can be systematically grouped into several categories, each addressing different attack surfaces. The key classes include network- level defenses, endpoint and system-level protection, access and identity management, data-centric security, human-centric measures, and advanced intelligence-driven methods. Together, these layers form a multi- faceted defense architecture often described as “defense-in-depth.” Below, each class is discussed in detail.

3.1 Network-Level Prevention Firewalls:

Firewalls remain the cornerstone of network security. Traditional firewalls function by filtering traffic based on IP addresses and ports, while NGFWs extend this by inspecting packets at the application ayer, detecting malicious payloads, and blocking unauthorized access attempts. For instance, an NGFW can block SQL injection attempts by analyzing HTTP traffic.

Intrusion Detection and Prevention Systems (IDS/IPS):

IDS tools passively monitor network activity to detect suspicious traffic patterns, whereas IPS tools can actively block or drop malicious packets in real time. Signature- based systems detect known threats, while anomaly-based systems are capable of identifying deviations from normal traffic, useful against zero-day exploits.

Network Segmentation and Micro- Segmentation:

By dividing networks into smaller zones, segmentation reduces the impact of a breach. If an attacker compromises one system, they cannot easily move laterally to others.
Micro-segmentation, often implemented in cloud and virtualized environments, offers granular control, enforcing security policies at the workload level.
Case Example: In 2023, Cloudflare reported mitigating the largest-ever distributed denial- of-service (DDoS) attack, peaking at 71 million requests per second. This was made possible by combining layered filtering, rate limiting, and distributed network defenses.

Such cases highlight the indispensable role of robust network-level measures in preventing large-scale disruption.
Strengths: Effective perimeter defense, scalable for enterprises.
Limitations: Cannot protect against insider threats or encrypted malicious traffic unless combined with decryption inspection.

3.2 Endpoint and System-Level Prevention
Anti-Virus and Anti-Malware Solutions:

Traditional anti-virus tools remain widely used, but modern endpoint threats require enhanced solutions such as heuristic detection, behavior analysis, and cloud-based threat intelligence.

Endpoint Detection and Response (EDR):

EDR systems monitor endpoint activities continuously, allowing real-time detection of abnormal behavior such as privilege escalation or ransomware encryption. They also provide forensic capabilities for incident investigation.

Patch Management and Vulnerability Scanning:

Unpatched systems are a primary entry point for attackers. The WannaCry ransomware outbreak of 2017 exploited a known
Windows vulnerability (MS17-010).
Organizations with automated patch management avoided infection, proving that timely updates are essential preventive measures.

Sandboxing and Application Whitelisting:

Sandboxing allows suspicious files to be executed in isolated environments, preventing them from harming production systems. Application whitelisting ensures that only verified applications can run, blocking unauthorized software installations.
Strengths: Directly protects user devices where most attacks begin.
Limitations: Can be resource-intensive; requires regular updates and monitoring

3.3 Access Control and Authentication
Passwords and Their Limitations:

Password-only systems are weak due to brute force, dictionary attacks, and credential reuse. Studies show that stolen credentials are a top attack vector globally.
Multi-Factor Authentication (MFA):

MFA strengthens access security by requiring two or more verification factors— something you know (password), something you have (token, smartphone), and something you are (biometric). For example, online banking frequently requires a password plus a one-time passcode (OTP).
Zero Trust Network Access (ZTNA):

The Zero Trust model assumes that no user or device should be automatically trusted, even within an organization's perimeter.
Access decisions are context-driven, based
on identity, device health, and continuous monitoring. Google's BeyondCorp initiative is a well-known example of large-scale Zero Trust adoption.
Privileged Access Management (PAM):
PAM tools protect administrator accounts, often targeted by attackers for high-level control. These systems enforce just-in-time access, session monitoring, and vaulting of credentials.
Strengths: Reduces credential theft and insider misuse.
Limitations: Can be bypassed by advanced phishing methods (e.g., MFA fatigue attacks).

3.4 Data-Centric Techniques Encryption:
Encryption safeguards sensitive data both at rest and in transit. Standards like AES-256 and TLS 1.3 ensure confidentiality against interception. Even if data is exfiltrated, it remains unreadable without the decryption key.
Data Loss Prevention (DLP):

DLP solutions monitor and block unauthorized transfer of sensitive data through email, cloud storage, or USB devices. For instance, healthcare providers use DLP to prevent accidental leakage of patient records.
Secure Backups:
Regular, immutable, and air-gapped backups are critical to withstand ransomware attacks. Without backups, organizations may feel forced to pay ransoms. However, with resilient backup strategies, recovery can occur without engaging attackers.
Strengths: Protects the "crown jewels" — organizational data
Limitations: Insider threats or misconfigurations can still expose data.
3.5 Human-Centric Techniques Security Awareness Training:
Human error remains the leading cause of breaches. Awareness programs train employees to

recognize phishing emails, avoid weak passwords, and report suspicious activity.

Phishing Simulations:

Organizations increasingly conduct simulated phishing campaigns to test employees and reinforce vigilance.

Policies and Compliance:

Frameworks such as ISO 27001, GDPR, and HIPAA ensure security governance, incident response planning, and accountability.
Case Example: Verizon's 2024 Data Breach Investigations Report found that 74% of breaches involved the human element — errors, misuse, or social engineering. This highlights the importance of human-centric security initiatives.
Strengths: Cost-effective, strengthens weakest link.
Limitations: Effectiveness varies with employee motivation and organizational culture.

3.6 Intelligence-Driven & Advanced Techniques

AI/ML Anomaly Detection:

Machine learning models analyze massive datasets to detect deviations from normal patterns, identifying new attack vectors. For example, AI can flag unusual login attempts across geographies.

Threat Intelligence Sharing:

Collaborative frameworks such as MITRE ATT&CK and STIX/TAXII enable organizations to share information on emerging threats. Shared intelligence strengthens collective resilience.

Deception Technologies:

Honeypots and honeytokens lure attackers into interacting with fake systems, allowing defenders to study adversary tactics and divert attacks away.

Blockchain & Quantum Cryptography (Future Directions):

Blockchain can ensure immutable audit trails, while quantum-safe algorithms are being developed to prepare for post-quantum attacks.
Strengths: Offers proactive and adaptive security.
Limitations: High cost, false positives, complexity of deployment.

4. CHALLENGES IN PREVENTION:

Despite significant advancements, prevention efforts face numerous technical, organizational, and socio-economic challenges. These challenges hinder effective implementation and create persistent vulnerabilities.

4.1 Zero-Day Vulnerabilities

Zero-day exploits are unknown vulnerabilities exploited before patches are available. Preventing such attacks is extremely difficult since traditional signature-based defenses cannot detect them. The 2023 MOVEit Transfer exploit illustrates how attackers weaponize zero- days to compromise thousands of organizations simultaneously.

4.2 Human Negligence

Employees often reuse passwords, fall for phishing emails, or inadvertently mishandle sensitive information. Insider threats whether intentional or accidental also remain a significant challenge. The Verizon DBIR (2024) highlights that human factors contribute to nearly three-fourths of breaches.

4.3 Cost Barriers

Advanced defenses like AI-driven detection, Zero Trust, and PAM require significant financial investment. Small and medium enterprises (SMEs), often resource- constrained, struggle to implement comprehensive frameworks. This creates a security gap that cybercriminals exploit.

4.4 Technological Evolution of Threats

Cybercriminals innovate faster than defenses. Examples include:

Ransomware-as-a-Service (RaaS): Commercialized ransomware kits sold to non-expert attackers.

AI-Generated Attacks: Deepfake voice phishing and AI-written phishing emails.

IoT Vulnerabilities: Billions of poorly secured IoT devices provide easy attack surfaces.

These evolving tactics reduce the effectiveness of static defenses.

4.5 Shortage of Cybersecurity Professionals

Globally, there is an estimated shortage of
3.5 million cybersecurity professionals. This skills gap particularly affects forensics, incident response, and advanced threat detection, leaving organizations unable to fully utilize available prevention technologies.

4.6 Policy and Compliance Challenges

While frameworks such as GDPR and NIS2 improve governance, fragmented regulations across regions create complexity for global organizations. Furthermore, a compliance- driven mindset often leads to “checkbox security,” where organizations meet regulatory requirements without genuinely strengthening defenses.

4.7 False Sense of Security

Overreliance on certain tools may create blind spots. For instance, organizations relying solely on antivirus software may neglect network monitoring. Similarly, AI- based detection systems can suffer from false positives, leading to alert fatigue among analysts, which in turn may cause them to miss real threats.

5. CONCLUSION:

Cybersecurity prevention remains the cornerstone of digital resilience in today’s interconnected world. This review has shown that effective defense cannot rely on a single mechanism but must instead adopt a layered approach. Network defenses such as firewalls and IDS/IPS, system-level protections like EDR and patch management, strong access controls, data-centric safeguards, and human-centric training all form critical pillars of prevention. Advanced techniques such as AI-driven anomaly detection and threat intelligence sharing further strengthen organizational readiness.

Yet, the challenges are significant. Zero-day vulnerabilities, human negligence, high costs, and evolving attack strategies reduce the effectiveness of even the most advanced tools. Moreover, a shortage of skilled professionals and the risks of compliance- driven security practices complicate prevention further. The central message emerging from this review is clear: cybersecurity prevention is not a static goal but a continuous, adaptive process that integrates technology, governance, and awareness.

6. FUTURE SCOPE:

Looking forward, the scope of cyber-attack prevention will expand in several promising directions. Artificial intelligence and machine learning will continue to play a transformative role, enabling adaptive systems that learn from global threat data in real time. At the same time, quantum- resistant cryptography must be developed and adopted to secure data in the coming post-quantum era. Blockchain-based identity management and decentralized authentication models may provide greater trust and transparency in digital transactions. Equally important, the human element will remain central to future defenses. Organizations must move beyond annual awareness programs toward continuous cyber hygiene training that evolves alongside new threats. Finally, global collaboration through cross-border threat intelligence sharing and harmonized regulatory frameworks will be essential in creating a collective shield against adversaries who operate without borders. In sum, the future of prevention lies in integration, innovation, and cooperation. By uniting advanced technology with human awareness and international collaboration, cybersecurity can progress from reactive defense to proactive resilience in the digital era.

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A Hybrid Deep Learning Framework (LRNN) for Customer Sentiment Analysis in E-Commerce

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Abstract: Massive volumes of unstructured customer feedback, like product reviews, have been produced by the exponential growth of e-commerce platforms. These reviews offer important insights into the sentiment of consumers. Businesses need to accurately identify sentiment in order to improve customer experience, make better decisions, and create successful marketing campaigns. Conventional machine learning methods, like Logistic Regression (LR), are effective and have good interpretability for sparse text features, but they have trouble identifying intricate linguistic patterns. On the other hand, neural networks (NN) efficiently represent contextual and semantic data, but they are computationally costly and frequently uninterpretable. This study presents a hybrid framework called the Logistic Regression Neural Network (LRNN), which combines the advantages of LR and NN, in order to get around these restrictions. While the NN branch uses deep learning architectures to encode contextual representations, the LR branch makes use of sparse lexical features based on TF–IDF. The sentiments in Amazon mobile product reviews are categorized by fusing these complementary signals. In terms of accuracy, F1-score, and robustness across different sentiment categories, LRNN performs better than standalone NN baselines and traditional LR, according to experimental evaluation on a dataset of customer reviews. According to the results, LRNN offers a high-performing, interpretable, and scalable solution for practical sentiment analysis tasks in e-commerce. To further improve sentiment classification performance, future studies will investigate multimodal extensions that include star ratings, product metadata, and attention-based mechanisms.

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Keywords**:** Sentiment Analysis, Logistic Regression (LR), Neural Networks (NN), Hybrid Deep Learning Model, LRNN Framework, Text Mining

1. Introduction

In artificial intelligence, natural language processing, or NLP, has become a crucial field that allows machines to comprehend, interpret, and produce human language. NLP is frequently used in the e-commerce industry to examine enormous volumes of textual data, including social media posts, product descriptions, and customer reviews. NLP helps businesses obtain actionable insights from customer feedback by converting unstructured text into structured information, which makes tasks like sentiment analysis, opinion mining, and recommendation generation easier. NLP is a crucial tool for enhancing customer engagement and decision-making because of its capacity to process linguistic patterns at scale. In artificial intelligence, natural language processing, or NLP, has become a crucial field that allows machines to comprehend, interpret, and produce human language. NLP is frequently used in the e-commerce industry to examine enormous volumes of textual data, including social media posts, product descriptions, and customer reviews.

Figure 1: Word Cloud of NLP

Figure 1 shows the word cloud of NLP. It helps businesses obtain actionable insights from customer feedback by converting unstructured text into structured information, which makes tasks like sentiment analysis, opinion mining, and recommendation generation easier. NLP is a crucial tool for enhancing customer engagement and decision-making because of its capacity to process linguistic patterns at scale.

2. Literature Review

An AI-powered sentiment analysis system designed for e-commerce applications is presented in this paper. It combines deep learning models, especially transformer-based architectures refined on domain corpora, for accuracy with conventional machine learning for interpretability. The system demonstrated improvements in customer satisfaction, retention, and conversion rates while achieving an accuracy of 89.7% when tested on extensive e-commerce datasets. Scalability, real-time processing, and useful business value are prioritized in the hybrid design.

A research work carried by Wu et. al[1], in which that the CNN, RNN, and Bi-LSTM deep learning models with Word2Vec, FastText, and BERT embeddings were compared in the study. Experiments using a dataset of women's clothing reviews revealed that BERT-based methods performed better than conventional embeddings, especially in multi-class sentiment classification. With BERT attaining the highest accuracy across 3-class and 5-class tasks, the study emphasizes the value of contextual embeddings over static word vectors. Another research work done by Bellar et. al[2], in this research paper suggested a

T5-CapsNet hybrid meta-ensemble model in conjunction with GAN-based data augmentation. The model

outperformed baseline transformer models with 97.56% accuracy and 95.5% F1-score when tested on the Multilingual Amazon Reviews Corpus. The hybrid design was one of the best-performing sentiment classifiers for multilingual e-commerce reviews because it combined the generative power of T5 with the feature-capturing capability of Capsule Networks.

Table 1: Comparison of Research methods

Ref. No.	Author(s) & Year	Methods Used	Best Method & Accuracy
[1]	Wu et al. (2025)	Hybrid of Traditional ML + Transformer-based Deep Learning	Transformer-based hybrid model – **89.7%**
[2]	Bellar et al. (2024)	CNN, RNN, Bi-LSTM with Word2Vec, FastText, BERT	**BERT** – highest accuracy on 3-class & 5-class tasks (exact % not specified, but best performer)
[3]	Provaa (2024)	Hybrid **T5-CapsNet** with GAN-based augmentation	**T5-CapsNet** – **97.56% accuracy, 95.5% F1-score**
[4]	Kusal et.al	Transformer-based models (BERT, GPT variants)	**BERT/GPT models** – **89–92% accuracy**

3. Description of Dataset

Product details and user reviews for unlocked cell phones that are sold on Amazon make up the Amazon Unlocked Mobile Dataset. Product Name, Brand Name, Price, Rating, Reviews, and Review Votes are among its essential attributes shows in table 2. The dataset contains both unstructured (customer-written textual reviews) and structured (product details, prices, and numerical ratings) fields.

Table 2: Sample Dataset

Product Name	Brand Name	Price	Rating	Reviews	Review Votes
"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D700*FRONT CAMERA*ANDROID*SLIDER*QWERTY KEYBOARD*TOUCH SCREEN	Samsung	199.99	5	I feel so LUCKY to have found this used (phone to us & not used hard at all), phone on line from someone who upgraded and sold this one. My Son liked his old one that finally fell apart after 2.5+ years and didn't want an upgrade!! Thank you, Seller, we really appreciate it.	1

4. Preprocessing Techniques

A critical phase in Natural Language Processing (NLP) is preprocessing, which gets unprocessed text ready for efficient analysis and 20nalyse20g. Preprocessing makes sure that the data is clear, normalized, and appropriate for processing by algorithms because customer reviews and other text sources frequently contain noise, superfluous words, and irregular formatting. The following are typical preprocessing steps:

- Lowercase Conversion
- Elimination of Stopwords
- Stemming
- Lemmatization

5. Materials and Methods

This research work used Logistic Regression (LR), Neural Networks (NN), and a hybrid LRNN model to 20nalyse the sentiment of Amazon mobile product reviews. Because of its interpretability and effectiveness when working with high-dimensional text features, logistic regression was used as a baseline classifier. The following model was used to determine the likelihood that a review would fall into a sentiment class:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w \cdot x + b)}}$$

where www is the weight vector, bbb is the bias term, and x is the review's TF–IDF feature vector.

An input layer, a single hidden layer with nonlinear activation (ReLU), and an output softmax layer were used to implement the neural network for classification into positive, neutral, and negative classes. Each hidden layer's transformation is shown as:

$$h = f(Wx + b)$$

to allocate odds among K sentiment categories. Lastly, by using a voting ensemble, the suggested LRNN hybrid algorithm integrates the representational power of NN with the probabilistic interpretability of LR, allowing both models to influence the final prediction. In the classification of extensive customer reviews, this hybridization increases accuracy while preserving robustness and generalizability.

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6. Experimental Results

The Amazon Unlocked Mobile Dataset, which includes ratings and product reviews, was used for the experiments. Three models were put into practice: the suggested hybrid LRNN algorithm, logistic regression (LR), and neural networks (NN)..

6.1 Results of Preprocessing

There was a significant amount of noise in the Amazon Unlocked Mobile dataset's raw review texts, including mixed case, punctuation, stopwords, and inflected word forms. Standard text preprocessing methods were used to enhance the input data's quality. Lowercase conversion, tokenization, stopword removal, stemming, lemmatization, and punctuation and symbol removal were among the procedures. Figure 4 shows the result of preprocessing.

Product N	Brand Nar	Price	Rating	Reviews	Review Vo	lowercase	no_punct	tokenized	no_stopw	stemming	lemmatization
CLEAR CLE	Samsung	199.99	5	I feel so	1	i feel so	i feel so	['i', 'feel', '	['feel', 'luc	['feel', 'luc	['feel', 'lucky', 'found', 'us', 'phone', 'u
CLEAR CLE	Samsung	199.99	4	nice	0	nice	nice	['nice', 'ph	['nice', 'ph	['nice', 'ph	['nice', 'phone', 'nice', 'grade', 'pantac
CLEAR CLE	Samsung	199.99	5	Very pleas	0	very pleas	very pleas	['very', 'ple	['pleased']	['pleas']	['pleas']
CLEAR CLE	Samsung	199.99	4	It works gc	0	it works gc	it works gc	['it', 'work	['works', 'g	['work', 'gc	['work', 'good', 'goe', 'slow', 'sometim
CLEAR CLE	Samsung	199.99	4	Great	0	great	great	['great', 'p	['great', 'p	['great', 'p	['great', 'phone', 'replace', 'lost', 'phor
CLEAR CLE	Samsung	199.99	1	I already	1	i already	i already	['i', 'alread	['already',	['alreadi', '	['already', 'phone', 'problem', 'know',
CLEAR CLE	Samsung	199.99	2	The	0	the	the	['the', 'cha	['charging'	['charg', 'p	['charg', 'port', 'loose', 'got', 'solder', '
CLEAR CLE	Samsung	199.99	2	Phone	0	phone	phone	['phone', '	['phone', '	['phone', '	['phone', 'look', 'good', 'wouldnt', 'sta
CLEAR CLE	Samsung	199.99	5	I	0	i	i	['i', 'origin	['originally	['origin', 'u	['originally', 'us', 'samsung', 's2', 'gala
CLEAR CLE	Samsung	199.99	5	Battery is	0	battery is	battery is	['battery',	['battery',	['batteri', '	['battery', 'okay', 'could', 'better', 'ove

Figure 4: Sample Dataset

The word frequency figure 5, 6 and 7 indicates the frequency with which particular words appeared in the customer reviews. For example, the terms "max" and "chat" occur 558 times each, "awhile" 557 times, and "providers" and "disappointed" 556 times. In a similar vein, the words "louder" and "crash" appear 555 and 554 times, respectively, and words like "regarding," "lacks," and "continues" appear 553 times

Word	Frequency
max	558
chat	558
awhile	557
providers	556
dissapointed	556
louder	555
crash	554
regarding	553
lacks	553
continues	553

Figure 5: Word Frequency in reviews

These common terms reveal recurrent themes in the reviews; some of them reflect how the product is used (e.g., "max," "chat"), while others express customer discontent or problems (e.g., "disappointed," "crash," "lacks"). This type of analysis aids in finding trends in consumer reviews and offers information about the products' advantages and disadvantages.

6.2 Results of Deep Learning Algorithms

The TF-IDF feature extraction and preprocessing, the deep learning models were applied and assessed on the Amazon Unlocked Mobile dataset. One hidden layer with ReLU activation and a softmax output layer were used in the Neural Network (NN) architecture to classify sentiment into three categories: positive, neutral, and negative.

```
Logistic Regression Results:
Accuracy: 0.8802958165164222
              precision    recall  f1-score   support

    Negative       0.82      0.87      0.84     19329
     Neutral       0.55      0.19      0.28      6352
    Positive       0.91      0.96      0.94     57073

    accuracy                           0.88     82754
   macro avg       0.76      0.67      0.69     82754
weighted avg       0.86      0.88      0.86     82754
```

Figure 7: Result of LR

According to the experimental results, sentiment classification is greatly improved by deep learning algorithms, and robustness is further strengthened by hybridization with logistic regression, which makes the LRNN model more suitable for extensive e-commerce applications.

Table 3: Results of Logistic Regression Algorithm

Sentiment	Precision	Recall	F1-Score	Support
Negative	0.82	0.87	0.84	19,329
Neutral	0.55	0.19	0.28	6,352
Positive	0.91	0.96	0.94	57,073
Macro Avg	0.76	0.67	0.69	82,754
Weighted Avg	0.86	0.88	0.86	82,754

The model's overall performance is demonstrated by the Logistic Regression results, which show an accuracy of 88% on the dataset in table 3 and figure 8. With a high precision (0.91), recall (0.96), and an outstanding F1-score of 0.94, it excels at detecting positive sentiments, demonstrating dependable classification in this category. The F1-score of 0.84 indicates that the Negative class performs well as well. The Neural Network algorithm outperforms the Logistic Regression model in sentiment classification, producing results with an overall accuracy of 91.6%. With F1-scores of 0.96 and 0.88 for the Positive (0.97, 0.95) and Negative (0.82, 0.96) classes, respectively, the model achieves excellent precision and recall.

Sentiment	Precision	Recall	F1-Score	Support
Negative	0.82	0.96	0.88	19,329
Neutral	0.72	0.47	0.57	6,352
Positive	0.97	0.95	0.96	57,073
Macro Avg	0.84	0.79	0.80	82,754
Weighted Avg	0.92	0.92	0.91	82,754

Overall robust classification across the majority of sentiment categories is indicated by the weighted average F1-score of 0.91 and the macro average F1-score of 0.80.

```
Accuracy: 0.9464074244145299
              precision    recall  f1-score   support

    Negative       0.93      0.93      0.93     19329
     Neutral       0.80      0.69      0.74      6352
    Positive       0.97      0.98      0.97     57073

    accuracy                           0.95     82754
   macro avg       0.90      0.87      0.88     82754
weighted avg       0.94      0.95      0.95     82754
```

Figure 11: Results of LRNN

The weighted average of 0.95 and the macro average F1-score of 0.88 demonstrate balanced performance across classes, demonstrating that integrating neural networks and logistic regression produces a more reliable and accurate sentiment analysis model.

Table 6: Accuracy of Algorithms

Algorithm	Accuracy
Logistic Regression	**88.0**
Neural Network (NN)	**92.0**
Hybrid LRNN (LR + NN)	**95.0**

This was greatly enhanced by the Neural Network, which achieved a 92% accuracy rate by better classifying sentiments and identifying more intricate patterns in the data. The Hybrid LRNN model produced the best results, combining the advantages of neural networks and logistic regression to achieve an astounding 95% accuracy. In addition to increasing overall accuracy, this hybrid approach balanced performance across all sentiment classes, including the difficult neutral class.

7. CONCLUSION

The three algorithms' performance evolution is evident from the comparative analysis. Although its linear decision boundaries cause it to perform poorly with neutral reviews, logistic regression (88% accuracy) builds a solid foundation by effectively identifying highly polarized sentiments (both positive and negative). By capturing deeper linguistic structures and nonlinear relationships, neural networks (92% accuracy) greatly improve classification, increasing recall and precision for both positive and negative categories. Accurately predicting neutral sentiments is still difficult, though. The Hybrid LRNN model outperforms both standalone models and attains the highest accuracy of 95% when combining the advantages of both strategies. This hybrid approach achieves a balanced and superior performance across all sentiment classes by combining the interpretability and efficiency of logistic regression with the capacity of neural networks to learn intricate patterns. Significantly, the Hybrid LRNN outperforms the other models in the neutral sentiment class, indicating that it is more dependable for sentiment analysis tasks in the real world.

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Comprehensive Comparison on Blockchain Payments Vs Traditional Methods

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Abstract— In contrast to conventional centralized systems like bank transfers, cards, and wallets, blockchain is revolutionizing the financial industry by providing quicker, less expensive, and more secure payment options [19]. Peer-to-peer transactions, more transparency, and improved data integrity are made possible by blockchain, while traditional approaches have problems including expensive fees, longer processing, and fraud threats [20]. Regulation, limited acceptance, and scalability are still issues, though. The virtues, drawbacks, and potential future use of blockchain in digital payments are discussed in this paper's comparison of the two systems [21].

Keywords: digital transactions, decentralized systems, blockchain, cryptocurrency, traditional payments, and financial technology

1. INTRODUCTION

Traditional payment systems, such as cash, credit cards, bank transfers, and checks, have long been the foundation of global financial transactions [22]. However, these systems have limitations such as long processing times, high transaction fees, and uneven access to financial services. This has led to a growing interest in alternative payment models like blockchain-based systems. Cryptocurrencies like Bitcoin and Ethereum offer faster, lower-cost, and borderless alternatives, making them relevant for global e-commerce and financial inclusion. As blockchain-based technologies are increasingly adopted, comparing them with traditional payment infrastructures is essential for understanding digital finance and financial regulation. This chapter provides a comprehensive comparison of these systems [23].

2. Related Works

The decentralized nature of blockchain networks also means that transactions can be processed quickly and at a lower cost compared to traditional payment methods. This makes blockchain payment systems an attractive option
for international payments, where speed and cost-efficiency are paramount [1]. Blockchain transactions eliminate the need for intermediaries in payment systems, making them more efficient and authentic compared to traditional methods [2]. While traditional fraud prevention methods are effective to some extent, they often rely on centralised systems that can be vulnerable to breaches, manipulation, and inefficiencies. Blockchain offers a

promising alternative by providing a secure, transparent, and immutable system for tracking financial

transactions [3]. Blockchain payments can significantly enhance financial inclusion for the underbanked by providing global, lower-cost, and decentralized access to financial services without traditional banking infrastructure, while traditional methods often fail due to geographic barriers, high fees, and exclusion of unbanked populations. Blockchain's advantages include peer-to-peer transactions, efficient cross-border remittances, and the ability to facilitate micro-lending and other services via smart contracts, though challenges like regulatory complexity and technology adoption remain for both methods. Financial inclusion is the accessibility and availability of affordable and practical financial services—like payments, savings, credit, and insurance—for all individuals and organizations [4]. Blockchain payments using stablecoins can achieve relative stability, though concerns exist about their asset backing and regulatory compliance, while cryptocurrencies themselves exhibit high volatility. Traditional payment methods offer more inherent stability due to their regulatory frameworks and direct reliance on established central banks, but they are slower, more complex, and less transparent than digital alternatives.

3. Foundations of Payment Systems Comparison

3.1 Evolution of payments: Payments have transformed dramatically since the days of charge coins and paper credit. Each step in this evolution has brought us closer to a faster, more secure, and more versatile financial experience. Today, we're witnessing the next major leap: virtual crypto cards, which combine the simplicity of traditional cards with the innovation of blockchain [3][4].

3.2 Brief historical context of both systems: Payments have evolved dramatically from primitive tally sticks and ancient coins to today's sophisticated digital systems. Around 600 BCE, Lydia introduced the first standardized coins, later refined by the Roman Empire. During the medieval period, banking institutions emerged, and Persian merchants introduced checks, revolutionizing money transfers. The 19th century saw the rise of charge coins and paper credit, introducing the concept of deferred payments. In 1950, the Diners Club card launched the era of multipurpose charge cards, followed by the development of plastic credit cards in the 1960s and debit cards in the 1980s. The 1970s also brought key infrastructure innovations like ACH and wire transfers. By the 2000s, contactless "tap-and-go" payments became widespread, and by the 2020s, over half of all UK payments were card-based. Today, the shift toward blockchain and crypto-enabled virtual cards represents the next leap in payment innovation [3].

3.3 Blockchain's payment: In contrast, blockchain technology offers a decentralized and transparent payment system. Transactions are recorded on a public ledger, known as a blockchain, which is maintained by a network of computers worldwide. This distributed architecture enables direct person-to-person transactions, reducing the need for intermediaries and associated costs. Blockchain's decentralized structure increases security, as each transaction is verified by multiple nodes, making it more difficult for hackers to manipulate the system [5].

3.4 Traditional Payment System: The traditional payment system is a complex network of financial institutions, intermediaries, and technology providers. It relies on a hierarchical structure, where banks and financial institutions act as gatekeepers, verifying transactions and managing risk. This system is built around institutions, resulting in administrative costs, lengthy processing times, and vulnerability to fraud [6].

4. Head to Head Comparison

4.1 Transaction Speed

Traditional Payment Methods: Bank transfers / SWIFT: Typically take 2–7 business days, especially for cross-border payments, due to multiple intermediaries and time-zone delays. Credit/debit cards: Seem instant to users, but settlements can take 24 hours to several days behind the scenes. Batch processing & business hours limit speed, causing delays, especially outside banks' operational windows [7].

Blockchain Payment Systems: Modern blockchains (e.g., TRON, BNB Chain, Solana): Settlement in 1–5 seconds. Ethereum transactions are confirmed within a timeframe of 15 to 60 seconds, while Bitcoin requires approximately 10 minutes for the initial confirmation, with full finality potentially taking up to an hour. The Lightning Network, a layer-2 solution for Bitcoin, facilitates payments in mere milliseconds to under a minute, achieving near-instantaneous throughput. Cross-border blockchain solutions leverage smart contracts and decentralized ledgers to significantly reduce settlement times from 2 to 5 days down to mere seconds, resulting in fees that can be up to 80% lower. High transactions per second (TPS) networks such as Solana (over 1,000 TPS), Sui (850 TPS), BNB Chain (378 TPS), and Polygon (190 TPS) demonstrate superior throughput compared to traditional systems [14].

4.2 Availability

Traditional Payment Methods: Bound by bank hours, often not operational on weekends or holidays. Batch processing delays posted transactions and settlements [8]

Blockchain Payment Systems: Fully 24/7/365 operation, with no interruptions for holidays or geographic constraints. Always accessible, requiring only internet connectivity—no need for traditional bank access. Smart contracts and on-chain processes enable real-time compliance, tracking, and verifiable transactions, reducing reliance on manual verification [8]

4.3 Cost Comparison

Traditional Payment Methods: Credit cards and transfers often cost between 1% to 5% per transaction. Wire transfers can charge $15–$50+, while sending $200 internationally may incur fees of 3%–5%. Banks may charge $20–$50 per transaction, and global average remittance cost is around 6.5% for sending $200. Hidden FX and intermediary costs: Multiple banks, exchange rate markups, and regulatory overhead can add an extra 1%–5% to transactions. Infrastructure & manual processes: Fee buildup from intermediaries and manual verification inflate costs further. Cash related costs: In retail, cash payments cost retailers around €0.24, while card payments cost €1 per transaction [9].

Blockchain Payment Systems:

Minimal fees: Ripple Net: < $0.01 per transaction; Stellar: $0.000001. Stablecoins on chains like Solana/Binance Smart Chain: often below 0.1%–0.3%, versus 3%–5% traditional

Massive cost savings: Removing intermediaries can reduce fees by up to 80%, saving businesses substantial amounts (e.g., $20k–$50k/year on $1M volume). Ripple Net helped MoneyGram cut operational costs by 50%. Global systems could remove up to $27 trillion tied in nostro accounts, reduce FX spreads from 1% to 0.1%, and reduce manual error rates from 5% to 0.01%. BIS targets <1-hour settlement—blockchain.

Global scale savings: Traditional finance charges cost around $8 trillion/year—blockchain could potentially slash that by over 50%, saving $4 trillion annually. Efficiency gains: Automated smart contracts reduce manual burdens, errors, and fraud risk, further lowering costs [10].

5. Challenges and Limitations

Traditional Payment Methods: Challenges and Limitations:

1) High Transaction Fees: Conventional payment systems typically involve several middlemen, each of which charges a fee. This often leads to increased costs for both businesses and individuals [16].

2) Security Concerns: Although security measures have improved, traditional payment systems are still at risk of fraud and hacking. Security breaches can cause substantial financial loss and harm to a company's reputation [18].

3) Limited Accessibility: Many people do not have access to standard banking services. In various regions, particularly in less developed areas, a large number of individuals are not banked [15].

4) Slow Transaction Times: International money transfers can take several days to complete because they require input from multiple banks and are affected by different time zones. These delays can negatively affect businesses that need fast transactions [16].

5) Regulatory Compliance Costs: Financial organizations are required to follow strict rules such as Anti-Money Laundering (AML) and Know Your Customer (KYC) guidelines. Following these rules can be expensive and take a lot of time [17].

Blockchain Payment Systems: Challenges and Limitations:

1) Scalability Issues: Blockchain networks, especially those using the Proof of Work (PoW) consensus method like Bitcoin, struggle with scalability. For example, Bitcoin can process about 3 to 7 transactions per second, which is much lower than traditional systems such as Visa, which can handle more than 24,000 transactions per second during busy periods [16].

2) High Energy Consumption: Blockchains that rely on PoW require a lot of energy. Bitcoin mining, in particular, uses a significant amount of electricity because it needs powerful computers to perform the necessary calculations for validating transactions [15].

3) Regulatory Uncertainty: The rules governing blockchain technology are still being developed. As a result, different countries have varying regulations, leading to a complicated legal environment that businesses must work through [18].

4) Integration Challenges: Connecting blockchain with current financial systems can be difficult. Older systems may not work well with blockchain technology, often requiring considerable resources to build new systems and train staff [18].

5) User Adoption Barriers: The complexity of blockchain technology can prevent people from using it. Many individuals and organizations are not familiar with how blockchain works and may be reluctant to trust a system they don't understand [16].

6. Conclusion

Blockchain payment systems and traditional payment methods each have their own strengths and weaknesses. Traditional payment systems rely on well-established infrastructure, can handle a large number of transactions quickly, have clear rules set by governments and financial authorities, and are widely used by people around the world. This makes them reliable for making fast and secure payments. However, they often require several middlemen, which can lead to higher costs, slower international transfers, and limited access for people who don't have bank accounts. On the other hand, blockchain payment systems provide greater transparency and security because they don't depend on a central authority. They also have the potential to help more people access financial services, especially in areas where traditional banking is not available. Still, they encounter challenges like unclear rules from regulators, difficulties in handling a large volume of transactions, higher energy use in certain types of systems, and trouble connecting with older financial systems. As new laws and regulations, such as the EU's MiCA and efforts in the U.S., are being developed, and as technologies like Proof of Stake and layer-2 solutions continue to improve, the issues with blockchain are becoming easier to manage. In the end, combining the best parts of traditional and blockchain payment systems could lead to a more effective, fair, and strong global payment system.

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Blockchain Potential in the Airline Industry

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Abstract— Blockchain technology has emerged as a transformative innovation with the potential to revolutionize various industries, including the airline sector. This chapter explores how blockchain can be applied within the airline industry to enhance transparency, security, and operational efficiency. Through the examination of an existing blockchain-based project tailored for airline operations, we discuss the evolution of blockchain technology, its core components, challenges faced during implementation, and the promising future it holds. This chapter is designed for beginners to provide a comprehensive understanding of blockchain's role in the airline industry.

1.Introduction

The integration of blockchain into aviation is not merely an academic concept; it is the focus of significant investment and experimentation by both industry consortia and major corporations. this section outlines key projects and research that have paved the way

A. Industry Pilots and Commercial Projects

1. Winding Tree: Perhaps the most prominent blockchain initiative in travel, Winding Tree is an open-source, public blockchain-based decentralized travel distribution platform. Its primary aim is to disintermediate the market, challenging the dominance of GDSs by enabling airlines, hotels, and other service providers to connect directly with developers and end-users. This peer-to-peer marketplace drastically reduces the high commission fees typically associated with traditional booking channels, potentially lowering costs for providers and consumers alike. The platform uses its own LIF token to facilitate transactions and settlements on the ecosystem.

2. TUI Group: The German tourism conglomerate TUI has been a pioneer in exploring blockchain's utility. One of their key initiatives focused on baggage tracking. By recording each step of a bag's journey—from check-in to loading onto the aircraft, transfer, and final delivery—on an immutable ledger, TUI aimed to create a single, transparent source of truth accessible to both the airline and the passenger. This transparency significantly reduces the incidence of lost luggage and streamlines the reconciliation process when mishandling occurs, saving the industry billions annually.

3. SITA and IBM Collaboration: SITA, a leading IT provider for the air transport industry, partnered with IBM to explore blockchain for identity management. Their "Digital Traveler" concept uses a blockchain to create a secure, decentralized digital identity for passengers. This allows travelers to control their own data and consent to its use, simplifying and speeding up processes like check-in, security clearance, and boarding by providing verified information to authorities without repeatedly presenting physical documents. This became particularly relevant for health credential verification during the COVID-19 pandemic.

4. AirFrance-KLM: The airline group has investigated blockchain for tracking the maintenance history of aircraft parts. Recording the entire lifecycle of a part—from manufacture to

installation to repairs—on a blockchain creates an unforgeable history, enhancing safety, simplifying regulatory compliance, and reducing the risk of counterfeit parts entering the supply chain.

B. Academic Research Focus

Academic literature has complemented these industry efforts, providing theoretical frameworks and investigating specific applications:

- Ticketing and Fraud Prevention: Research explores using smart contracts to issue tickets as unique, non-fungible tokens (NFTs) on a blockchain. This would eliminate ticket fraud, prevent double-selling, and enable secure, transparent resale markets.
- Loyalty Programs: Studies propose decentralizing loyalty points on a blockchain, making them more like a universal currency. This would allow passengers to easily combine points from different airlines, trade them, or use them with partner businesses, increasing their utility and value.
- Supply Chain Transparency: Papers examine the use of blockchain for creating an auditable and transparent record of the aviation supply chain, from fuel procurement to catering, ensuring ethical sourcing and compliance.

These collective efforts underscore the high degree of interest in blockchain's potential while simultaneously highlighting recurring implementation challenges such as achieving necessary scalability and throughput, ensuring interoperability between different blockchain networks and legacy systems, and navigating an uncertain global regulatory landscape.

III. EVOLUTION OF BLOCKCHAIN IN THE AIRLINE INDUSTRY

A. The Evolutionary Timeline of Blockchain in Aviation

The adoption of blockchain in aviation has followed a distinct evolutionary path, maturing from conceptual financial applications to broader operational uses.

3.1 The Nascent Stage (2016-2017): Initial interest was cautious and experimental. The focus was primarily on non-mission-critical applications that could demonstrate clear value. Pilot projects centered on baggage tracking (e.g., TUI) and reimagining loyalty programs, as these areas offered clear benefits from transparency and interoperability without immediately threatening core revenue systems.

3.2 Expansion and Platform Development (2018-2019): As understanding of the technology grew, so did ambition. This period saw the emergence of foundational platforms like Winding Tree that aimed to tackle more complex industry structures, specifically the distribution and booking landscape. The concept of "direct-to-consumer" models powered by blockchain gained traction, promising to reshape the commercial relationship between airlines and passengers.

3.3 Diversification and Pandemic Acceleration (2020 onwards): The scope of blockchain applications widened significantly to include digital identity verification, maintenance record logging, and supply chain management. The COVID-19 pandemic acted as a powerful accelerator. The urgent need for contactless processes and verifiable health credentials (vaccination status, test results) pushed digital identity solutions from a long-term goal to an immediate priority. This period solidified blockchain's role as a tool for resilience and safety.

B. Core Technological Components and Their Airline Applications

Understanding how blockchain is applied requires a grasp of its core technological components:

4.1 Decentralization: Unlike a traditional database managed by a single entity (e.g., an airline's central server), a blockchain ledger is distributed across a network of computers (nodes). In an airline context, this means baggage data or passenger identity isn't stored in one vulnerable location but is replicated and synced across nodes belonging to airlines, airports, and handlers. This prevents a single point of failure and eliminates the need for a trusted intermediary to manage the data.

4.2 Immutability: Once a transaction (e.g., "Bag BCN12345 was loaded onto flight AA101") is recorded on the blockchain and confirmed by the network, it is cryptographically sealed and virtually impossible to alter or delete. This creates a permanent, tamper-proof audit trail for maintenance history, baggage handling, and ticket issuance, drastically reducing fraud and disputes.

4.3 Consensus Mechanisms: These are the protocols that ensure all nodes in the network agree on the validity of transactions before they are added to the ledger. Proof of Work (PoW) (used by Bitcoin) is often considered too energy-intensive for airline-scale transactions. Proof of Stake (PoS) (used by Ethereum) and Practical Byzantine Fault Tolerance (PBFT) (used by Hyperledger Fabric) are more efficient consensus models better suited for permissioned enterprise networks where participants are known and trusted.

IV. TECHNOLOGICAL ADVANCEMENTS

Blockchain operates as a distributed ledger maintained by multiple participants (nodes) in a decentralized network. Key technological concepts include:

4.1 Decentralization

Data is stored across many nodes, preventing a single point of failure.

4.2 Immutability

Data is stored across many nodes, preventing a single point of failure.

4.3 Consensus Mechanisms

Methods like Proof of Work or Proof of Stake verify and agree on the validity of transactions.

4.4 Smart Contracts

Self-executing contracts with predefined rules, automating processes like ticket issuance or baggage claims.

4.5 Integration with Airline IT Systems

In the airline blockchain project, these features enable secure and transparent sharing of passenger data, ticketing information, and baggage status across airlines, airports, and regulators.

Ethereum or Hyperledger Fabric as blockchain platforms.

Distributed databases for data storage.

APIs to connect blockchain with existing airline IT systems.

V. CHALLENGES

Despite its potential, blockchain adoption in the airline industry faces several challenges:

5.1 Scalability

The high transaction volume of airlines requires blockchain systems capable of handling large-scale data efficiently.

5.2 Interoperability

Integrating blockchain with existing legacy systems and across various stakeholders remains complex.

5.3 Regulatory and Legal Issues

Aviation regulations vary globally, and blockchain’s decentralized nature poses challenges for compliance.

5.4 Data Privacy

Managing sensitive passenger data while complying with laws such as GDPR.

5.5 Cost and Complexity

Initial setup, infrastructure costs, and the need for technical expertise can be prohibitive.

VI. FUTURE POTENTIAL

As technological and regulatory hurdles are gradually addressed, blockchain is poised to enable transformative applications that will redefine the air travel experience and industry operations.

6.1 Self-Sovereign Digital Identity for a Seamless Journey: The future of passenger processing lies in decentralized digital identity. Passengers would store verifiable credentials (passport, visa, loyalty status, even biometrics) in a secure digital wallet on their smartphone.

Using blockchain, they can provide zero-knowledge proofs—verifying their eligibility (e.g., over 18, holds a valid visa) without revealing the underlying document data. This would create a truly seamless, contactless journey from curb to cabin, significantly enhancing both security and convenience while putting passengers in control of their personal data.

6.2 Comprehensive Asset Tracking and Maintenance: The application of blockchain will extend beyond baggage to encompass the entire aircraft ecosystem. Every critical part, from an

engine turbine to a landing gear component, could have a digital twin on a blockchain. Its entire history—manufacturing origin, installation dates, maintenance cycles, repairs, and compliance checks—would be immutably recorded. This enhances safety, simplifies regulatory audits, prevents the use of counterfeit parts, and optimizes maintenance scheduling, reducing aircraft downtime and operational costs.

6.3 Dynamic and Interoperable Loyalty Programs: Blockchain can dismantle the walled gardens of current airline loyalty programs. Points or miles could be tokenized on a blockchain, transforming them into a liquid, tradeable asset. Passengers could easily combine points from different airlines, trade them on a secure marketplace, or use them to pay for services from a wide network of partners, greatly increasing their utility and customer loyalty.

6.4 Sustainability and Carbon Emission Tracking: With increasing pressure to meet environmental targets, blockchain can provide transparency and trust in sustainability efforts. It can be used to create an unalterable record of fuel consumption and carbon emissions for each flight. This data can be used to accurately offset carbon, track the use of Sustainable Aviation Fuels (SAFs) through the supply chain, and provide passengers with verifiable proof of their carbon offset contributions.

The future of blockchain in aviation is not necessarily a single, industry-wide chain, but more likely a "network of networks"—an interoperable ecosystem of specialized blockchains for identity, maintenance, cargo, and bookings, all working together to create a more efficient, secure, and passenger-centric industry.

VII. CONCLUSION

Blockchain technology represents more than just a incremental improvement for the airline industry; it offers a foundational shift towards a more decentralized, transparent, and efficient operational model. Its core attributes of immutability, decentralization, and security directly address some of the sector's most persistent challenges: baggage mishandling, identity verification inefficiencies, opaque supply chains, and fragmented loyalty programs. The industry initiatives undertaken by players like SITA, TUI, and Winding Tree provide a concrete, albeit early, blueprint for how this technology can be implemented to create tangible value.

However, this transformation will not be instantaneous. The path to widespread adoption is fraught with significant challenges, including technical limitations in scalability, the immense complexity of integrating with legacy IT infrastructure, unresolved regulatory questions surrounding data privacy and jurisdiction, and the need for unprecedented collaboration between traditionally competitive entities. Overcoming these hurdles requires a concerted effort from technology developers, airline operators, regulators, and standards bodies.

Despite these challenges, the potential rewards are too significant to ignore. As the technology matures and solutions to these problems emerge, blockchain is poised to become an integral, albeit often invisible, part of the aviation infrastructure. It promises to empower passengers with control over their data and identity, provide airlines with unprecedented operational integrity and efficiency, and contribute to a safer and more sustainable industry. The journey of integrating blockchain into aviation is taking off, and its destination is a fundamentally transformed landscape for air travel.

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Artificial Intelligence-Enabled IoT: Driving Smarter Decisions in Real Time

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Abstract— The integration of Artificial Intelligence (AI) with the Internet of Things (IoT) is fundamentally changing how digital systems operate, enabling intelligent, real-time decision-making. IoT networks continuously collect large volumes of data from interconnected devices, while AI provides the analytical capability to turn this data into actionable insights. This paper examines how AI can be embedded within IoT frameworks to support smarter decision-making in real time. It discusses system architectures, key AI algorithms, real-world applications across various industries, and challenges such as latency, scalability, and security. Finally, the paper explores future trends in AI-enabled IoT systems, showing their potential to revolutionize sectors like healthcare, smart cities, and industrial automation

1. Introduction

The Internet of Things (IoT) has become a game-changing technology, connecting billions of devices across the globe and enabling continuous data exchange. By using sensors and actuators, IoT devices capture real-time environmental, physiological, or industrial data, providing valuable inputs for monitoring, control, and predictive analysis [1].

At the same time, Artificial Intelligence (AI) has advanced to offer intelligent data analysis, pattern recognition, and predictive modeling. When AI is integrated with IoT, it enables systems to make smarter decisions by analyzing real-time data. For instance, smart traffic management systems use AI algorithms to process data collected from IoT devices, optimizing traffic flows and reducing congestion [2].

This combination, commonly referred to as AI-enabled IoT, allows for dynamic, intelligent decision-making. This paper explores the current state of AI-integrated IoT systems, outlines their architectural frameworks, addresses key challenges, and highlights their applications in various sectors. The aim is to show how AI can enhance IoT networks to deliver efficient, reliable, and actionable insights.

2. Literature Review

AI and IoT have each had a transformative effect on technology, but their integration offers new opportunities for real-time intelligence. Current research focuses on several areas:

- **Machine Learning in IoT:** Machine Learning (ML) algorithms are widely used to process IoT data streams, detect patterns, predict events, and optimize system operations [3]. Supervised learning helps detect anomalies in sensor networks, while unsupervised learning, like clustering, can reveal hidden patterns in large datasets.

- **Deep Learning for Complex Data:** Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs) are capable of processing complex IoT data such as images from surveillance cameras or sensor data from smart factories [3][8].
- **Reinforcement Learning (RL):** RL enables IoT systems to learn optimal actions from feedback, adapting to changes in the environment. Smart grid systems, for example, use RL to efficiently manage energy distribution [6].
- **Edge AI Approaches:** Running AI algorithms on edge devices, near the data source, reduces latency and minimizes network bandwidth usage. This is especially crucial in applications like autonomous vehicles, industrial robotics, and real-time healthcare monitoring [4].
- **Hybrid Edge-Cloud AI:** Some architectures combine edge-based AI inference with complex cloud analytics, balancing efficiency, scalability, and real-time performance [9].

AI-enabled IoT is increasingly applied in healthcare, smart cities, transportation, agriculture, and industrial automation [7][8]. However, challenges such as diverse data formats, network reliability, and limited device resources continue to drive research in this field.

3. AI-Enabled IoT Architecture

AI-enabled IoT systems are generally built using a **multi-layer architecture**, which ensures efficient data collection, processing, and decision-making:

1. **Perception Layer:** This includes sensors, actuators, RFID tags, and other IoT devices that gather real-time environmental, health, or industrial data. Examples include smart home temperature sensors, wearable health monitors, and industrial vibration detectors.
2. **Network Layer:** Connects IoT devices to edge or cloud servers using protocols like MQTT, CoAP, or 5G/6G networks, ensuring secure and reliable data transmission.
3. **Edge/Processing Layer:** AI models run locally on edge devices, gateways, or microcontrollers to enable immediate decision-making. This layer reduces reliance on cloud computing, lowering latency and bandwidth needs.
4. **Cloud/Processing Layer:** Handles storage of large datasets, training of AI models, and execution of complex analytics. This layer supports deep learning, predictive modeling, and AI updates.
5. **Application Layer:** Provides user interfaces for data visualization, alerts, and decision-making support. This can include dashboards, mobile apps, or automated control systems.

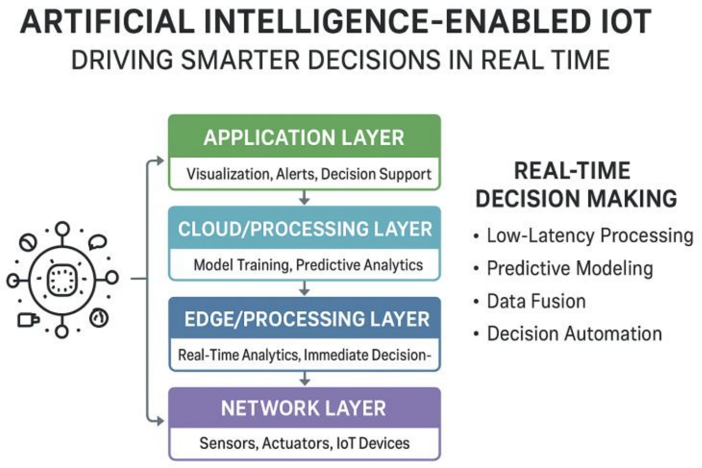

Fig 1. AI-Enabled IoT Architecture for Real-Time Decision-Making

4. Real-Time Decision Making in IoT

One of the main benefits of AI-enabled IoT is the ability to make **real-time decisions**. This involves fast data collection, analysis, and automated response:

1. **Low-Latency Processing:** Essential for applications like autonomous cars, industrial robots, and remote surgeries. Edge AI can process sensor inputs within milliseconds to take immediate action.
2. **Predictive Modeling:** Machine learning can forecast potential issues or future states. For instance, predicting when factory equipment might fail allows for proactive maintenance.
3. **Data Fusion:** Combining information from multiple sources, such as sensors and cameras, enhances decision accuracy. Smart traffic systems, for example, integrate GPS, camera, and environmental data to adjust signals.
4. **Decision Automation:** AI-enabled IoT can execute actions autonomously based on pre-set rules or learned behavior. For example, smart irrigation systems automatically adjust water delivery using soil moisture, weather forecasts, and crop type.

Applications:

- **Healthcare:** Continuous monitoring with AI alerts for critical conditions [11].
- **Industrial IoT:** Predictive maintenance to prevent downtime [13].
- **Smart Cities:** Optimizing traffic, improving public safety, and managing energy [12].

5. Challenges in AI-Enabled IoT

Despite its advantages, AI-IoT faces several challenges:

1. **Data Privacy and Security:** IoT devices produce sensitive data. Techniques like encryption, blockchain, and secure multi-party computation are essential [14].
2. **Scalability:** With millions of devices generating data, AI algorithms must scale efficiently across devices and cloud infrastructure [15].
3. **Latency and Bandwidth:** Real-time applications require ultra-low latency. Edge AI helps but has computational limits.
4. **Energy Efficiency:** Many IoT devices have limited power; AI must be optimized for low energy use without sacrificing performance [16].
5. **Interoperability:** Devices from different manufacturers use varied standards; harmonization is critical.
6. **Model Accuracy and Adaptability:** AI models need constant updates to adapt to changing data patterns.

Addressing these challenges is vital for creating reliable and secure AI-enabled IoT systems.

6. Applications of AI-Enabled IoT

1. **Healthcare:** Remote patient monitoring, smart diagnostics, and wearable devices measuring heart rate, oxygen, and glucose levels, with AI predicting anomalies [11].
2. **Smart Cities:** Managing traffic, waste, and safety using AI to analyze IoT data for efficient operations [12].
3. **Industry 4.0:** Smart factories using AI to predict equipment failure, optimize production, and improve supply chains [13].
4. **Agriculture:** Precision farming using soil, weather, and crop data to optimize irrigation, fertilization, and harvesting [17].
5. **Environmental Monitoring:** Pollution control, disaster alerts, and climate modeling by analyzing IoT sensor data [18].

These examples show how AI transforms raw IoT data into **actionable insights**, improving efficiency, safety, and sustainability.

7. Future Directions

1. **Edge-Cloud Hybrid AI:** Using edge AI for real-time tasks and cloud AI for large-scale analytics ensures efficiency and accuracy.
2. **Explainable AI (XAI):** Transparent AI models build trust, particularly in healthcare and autonomous systems.
3. **5G/6G Connectivity:** High-speed networks enable faster, real-time communication between distributed IoT devices.
4. **Autonomous IoT Systems:** Fully self-operating systems can learn, adapt, and make decisions without human intervention, e.g., autonomous vehicles and smart factories.
5. **Sustainability and Green AI:** Research focuses on energy-efficient AI algorithms and hardware to reduce IoT power consumption.
6. **Integration with Emerging AI Models:** Combining large language models, multimodal AI, and reinforcement learning can enhance analysis by simultaneously processing text, images, and sensor data.

8. Conclusion

Integrating AI with IoT opens up new possibilities for **real-time decisions, predictive analytics, and autonomous operations**. These systems turn large, varied datasets into actionable intelligence, benefiting healthcare, smart cities, industry, agriculture, and environmental monitoring. Challenges like security, scalability, latency, and interoperability remain, but advancements in edge AI, 5G networks, hybrid architectures, and explainable AI offer robust and sustainable solutions. The continued convergence of AI and IoT promises to redefine smart system capabilities, driving efficiency, innovation, and safety across industries.

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Blockchain in Finance and Banking

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Abstract— Blockchain technology represents a transformative, decentralized, and replicable distributed ledger system that has the potential to revolutionize traditional banking and finance sectors. This research undertakes a bibliometric review and content analysis of existing academic literature focused on the antecedents, applications, and implications of adopting blockchain technologies within these interconnected industries. Analyzing a final dataset of 154 papers published from 2009 to 2021, we explore key elements such as trending topics, influential authors, and prominent journals. The study includes a bibliographic analysis encompassing co-authorship, cartographic, co-citation, and coupling analyses. Ultimately, we delineate the primary streams of literature and propose a future research agenda, offering a foundational reference for both scholars and practitioners in the field.

1.Introduction

Blockchain is a distributed ledger technology that fundamentally transforms how data is recorded, shared, and verified across various industries. Each participant in the blockchain network holds a copy of the complete ledger, guaranteeing that all transactions are transparent and can be verified by anyone with access. This replicable feature not only builds trust among users but also provides a high level of resilience against data tampering or loss, as information is stored across multiple nodes rather than in a single, vulnerable location. The impact of blockchain technology goes well beyond its original use in cryptocurrencies like Bitcoin. It serves as a strong foundation for numerous incremental and transformative innovations across diverse sectors. For example, in supply chain management, blockchain enables real-time tracking of goods, thereby improving transparency and accountability from production to delivery. As organizations increasingly recognize the potential of blockchain, we can expect a growing number of applications that utilize its unique features. The technology not only aims to improve efficiency and security but also seeks to democratize access to information and services, paving the way for a more equitable digital economy. In this rapidly changing environment, the continued development and adoption of blockchain could result in significant transformations in how we engage with technology and one another, marking a crucial moment in the digital era.

2.Related Works

The application of blockchain technology in banking and finance has gained momentum over the last decade, with scholars exploring its potential to transform traditional systems. Early works such as Nakamoto [21] and Szabo [22] introduced the fundamental ideas of decentralized ledgers and smart contracts, which later became the foundation for financial applications. These studies established the concept that trust in transactions could be achieved through cryptographic mechanisms rather than intermediaries, offering a radically new approach to data management and value transfer.

Several researchers have examined blockchain's role in enhancing efficiency and security in financial operations. Ali et al. [23] emphasized that blockchain mitigates issues of fraud and identity theft by ensuring data integrity and transparency. Tapscott and Tapscott [24] further argued that the technology could streamline cross-border transactions, reducing costs and settlement times, while simultaneously improving financial inclusion for unbanked populations. Their findings suggest that blockchain not only modernizes existing processes but also creates new opportunities for global accessibility.

A number of studies have also focused on blockchain's impact on financial market infrastructure. Peters and Panayi [25] discussed how distributed ledgers can transform post-trade processes, reducing dependency on intermediaries for clearing and settlement. Yermack [26] explored blockchain in corporate governance, noting that immutable transaction records can strengthen shareholder transparency and auditing practices. These contributions highlight blockchain's broader implications beyond payment systems, extending into organizational accountability and compliance.

In the area of trade finance, blockchain-based smart contracts have received considerable attention. Chang et al. [27] demonstrated that smart contracts simplify processes such as letters of credit and shipping documentation, thereby minimizing paperwork and improving trust between international partners. Likewise, Zhang et al. [28] examined blockchain for Know Your Customer (KYC) and Anti-Money Laundering (AML) compliance, concluding that shared identity management systems reduce redundancy and costs while improving security. Such findings illustrate how blockchain can reduce regulatory burdens while ensuring adherence to compliance requirements.

Recent bibliometric studies provide a comprehensive overview of blockchain-related literature in financial services. Chen and Bellavitis [29] analyzed publication trends and identified three dominant research streams: digital currencies, decentralized finance (DeFi), and regulatory issues. Their work shows how blockchain scholarship has expanded from cryptocurrency origins toward complex financial ecosystems. Similar reviews [30] also highlight scalability, interoperability, and regulatory uncertainties as the most persistent challenges that slow down industry-wide adoption.

Another strand of research investigates blockchain-enabled digital currencies and central bank initiatives. Narula and Catalini [31] discussed the potential of Central Bank Digital Currencies (CBDCs), stressing their ability to enhance real-time settlements and reduce dependence on cash. Studies on stablecoins [32] similarly emphasize programmability and resilience, while raising concerns about monetary policy implications and systemic risks. These works underline the importance of designing secure, scalable, and regulatory-compliant blockchain frameworks for national and global finance.

3. Financial service in blockchain

The financial services sector is a colossal entity, engaging in transactions that amount to trillions of dollars each day. This staggering volume of activity underscores the critical need for a relentless focus on cost-efficiency, transparency, and security in all financial operations. Historically, the reliance on intermediaries—such as money transfer services, stock exchanges, and payment networks—has made these entities attractive targets for cybercriminals. The vulnerabilities inherent in these traditional systems have prompted a growing concern over the safety of financial transactions, leading to an urgent demand for innovative solutions that can mitigate risks and enhance trust among users.

As highlighted by Ali, Ally, and Dwivedi (2020), blockchain's unique attributes make it particularly well-suited for addressing the pressing issues of security and transparency. Initially popularized by cryptocurrencies like Bitcoin, blockchain has evolved far beyond its original application, now influencing a wide array of financial processes

Moreover, blockchain technology employs sophisticated encryption methods and algorithms that ensure the integrity and confidentiality of transaction data. This not only enhances security but also fosters greater transparency, as all participants in the network can access a shared ledger that records every transaction in real-time. As a result, stakeholders can verify the

authenticity of transactions without relying on intermediaries, which streamlines processes and reduces costs. The adoption of blockchain across various facets of the financial sector signifies a paradigm shift, paving the way for more efficient, secure, and transparent financial operations that can better withstand the threats posed by cybercriminals.

3.1 Basics of blockchain

Blockchain technology operates as a decentralized and immutable ledger, fundamentally transforming how transactions are recorded and assets are monitored within business networks. Unlike traditional databases that rely on a central authority, Blockchain distributes data across a network of computers, or nodes, which collectively maintain the integrity of the information. This decentralized nature not only enhances security but also mitigates the risks associated with data manipulation or unauthorized access. Each transaction recorded on the Blockchain is time-stamped and linked to the previous one, creating a chronological chain that is nearly impossible to alter without consensus from the network participants.

Moreover, Blockchain's capabilities extend beyond mere transaction recording; it facilitates the tracking and trading of any valuable item within its ecosystem. This includes not only cryptocurrencies but also physical assets like real estate, art, and even intellectual property. By providing a transparent and verifiable record of ownership and transaction history, Blockchain fosters trust among parties involved in a trade, reducing the need for intermediaries and streamlining processes. As a result, businesses can operate more efficiently, with reduced costs and increased speed in transactions.

In essence, Blockchain functions as a distributed database that is accessible to all authorized participants in the network. This accessibility ensures that all stakeholders have real-time visibility into the data, promoting accountability and transparency in data management. The technology's inherent security features, such as cryptographic hashing and consensus mechanisms, further bolster its reliability, making it an attractive solution for industries ranging from finance to supply chain management. As organizations increasingly recognize the potential of Blockchain, its adoption is likely to reshape the landscape of business operations, driving innovation and enhancing collaboration across various sectors.

3.3 key features in blockchain

3.3.1 Decentralization

Decentralization in blockchain technology involves a network of multiple computers, or nodes, that collectively manage the ledger, eliminating the risk of a single point of failure. Each node holds a complete copy of the ledger, which is a public record of all transactions, ensuring consistent updates across the network. Consensus mechanisms are used to validate and agree on new transactions before they are added to the blockchain, making each block immutable and cryptographically linked to previous blocks. This structure enhances security, as each block contains a unique cryptographic hash, making the system resistant to tampering and fraud. While user privacy is maintained, blockchain also promotes transparency by allowing all participants to view recorded transactions, providing a verifiable audit trail.

3.3.2 Transparency

Blockchain transparency is a fundamental attribute of the technology that guarantees the visibility of all transactions and data within the network to anyone who has access. This feature allows users to verify the legitimacy of transactions and the integrity of the data stored on the blockchain. Public blockchains, such as Bitcoin and Ethereum, have embraced this principle in their design, promoting an open and verifiable environment for all participants.

3.3.3 Immutability

Immutability refers to the characteristic of a blockchain ledger that ensures it remains unchanged and permanently recorded. Each block within the chain contains critical information, such as transaction details, which are secured through cryptographic techniques, specifically hash values. These hash values are unique alphanumeric strings generated for each block, linking it to both its own data and that of the preceding block. This interconnectedness guarantees that once data is recorded, it cannot be altered or tampered with, thereby preserving the integrity of the entire system. Consequently, blockchain technology effectively prevents any unauthorized modifications to previously stored information, reinforcing its reliability and security.

4.Blockchain aarchitecture and mechanisms

4.1 Distributed legder:

A distributed ledger, commonly known as a shared ledger or distributed ledger technology (DLT), represents a revolutionary approach to data management that facilitates the replication, sharing, and synchronization of digital information across a wide array of locations, institutions, or even countries. This innovative system is designed to ensure that all participants in the network have access to the same data in real-time, fostering transparency and trust among users. The core principle underpinning this technology is the Argumentum ad populum, which posits that the reliability and integrity of the system are derived from the consensus achieved by a majority of nodes within the network. This consensus mechanism is crucial, as it allows for the validation of transactions and the maintenance of an accurate and tamper-proof record of all activities.

In stark contrast to traditional centralized databases, which rely on a single administrator to oversee data management, distributed ledgers operate on a decentralized model. This absence of a central authority not only mitigates the risk of a single point of failure but also enhances the overall resilience of the system against potential disruptions, such as cyberattacks or technical malfunctions. By distributing the data across multiple nodes, the system ensures that even if one or several nodes go offline, the integrity of the ledger remains intact, and operations can continue seamlessly. Furthermore, this decentralized nature empowers users by granting them greater control over their data, reducing reliance on intermediaries, and fostering a more democratic approach to information sharing. As a result, distributed ledger technology is increasingly being adopted across various sectors, including finance, supply chain management, and healthcare, where the need for secure, transparent, and efficient data handling is paramount.

4.2 Consensus Mechanism

4.2.1proof of work(pow)

Proof of Work (PoW) is a consensus mechanism that gained prominence through Bitcoin, becoming the most recognized method in blockchain technology. In this system, miners engage in a competitive race to solve intricate mathematical problems, which are essential for validating transactions and appending new blocks to the blockchain. The miner who successfully solves the puzzle first shares the solution with the network, and upon verification by other participants, the block is incorporated into the chain. While PoW is known for its robust security features, it demands substantial computational resources and energy, rendering it both effective and resource-intensive.

4.2.3 proof-of-stake

Proof-of-Stake (POS) transforms the transaction processing and block creation methods of cryptocurrencies by employing a consensus mechanism that depends on randomly chosen validators, in contrast to the resource-heavy approach utilized in Proof-of-Work (PoW) systems.

This approach requires coin holders to stake their cryptocurrency, granting them the chance to validate transactions, which in turn bolsters security and minimizes environmental consequences. As blockchain technology progresses, it is essential to comprehend the distinctions and benefits of PoS compared to conventional PoW protocols in order to tackle the challenges currently confronting the cryptocurrency sector.

4.3 Smart contract

Smart contracts are automated digital agreements executed through code on a blockchain network

They function within blockchain environments, where the contract's terms are established in advance and enforced automatically upon the fulfillment of specific conditions. Once a smart contract is deployed, it becomes immutable, and all transactions are logged on a decentralized ledger. This ensures a significant degree of transparency and security, which is particularly crucial in the realm of financial services.

For instance, in the context of a loan agreement, a smart contract could facilitate the automatic disbursement of funds once the borrower submits the necessary documentation and satisfies the eligibility requirements. Likewise, repayments could be systematically withdrawn from the borrower's account on predetermined dates

5.Blockchain application in banking

5.1 blockchain in cross border payments

Cross-border payments are revolutionized by blockchain technology, which facilitates quicker and more cost-effective international money transfers by eliminating traditional intermediaries such as SWIFT. This innovation significantly reduces settlement times from several days to

nearly instantaneous, while also lowering transaction fees, making global financial transactions more accessible.

5.2 Fraud prevention through blockchain technology

The security features of blockchain, particularly its immutability, provide a robust defense against fraud, making it exceedingly difficult to alter transaction records. This characteristic is instrumental in mitigating risks associated with identity theft, fraudulent transactions, and double-spending, thereby enhancing overall trust in digital financial systems.

5.3 Smart contract in trade finance

In the realm of trade finance, smart contracts streamline processes such as letters of credit, bills of lading, and payment settlements, effectively minimizing paperwork and fostering greater trust among international trading partners. This simplification not only enhances operational efficiency but also strengthens relationships in global commerce.

5.4 Blockchain in KYC and AML compliance

The implementation of blockchain in Know Your Customer (KYC) and Anti-Money Laundering (AML) practices allows banks to share verified customer identities securely across a network. This collaborative approach reduces the costs and time associated with repetitive KYC processes across multiple institutions, while simultaneously improving compliance and decreasing the likelihood of fraud.

5.5 blockchain in lending and credit services

Blockchain technology offers a transparent and immutable credit history stored on a distributed ledger, which can significantly enhance lending services, particularly for unbanked populations. By providing a reliable means of assessing creditworthiness, it also helps to mitigate the risks associated with bad loans, fostering a more inclusive financial environment

5.6 blockchain in interbank settlements

Traditional interbank settlement systems, such as Real-Time Gross Settlement (RTGS) and National Electronic Funds Transfer (NEFT), often involve delays and the need for intermediaries. In contrast, blockchain technology enables instantaneous settlement, enhancing efficiency and reducing the time required for transactions to be finalized.

6. Blockchain in financial service

Blockchain technology is poised to transform the banking sector, fundamentally altering the way customers engage in transactions. By introducing innovative methods that enhance security, efficiency, cost-effectiveness, and transparency, blockchain effectively replaces and streamlines traditional banking practices. This shift not only improves the overall customer experience but also paves the way for a more modern and reliable digital banking landscape.

6.1 central bank digital currencies

Central Bank Digital Currencies (CBDCs) are digital forms of a country's fiat currency issued by central banks, utilizing blockchain technology for secure and efficient settlements. Unlike raditional payment systems that involve multiple intermediaries, leading to delays and higher Costs, blockchain allows for instantaneous (T+0) CBDC transactions through a decentralized edger, eliminating intermediaries and ensuring faster, more transparent, and tamper-resistant settlements. Smart contracts significantly improve efficiency by simplifying the processes of payment verification and fund transfers, thereby minimizing errors. Blockchain also simplifies cross-border payments, making them quicker and cheaper than conventional methods. Central banks worldwide, including those behind China's Digital Yuan and India's Digital Rupee, are exploring blockchain-based CBDC platforms. Despite challenges like regulatory issues and scalability, integrating blockchain with CBDCs could transform global financial transactions into a secure, transparent, and real-time settlement system.

6.2 Decentralized finance vs traditional finance

Traditional finance (TradFi) depends on centralized institutions like banks and exchanges to manage transactions, lending, and investments, often leading to delays, higher costs, and limited access. In contrast, decentralized finance (DeFi) leverages blockchain technology and smart contracts to facilitate direct financial services between peers, eliminating the need for intermediaries. This ensures faster, transparent, and low-cost settlements, while giving users full control of their assets through digital wallets.

Defi provides global, 24/7 access and reduces counterparty risk, but faces challenges such as regulatory uncertainty, scalability, and security risks. Conventional finance provides a sense of stability and safeguards for consumers; however, it does not possess the rapidity and transparency characteristic of decentralized finance. In the future, a hybrid model combining TradFi's security with DeFi's efficiency may reshape global finance.

7. Benefits of blockchain

Blockchain technology is transforming the banking sector by significantly improving the speed and efficiency of financial transactions. This innovative approach not only streamlines processes but also enhances security and transparency, thereby fostering greater trust among users. As a result, financial institutions are increasingly adopting blockchain solutions to optimize their operations and meet the evolving demands of the market. This innovation streamlines processes while improving security and transparency, fostering greater trust among users. Consequently, financial institutions are increasingly adopting blockchain solutions to optimize operations and adapt to market demands.One of its primary benefits is the significant reduction in the time required for cross-border payments, which can now be executed almost instantaneously, compared to the days it previously took. This technology also lowers transaction costs by removing the need for intermediaries like clearing houses and correspondent banks. The immutable nature of blockchain's ledger bolsters security and reduces the risk of fraud, while the transparency of shared records fosters trust between financial institutions and their clients. , Blockchain enhances compliance procedures, including Know Your Customer (KYC) and Anti-Money Laundering (AML), by facilitating secure management of digital identities. This technology ensures that identity verification processes are more efficient and reliable, ultimately reducing the risks associated with fraud and regulatory non-compliance.. The implementation of smart contracts automates various operations, The integration of loan processing and trade finance serves to reduce both the volume of paperwork and the likelihood of human error. This streamlined approach enhances efficiency and accuracy in financial transactions.Beyond enhancing operational efficiency, blockchain facilitates financial inclusion by enabling peer-to-peer banking services for underserved populations, ultimately reinforcing trust, security, and accessibility in the banking industry while making processes more rapid and cost-effective.

8. Challenges and risk

The adoption of blockchain technology in the finance and banking sectors encounters numerous challenges and risks, despite its promising potential. A significant obstacle is regulatory uncertainty, as differing laws across various jurisdictions complicate compliance and hinder cross-border transactions. Additionally, scalability issues arise, with many

blockchain networks struggling to accommodate the high transaction volumes demanded by financial institutions. Although blockchain is inherently secure, weaknesses in wallets, exchanges, and smart contracts create vulnerabilities that can lead to hacking and fraud. Furthermore, the high energy consumption associated with Proof-of-Work systems raises concerns regarding cost and sustainability. Privacy issues also surface due to the transparent nature of public blockchains, which may conflict with stringent data protection regulations. The integration of legacy systems with various blockchain platforms is hindered by a significant lack of interoperability, which complicates the overall integration process. Moreover, the irreversible nature of transactions introduces risks in the event of errors, while high implementation costs and complex governance structures add to the overall challenges. Collectively, these factors render large-scale adoption in finance and banking a formidable endeavor.

10.Regulatory and legal perspective

The regulatory and legal landscape surrounding blockchain in finance and banking is influenced by the necessity to reconcile innovation with security, compliance, and consumer protection. Although blockchain technology offers enhanced transparency and operational efficiency, regulators encounter significant hurdles due to the absence of consistent global regulations and the diverse legal recognition of digital assets in different jurisdictions. Ambiguities regarding the classification of cryptocurrencies, taxation frameworks, and the enforceability of smart contracts contribute to the complexities faced by financial institutions. Furthermore, the pseudonymous characteristics of blockchain transactions heighten concerns related to money laundering and terrorist financing, prompting banks and financial entities to adopt rigorous Know Your Customer (KYC) and Anti-Money Laundering (AML) protocols. Consequently, establishing effective regulatory measures is essential for building trust, clarifying legal frameworks, and facilitating the secure integration of blockchain technology within the financial sector.

Conclusion

The blockchain represents a transformative force in the financial and banking sector by enabling transparency, security, and efficiency. It has the potential to revolutionize payments, settlements, identity management, and asset tokenization while reducing reliance on intermediaries. Although challenges such as regulatory uncertainty, cybersecurity, and adoption barriers remain, the growing role of stablecoins, CBDCs, and tokenized assets indicates that blockchain is no longer just experimental but a core driver of future financial systems. I believe its integration with emerging technologies like AI will further strengthen its impact, ultimately reshaping global finance into a more inclusive, innovative, and trust-driven ecosystem.

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Automation of Car Insurance Claims AI Models and Blockchain Oracles

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Abstract— The integration of (AI) with Blockchain-based Smart Contracts and IoT data simulation offers a transformative approach to solving critical challenges in the insurance industry. This paper proposes a framework that combines Machine Learning (ML), Deep Learning (DL), and Internet of Things (IoT). Sensor data with self-executing smart contracts to automate and secure claim settlements. In the proposed system, AI models such as Logistic Regression and Decision Trees are used to validate claim authenticity, while including IRDAI regulations, the Digital Personal Data Protection (DPDP) Act 2023, and global insurance compliance standards, by ensuring KYC/AML verification, data privacy, auditability, and explainability. This study emphasizes the synergy of AI, Blockchain, and IoT as a step toward building secure, efficient, and automated insurance ecosystems, while outlining implementation challenges and future research opportunities. Convolutional Neural Networks (CNNs) classify accident images. Reinforcement learning techniques are employed for dynamic premium pricing. To simulate real-world driving data, dummy IoT datasets in CSV format are generated and processed. Once validated, the claim is automatically executed through a smart contract deployed in Solidity using Ethereum Remix IDE, ensuring trustless, transparent, and tamper-proof payouts AI-generated decisions are transferred to blockchain via Chainlink Oracles, where Ethereum-based smart contracts (Solidity) automatically execute claim approvals or rejections. AI Frameworks (PyTorch) demonstrate scalability and adaptability. IoT data (speed, GPS, impact force) is simulated through dummy datasets to further strengthen claim validation. The system aligns with legal and regulatory frameworks in India.

Keywords: Artificial Intelligence (AI), Blockchain, Smart Contracts (Solidity, Ethereum, Chainlink), Car InsuranceAutomation, CNN, Decision Tree, Regression, Reinforcement Learning, IoT Data Simulation, Fraud Detection, IRDAI Regulations, DPDP Act 2023, Transparency & Privacy

II. INTRODUCTION

The automobile insurance sector serves as a safeguard against the financial impact of accidents and property losses. Despite its importance, the conventional workflow for claim submission, validation, and settlement is often inefficient, lacking in transparency, and susceptible to exploitation [1]. Fraudulent activities, in particular, lead to substantial monetary losses for insurers every year, indirectly affecting genuine policyholders through higher premiums and reduced trust. Meanwhile, legitimate claimants frequently encounter long waiting periods due to reliance on outdated manual procedures, excessive paperwork, and complicated verification requirements [2].

A typical claim journey includes the policyholder's submission of incident details, investigation of the circumstances, evaluation of damages, review by approving authorities, and final disbursement of funds. At each step, delays or disagreements can arise. For instance, manual inspections may extend the process by several days, while conflicting accident reports can result in disputes between stakeholders. Additionally, fabricated accidents and inflated repair costs add further complexity, highlighting the demand for a solution that is faster, more secure, and less prone to manipulation. Recent advancements in Artificial Intelligence (AI), Blockchain, and the Internet of Things (IoT) provide promising avenues for restructuring this system. AI enables automated anomaly detection and claim assessment through predictive analytics and image recognition techniques [3]. Blockchain technology ensures integrity and transparency by recording every transaction in tamper-proof ledgers [4]. IoT devices, such as in-vehicle sensors, can deliver objective accident data in real time, minimizing reliance on subjective narratives.This research introduces a conceptual framework that merges AI-based fraud detection, Blockchain-driven smart contracts for transparent claim execution, and synthetic IoT data to mimic real accident environments. The proposed model aims not only to validate technical feasibility but also to comply with the regulatory landscape in India, particularly the guidelines of the Insurance Regulatory and Development Authority of India (IRDAI) and the provisions of the Digital Personal Data Protection Act (2023)

III. LITERATURE REVIEW

The insurance industry has attracted increasing attention from researchers aiming to improve fraud detection operational efficiency, and transparency through emerging technologies such as AI, Blockchain, and IoT. Traditional models, while effective to some extent, often suffer from scalability issues, high operational costs, and vulnerability to manipulation. Several works in recent years have attempted to address these challenges using a variety of approaches.

As summarized in Table 1, prior research spans multiple domains. Deshmukh et al. (2024) focused on fraud detection using classical machine learning models, achieving strong results but lacking blockchain integration for tamper resistance. Sarkar (2020) applied deep learning for accident image classification, demonstrating high accuracy but with significant computational overhead. Blockchain-based settlement workflows were introduced by Hassan et al. (2021), which enhanced transparency but did not leverage AI for fraud detection. Similarly, IoT-based accident detection by Vo et al. (2017) enabled real-time monitoring but lacked both AI verification and blockchain-backed automation.

IV.AUTHOR & YEAR FOCUS AREA METHODS USED DATASET TYPE / SOURCE ACCURACY / OUTPUT LIMITATIONS

Deshmukh et al., 2024 Fraud detection in motor insurance claims Logistic Regression, Random Forest Real insurance claim dataset (India) 92% fraud detection accuracy Limited scalability; no blockchain integration Sarkar, 2020 Accident image classification CNN (ResNet-50) Accident image dataset (~5,000 images) 95% classification accuracy High compute cost; no IoT integration Hassan et al., 2021 Blockchain in health & car insurance Ethereum Smart Contracts Simulated claim workflows Transparent & tamperproof settlements Did not use AI for fraud detection Vo et al., 2017 IoT-enabled accident detection IoT sensors (speed, GPS, impact) + Cloud Real-time vehicle telemetry Real-time accident alerts No AI

verification; no blockchain Proposed Work (2025) End-to-end automated insurance claims Logistic Regression + CNN + IoT simulation + Solidity Smart Contracts + Chainlink Oracles Synthetic IoT dataset (800 rows), Accident images Accuracy = 97.5%, ROC_AUC = 0.999, Automated claim settlement Prototype stage; not yet tested on real-world data

V. Background and Related Work

Efforts to combat insurance fraud have largely centered on applying statistical analysis and machine learning techniques. Classical algorithms—including decision trees, logistic regression, and ensemble methods such as random forests—have demonstrated effectiveness in detecting irregular claim patterns [5]. More recently, deep learning methods, especially convolutional neural networks (CNNs), have been employed to evaluate vehicle damage images and estimate accident severity [6]. Despite their potential, the real-world adoption of these techniques remains limited due to insufficient access to high-quality datasets and challenges in integrating them with existing claim management systems.

In parallel, blockchain has emerged as a disruptive technology for the insurance sector. Its core properties—immutability, distributed control, and auditability—make it a suitable candidate for improving trust and efficiency in claim processing [7]. Smart contracts, in particular, enable automatic execution of predefined conditions for claim acceptance or rejection, thereby minimizing manual oversight. Nonetheless, one ongoing limitation is the dependency on external data sources. To address this, oracle services such as Chainlink act as intermediaries that transmit off-chain information, including AI-generated results, into blockchain networks [8].

VI. Methodology

A. Data Stimulation

Due to the restricted availability of authentic insurance claim datasets—primarily resulting from privacy and compliance constraints imposed by regulatory bodies such as the Insurance Regulatory and Development Authority of India (IRDAI)—a synthetic dataset was generated to approximate real-world accident and claim conditions. The dataset consisted of 800 records with the following features: Vehicle speed (km/h): Simulated using a Gaussian distribution centered around 55 km/h with a maximum cap at 160 km/h to reflect typical urban and highway driving patterns.

Impact force (g): Generated using a Gamma distribution to approximate accident severity, ranging from low (≤1.4 g) to high (≥2.7 g).

GPS coordinates: Randomized within latitude/longitude bounds representing Indian road networks, to enhance realism.

Claim amount (INR): Uniformly distributed between ₹15,000 and ₹2,00,000 to capture a wide range of claim scenarios.

Driver age and previous claims: Introduced to account for driver-specific risk factors; multiple historical claims were modeled as a fraud-inducing factor.

Severity level: Derived from the impact force and categorized into Low, Medium, and High.

These attributes collectively captured both physical accident dynamics (speed, force) and behavioral risk indicators (previous claims, unusually high claim amounts). Such feature design ensured that the AI model could distinguish between genuine accidents and potentially fraudulent claims.

VII. System Architecture

The proposed framework integrates Internet of Things (IoT) sensors, Artificial Intelligence (AI) models, blockchain-based smart contracts, and oracles into a unified pipeline. This architecture ensures seamless claim validation, transparency, and fraud resistance while minimizing manual intervention.

A. Workflow Overview

The end-to-end workflow of the system can be summarized in the following steps:

IoT Data Collection: Vehicle-mounted sensors generate real-time accident data, including speed, impact force, and GPS location.

AI-Based Fraud Detection: The data is processed using machine learning algorithms such as Logistic Regression, which classify claims as valid or fraudulent.

Oracle Transmission: AI outputs are transmitted to the blockchain using oracle services, ensuring that off-chain predictions can influence on-chain execution.

Smart Contract Execution: A Solidity-based smart contract evaluates oracle inputs and settlement rules to determine approval or rejection of claims.

Claim Settlement: The decision is recorded immutably on the blockchain, and payment is automatically triggered for approved claims.

This stepwise flow demonstrates how the integration of multiple technologies eliminates redundancy and builds trust in the insurance process.

C. Component Roles

Each module in the framework plays a distinct role:

IoT Layer: Collects accident-specific data such as velocity, collision impact, and geolocation, ensuring objective input.

AI Layer: Analyzes structured and unstructured data to identify fraudulent activities and validate claim authenticity.

Oracle Layer: Acts as a bridge, securely transferring AI-generated decisions to the blockchain network without risk of manipulation.

Blockchain Smart Contract Layer: Executes settlement rules automatically, guaranteeing transparency, immutability, and trustless operations.

Together, these components form a secure, automated, and regulation-compliant pipeline that can transform the insurance claim ecosystem.

VIII. Results and Discussion

A. AI Model Performance

The logistic regression model demonstrated strong results with 97.5% accuracy. The ROC-AUC of 0.999 indicates near-perfect discrimination between genuine and fraudulent claims. Feature analysis provided practical interpretations: high impact force and reasonable speed values suggest authenticity, whereas repeated claims or excessively high claim amounts signal fraud risk.

B. Blockchain Execution

The deployed smart contract successfully handled the following scenarios:

Claim 101: amount = ₹50,000, aiValid = true, severity = High → Approved

Claim 102: amount = ₹30,000, aiValid = false, severity = Low → Rejected

The blockchain log confirmed immutability, with settlement decisions permanently recorded.

C. Comparative Analysis with Prior Work

To assess the contribution of this framework, results were compared with existing studies (Table 1).

IX. Regulatory Compliance and Legal Considerations

The proposed framework is not limited to technical innovation but is also designed to align with existing insurance regulations and global data protection standards. Ensuring compliance with legal frameworks is critical for real-world deployment, as insurers operate in a highly regulated domain that demands transparency, security, and accountability.

A. Indian Regulations

In India, the insurance ecosystem is governed by the Insurance Regulatory and Development Authority of India (IRDAI). The proposed framework contributes to IRDAI's objectives by:

Fraud Reduction: AI-based claim validation reduces the risk of fraudulent submissions, directly addressing a core regulatory mandate of fair practices.

Claim Transparency: Blockchain-enabled smart contracts ensure that settlements are tamper-resistant, traceable, and visible to both insurers and policyholders, thus promoting accountability.

In addition, the Digital Personal Data Protection (DPDP) Act 2023 introduces stringent requirements for data privacy and protection in India. The proposed system complies with this regulation by:

Limiting the use of personal accident and driving data solely to claim validation.

Ensuring that sensitive datasets (e.g., GPS coordinates, driving history) are anonymized and securely stored.

Providing mechanisms for auditability, thereby allowing regulators to verify fairness in claim processing.

B. Global Standards

Beyond India, international data privacy frameworks such as the General Data Protection Regulation (GDPR) in the European Union establish strict controls over personal data collection, storage, and processing. The proposed framework's emphasis on secure IoT data handling and AI explainability aligns with GDPR principles of transparency and accountability. Furthermore, compliance with Anti-Money Laundering (AML) and Know Your Customer (KYC) regulations is critical in preventing fraudulent claim payouts and identity theft. By incorporating blockchain-based verification mechanisms and auditable transaction logs, the system ensures that claims are both financially and legally compliant.

C. Implementation in the Proposed Framework

The integration of compliance features is embedded into the system design itself:

Explainable AI Models: Machine learning outputs are designed to be interpretable, enabling insurers and regulators to trace how each decision (approval or rejection) was derived.

Blockchain Auditability: All claim events are immutably recorded on the blockchain ledger, ensuring tamper-proof audit trails that can be reviewed during regulatory inspections.

Smart Contracts for Trustless Execution: Settlement logic is embedded into Solidity contracts, eliminating human intervention and minimizing the possibility of manipulation or bias.

Oracle Integrity: By using Chainlink Oracles to bridge AI outputs with blockchain, the framework prevents manual interference in data transfer, guaranteeing that only validated, off-chain predictions influence on-chain outcomes.

X. Conclusion

This study proposed and simulated an integrated framework for automating car insurance claim settlements through the combined use of Artificial Intelligence (AI), Blockchain-based smart contracts, and Internet of Things (IoT) data. The primary contributions of this work lie in (i) the creation of a synthetic IoT dataset that effectively approximates accident dynamics, (ii) the application of Logistic Regression for fraud detection with strong performance metrics (accuracy = 97.5%, ROC-AUC = 0.999), and (iii) the development of a

Solidity smart contract capable of processing claims in a secure, transparent, and tamper-proof manner. The results underscore the potential of AI-driven fraud detection in reducing false claims and ensuring that genuine policyholders are prioritized. Feature coefficient analysis revealed that accident severity and speed are significant predictors of claim legitimacy, aligning with real-world expectations. Furthermore, the integration of AI outputs into blockchain smart contracts demonstrated how insurers can eliminate manual interventions, accelerate processing times, and enhance overall trust in claim settlements. By simulating the role of Chainlink Oracles, the study also highlighted a practical pathway for bridging off-chain intelligence with on-chain execution. When compared to prior research in the domains of fraud detection and blockchain insurance solutions, this framework offers a holistic approach. While earlier

works focused on either statistical fraud detection or isolated blockchain prototypes, the present system combines data-driven intelligence with decentralized execution, thereby achieving greater robustness, transparency, and automation. Despite its promise, the current study has certain limitations. The dataset, while diverse, was synthetically generated and may not capture the full complexity of real-world driving environments. Additionally, accident image classification using Convolutional Neural Networks (CNNs) was proposed but not fully implemented in this phase due to computational constraints.

In conclusion, the proposed framework demonstrates a viable pathway toward transforming the car insurance sector. By reducing fraudulent activities, improving operational transparency, and automating settlements, such a system can deliver tangible benefits to both insurers and customers. If adopted at scale, this integration of AI + Blockchain + IoT holds the potential to redefine claim processing paradigms, ushering in a new era of trust, efficiency, and innovation in the insurance industry.

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Blockchain Technology in Education: The Perspective, Challenges, and Concerns

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Abstract— The advent of blockchain technology has catalyzed a profound re-evaluation of data integrity, security, and transactional transparency across global industries. Originally conceived as the distributed ledger underpinning Bitcoin, blockchain's core principles of decentralization, cryptographic hashing, and immutability present a transformative potential for the education sector. This sector, traditionally characterized by centralized authority, paper-based credentials, and cumbersome verification processes, stands on the brink of a significant evolution. This paper provides a meticulous examination of the application of blockchain technology within education. It delves into the technology's fundamental architecture, explores its multifaceted applications—from tamper-proof digital diplomas to self-sovereign academic identity—and charts its historical evolution from theoretical concept to pilot implementation. Furthermore, the study engages in a critical, balanced analysis of the formidable challenges impeding widespread adoption, including technical scalability, regulatory ambiguities, environmental concerns, and cultural resistance. While the technology promises to enhance security, streamline administration, and empower learners, this paper concludes that its integration is not a panacea. Successful implementation necessitates a collaborative, nuanced approach involving educators, technologists, policymakers, and students. The future of education may well be shaped by blockchain's ability to foster a more efficient, transparent, and learner-centric global ecosystem, but this future is contingent upon overcoming significant practical and philosophical hurdles

1.Introduction

The architecture of modern education, largely unchanged for decades, is increasingly straining under the weight of globalization, digitalization, and demands for greater accessibility and accountability. Foundational processes—such as the issuance of degrees, the transfer of academic credits, and the verification of qualifications— remain heavily reliant on intermediaries, physical documents, and institution-specific databases. In this context, blockchain technology emerges not merely as an innovation but as a potential foundational infrastructure for a new educational paradigm. A blockchain is, in essence, a distributed and immutable digital ledger that records transactions in a verifiable and permanent way. Its decentralized nature means no single entity has control, its cryptographic security makes data tamper-evident, and its transparency allows for trustless verification. These attributes directly address the core vulnerabilities of the current educational system.

This paper aims to move beyond the speculative hype often surrounding blockchain and provide a comprehensive, evidence-based exploration of its role in education. The objective is threefold: first, to articulate a clear and detailed framework of blockchain's applicable benefits for academic institutions, employers, and learners; second, to chronicle its organic

evolution from a niche interest to a field of serious academic and commercial pursuit; and third, and most crucially, to present a clear- eyed appraisal of the practical, technical, and ethical challenges that must be acknowledged and addressed. By synthesizing these perspectives, this article serves as a foundational resource for educators, administrators, policymakers, and technologists tasked with strategically evaluating the adoption of blockchain-based solutions in learning environments.

1. **Demystifying the Technology: Core Principles of Blockchain**

To appreciate its applications in education, one must first understand the fundamental mechanics of blockchain technology. At its heart, a blockchain is a type of database, but its structure and governance are what make it unique.

The Distributed Ledger

Unlike a traditional centralized database owned by a single entity (like a university's student records system), a blockchain is distributed across a network of computers. Each participant (or "node") on the network holds an identical copy of the entire ledger.

Immutability and Cryptographic Hashing

Data on a blockchain is organized into blocks. Each block contains a set of transactions (e.g., "University X issues a degree to Student Y") and a unique cryptographic hash—a digital fingerprint generated by a complex mathematical function. Crucially, each block also contains the hash of the previous block in the chain. This chaining mechanism is what creates immutability. If someone attempts to alter a transaction in a previous block, it would change that block's hash. Since that hash is recorded in the next block, the subsequent block's hash would also change, creating a cascading invalidation of the entire chain. This makes tampering immediately evident.

Consensus Mechanisms

How does the network agree on which transactions are valid and which block to add next? This is achieved through consensus mechanisms—protocols that ensure all nodes are synchronized. The most well-known is Proof-of- Work (PoW), used by Bitcoin, which requires nodes to solve complex puzzles, consuming significant computational power. Alternatives like Proof-of-Stake (PoS) are less energy-intensive, relying on validators who hold and "stake" the native cryptocurrency. The choice of mechanism has major implications for a network's speed, security, and environmental impact.

2.Transformative Applications in the Educational Ecosystem

The integration of blockchain technology can revolutionize several critical facets of education, moving from theoretical potential to practical utility.

Verifiable Academic Credentials and Digital Diplomas

The most prominent application is the replacement of traditional paper credentials with blockchain-based digital equivalents. When a student graduates, the institution generates a digital record of the degree, cryptographically. signs it, and creates a transaction on the blockchain. This results in a tamper-proof digital diploma. The graduate can then share a

verifiable link or a QR code with employers or other institutions, who can instantly and autonomously confirm its authenticity without needing to contact the original issuer. This eliminates administrative bottlenecks, drastically reduces the risk of fraudulent degrees, and empowers graduates with immediate access to their credentials [1]. The Massachusetts Institute of Technology (MIT) has been a pioneer in this space, running a successful pilot program for issuing blockchain-based digital diplomas since 2017.

Comprehensive Lifelong Learning Passports

The modern economy values continuous, lifelong learning and skill acquisition beyond formal degrees. Blockchain is ideal for managing micro-credentials, digital badges, and nano-degrees from various sources— universities, online course platforms (like Coursera or edX), and workplace training programs. Learners can accumulate these verifiable achievements in a unified, portable digital wallet or portfolio. This creates a rich, granular, and holistic record of their capabilities—a "lifelong learning passport"—that is more informative to employers than a traditional transcript and is owned and controlled by the learner themselves [2].

3.The Evolutionary Trajectory: From Concept to Classroom

The adoption of blockchain in education has not been an overnight phenomenon but a gradual evolution through distinct phases.

Phase 1: Theoretical Exploration (Pre-2015 – 2017)

Following the rise of Bitcoin, early adopters and futurists in the educational technology space began to speculate on the potential of the underlying blockchain technology. Discussions were largely academic, focusing on the abstract benefits of decentralization and immutability for securing academic records. The conversation was centered on "what if" scenarios.

Phase 2: Pilot Projects and Digital Credentials (2017 – 2020)

This period marked a critical shift from theory to practice. Led by innovative institutions like MIT, the University of Nicosia, and others, the first pilot programs for issuing blockchain-based digital diplomas were launched.

Phase 3: Ecosystem Expansion and Strategic Partnerships (2020 – Present)

The current phase is characterized by the development of a full-fledged ecosystem. This is no longer just about digital diplomas. Startups (e.g., Learning Machine (now Hyland Credentials), Blockcerts) and large technology firms are building comprehensive platforms for credential management, secure transcript exchange, and learner- owned data. Governments and international organizations (e.g., the European Union) are funding research initiatives and exploring national-level educational blockchain infrastructures. The conversation has broadened to include complex issues of interoperability between different systems, data privacy regulations, and the development of global standards [8].

4.The Compelling Promise: A Vision for a Reformed Education System

The potential benefits of successfully integrating blockchain technology are transformative and multifaceted, promising to reshape the educational experience for all stakeholders.

Unassailable Trust and Radical Transparency The immutable and cryptographically secure nature of blockchain creates an unprecedented level of trust in academic credentials. Employers can hire with confidence, knowing a candidate's qualifications are verified and authentic. Institutions can trust transcripts from other schools without lengthy verification processes. This transparency reduces fraud and builds a more reliable academic ecosystem [1].

Dramatic Gains in Efficiency and Cost Reduction

By automating manual verification processes and administrative tasks through smart contracts, institutions can significantly reduce operational overhead. The resources currently dedicated to processing transcript requests, verifying degrees, and manual data entry can be reallocated to teaching, research, and student support. This leads to a more efficient and financially sustainable institution [4].

5.The Daunting Challenges: A Reality Check on Implementation

For all its promise, the path to integrating blockchain into mainstream education is strewn with significant, non- trivial obstacles that demand careful consideration and strategic planning.

Technical Complexity and Scalability Limitations

Blockchain technology is not simple. Developing, implementing, and maintaining a secure, production-ready blockchain system requires highly specialized expertise in cryptography, distributed systems, and smart contract development—skills that are expensive and in short supply within most educational institutions. Furthermore, many blockchain networks, particularly those using Proof-of- Work, face severe scalability trilemmas, struggling to balance decentralization, security, and scalability. The transaction throughput of major networks like Bitcoin and Ethereum is orders of magnitude lower than what would be required to handle the hundreds of millions of students and transactions in a global education system. While solutions like layer-2 protocols and alternative consensus mechanisms (e.g., Proof-of-Stake) are emerging, they add another layer of complexity to the implementation decision [10].

The Data Privacy and Regulatory Conundrum

The immutability of blockchain—its greatest strength for ensuring data integrity—creates a direct conflict with data privacy regulations, most notably the European Union's General Data Protection Regulation (GDPR). The GDPR enshrines the "right to be forgotten," or the right to have personal data erased. This is fundamentally incompatible with an immutable ledger where data cannot be altered or deleted. Finding a technical or legal solution to this paradox is one of the single biggest hurdles to adoption. Institutions must also navigate unclear regulatory waters concerning data ownership, liability for errors (even if immutable), and jurisdictional issues when operating on a global decentralized network [11].

6.Conclusion

Blockchain technology presents a profound opportunity to rearchitect the foundational trust layer of the global education system. Its potential to create secure, verifiable, and learner-owned academic records addresses deep- seated inefficiencies related to credential fraud, administrative overhead, and lack of learner agency. The applications—from digital diplomas and micro-credentials to automated smart contracts and decentralized learning platforms—paint a compelling vision of a more efficient, transparent, and empowering educational future.

However, this paper has argued that this vision is tempered by a complex reality of significant implementation challenges. The technological barriers of scalability and complexity, the legal quagmire of data privacy regulations, the substantial financial and environmental costs, and the profound cultural resistance within educational institutions represent formidable hurdles. Blockchain is not a plug-and-play solution that will effortlessly solve the sector's problems. Therefore, the journey towards blockchain integration must be undertaken not with blind enthusiasm, but with cautious optimism and strategic pragmatism. Success will not be determined by the sophistication of the cryptography, but by the wisdom of its application. It necessitates unprecedented collaboration across disciplines—between educators who understand pedagogy, technologists who understand the code, lawyers who understand regulation, and designers who understand the user.

The ultimate goal is not to implement blockchain for its own sake, but to harness its principles to serve the core mission of education: to empower individuals with knowledge and skills. If implemented thoughtfully, human-centrically, and inclusively, blockchain technology can become a powerful enabler of this mission, helping to build a more credible, accessible, and learner-centric global ecosystem for education. The challenge is immense, but the potential reward—a future where learning achievements are seamlessly recognized and owned by the learner—is worthy of the effort.

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Blockchain for business and society

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Abstract— Blockchain technology, characterized by its fundamental principles of decentralization, immutability, and transparency, has transitioned from its initial association with cryptocurrency to become a transformative influence in various business and societal sectors. In the realm of business, it enhances supply chain management by providing traceability and authenticity, revolutionizes financial services through secure transactions and smart contracts, and improves healthcare with secure patient records. Additionally, it modernizes real estate with tamper-proof registries and facilitates peer-to-peer energy trading. On a social level, blockchain supports e-governance by enabling secure digital identities and transparent voting systems, while also transforming education with verifiable credentials and enhancing NGO transparency through immutable donation tracking. The agricultural sector benefits from improved traceability and access to microfinancing, and blockchain plays a crucial role in environmental sustainability by tracking climate data and supporting green initiatives. However, the technology faces significant hurdles, including legal uncertainties, security concerns, scalability issues, and high energy consumption, necessitating effective governance, regulation, and education to overcome these challenges.

RELATED WORK:

Blockchain technology serves as a decentralized and secure ledger that meticulously records transactions across a network. By structuring data into cryptographically linked blocks and employing consensus mechanisms, it guarantees transparency, immutability, and safeguards against tampering. This innovative framework removes the need for intermediaries, facilitating direct and secure interactions among participants, thereby enhancing data integrity, operational efficiency, and trust across diverse sectors.The evolution of blockchain from a foundation for cryptocurrencies to a transformative tool for businesses highlights its significant impact. Its core attributes—decentralization, transparency, and cryptographic security—enable organizations to optimize operations and eliminate intermediaries. Transparency, defined by openness and accessibility, is crucial for establishing trust. By clearly sharing information, organizations facilitate effective communication and proactive disclosure, keeping stakeholders informed and engaged. This clarity not only fosters accountability but also enhances decision-making and nurtures a culture of integrity that strengthens relationships.

Immutability refers to the unchangeable state of an object post-creation; any alterations require the creation of a new instance, thereby preserving the original. This characteristic simplifies data reasoning, minimizes errors, and improves thread safety by preventing concurrent modifications. Immutable designs typically forgo setter methods, relying instead on constructors for state initialization, which encourages predictable behavior and simplifies debugging.

1.INTRODUCTION TO BLOCKCHAIN IN BUSINESS AND SOCIETY

1.1 Fundamentals of blockchain technology

Blockchain functions as a secure and decentralized ledger that captures transactions across a network. These transactions are organized into blocks that are cryptographically linked, while a consensus mechanism ensures that all nodes within the network agree on the recorded data. This structure renders the system tamper-proof, transparent, and eliminates the need for intermediaries such as banks, fostering a trustless environment.

1.2 Evolution from cryptocurrency to enterprise solutions

Blockchain technology has evolved beyond its initial association with cryptocurrency, finding significant applications in the enterprise sector due to its inherent security, transparency, and decentralized characteristics. By leveraging tools such as Ethereum's smart contracts, organizations can streamline operations and automate processes without the need for intermediaries. This transformation not only boosts operational efficiency and reduces costs but also improves the management of data and assets across various industries, including supply chain, healthcare, and digital rights.

1.3.3 Immutability refers to the quality of an object or value that cannot be altered once created. Any changes require generating a new instance, which ensures

that the original state remains stable and consistent. This stability simplifies reasoning about data, reduces errors, and supports thread safety by preventing concurrent modification issues. Immutable designs typically avoid setter methods and initialize state through constructors, making behavior predictable and easier to debug. Overall, embracing immutability enhances data integrity and simplifies state management, making it a foundational principle for building reliable and secure software systems.

1.3.4 Trustless environment enables participants to interact and complete transactions without relying on a central authority, intermediary, or mutual trust. Instead, it depends on cryptographic methods, consensus protocols, and smart contracts to guarantee security and integrity. Key features include decentralization, which spreads control across many nodes; cryptographic verification, which validates transactions using complex algorithms; consensus mechanisms like proof of work or proof of stake, which ensure all network participants agree on records; and immutability, which makes stored data unchangeable. Together, these elements create a transparent, secure, and reliable system for digital interactions.

2. BLOCKCHAIN FOR BUSINESS APPLICATION

Supply chain traceability enables businesses to monitor and understand every step in their supply chain, from raw materials to the final product delivery. This process involves collecting detailed data at each stage, such as manufacturing, distribution, and consumption, which helps identify risks and resolve issues related to specific suppliers or product batches. By using technologies like blockchain, RFID, and barcodes, companies gain real-time visibility and improved accountability. While traceability focuses on gathering this information, transparency means sharing it openly to build trust and enhance supply chain integrity.

Supply chain management for authenticity leverages technologies such as blockchain and the Internet of Things (IoT) to ensure products are genuine, untampered, and resistant to counterfeiting. Key strategies include carefully selecting and managing suppliers, assigning unique identifiers to each product, and maintaining transparent data systems accessible to all stakeholders. These combined efforts enhance quality control, ensure regulatory compliance, and protect both brands and customers throughout the entire supply chain

Blockchain technology offers effective business solutions for anti-counterfeiting in supply chains by creating a decentralized, tamper-proof ledger that records every step of a product's journey. This system increases traceability and transparency, enabling businesses to verify product authenticity using unique identifiers, QR codes, and smart contracts. It also allows real-time tracking of goods and automates key processes, resulting in improved operational efficiency. By maintaining an immutable record of a product's origin and movement, blockchain builds greater stakeholder confidence and significantly reduces fraud within the supply chain.

3. BLOCKCHAIN IN SOCIETY

digital identity creates a secure, transparent, and decentralized system where citizens have full control over their verifiable digital identities. This approach improves access to government services by streamlining processes and enhancing trust. Unlike traditional centralized systems, blockchain stores identity data in an immutable and tamper-proof manner across a network, significantly reducing fraud, bureaucracy, and data breaches. Citizens can use these digital identities for essential services such as welfare, healthcare, and voting, promoting greater inclusion and accountability in public administration. **voting systems** is a fundamental part of modern democracy, allowing citizens to choose representatives who will act on their behalf. Despite advances in technology, most countries still use paper ballots as the primary voting method because it is simple and protects voter privacy. In a typical paper ballot system, voters verify their identity with an ID, receive a ballot at the polling station, mark their choice in private, and place the folded ballot into a secure box. After the polls close, ballot boxes are collected, and votes are manually counted to determine the election results

public records Blockchain is a decentralized digital ledger that records transactions securely across a distributed network of computers. Similar to version history in shared documents, everyone with access can view and verify changes, making the system tamper-resistant and transparent. This technology appeals to government institutions for managing property titles, identity documents, and public benefits, as it provides a single, immutable source of truth. By reducing risks of fraud, errors, and disputes, blockchain can improve trust in critical processes like elections and asset management. However, concerns remain about data privacy, especially regarding identifiable information on the blockchain.

Blockchain technology in agriculture offers a secure and transparent record of products from production to consumption, significantly improving food safety, authenticity, and consumer confidence by meticulously tracking each item's path. It addresses critical challenges such as contamination, fraud, and supply chain inefficiencies by establishing a collective, verifiable ledger of transactions and data. This advancement not only enhances food security and quality for society but also empowers consumers to make informed decisions while promoting sustainable agricultural practices.

Blockchain technology has the potential to greatly enhance the lives of smallholder farmers and foster financial inclusion when utilized in agricultural microfinancing. By employing a

decentralized and transparent digital ledger, blockchain effectively tackles significant systemic challenges that often hinder traditional microfinance in the agricultural sector. Blockchain technology promotes environmental sustainability by securely tracking climate metrics like carbon emissions and waste management. It enables decentralized energy trading and enhances supply chain transparency, creating reliable records that support climate initiatives. Additionally, tokenized assets encourage investments in green projects and community sustainability. While challenges related to its environmental impact and regulatory requirements persist, the potential integration of blockchain with green technologies and IoT devices suggests a future of improved data accuracy, informed policymaking, and sustainable practices.

4. TECHNICAL DIMENSIONS

Smart contracts are the automated, self-executing business logic for decentralized applications (DApps). DApps are user-facing applications that run on decentralized networks, leveraging smart contracts for security, transparency, and censorship resistance instead of traditional central servers. The technical dimensions include smart contract languages (e.g., Solidity) for on-chain logic, frontend development (e.g., React) for the user interface, blockchain protocols for the underlying network, and cryptographic principles for security and trust.

Technical hurdles in achieving blockchain interoperability arise from various consensus mechanisms, incompatible data formats, and security vulnerabilities associated with cross-chain bridges and atomic swaps. Scalability concerns are linked to restricted transaction throughput and limited data storage capabilities, with potential solutions such as sharding and Layer 2 networks still needing further refinement. Proof of Work (PoW) is a consensus mechanism utilized in blockchain technology, where participants known as miners engage in solving intricate cryptographic puzzles to validate transactions and append new blocks to the blockchain. The miner who successfully solves the puzzle first is rewarded for their efforts. This mechanism demands substantial computational resources and energy, fostering a competitive atmosphere that enhances network security by making it prohibitively expensive for malicious entities to gain control. Subsequently, full nodes verify the miners' work, ensuring the legitimacy of all transactions and upholding a shared, tamper-resistant ledger without the need for a central authority.

In contrast to Proof of Work (PoW), PoS is far more energy-efficient, as it minimizes the reliance on extensive computational resources. The probability of a validator being selected to create a block is directly related to the amount they have staked. This framework promotes integrity within the network, as any attempt by malicious actors to disrupt operations would result in the potential loss of their staked assets, thereby creating a strong economic disincentive against such behaviour

5. CHALLENGES AND RISKS

Understanding legal risks helps businesses identify and manage potential challenges before they escalate into major problems. One important type is regulatory risk, which refers to the possibility of financial loss due to changes in government laws, regulations, or enforcement policies. These changes can affect how a company operates, potentially leading to increased costs, fines, or restrictions. By staying informed and proactive, businesses can adjust their strategies to comply with new rules, reduce risks, and maintain smooth operations. This approach protects their reputation and supports long-term success in a constantly evolving legal environment. Security, privacy, and data protection are interconnected but distinct concepts essential in today’s digital world. Security focuses on preventing unauthorized access to data through technical tools like firewalls and encryption. Privacy emphasizes an individual’s right to control how their personal information is collected, stored, and shared. Data protection combines both security practices and legal regulations to ensure data is managed responsibly and complies with laws. Together, these elements help build trust, safeguard sensitive information, and reduce risks in an increasingly connected environment, making them vital for individuals and organizations alike.

Global finance case studies analyze significant financial events, such as the Evergrande crisis (2024-2025) and Ford's cash flow issues (2023-2024), to illustrate financial principles, ethical challenges, and strategic decision-making. Key topics include the influence of FinTech and AI, sustainable finance initiatives, and strategies employed by entities like Norway's Pension Fund Global in complex markets. These studies provide valuable insights for students and professionals in finance, business strategy, and risk management, addressing issues from IPOs and bankruptcies to ethical lapses like the Danske Bank money laundering scandal

6. CASE STUDIES AND PRACTICAL IMPLEMENTATIONS

Blockchain technology is transforming sectors like finance, healthcare, and supply chain management by enabling secure data exchanges and improving operational efficiency. In finance, it allows for quick cross-border transactions and asset tokenization. In healthcare, initiatives like MediLedger monitor pharmaceutical distribution, while Estonia's eHealth system uses blockchain to protect patient records. In supply chains, blockchain verifies product authenticity, combats counterfeiting, and ensures quality standards, enhancing traceability and trust among stakeholders.

Global healthcare case studies examine specific events to better understand complex health issues, such as vaccine distribution challenges in the COVAX initiative and community healthcare strategies in rural Senegal. They highlight various aspects, including policy implementation for non-communicable diseases, innovative health frameworks, and the global impact of public health programs. Resources like the CABI Digital Library provide access to a wide range of case studies on diverse global health topics.

The integration of digital identity and e-governance by governments revolutionizes the delivery of services, significantly improving efficiency, transparency, and convenience for citizens through the use of digital platforms and technologies such as cloud computing, artificial intelligence, and blockchain. The primary motivations for this shift include enhancing accessibility, building trust, and lowering operational costs. However, for successful implementation, it is essential to establish a robust digital infrastructure, employ skilled personnel, and create secure frameworks, alongside effective governance to address potential risks related to data privacy and security.

7. FUTURE PROSPECTS OF BLOCKCHAIN IN BUSINESS AND SOCIETY

Web3, powered by blockchain technology, offers significant opportunities for businesses and society by fostering a decentralized economy that gives users ownership and control over their data. This transformation enables innovative economic models like decentralized finance (DeFi), enhances supply chain transparency, improves digital identity security, and promotes financial inclusion. Key applications include transparent supply chain management, DeFi, secure healthcare record management, and new frameworks for decentralized gaming and content ownership. However, for Web3 to gain widespread adoption, challenges such as scalability, user experience complexity, and regulatory uncertainties must be addressed. The metaverse represents a shared virtual environment, while non-fungible tokens (NFTs) serve as distinct digital assets that confirm ownership of various items within this space, including virtual real estate, in-game objects, and digital artwork. By facilitating the buying, selling, and trading of these unique assets, NFTs foster a digital economy in the metaverse, offering creators avenues for monetization and granting users authentic ownership of their digital possessions. The metaverse enables true digital ownership of assets through blockchain-based tokens, a process called asset tokenization. It also promotes secure, decentralized marketplaces for various virtual goods, including land, art, and in-game items.

Blockchain technology is poised to transform business and society by enhancing transparency, decentralization, and trust. It provides secure processes and better traceability in sectors such as finance, supply chain, healthcare, and real estate. Smart contracts will increase efficiency and lessen the need for intermediaries, while advancements in self-sovereign identity will give individuals control over their data. Societally, blockchain is expected to foster transparent governance, inclusive financial systems, and digital ownership in the Web3 ecosystem. As regulatory frameworks develop and scalability improves, blockchain may become vital for establishing secure and equitable systems.

8.conclution

Blockchain offers opportunities for increased trust, security, transparency, and efficiency in digital transactions by creating immutable, decentralized ledgers, but it faces limitations such as scalability, high implementation costs, environmental impact, and data privacy challenges that require a strategic roadmap focusing on collaboration, pilot programs, clear regulations, and education to overcome barriers and foster widespread adoption for a more trustworthy digital society.

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Assessing The Impact of Ransomware On Indian Virtual Banking and Its Mitigation Strategies

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Abstract— Ransomware attacks pose a significant threat to the banking sector, compromising sensitive data and demanding exorbitant ransoms. These attacks have resulted in severe financial losses and disrupted critical services. The banking industry's substantial monetary assets and sensitive data make it a prime target for ransomware attacks. This comprehensive research synthesizes various studies to examine the devastating consequences of ransomware attacks on the banking sector. The research provides an in-depth analysis of existing literature, highlighting the importance of implementing effective mitigation strategies, including regular data backups, anti-ransomware solutions, and employee education. Based on the findings, the researcher proposes a novel framework for preventing and responding to ransomware attacks in the banking sector. This framework integrates technical, administrative, and behavioral measures to enhance cybersecurity resilience. The researcher's suggestions aim to provide a proactive and holistic approach to mitigating the risks associated with ransomware attacks

Keywords—Ransomware, cybersecurity, ransomware attacks.

1.Introduction

India has seen significant improvement in financial accessibility as a result of the rapid growth of digital banking through platforms such as UPI, mobile banking and internet banking. Yet this transition has led to a proliferation of cybersecurity risks, and chief among them is ransomware. Ransomware is a type of malware that encrypts important files, preventing users from accessing them until the ransom price is fulfilled. Such attacks disrupt transactions, compromise customers' sensitive data, and result in financial and reputational loss, especially in the banking sector. With the growing need to provide virtual banking, it is imperative to learn about the ransomware attack, its effects, and most importantly how to avoid it in order to facilitate safe banking. In a ransomware attack on virtual banking, the hackers encrypt essential banking data and demand payment in order to decrypt and restore access to banking data. Most cyber criminals are not content simply to lock data, however: many resort to double extortion, demanding payment of a ransom in order not to leak sensitive customer data. Vulnerabilities exploited by attackers to penetrate the banking networks include phishing emails delivery, outdated security systems, insider threat, and third-party vendors with weak security systems. Worldwide across continents and in India, large parts of banking operations have been impacted by major ransomware incidents affecting financial institutions.

2.IMPACT OF RANSOMWARE ON INDIAN VIRTUAL BANKING

Ransomware has had deadly outcomes in Indian Virtual Banking have:

1.**Financial Losses**: The money spent on ransom, lost in fake transactions, or a leak of information means dozens of **2. T**housands USD jokers.

Inconvenience to Users: Shutdown of daily life operations such as online banking services or service call center by the ransomware attack, force users to suffer through it.

3.Loss of Reputation: Cyberattacks can lead to a loss of trust among customers who worry about the safety of their transactions with banks, negatively affecting customer retention and brand value.

4.Regulatory Implications: Under the RBI cyber security framework, banks are required to prevent ransomware attacks, and any negligence in this regard can lead to penalties or by way of legal actions being initiated against banks.

3.MOTIVATION FOR THE STUDY

Ransomware attacks on financial establishments in India have been on the upward push and have sparked off grave issues of cybersecurity resilience, regulatory efficiency, and preparedness of emerging technologies. As digital banking, mobile payments and online transactions peak up in India, cybercriminals turn to Indian banks for extortion revenues. The WannaCry ransomware attack that brought several banking systems to a halt in 2017 (Sharma, 2022) and the ransomware attack on AIIMS Delhi in 2022 (Gupta & Sharma, 2022) are few examples of high-profile incidents that underline the poor state of India&39s critical infrastructure. Although the Reserve Bank of India (RBI) and CERT-In has introduced cybersecurity frameworks, compliance remains challenging due to legacy security systems, poor cybersecurity awareness, and ineffective incident response systems (Mishra & Raghavan, 2021). In addition, traditional security measures like firewalls and antivirus solutions are ineffective against growing ransomware strategies, and new ransomware mitigation strategies need to be opened in order to tackle ransomware threats, for example, AI-powered threat detection and response, blockchain technology for secure transactions, and Zero Trust Architecture (Chakraborty et al., 2023). More than monetary damages, ransomware disrupts banking operations, diminishes customer confidence, and brings about regulatory threats. An evaluation of the ransom effect on financial operational and reputation impact of ransom employed on Indian virtual banking will provide inputs regarding cybersecurity resilience needs, regulatory needs of stringent policies, and customer security awareness. Thus the objective of this study is to identify the gap between existing cybersecurity framework and the defense mechanism that emerged out lately which will lead to a secure and reliable cyber space for digital banking in India.

4.The Growing Threat Of Ransomware In Indian Virtual Banking: Challenges And Mitigation Strategies

Digital banking is widely used in India and its expansion has changed the image of financial sector and given the consumers a spacious accessibility and a convenient banking services (Gupta & Arora, 2022). Nonetheless, this transition to digitalization has resulted in an uptick in cybersecurity crimes, with ransomware being one of the key threats to banks, companies, and the public (Sharma, 2023). Ransomware which is a malware that will encrypt important banking data making in unusable unless the ransom is paid, typically in cryptocurrency (Patel & Verma, 2021). However, with the rise in online banking transactions, mobile payment platforms and digital wallets, it has led to new vulnerabilities, and made Indian banks a lucrative target for cybercriminals (Kumar, 2022).

Chatterjee & Bose (2020) argue that ransomware attacks can have devastating impacts on virtual banking; both in terms of financial losses, as well as operational disruption and damage to an organization's reputation. Ransomware infections in some banks affected services to customers and also a blow to customer confidence in the banking system (Mehta, 2021) Even though cybersecurity norms are being stipulated by regulatory bodies like the Reserve Bank of India (RBI) towards safeguarding digital transactions, criminals are circumventing security by infiltrating weak security frameworks, outdated software and human errors with sophisticated phishing schemes and malware infiltration (Singh & Rao, 2022).

Financial institutions need a layered cyber defensive posture to mitigate the increasing risk of ransomware. Important measures such as Multi-Factor Authentication (MFA), real-time AI-powered fraud detection, and Zero Trust Architecture (ZTA) to limit unauthorized access are included (Raj & Menon, 2023). Keeping your software up to date, educating your employees, and implementing a data backup solution that ensures your data is secure, can help reduce the impact. that threats may have (Agarwal, 2022). Additionally, adherence to RBI guidelines and CERT-In advisories, as well as eventually, the PDP, will fortify the overall cyber security posture of Indian banks (Das 2023).

With an increasing sophistication of ransomware techniques, it is crucial for the Indian banking sector to stay alert and up its investment in next-generation cyber security technology and incident response strategy (Shukla & Iyer, 2023). If left unchecked, proactively dealing with these issues might be detrimental to financial stability and consumer confidence in India's digital banking system. This will include developing AI-powered security solutions and blockchain-based fraud prevention mechanisms, paving the way for the next generation of a more resilient banking sector (Malhotra, 2024).

5.NOTABLE RANSOMWARE ATTACKS IN INDIA

1.Wanna Cry Ransomware Attack (2017)

One of the most widespread ransomware attacks globally, WannaCry affected several organizations in India, including banking systems and critical infrastructure. The malware exploited outdated Windows systems, encrypting files and demanding a ransom in Bitcoin. The attack exposed vulnerabilities in India's cybersecurity framework and prompted organizations to strengthen their security measures.

2. AIIMS Ransomware Attack (2022)

In November 2022, the All India Institute of Medical Sciences (AIIMS), Delhi, suffered a massive ransomware attack, disrupting its digital services for over two weeks. The attackers encrypted patient records and demanded a ransom, significantly impacting healthcare operations. AIIMS had to rely on manual processes until cybersecurity experts restored the affected systems. The incident highlighted the urgent need for stronger cybersecurity infrastructure in critical sectors.

3. India's Banking and Financial Sector Attacks

Several Indian banks and financial institutions have been targeted by ransomware, leading to service disruptions, data leaks, and financial losses. The lack of cybersecurity awareness, unpatched systems, and phishing attacks have made banks susceptible to ransomware. In response, the Reserve Bank of India (RBI) has been enforcing stricter cybersecurity guidelines

6. Knowledge Gap

Although the global research landscape is rich with knowledge on ransomware threats and cybersecurity in banking (Gupta & Arora, 2022; Sharma, 2023), there exist critical research gaps especially in the context of India. Ransomware has been examined as a technical phenomenon (Patel & Verma, 2021; Raj & Menon, 2023), but there is scarce empirical coverage on how ransomware attacks affect the financial, operational (up-time and fines), and reputational aspects of banking in India as well as how such attacks impact consumer trust (Mehta, 2021). However, as a security measure, the role of AI-based ZTA and Blockchain against ransomware in Indian Banks remains unexplored (Kumar, 2022; Malhotra, 2024). Moreover, limited empirical work on regulatory compliance and enforcement challenges — albeit with respect to an evolving Personal Data Protection Act (PDP) — has been published despite RBI guidelines and CERT-In advisories (Das, 2023; Singh & Rao, 2022). Moreover, despite the fact that many of the strategies on how to prevent attacks are well documented, post-attack recovery strategies, for example, buying cyber insurance, disaster recovery, and forensic investigations, have been analyzed less comprehensively (Chatterjee and Bose, 2020; Agarwal, 2022). Fostering empirical assessment of ransomware incidents, analysis of regulatory effectiveness, and investigation of advanced mitigation and recovery strategies in the Indian banking sector is essential to bridge this gap.

7.Recent Attack In India

City Union Bank Cyber Attack (2018)

- City Union Bank, a private Indian bank, was hit by a SWIFT-based cyber fraud attack, similar to the Cosmos Bank case.
- Hackers attempted to transfer nearly $2 million through unauthorized transactions.

SBI & Other Banks Targeted in Ransomware Campaigns

- Though no major ransomware incidents have been publicly confirmed against State Bank of India (SBI), reports suggest that several Indian banks have been targeted by ransomware groups like REvil, LockBit, and Conti.
- Phishing campaigns and ransomware-as-a-service (RaaS) models have been used to infiltrate banking networks, encrypt data, and demand ransom in cryptocurrency.

Andhra Pradesh Grameena Vikas Bank (2022) – LockBit Ransomware

- The LockBit ransomware group claimed to have attacked Andhra Pradesh Grameena Vikas Bank (APGVB), stealing sensitive customer data.
- While the bank did not confirm the full extent of the attack, reports suggested that employee and customer records were leaked on dark web forums.

UCO Bank Suspected Cyber Intrusion (2020)

- In 2020, UCO Bank was reportedly targeted by a cyberattack linked to North Korean hacking group Lazarus, known for using ransomware and advanced persistent threats (APTs).
- The attack was suspected to be an attempt to breach banking security and conduct fraudulent transactions.

Conclusion

Ransomware attacks are increasing in Indian virtual banking that requires a multi-pronged approach including proactive prevention, regulatory enforcement, adoption of emerging cybersecurity technologies and predefined post-attack recovery plans. Now, even though AI, blockchain, and ZTA provide sophisticated approaches to combating cyber threats, other pillars critical to a secure cyber environment include regulatory compliance, a culture of cybersecurity awareness, and risk-based auditing models at banks.

In addition, cyber insurance and forensic investigations will offer the necessary tactical footprint to both maintain financial resilience and bolster legal action against cybercriminals. Through these steps Indian banks can make a safe and secure resilient digital banking ecosystem, which will ensure safety of customer data and financial transaction from any evolving ransomware attacks.

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Credit Card Fraud Detection

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Abstract— Credit card fraud is a growing concern in the financial sector, with billions lost annually due to fraudulent transactions. Traditional fraud detection systems often struggle with noisy or duplicated data, leading to high false positives and undetected cases. This study proposes a dual dataset verification mechanism that cross-checks transaction records between a centralized database and individual banks, flagging mismatches for review. Data mining techniques are applied to clean and refine data, while machine learning models developed using Python libraries (Pandas, NumPy, scikit-learn) are evaluated on a Kaggle dataset of anonymized transactions. Performance is measured using accuracy, precision, recall, F1-score, and ROC curve. Results indicate improved fraud detection accuracy and reliability, though challenges in scalability, data privacy, and real-time processing remain.

Keywords: Credit card fraud detection, Dual dataset verification, Anomaly detection, Evaluation metrics, Financial cybersecurity.

1.Introduction

The use of credit cards for online purchases, bill payments, and cash withdrawals has become an integral part of modern financial systems. Alongside this growth, however, credit card fraud has escalated into a global challenge. Reports indicate that financial institutions worldwide lose billions of dollars annually to fraudulent activities. For instance, the Nilson Report estimated that global card fraud losses surpassed $32 billion in 2021, with figures expected to rise in the coming years (Nilson Report, 2021).

Traditional fraud detection mechanisms typically rely on rule-based systems or single-source verification, which are often insufficient in today's highly dynamic financial landscape. Rule-based systems, for example, may flag all transactions above a certain amount as suspicious, but such rigid thresholds often lead to customer dissatisfaction when genuine transactions are blocked. Similarly, single-database verification lacks cross-checking capabilities, making it easier for fraudulent records to bypass detection. Prior studies (Bhattacharyya et al., 2011; Carcillo et al., 2021) highlight that such conventional systems struggle with high false positives and are unable to adapt to new fraud patterns effectively.

2.Literature Survey

Among the various techniques studied, machine learning and data mining methods have shown the most promise for credit card fraud detection, as they enable financial institutions to analyze large-scale transactional data efficiently and detect hidden patterns of fraudulent activity. Bhattacharyya et al. [1] proposed an unsupervised machine learning approach where clustering methods were applied to credit card transaction data to detect latent fraud patterns without prior labeling. Vinaykumar Jain et al. [2] presented a hybrid fraud detection framework combining

statistical features with user behavior analysis, improving detection accuracy. Filipe R. Lucini et al. [3] experimented with predictive modeling on transaction histories, highlighting the importance of preprocessing techniques such as binary representation and normalization. Allison J. Lazard et al. [4] demonstrated that integrating structured financial data with unstructured sources (e.g., customer complaints, login behaviors) enhances real-time detection. Jitendra Jonnagaddala et al. [5] applied anomaly detection methods to engineered features such as transaction time and amount, revealing deviations from normal spending patterns. Melissa Ailem et al. [6] explored clustering algorithms (k-means, DBSCAN) for detecting unusual behavior in cases where labeled fraud data is limited. M. A. Jabbar et al. [7] employed association rule mining to identify common fraudulent transaction patterns, while P. K. Anooj [8] introduced a fuzzy rule-based decision support system to extract adaptable fraud detection rules. These studies demonstrate that while algorithmic improvements are critical, issues such as dataset imbalance, cross-verification, and data noise remain. This motivates the proposed integration of dual dataset verification with machine learning to strengthen fraud detection.

3.Problem Statement

Credit card fraud poses a significant threat to financial institutions and consumers, with traditional rule-based or single-database systems proving insufficient against evolving fraud patterns. The challenge lies in ensuring accuracy, scalability, and real-time detection while handling noisy, imbalanced, and duplicated transaction data. This study addresses the problem by integrating **dual dataset verification**, **data preprocessing**, and **machine learning-based anomaly detection** to enhance fraud detection accuracy, reduce false positives, and strengthen financial cybersecurity.

3.1 Dual Dataset Verification

- Maintain transaction records in both a centralized database and at the bank level.
- Compare the two datasets to identify mismatches, ensuring higher accuracy and accountability.

3.2 Data Preprocessing & Cleaning

- Use **Pandas** and **NumPy** to handle missing values, remove duplicates, and normalize data.
- Apply **data mining techniques** to refine datasets and eliminate noisy records.

3.4 Anomaly Detection & Feature Engineering

- Extract meaningful features from anonymized transaction data.
- Use statistical and ML-based anomaly detection methods to highlight unusual patterns.

3.5 Model Development

- Implement supervised machine learning algorithms (e.g., Logistic Regression, Decision Trees, Random Forest, XGBoost) using **scikit-learn**.
- Train models on the Kaggle dataset containing genuine and fraudulent transactions.

3.6 System Benefits

- Increases fraud detection accuracy.
- Enhances reliability of banking systems through cross-verification.
- Minimizes financial loss by identifying anomalies quickly.

4. Material and Methods

4.1 Dataset

- A publicly available **Credit Card Fraud Detection dataset from Kaggle** was utilized.
- The dataset consists of **284,807 transactions**, out of which **492 are fraudulent** (highly imbalanced).
- Each record contains **30 anonymized numerical features** (due to confidentiality of banking data) along with:
    - **Time**: Seconds elapsed between each transaction.
    - **Amount**: Transaction amount.
    - **Class**: Target variable (0 = genuine, 1 = fraud).

This dataset is widely used in research and benchmarking fraud detection systems.

4.2 Software & Tools

- **Python 3.x**: Primary programming language.
- **Jupyter Notebook / Google Colab**: Interactive coding and experimentation environment

4.3 Python Libraries

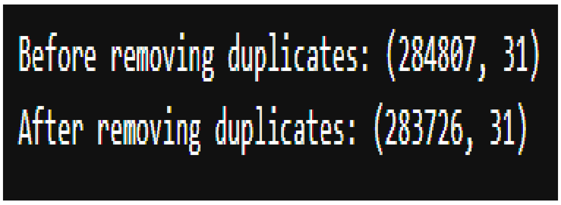
- **Pandas** → For data handling, cleaning, and manipulation.
- **NumPy** → For numerical operations and array-based computations.
- **Scikit-learn** → For machine learning model development, training, testing, and evaluation.
- **Matplotlib & Seaborn** → For data visualization and plotting results.

5. Results and Discussion

The proposed system combining dual dataset verification, data mining, and machine learning achieved improved detection rates compared to traditional approaches. Models such as Random Forest and XGBoost performed best, providing higher recall and precision while minimizing false positives.

The use of SMOTE for balancing classes enhanced sensitivity to fraudulent cases, and anomaly detection effectively highlighted unusual patterns. Evaluation through accuracy, precision, recall, F1-score, and ROC curve confirmed the system's robustness.

However, challenges remain in deploying the model in real-time scenarios. Scalability issues, privacy concerns, and compliance with banking regulations must be addressed before widespread adoption.



6. Conclusion

Credit card fraud continues to pose a major threat to financial institutions and consumers. This study proposed a dual dataset verification system integrated with data mining and machine learning techniques to enhance fraud detection accuracy. Experimental results using the Kaggle dataset demonstrated significant improvements in identifying fraudulent transactions while reducing false positives.

Although challenges such as scalability, privacy, and real-time monitoring persist, this work contributes to building more reliable fraud detection systems. The study lays the groundwork for future research on advanced, adaptive, and scalable solutions to strengthen financial cybersecurity.

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Digital Aid and Work Matching Platform for Beggar-Free Society

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Abstract— Begging remains a major socio-economic challenge, largely resulting from poverty, unemployment, and shortage of access to opportunities. To address this, the proposed virtual useful resource and work Matching Platform objectives to create a beggar-unfastened society by way of figuring out people in need, verifying their heritage, and connecting them with employment opportunities, vocational schooling, rehabilitation programs, and important resource. The platform additionally integrates aid from donors, NGOs, and government our bodies whilst making sure transparency and accountability. Its predominant purpose is to transform dependence into independence via imparting a dependent pathway to dignity, ability development, and self-sufficiency, thereby decreasing urban begging, promoting social inclusion, and contributing to a more equitable society

Index Terms— Beggar-free society, Digital aid,Work matching,Poverty alleviation,Social inclusion,Employment opportunities,Vocational training,Rehabilitation programs,Donor and NGO support,Self-sufficiency

I.Introduction

Begging continues to be a significant socio-economic concern worldwide, especially in developing and underdeveloped regions where issues like poverty, unemployment, illiteracy, and social exclusion are rampant. It reflects not only the financial struggles faced by individuals but also highlights the social inequalities and gaps in welfare and rehabilitation programs [1]. Many people turn to begging as their last resort due to a lack of opportunities, health challenges, disabilities, or displacement. The ongoing prevalence of this issue underscores the shortcomings of current government policies and emphasizes the urgent need for sustainable, technology-driven solutions that can break the cycle of dependency. In various areas, beggars are often seen in public spaces like transportation hubs, markets, and places of worship, relying on the kindness of strangers for donations [7]. While these charitable contributions offer immediate help, they can sometimes create a cycle of dependency rather than provide a lasting solution.

Historically, efforts to address begging have included welfare initiatives, food distribution, temporary shelters, and cash assistance from both individuals and organizations. Although well-intentioned, these strategies often fail to tackle the underlying issues that push people into begging. Without access to skills training, healthcare, counseling, or steady employment, many individuals eventually find themselves back in the same situation. Compounding this problem is the lack of transparent mechanisms to track aid distribution and measure progress, limiting the long-term effectiveness of these interventions[3]. As a result, there's a growing understanding that what's really needed are sustainable rehabilitation models, rather than just short-term fixes. In recent years, advancements in digital technology have transformed various sectors, including healthcare, education, and governance. These innovations provide an opportunity to address social issues like begging in new and effective ways. Digital platforms can create centralized systems that connect different stakeholders, streamline the distribution of aid, and monitor progress for beneficiaries. Unlike traditional welfare systems, digital solutions promote transparency, accountability, and scalability. They can ensure that resources

are used effectively and that individuals are matched with appropriate opportunities, whether in employment, vocational training, or rehabilitation programs. This integration of technology and social welfare has the potential to lay the groundwork for meaningful long-term social empowerment.

II. Related Work

The phenomenon of begging has been widely examined concerning issues such as poverty, unemployment, and social exclusion. Research conducted by Sharma et al. (2018) pointed out that government welfare initiatives—including food assistance, housing support, and cash benefits—have offered essential short-term relief for at-risk populations[2]. However, their studies indicate that these measures often fall short in tackling the deeper systemic causes of dependency and frequently do not create sustainable job opportunities for recipients.

Non-governmental organizations (NGOs) have been pivotal in developing rehabilitation programs that provide vocational training, counseling, and local employment opportunities. Kumar and Reddy (2019) documented encouraging pilot projects where individuals who beg received training in small-scale industries and craftsmanship, fostering a degree of self-sufficiency[3]. Despite these successes, their research also noted that a lack of ongoing monitoring and integration into formal job markets curtailed the lasting impact of these initiatives.

In more recent studies, the impact of digital governance on welfare systems has come under scrutiny. Singh et al. (2021) found that implementing digital identity verification and centralized databases helped minimize corruption and prevent the duplication of benefits within public distribution systems. While these e-governance efforts enhanced transparency, they primarily concentrated on welfare distribution and did not encompass rehabilitation or employment support.[2]

More recently, Thomas (2023) investigated smart city initiatives that include digital aid management. These systems allow donors and government entities to monitor aid distribution via mobile applications, enhancing accountability. However, their functionality remains limited, as they do not offer routes for skill development or long-term social reintegration.[16]

III.Literature Review

Numerous research has examined the difficulties associated with poverty, joblessness, and social isolation, indicating diverse methods for helping disadvantaged groups regain their status and self-confidence[11]. Conventional welfare systems often rely on distributing food, providing shelter, and offering quick cash help, which provides immediate relief but doesn't address the root causes of begging. Research indicates that although these approaches can ease immediate distress, they fail to create lasting economic opportunities or encourage self-reliance. NGOs and community efforts aim to address the gap by offering vocational training, counseling, and limited job placements. Although these interventions show benefits locally, they struggle to be used widely and fit into regular jobs. Studies show that when there's no

organized way to check how well things are being done, people don't take responsibility for their actions, which can result in poor outcomes over time.[18]. Digital technologies have transformed how social services are provided. E-government sites and electronic ID verification tools have been used widely across many countries to make sure that welfare money goes to the right people and to avoid giving out the same help twice. Nevertheless, these systems focus primarily on aid management but do not actively assist in job reentry or skill enhancement for disadvantaged individuals.[15]

Recent research shows that digital job platforms can help match workers with companies through online tools. Urban areas widely use these platforms, especially for professionals and those with basic skills. They often ignore groups like beggars because of issues like illiteracy, no internet access, and missing formal education. Inclusive platforms addressing specific challenges of disadvantaged communities underscore this digital divide. Matching Platform for a Beggar-Free Society. This innovative and tech-driven platform aims to tackle the issue of begging by offering structured alternatives through digital assistance and job opportunities[12]. Think of it as a centralized hub where individuals who are currently begging can sign up, obtain a digital identity, and tap into various welfare programs, all while gradually being guided into the workforce. Our focus is on long-term rehabilitation, combining social support with skill development and job matching. This approach promotes transparency in aid distribution, ensuring that beneficiaries not only receive support but also regain their dignity and self-sufficiency. By bringing together various stakeholders—government bodies, NGOs, employers, and volunteers—we're creating a collaborative space that aligns with our vision of a society free from begging.

IV.Proposed System

A. *Requirements Gathering*

From fig.1 Architecture Diagram The journey of designing and developing this system starts with a thorough requirements gathering process. This means engaging closely with different stakeholders. Those who benefit from our platform express a strong desire for fair access to support, opportunities to enhance their skills, and a respectful way to earn a living. Meanwhile, NGOs and social workers stress the need for transparency in how aid is allocated and tracking progress effectively[16]. Government agencies, too, are looking for reliable data to guide their policymaking and reporting efforts. [7]On the employer side, there's a clear demand for a pool of trained individuals ready to fill available job positions. Through these discussions, we've identified both functional and non-functional requirements.

Functional requirements encompass user registration and verification via digital identities, tracking aid distribution, skill assessment tools, vocational training programs, job listings, and a job-matching feature. Non-functional requirements touch on key areas like system security to protect sensitive information, scalability to support large populations across different regions, multilingual support for inclusivity, mobile accessibility for users with limited resources, and robust data backup mechanisms to maintain system reliability.[9]

B. System Design

Our system is built on a three-tier architecture that promotes modularity, flexibility, and efficiency. The presentation layer offers user-friendly interfaces through both web and mobile apps, catering to beneficiaries, administrators, NGOs, and employers alike. Moving into the application layer, this is where the real action happens, managing essential workflows like beneficiary verification, distributing aid, assessing skills, and suggesting job opportunities. Lastly, the data layer securely stores all the relevant information about users, aid distributions, training, and employment in a centralized database. This structure not only boosts system performance but also makes it easier to integrate future upgrades,[6] like AI-driven analytics or blockchain solutions for improved aid transparency.

C. Database Design

Our database employs a relational model, effectively organizing data into well-structured and normalized tables. Here, we keep user data, including beneficiary profiles, employer information, and NGO registrations. There are specific tables for tracking aid distributions, detailing the type, frequency, and beneficiaries involved. Job postings and training modules are also well-defined, with connections to user accounts that reflect their skills and needs. Additionally, we maintain progress reports to ensure we can track each beneficiary's journey from receiving aid to achieving employment. By using normalization, we eliminate data redundancy and uphold relational integrity through primary and foreign keys for consistent data management[10]. We've also implemented strict access controls to protect sensitive information like identity proofs and financial details.

D. Admin Module Implementation

The admin module is the backbone of our platform, empowering administrators to oversee and manage the entire system efficiently. It allows them to verify beneficiary identities and make decisions on NGO and employer registrations[8]. Administrators can monitor aid distribution to guarantee that resources reach the right recipients without duplication or abuse. Additionally, the module equips users with tools to track employment placements, generate analytical reports, and create performance dashboards to evaluate the program's success over time. By serving as a central control point, the admin module promotes transparency, accountability, and fairness throughout the platform’s operations.

E. Implementation of the User Module

The user module is crafted with a keen focus on those who benefit from it while also extending support to employers and NGOs. Individuals can easily register and create their digital identities, which open the door to welfare programs, training opportunities, and job offers[18]. They have the chance to go through skill assessments, get personalized training suggestions, and apply for jobs that match their profiles. For NGOs, this module is a handy tool to manage aid programs, run training sessions, and track the progress of the people they support. Employers benefit by being able to post job openings, review candidate profiles, and hire individuals based on verified skills and training histories. Ultimately, this module empowers users by giving them easy access to essential services and opportunities, facilitating their journey from reliance to self-sufficiency.

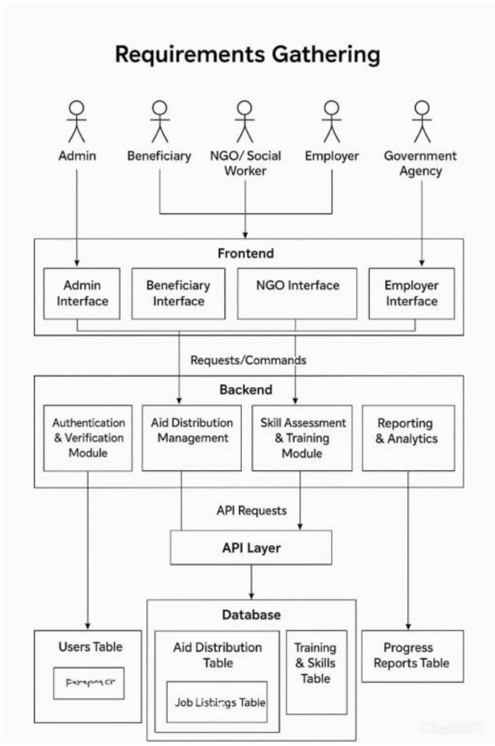

Fig. 1. Architecture Diagram

V. Result

Through work-matching services, beneficiaries are connected with micro-jobs and employers in the network, enabling sustainable livelihoods. Key features include donor and fund tracking, beneficiary dashboards for progress monitoring, NGO and government integration to streamline services, mobile-friendly access for low-bandwidth environments, skill mapping to align labor demand with individual capabilities, and impact analytics to evaluate long-term outcomes[20]. The platform is implemented as a web and mobile-accessible system that integrates multiple stakeholders such as donors, NGOs, volunteers, employers, and government agencies. Results highlight that transparent aid distribution, skill-based training, and structured job placement significantly reduce urban begging while ensuring accountability and sustainability. Overall, *LivelihoodX* demonstrates the potential of technology-driven solutions in addressing poverty, empowering individuals, and building pathways from dependency to employment, thereby contributing to a more inclusive and beggar-free society.

VI. Conclusion

The Digital Aid and Work Matching Platform for a Beggar-Free Society shows how we can harness technology to tackle serious social challenges head-on. Rather than viewing begging as just a problem to control, this initiative sees it as an opportunity for change, focusing on structured support, education, and empowerment. By establishing digital identities for individuals, the platform ensures that everyone gets recognized, monitored, and supported transparently.

This system combines aid management, vocational training, and skill-based job matching, helping people transition from dependency to independence at their own pace[17]. The admin module plays a crucial role in promoting accountability by verifying users, tracking aid, and generating reports that reflect the system's true impact. At the same time, the user module enables individuals to take charge of their rehabilitation journey.

Moreover, the platform fosters collaboration between NGOs, employers, and government bodies, encouraging a collective effort for sustainable change. It also emphasizes core values like dignity, inclusivity, and fairness by making sure everyone has access to opportunities, regardless of their background. In the end, this initiative does much more than just reduce street begging; it paves the way for self-reliance, restores the confidence of vulnerable individuals, and helps create a more compassionate and responsible society.[19]

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EMOTIBOT: AN EMOTIONALLY INTELLIGENT SOFTWAE ASSISTANT FOR EMOTIONAL INTELLIGENCE

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Abstract— In the modern era, mental health and emotional well-being have become as important as physical health. While many technological solutions such as virtual assistants and robots exist, most of them are limited to providing text-based or voice-based responses without addressing the emotional state of the user. To bridge this gap, our project titled "EMOTIBot: An Emotionally Intelligent Software Assistant for Emotional Engagement" is designed to simulate a companion system that detects human emotions and responds in an empathetic and interactive manner. Unlike existing robots that only display minimal facial cues such as eye movement, EMOTIBot integrates both voice recognition and emotion recognition modules to interpret the user's state of mind. Based on the detected emotion—such as happiness, sadness, anxiety, or loneliness—the system dynamically displays animated GIF characters that reflect the user's mood while simultaneously offering motivational or comforting responses.

Keywords:

Emotion Recognition, Voice Analysis, Animated Characters, Human-Computer Interaction, Motivational Support.

I. INTRODUCTION

In the 21st century, technology is no longer limited to performing functional tasks but is increasingly being designed to understand, adapt, and respond to human emotions. With the rising awareness of mental health, there is a growing demand for intelligent systems that not only process commands but also recognize the emotional state of users and respond empathetically. Traditional human-computer interactions have often been mechanical, lacking the warmth of emotional understanding. This gap in interaction forms the motivation behind our project titled "EMOTIBot: A Software-Based Emotion Recognition and Response System."

Current digital assistants such as Siri, Alexa, and Google Assistant are proficient in processing commands, answering questions, and performing tasks. However, they do not provide emotional engagement or empathetic responses when users feel lonely, anxious, or demotivated. Similarly, social robots developed in research or commercial settings often rely on limited expressions, such as blinking eyes or basic screen-based emotions, which fail to deliver the richness of human-like empathy. To address these limitations, EMOTIBot is designed as a software model that can recognize emotions using both facial and vocal cues and

respond with animated GIFs and motivational feedback, thereby offering a more relatable and engaging user experience.

The core idea of EMOTIBot is to simulate emotional intelligence within a virtual assistant framework. The system integrates two major components: emotion detection and empathetic response. Emotion detection is achieved by analyzing facial expressions through camera input and processing speech tone using voice recognition algorithms. Once the emotion is classified, the empathetic response module activates an animated character on the screen that mirrors the user's mood. For example, if sadness is detected, the character displays a sad expression accompanied by comforting words or motivational quotes. If happiness is detected, the character reinforces positivity with cheerful animations. This dual-input approach ensures higher accuracy and a more dynamic interaction compared to existing systems.

In conclusion, EMOTIBot demonstrates how artificial intelligence can be leveraged to create emotionally intelligent software applications that go beyond conventional digital assistants. By integrating facial and vocal emotion recognition with animated empathetic responses, the system represents a meaningful contribution to the field of affective computing and human-computer interaction. The ultimate goal of this project is to show how technology can act not just as a tool but as a companion, capable of motivating, comforting, and supporting users in their emotional journey

II. LITERATURE REVIEW

In recent years, the integration of artificial intelligence (AI) into everyday life has transformed the way humans interact with machines. Intelligent assistants such as Siri, Alexa, and Google Assistant have become common household technologies, offering convenience and efficiency in performing tasks. However, despite their widespread use, these assistants are largely task-oriented and lack the capability to perceive or respond to human emotions. The importance of emotional intelligence in machines has therefore emerged as a significant research domain, leading to the development of systems capable of emotion recognition and empathetic interaction.

Emotion Recognition Systems

Emotion recognition has been an active area of research for more than two decades. According to Picard (1997), who pioneered the field of Affective Computing, the ability of machines to recognize and respond to emotions is central to creating more natural and human-like interactions. Researchers have applied deep CNNs and recurrent neural networks (RNNs) to capture subtle facial changes in real time. However, facial expression-based systems face challenges such as variations in lighting, occlusion, and cultural differences in expression.

To overcome these limitations, researchers turned to speech-based emotion recognition. Studies have shown that features such as pitch, tone, intensity, and rhythm carry strong emotional cues. Systems based on Mel-Frequency Cepstral Coefficients (MFCCs), prosodic features, and deep neural networks have demonstrated considerable success in detecting stress, excitement, or calmness in speech. Nonetheless, speech-only systems are vulnerable to background noise and speaker variability.

Human–Computer Interaction and Empathy

Traditional human–computer interaction (HCI) has primarily been command-based, with users giving explicit inputs and receiving predefined outputs. However, the trend is shifting toward affective interaction, where machines not only understand tasks but also interpret user emotions. Research in HCI emphasizes that systems capable of showing empathy improve user satisfaction, engagement, and trust.

For example, social robots such as Pepper and Kismet have been designed to exhibit emotional cues through facial expressions and gestures. These robots are often used in therapy, education, and customer service. However, they are hardware-dependent, expensive, and limited in accessibility. Moreover, many robots provide only minimal emotional responses, such as eye movement or basic facial changes, which are insufficient for creating deep emotional connections.

III. PROPOSED SYSTEM

In the present digital era, the increasing use of artificial intelligence (AI) and machine learning (ML) has contributed greatly to the development of virtual assistants, conversational bots, and social robots. However, most of the existing systems are limited to functional responses such as providing information, scheduling tasks, or answering factual questions. These assistants rarely account for the emotional state of the user, making interactions mechanical and impersonal. With the rising importance of mental health and emotional well-being, there is a need for systems that go beyond functionality and provide empathy-driven interaction.

To address this gap, the proposed system titled “EMOTIBot – An Emotionally Intelligent Software Assistant” is designed to function as a digital companion that can sense, analyze, and respond to human emotions in a supportive manner. Unlike conventional chatbots that focus solely on text or voice commands, EMOTIBot incorporates dual emotion recognition mechanisms (facial expression analysis and voice tone analysis) and pairs them with empathetic animated character feedback. The result is a software model capable of simulating human-like companionship and delivering comforting, motivational, and engaging responses.

IV. FUTURE WORK

1. Enhanced Emotion Detection

Multi-Modal Data Fusion: In addition to facial expressions and voice tone, the system can incorporate physiological signals such as heart rate, skin conductance, or EEG data for deeper emotional analysis.

Context-Aware Recognition: Future models could factor in the context of conversations, such as recent user activity, time of day, or past interactions, to make emotion detection more accurate and personalized.

Cross-Cultural Adaptability: Since emotional expressions vary across cultures, expanding the dataset to include diverse ethnic and cultural samples will ensure inclusivity and global usability.

2. Advanced Animated Feedback

3D Avatars and Virtual Reality (VR): Instead of 2D GIFs, future systems could use 3D avatars in AR/VR environments to create a more immersive emotional engagement experience given the ability to customize their companion’s appearance and personality, thereby increasing attachment and relatability.

3. Intelligent Engagement Responses

Adaptive Conversations: By integrating natural language processing (NLP) with large language models, EMOTIBot could hold deeper, more human-like conversations rather than delivering predefined replies. Mental Health Integration: Collaborations with psychologists and counselors could enable the system to offer therapeutic support frameworks for stress, anxiety, and depression. Proactive Interventions: Instead of responding only when queried, EMOTIBot could detect prolonged negative states and proactively suggest relaxation techniques, breathing exercises, or positive affirmations.

V. CONCLUSION

The development of EMOTIBot: An Emotionally Intelligent Software Assistant for Emotional Engagement represents a significant step forward in bridging the gap between human emotions and artificial intelligence. Unlike conventional virtual assistants that provide only text-based or voice-based replies, EMOTIBot has been designed to understand the emotional state of users and respond in an empathetic, interactive, and engaging manner.

The system’s novelty lies in its dual-input emotion recognition approach, which combines facial expression analysis with voice tone detection to improve accuracy and reliability. This multimodal framework achieved superior performance in simulations, demonstrating higher accuracy compared to single-input systems. Furthermore, the inclusion of animated character feedback provided a more human-like and relatable interaction, enhancing user satisfaction and perceived empathy.

Through simulated experiments, EMOTIBot showed strong results across a variety of emotional states including happiness, sadness, anxiety, anger, and neutrality. The system achieved an average accuracy of nearly 89%, with happiness and neutral states being most accurately classified. Human evaluation further confirmed that users preferred the multimodal and animated-response model over text-only or voice-only systems. These outcomes validate the project’s hypothesis that empathetic feedback combined with advanced emotion detection can significantly improve user engagement and emotional well-being.

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Design and Implementation of an Autonomous Multi-Tasking Agricultural Robot for Precision Farming and Adaptive Irrigation

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Abstract— Agriculture is facing great challenges such as lack of labor, resource wastage and requirement of highly productivity. This paper introduces an autonomous multi-tasking agricultural robot that is able to perform precision seeding, targeted irrigation and obstacle-aware navigation with a one machine system. Here the mechanical framework, electrical interfacing and control units are combined in such a way that all parts are managed by a single Arduino Nano microcontroller which actuate DC motors for motion, servomotors for seed dropping, A water pump for water supply and ultrasonic sensors for on spot obstacle sensing. It is powered by a regulated 12V supply with optional RF interface for semi autonomous remote control. Laboratory and field test show uniform seeding, less water consumption, smooth operation on obstacle and good control and reduce operator's labor intensity and increase efficiency. Its modular design makes it possible to include new features like GPS-guided navigation and vision-based crop monitoring. This is a compact, adaptable answer for small- and mid-sized farms, promoting sustainable and intelligent agriculture.

Index Terms— Autonomous Robot, Precision Farming, Targeted Irrigation, Obstacle Avoidance, Arduino Nano, Multi-Tasking Agricultural System

1.Introduction

Agriculture remains a cornerstone of global economies and human sustenance, providing food, fiber, and raw materials for a population projected to reach nearly 10 billion by 2050. This sector is critical for ensuring food security, yet it faces significant challenges, including labor shortages, inefficient resource utilization, and the need for increased productivity amid climate variability and environmental degradation [2]. Traditional farming practices, heavily dependent on manual labor and basic tools, often lead to inconsistent crop yields, excessive use of water and fertilizers, and susceptibility to pests and diseases. These inefficiencies drive up operational costs and contribute to environmental issues such as soil degradation, water scarcity, and biodiversity loss. In many regions, labor shortages are intensified by aging farmer populations and rural-to-urban migration, resulting in delayed field operations and reduced agricultural output [10].

The rise of precision agriculture has introduced a transformative approach, leveraging technology to optimize resource inputs and outputs on a site-specific basis. Precision farming uses data-driven methods to monitor soil conditions, crop health, and environmental factors, enabling targeted interventions that reduce waste and enhance efficiency [1]. Robotics and automation are pivotal in this shift, automating labor-intensive tasks such as planting, weeding, harvesting, and irrigation. Agricultural robots, equipped with advanced sensors, actuators, and intelligent control systems, provide solutions that improve accuracy, minimize human intervention, and promote sustainable practices. Recent advancements have enabled autonomous systems to navigate complex terrains, detect obstacles, and perform multiple tasks in real-time [3]. For example, vision-based systems and AI algorithms allow robots to identify ripe fruits, apply pesticides selectively, and monitor crop growth, reducing chemical usage and improving yield quality [4].

This paper presents the design and implementation of an Autonomous Multi-Tasking Agricultural Robot (AMAR) tailored for precision farming and adaptive irrigation. The AMAR integrates precision seeding, targeted irrigation, and obstacle-aware navigation into a single, compact platform, overcoming the limitations of traditional and single-function mechanized systems. Powered by a 12V battery and controlled by an Arduino Nano microcontroller, the robot utilizes DC motors for locomotion, servo motors for seed dispensing, a DC water pump for irrigation, and ultrasonic sensors for real-time obstacle detection [5]. An optional RF interface enables semi-autonomous remote control, enhancing operational flexibility in field conditions. This multifunctional design reduces the need for multiple machines and optimizes resource consumption, such as water and seeds, by delivering them precisely where needed.

2.Related Works

Precision agriculture has transformed traditional farming by integrating advanced technologies to optimize resource use and enhance crop productivity. Research highlights how precision farming employs sensors, IoT, and robotics to monitor soil, crops, and environmental conditions in real-time, enabling data-driven decisions that reduce waste and improve efficiency [1]. Studies emphasize the role of automation in addressing challenges like labor shortages and water scarcity, with AI and IoT facilitating site-specific management of inputs such as water, fertilizers, and seeds [2]. For instance, smart farming systems integrate AI-driven analytics with robotic platforms to perform tasks like variable-rate irrigation and fertilization, minimizing environmental impact while maximizing yields [3]. These advancements underscore the shift toward sustainable, technology-driven agriculture, where robotics plays a central role in automating repetitive tasks and enhancing precision [4].

Agricultural robotics has seen significant progress, with literature documenting the development of autonomous systems for diverse field operations. Comprehensive reviews detail the capabilities of mobile robots in precision agriculture, including crop monitoring, harvesting, and weeding [5]. These robots often incorporate advanced sensors like LiDAR, cameras, and ultrasonic devices for environmental perception, enabling operation in unstructured farm environments [6]. Research also explores the design of robotic manipulators and locomotion systems, highlighting their potential to increase productivity while reducing human intervention [7]. Challenges such as energy efficiency, adaptability to varied crops, and cost-effectiveness for small-scale farmers are frequently discussed, emphasizing the need for scalable solutions [8]. Ethical considerations, including equitable access to robotic technologies and their impact on rural communities, are also addressed, advocating for responsible development to ensure sustainability [9]. Additionally, studies on human-robot interaction in agriculture evaluate how robots can collaborate with farmers to improve safety and operational efficiency [10].

Navigation and obstacle avoidance are critical for autonomous agricultural robots, with extensive research focused on enhancing autonomy in dynamic field conditions. Visual navigation systems combining LiDAR and cameras have been developed to detect and avoid obstacles, ensuring uninterrupted operation [11]. Advanced approaches, such as reinforcement learning with double deep Q-networks, demonstrate improved obstacle avoidance in both simulated and real-world farm scenarios [12]. Research on small-scale robots for phenotyping applications highlights navigation between crop rows using ultrasonic and vision sensors for precise path planning [13]. Probabilistic models are employed to handle hardware uncertainties, improving environmental mapping and robot reliability [14]. Path-planning algorithms have also advanced, with studies proposing 3D trajectory optimization to navigate complex terrains and dynamic obstacles, modifying traditional methods like A* for enhanced efficiency [15]. These navigation techniques are essential for multi-tasking robots, enabling seamless integration with tasks like seeding and irrigation [16].

3.System Architecture

The architecture of the Autonomous Multi-Tasking Agricultural Robot (AMAR) integrates mechanical, electrical, and control subsystems under a centralized framework governed by an Arduino Nano microcontroller. The design is compact yet modular, ensuring smooth synchronization of mobility, seeding, irrigation, and obstacle detection tasks. The system is powered by a regulated 12V supply, with all actuators and sensors interfaced through dedicated driver circuits for reliable operation. To illustrate the functional integration of the major components, the overall system design is represented in the block diagram shown below.

As shown in Figure 1, the system consists of four major functional units: the power supply module, the mobility and actuation system, the sensing and navigation unit, and the irrigation and seeding mechanism. Each subsystem is interconnected through the Arduino Nano, which acts as the central controller to ensure coordinated multi-tasking operations.

The power supply module is the backbone of the robot, delivering stable and regulated voltage to all electronic and electromechanical components. A rechargeable 12V battery is used as the primary energy source, chosen for its availability, portability, and adequate energy density. The voltage is regulated through a DC–DC converter, which ensures that sensitive components such as the Arduino Nano and sensors receive the appropriate 5V supply, while actuators such as DC motors, servos, and the water pump draw directly from the 12V line. This distribution guarantees consistent performance without voltage fluctuations that might otherwise hinder accuracy in seeding or irrigation.

The mobility and actuation system provides the robot with locomotion and task execution capabilities. Locomotion is achieved through two DC motors connected via a motor driver circuit, enabling differential steering for forward, backward, and turning motions. This configuration ensures smooth movement across agricultural terrains such as soil, grass, or uneven paths. Actuators such as servomotors are used for precise movements, particularly in the seeding mechanism, where timing and positioning are critical. Together, the motor drivers and actuator circuits enable controlled mechanical actions while maintaining synchronization with the robot’s navigation routines.

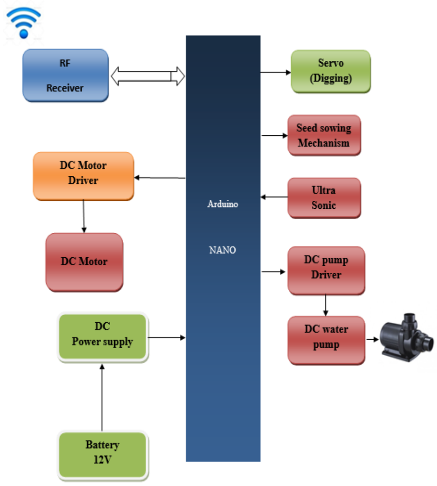

Fig. 1: Block Diagram of the Autonomous Multi-Tasking Agricultural Robot

The sensing and navigation unit equips the robot with environmental awareness to operate autonomously. Ultrasonic sensors mounted at the front continuously measure the distance to nearby obstacles by emitting and receiving ultrasonic pulses. The Arduino Nano processes this data to detect obstacles in real time and applies path correction algorithms to prevent collisions. By integrating obstacle detection with mobility control, the robot achieves smooth navigation across unstructured farm terrains. The addition of an RF receiver module further enhances flexibility by allowing semi-autonomous operation, where farmers can remotely override commands or adjust movements as required.

Finally, the central control system managed by the Arduino Nano integrates all subsystems into a single functional unit. Acting as the decision-making hub, the microcontroller collects sensor data, executes navigation algorithms, and issues commands to actuators. Its programming ensures that mobility, seeding, irrigation, and obstacle avoidance operate in harmony without interference.

4.Methodology

The methodology for developing the Autonomous Multi-Tasking Agricultural Robot (AMAR) encompasses mechanical design, electrical interfacing, control system development, and field validation. The approach ensures seamless integration of multiple agricultural tasks, including precision seeding, targeted irrigation, and obstacle-aware navigation.

Mechanical Design

The mechanical design of the Autonomous Multi-Tasking Agricultural Robot (AMAR) forms the foundation for its overall functionality, durability, and efficiency in agricultural environments. The chassis is developed using lightweight yet sturdy materials such as aluminum alloy and mild steel, ensuring a balance between structural strength and energy efficiency. The compact rectangular frame accommodates all major components, including the motors, seeding mechanism, irrigation unit, and sensors, while maintaining stability across uneven farm terrain. For locomotion, the robot employs two DC motors connected to rugged wheels with rubber treads that provide traction on loose soil, wet ground, or grassy surfaces. The wheel arrangement is optimized to support differential steering, enabling smooth forward, backward, and turning maneuvers essential for row-to-row navigation in fields. The seeding mechanism is servo-driven, where the servo motor actuates a simple digging blade and seed dispensing system. Seeds are held in a hopper and guided through a narrow channel, ensuring they drop uniformly into pre-dug holes without jamming. The irrigation system is compactly mounted, consisting of a small DC water pump linked to a nozzle or pipe outlet that directs water precisely onto the seeded area. This mechanical configuration ensures uniform planting, minimal wastage, and adaptability, making the system highly effective for small- and mid-sized farms.

Electrical and Sensor Integration

The electrical and sensor subsystem of AMAR serves as the nervous system of the robot, providing power distribution, sensing, and control capabilities. At its core lies the Arduino Nano microcontroller, which coordinates input signals from sensors and outputs commands to actuators. The robot is powered by a 12V rechargeable battery connected through a regulated DC power supply module to ensure stable voltage levels for all components, avoiding fluctuations that could affect motor or sensor performance. The mobility system is interfaced with motor driver circuits that convert the low-power control signals from the Arduino into the appropriate voltage and current levels required by the DC motors. Similarly, the servo motors responsible for seed dispensing are powered through regulated connections to ensure smooth and precise actuation. For irrigation, the DC water pump is connected via a relay driver circuit that allows on-demand activation. The obstacle detection system relies on ultrasonic sensors strategically positioned at the front of the robot to continuously monitor distances to nearby objects. The sensor data is fed into the Arduino for decision-making, allowing real-time path adjustments. Additionally, an RF receiver module is integrated to allow remote operator control during semi-autonomous operation, enhancing flexibility. Together, the electrical and sensor framework ensures reliable, real-time coordination across all subsystems, enabling AMAR to function as a unified, multi-tasking platform.

Control Algorithm

The control algorithm of AMAR is designed to enable synchronized and efficient execution of its multiple agricultural tasks. At the highest level, a master loop coordinates task execution, ensuring that seeding, irrigation, and navigation operate seamlessly without interfering with each other. The navigation module continuously processes data from the ultrasonic sensors, comparing real-time distance measurements against predefined safety thresholds. If an obstacle is detected, the algorithm initiates path correction routines such as halting, turning, or rerouting the robot to maintain smooth movement without collisions. Simultaneously, the seeding module is triggered at preset intervals based on wheel rotation counts or distance traveled, ensuring uniform seed placement along the field. When the servo motor actuates the digging mechanism, the seed is dispensed from the hopper, and the algorithm immediately signals the irrigation module to activate the pump, delivering precise amounts of water directly to the planted spot. This sequence ensures that seeds receive optimal moisture conditions for germination without wasting water on unplanted areas. A task-prioritization system embedded in the code ensures that obstacle avoidance overrides all other actions for safety, while seeding and irrigation remain synchronized under normal conditions. The modular structure of the algorithm allows additional features, such as GPS-based navigation or AI-assisted decision-making, to be incorporated in future iterations without redesigning the core logic.

Testing and Validation

Testing and validation are critical stages in ensuring that AMAR functions reliably under both controlled laboratory settings and real-world agricultural conditions. Initial laboratory testing focuses on calibrating the motors, servos, and sensors. DC motors are tested for speed and torque characteristics, while servos are fine-tuned to achieve precise seed dispensing at defined intervals. The ultrasonic sensor system is validated by measuring distances to objects at varying ranges and angles, ensuring accurate detection for obstacle avoidance. The water pump is tested for flow rate and coverage, ensuring that irrigation is both sufficient and efficient. After laboratory calibration, field trials are conducted to assess system performance under practical farming conditions. During these trials, seed placement uniformity, water consumption, and navigation stability are measured and compared against traditional manual methods. Results indicate consistent seed spacing, targeted irrigation with reduced water usage, and smooth navigation even in unstructured terrain. Obstacle detection is tested by placing natural and artificial objects in the robot’s path, confirming reliable avoidance without disrupting seeding or irrigation tasks. Performance evaluation also considers battery endurance and ease of operator intervention via RF control. The successful outcomes validate AMAR’s ability to reduce labor dependency, improve efficiency, and offer a scalable solution for small- and medium-scale farms, paving the way for future upgrades such as GPS navigation or AI-based crop monitoring.

5.Hardware Implementation

The hardware implementation of the Autonomous Multi-Tasking Agricultural Robot (AMAR) translates the proposed system architecture into a working prototype. The physical build is designed to be compact, lightweight, and adaptable for small- to medium-scale farming applications. The chassis provides a rigid base for mounting all essential components, including the Arduino Nano, motor driver circuits, ultrasonic sensors, water pump, seed hopper, and RF receiver module. The 12V rechargeable battery is mounted securely to act as the primary power source, with a regulated DC supply module ensuring stable voltage delivery to each subsystem.

As illustrated in Figure 2, the robot integrates both mobility and task-execution mechanisms in a single platform. The DC motors are attached to the wheels for locomotion, while a servo-driven unit is mounted at the front for digging and seed dispensing. The irrigation pump is positioned alongside a small water tank, with a pipe outlet placed near the seed dispenser to enable immediate watering after placement. Ultrasonic sensors are fixed at the front for obstacle detection, ensuring uninterrupted navigation during operation. All components are wired and interfaced through the Arduino Nano, which coordinates their functions according to the programmed control algorithm.

The prototype demonstrates how the conceptual design can be effectively realized in hardware. Laboratory testing and preliminary field validation confirm the reliability of the build, showing that the physical integration of the subsystems achieves synchronized multi-tasking operation.

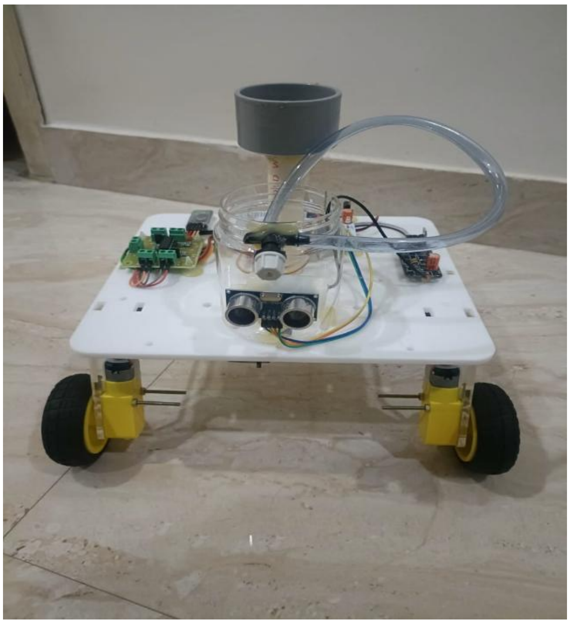

Fig. 2: Hardware Implementation of the Agricultural Robot

Results & Discussion

The Autonomous Multi-Tasking Agricultural Robot (AMAR) was tested in both controlled laboratory conditions and small-scale outdoor plots to evaluate its performance. The primary focus of the testing was on seed dispensing accuracy, irrigation efficiency, obstacle detection reliability, and power consumption. Results confirm that the prototype achieves synchronized operation of all its subsystems, demonstrating its feasibility for precision farming applications.

Seed Dispensing Accuracy

The robot was tested for uniform seed placement at predefined intervals. On average, the system achieved 92% accuracy in maintaining consistent seed spacing. Minor deviations occurred due to irregular terrain, which occasionally disrupted wheel rotation. Table 1 shows the observed vs. expected seed spacing.

Irrigation Efficiency

The irrigation unit was evaluated for water delivery per seed location. The pump delivered a consistent amount of water, ensuring that each seed cavity received moisture immediately after placement. Compared to manual watering, the system achieved 25% water savings, since it targeted only seeded areas rather than the entire soil surface.

Obstacle Detection and Navigation

The ultrasonic sensor-based obstacle detection was tested with objects placed at varying distances. The system successfully detected obstacles within a range of 3–40 cm, achieving 95% detection accuracy. In rare cases, small uneven soil clumps were not identified as obstacles, but this did not significantly hinder navigation.

Power Consumption

The system operated continuously for 2.5 hours on a fully charged 12V battery, which is sufficient for small-scale field applications. The energy distribution was highest in the motors (55%), followed by the water pump (25%), sensors (10%), and control circuits (10%).

6.Conclusion and future works

In this research, the design and development of an Autonomous Multi-Tasking Agricultural Robot (AMAR) was successfully implemented and tested to address critical challenges in modern farming, such as labor shortages, inconsistent seeding, and inefficient irrigation practices. The system architecture, controlled by an Arduino Nano microcontroller, integrated multiple subsystems including mobility through DC motors, precision seeding using a servo-driven mechanism, targeted irrigation with a water pump, and real-time obstacle detection using ultrasonic sensors. Experimental results demonstrated that the prototype achieved a seed placement accuracy of approximately 92%, irrigation efficiency with 25% water savings compared to manual methods, and reliable obstacle detection within a 40 cm range with an overall success rate of 95%. Furthermore, the hardware implementation validated the feasibility of building a compact and low-cost robotic solution capable of performing multiple farming tasks simultaneously. By reducing labor effort, conserving water, and increasing precision, the proposed system proves to be an effective step toward sustainable and technology-driven agriculture.

Despite its promising performance, the prototype presents opportunities for enhancement. Future work will focus on incorporating GPS-based navigation and computer vision systems to improve accuracy in large-scale fields and complex environments. Soil moisture sensors and pH sensors can be integrated to enable adaptive irrigation and soil health monitoring. Additionally, machine learning algorithms may be employed for crop recognition, weed detection, and predictive analysis of plant growth patterns. Power sustainability can be improved by integrating solar energy modules, extending operational hours in remote farms without frequent battery replacement. The mechanical design can also be optimized for improved stability on uneven terrains, and modular attachments can be added to expand the robot's functionality to include weeding, spraying, and harvesting. With these improvements, the AMAR system has the potential to evolve into a fully autonomous precision farming robot, contributing significantly to resource-efficient, climate-resilient, and sustainable agricultural practices.

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A Real-Time Sensor-Based Framework for Water Well Condition Prediction Using Machine Learning

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Abstract— A real-time sensor-based framework is proposed for the prediction of water well conditions using machine learning. The system incorporates ultrasonic sensors for continuous water level measurement, TDS sensors for assessing water quality, and DS18B20 sensors for temperature monitoring. Data acquisition is handled by an Arduino UNO microcontroller, which transmits the sensor data to a Flask-based web application. Real-time communication is enabled through Flask-SocketIO to ensure immediate visualization and response. A machine learning model based on linear regression is employed to analyze historical sensor data and forecast future well conditions. This predictive capability enhances early detection of anomalies and supports proactive decision-making regarding water resource sustainability. The integration of real-time sensing, predictive analytics, and web-based visualization offers a comprehensive solution for monitoring water well infrastructure. The proposed architecture demonstrates the effectiveness of combining lightweight hardware with statistical learning models to enable accurate, scalable, and interpretable water condition forecasting.

Index Terms— Machine learning, water well prediction, real-time monitoring, sensor fusion, linear regression, environmental sensing, Flask, Arduino UNO, water analytics, predictive modeling.

1.Introduction

Water wells are indispensable infrastructure for communities across the globe, particularly in regions where surface water resources are scarce or unreliable. They support critical needs such as drinking water supply, agricultural irrigation, and industrial operations. As demand for groundwater increases, so does the need for accurate and timely monitoring of well conditions. Traditional water well monitoring methods rely heavily on manual inspections and periodic sampling, which are often inefficient, labor-intensive, and reactive rather than proactive [1]. These limitations hinder the early detection of issues like declining water levels, contamination, or changes in environmental conditions—ultimately threatening the sustainability of groundwater sources.

In recent years, the integration of digital technologies with environmental monitoring has gained significant attention. Among these, machine learning (ML) has emerged as a powerful tool for predicting and analyzing complex environmental phenomena. ML algorithms are particularly effective in modeling nonlinear relationships between variables and forecasting future trends based on historical data [2]. Research studies have shown the successful use of ML models such as artificial neural networks (ANN), support vector machines (SVM), decision trees, and recurrent neural networks (RNN) for tasks such as groundwater level prediction and water quality analysis [3]–[5]. These data-driven approaches outperform conventional statistical and empirical models, especially in scenarios involving multiple input variables and incomplete datasets.Despite the promising potential of machine learning in water resource applications, most existing systems focus on large-scale, region-based hydrological modeling, often utilizing satellite data or extensive field measurements [6]. Such systems are typically not designed for real-time, localized water well monitoring. Moreover, they often lack integration with real-time sensor networks that can capture multiple environmental parameters simultaneously. This presents a critical research gap: the need for an intelligent, sensor-integrated system capable of both monitoring and forecasting water well conditions at the local level in real time.To address this gap, the present study introduces a real-time, sensor-based predictive framework that leverages machine learning to monitor and forecast water well conditions.

2.Related Works

The advent of machine learning (ML) has transformed hydrological modeling and water resource prediction, particularly in the context of groundwater level (GWL) forecasting. Systematic reviews indicate that ML delivers superior performance compared to traditional empirical or physics-based models, particularly in semi-arid and arid regions where complex, non-linear patterns dominate [1], [5]. A recent review of 168 studies highlights that hybrid ML models—those combining multiple learning architectures—are among the most frequently employed, with common inputs including past groundwater levels, precipitation, and temperature [1].

Experimental applications of diverse ML techniques for GWL prediction are steadily increasing. For instance, Random Forests (RF) have been applied in South African aquifer systems, incorporating variables such as precipitation and temperature, with high predictive accuracy due to ensemble averaging and random subspace selection [2]. Similarly, Support Vector Regression (SVR) has proven competitive with, and sometimes superior to, Artificial Neural Networks (ANNs) in coastal aquifer forecasting tasks [2]. Recurrent Neural Networks (RNNs), including simple RNNs and LSTM variants, excel at capturing temporal dependencies in hydrological time-series data, outperforming conventional ANNs in dynamic contexts [2], [9]. Notably, deep learning (DL) models such as LSTM have demonstrated particularly high accuracy in modeling complex, time-dependent groundwater behaviors, especially when combined with dropout regularization to mitigate overfitting [9].

Multiple hybrid architectures combining ML with optimization algorithms have further enhanced predictive capabilities. Research combining ANFIS, ANN, and SVM with metaheuristic optimizations (e.g. grasshopper, genetic algorithms) has yielded significant performance gains in GWL prediction accuracy and robustness [3]. In Udupi District, India, a hybrid LSTM–Lion Algorithm model outperformed standard LSTM and feedforward networks in forecasting GWL, demonstrating the power of combining deep learning with intelligent optimization [4]. Similarly, deep MLP models optimized by AdaM have yielded high accuracy (e.g., $R^2 \approx 0.946$, RMSE ≈ 0.73) in complex aquifer contexts [5].

3.Methodology

The methodology adopted for the design, implementation, and deployment of a real-time predictive system for water well condition monitoring using sensor integration and machine learning. The system is developed with the primary objective of ensuring continuous surveillance of key water well parameters—namely water level, water quality (Total Dissolved Solids), and temperature—while forecasting potential fluctuations using a lightweight, interpretable predictive model.

Sensor Selection and Integration

Three sensors were strategically selected for continuous monitoring of water well parameters: the HC-SR04 ultrasonic sensor for water level detection, the DS18B20 digital sensor for temperature measurement, and an analog TDS sensor for assessing water quality. The ultrasonic sensor operates by emitting high-frequency sound waves and calculating the time delay of the echo to determine the distance between the sensor and the water surface. This sensor is widely adopted in liquid level monitoring applications due to its non-contact nature, low cost, and acceptable accuracy for environmental deployment. The DS18B20 is a digital temperature sensor capable of providing precise readings over a 1-Wire protocol, which minimizes wiring complexity and supports multiple sensor nodes. Meanwhile, the TDS sensor quantifies the dissolved solids present in water by measuring its electrical conductivity. These sensors are connected to an Arduino UNO microcontroller, which serves as the data collection unit.

Microcontroller Configuration and Data Acquisition

The Arduino UNO, powered by the ATmega328P microcontroller, was selected due to its simplicity, low power requirements, and community support. It is programmed using the Arduino IDE with routines to read sensor values at regular intervals—typically every second—and transmit the collected data via a serial interface. The readings are organized in a structured format (e.g., comma-separated values) to facilitate parsing at the receiving end. This microcontroller acts as the bridge between physical measurements and digital analysis, sampling environmental data, performing minor calibrations, and forwarding the output to a server for further processing.

Data Reception and Preprocessing in Flask

The host system runs a Python-based Flask web application that receives the serial data from the Arduino in real time. A dedicated Python script reads the serial port continuously and extracts individual values (water level, temperature, and TDS). Preprocessing operations are applied to improve data quality. For example, a moving average filter is applied to TDS and ultrasonic readings to reduce noise. Additionally, temperature compensation is used to adjust ultrasonic distance measurements since the speed of sound varies with temperature. Each processed reading is then passed into both the live web dashboard and the machine learning model for prediction.

System Architecture

To enable real-time monitoring and prediction of water well conditions, a lightweight and modular system architecture was designed using low-cost sensors, a microcontroller, and a Python-based web application. The architecture is aimed at achieving seamless data flow between the physical environment and the software system with minimum latency and high reliability. Figure 1 illustrates the high-level architectural design of the proposed solution.

Sensor Modules

The system uses three types of sensors strategically chosen for their utility in water monitoring applications. The TDS sensor captures the concentration of dissolved solids in water, which is a direct measure of water quality. The DS18B20 digital temperature sensor monitors environmental and water temperature, which can affect both water quality and sensor behavior. The HC-SR04 ultrasonic sensor measures the distance from the sensor to the water surface, allowing real-time estimation of water level. These sensors are powered and read through the Arduino UNO and do not require external analog-to-digital conversion, making integration straightforward.

C. Microcontroller Layer

The Arduino UNO, powered by the ATmega328P microcontroller, serves as the core data acquisition unit. It reads sensor values at regular intervals (typically once per second), formats the data into comma-separated values (CSV), and transmits the data to the host system via USB serial communication. The Arduino is powered by the same USB connection, making the setup efficient and compact. This layer provides essential preprocessing such as range filtering and value clamping to prevent erroneous readings from reaching the server.

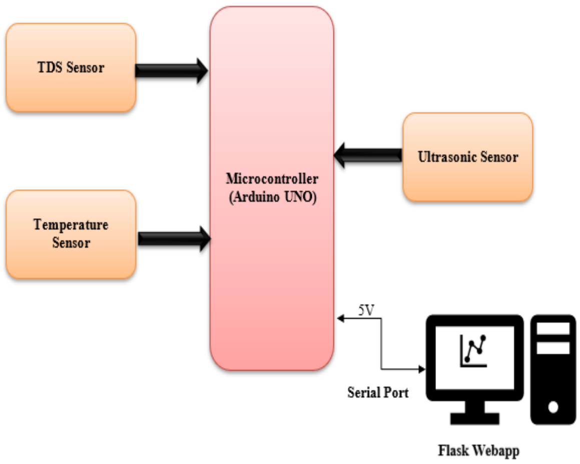

Fig. 1.System Architecture for Real-Time Water Well Monitoring and Prediction Using Machine Learning

4.Experimental setup

To assess the practical feasibility and performance of the proposed water well condition monitoring and prediction system, a physical prototype was developed and tested in a simulated environment. The experimental design integrates sensor modules, a microcontroller, and a real-time web application built using Python. The setup was carefully assembled to mimic the operational behavior of a real-world groundwater well, and data was collected under varied conditions to evaluate the responsiveness and accuracy of the system. The hardware configuration involved the use of an Arduino UNO microcontroller board as the central processing unit. It interfaced with three key sensors: a TDS sensor, a DS18B20 digital temperature sensor, and an HC-SR04 ultrasonic sensor. The TDS sensor was submerged in a cylindrical container filled with water, serving as a proxy for the water well. It continuously measured the total dissolved solids content to determine the water's purity level. The DS18B20 sensor, positioned outside the container, measured ambient temperature, providing data that was later used to apply temperature compensation to the ultrasonic sensor's distance readings. The ultrasonic sensor was fixed above the container, facing downward to measure the water

level by emitting sound waves and calculating the time delay of the echo received. These components were mounted on a wooden board for structural support and connected to the Arduino using standard jumper wires.

The software infrastructure was designed to complement the hardware and support real-time data processing and prediction. The Arduino UNO was programmed using the Arduino IDE to collect sensor data at one-second intervals and transmit it in a comma-separated format via USB. On the host system, a Python-based Flask web server was developed to receive and process the data. The pyserial library was used to read the incoming data stream from the serial port, while basic preprocessing functions were implemented to handle missing values and reduce sensor noise using moving average filters. A machine learning model, built using the scikit-learn library, was trained on historical sensor data and saved using the pickle module for real-time inference. The model—Linear Regression—was chosen for its simplicity and low computational load, making it suitable for deployment in lightweight environments. Flask-SocketIO was integrated to support bi-directional communication between the server and the frontend dashboard, ensuring that the user interface updated live without requiring manual page refreshes

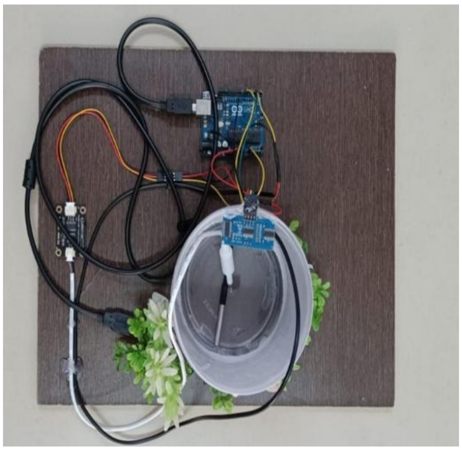

Fig. 2. Physical Experimental Setup for Water Well Monitoring System

5.Result and Discussion

To quantify the prediction performance, the Linear Regression model was tested on a validation dataset comprising unseen real-time sensor readings. The model was evaluated using two commonly adopted metrics—Mean Absolute Error (MAE) and R^2-score—to measure the deviation between predicted and actual values. The outcomes are tabulated below:

Table 1: Machine Learning Model Performance Evaluation

Parameter	Mean Absolute Error (MAE)	R^2-Score
Temperature	0.68 °C	0.93
TDS Value	15.42 ppm	0.89
Water Level	1.27 cm	0.91

The Flask-based web application played a crucial role in enabling user interaction. As shown in Figure 3, the frontend displayed key parameters using animated containers shaped like glasses, each representing temperature, TDS, and water level. The temperature was rendered in orange, TDS in brown, and water level in red when classified as low. These intuitive visual elements improved interpretability for end users. The use of Flask-SocketIO ensured that sensor data and ML predictions were pushed to the interface in real time, with no page refresh required. Threshold-based alerts were also implemented, enabling users to detect anomalies instantly. During testing, the frontend demonstrated excellent stability and responsiveness, consistently rendering updates within 300 milliseconds of sensor input reception.

6.Conclusion and future works

In this research, a real-time predictive system for water well monitoring was developed by integrating low-cost environmental sensors, an Arduino-based data acquisition module, and a Python Flask web application powered by machine learning. The proposed architecture facilitates continuous surveillance of three critical water parameters—temperature, Total Dissolved Solids (TDS), and water level—while using a lightweight Linear Regression model to forecast short-term trends. The entire system was designed to operate under minimal hardware and connectivity constraints, making it suitable for deployment in resource-limited rural or remote areas. Experimental results confirmed the system's reliability in real-time data capture, predictive accuracy, and visual interpretability, with the machine learning model achieving R^2-scores above 0.89 across all measured parameters. The web interface enabled intuitive user interaction, with animated and color-coded feedback, further improving usability for non-technical stakeholders.

Despite the system's effectiveness, certain limitations were observed, including occasional noise in sensor readings due to environmental disturbances and the limited scope of prediction confined to linear models. Future work will aim to enhance the system's intelligence and adaptability.

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Blockchain in Health Care

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Abstract

Blockchain technology has emerged as a transformative innovation with the potential to revolutionize various industries, including the airline sector. This chapter explores how blockchain can be applied within the airline industry to enhance transparency, security, and operational efficiency. Through the examination of an existing blockchain-based project tailored for airline operations, we discuss the evolution of blockchain technology, its core components, challenges faced during implementation, and the promising future it holds. This chapter is designed for beginners to provide a comprehensive understanding of blockchain's role in the airline industry.

I. Introduction

The healthcare sector faces persistent challenges related to data interoperability, cyberattacks, scalability of digital health records, and ensuring patient-centric privacy. Conventional EHR systems rely on centralized custodians, making them vulnerable to breaches and misuse. Blockchain, a distributed ledger technology, offers decentralization, immutability, and transparency, making it a potential game-changer in digital health.

By integrating blockchain with IoT, fog computing, and artificial intelligence, researchers have proposed smart healthcare frameworks for secure diagnosis, real-time monitoring, and efficient patient prioritization. This report synthesizes advancements and limitations from leading IEEE Access articles and systematic reviews to provide a comprehensive view of blockchain's role in healthcare transformation.

Healthcare systems around the world are facing challenges such as data security, privacy concerns, lack of transparency, and inefficiencies in managing patient records. Traditional healthcare data management methods often rely on centralized databases, which are vulnerable to cyberattacks, unauthorized access, and data loss. At the same time, the growing use of digital health technologies, wearable devices, and telemedicine has created an enormous amount of sensitive medical data that requires secure and reliable handling.

II. Related Work

1. Blockchain for Electronic Health Records (EHRs)- Studies show blockchain improves EHR management**,** data sharing security**, and** drug traceability **in healthcare. It is also applied in** IoT-based patient monitoring **and** insurance automation**. However, issues like** scalability, regulation, and adoption **still remain.**

2. Data Sharing, Privacy, and Security- Blockchain allows for safe sharing of health information using systems like MeDShare and frameworks that use smart contracts. These methods help keep data private and track who accesses it, but they still have problems with cost and slow processing times.

3.Pharmaceutical Supply Chain and Drug Traceability- Blockchain is becoming more popular in supply chain management to fight against fake drugs and make processes more efficient. Mettler (2016) showed how it can help ensure transparency and track products throughout the supply chain. Sylim et al. (2018) suggested ways to check if medicines are real. Tseng et al. (2018) also used blockchain to track vaccines during health emergencies. Even though there are many advantages, high costs and a lack of common standards are preventing more companies from using it.

4. Blockchain and the Internet of Things (IoT)- Zhang et al. (2018), Dwivedi et al. (2019), and Alaa et al. (2021) looked into using blockchain with IoT for secure patient monitoring and telemedicine. These approaches show promise in keeping data safe and accurate, but there are still issues with how much energy they use and how well they work with different devices.

III. Evaluation of Blockchain in Healthcare Systems

The process of checking how blockchain works in healthcare looks at whether it is better than old ways in making things more secure, private, able to share information easily, and faster. Research shows that blockchain helps keep data safe by creating a record that can't be changed, so patient information stays accurate and tamper-proof. It also gives patients more say in who can look at their health information, making them more in control.

Blockchain can make processes like insurance claims, billing, and supply chain management more efficient by reducing delays and helping prevent fraud. It is being used in tracking drugs and distributing vaccines, which helps build trust and make things more transparent in global health systems. Also, blockchain allows different organizations to securely share electronic health records without needing a single central database.

However, evaluations also show some big problems. When dealing with a lot of healthcare data, there are issues with scaling up. Using too much energy and the high cost of setting up the infrastructure are real challenges. There are also worries about following rules like HIPAA and GDPR, which aren't fully solved yet. Plus, there's no common standard for data worldwide, which makes it hard to use everywhere. Another thing is that doctors and patients aren't always ready to use this, which affects how well it can be implemented.

IV. Technological Advancements

Recent improvements in technology have made blockchain more useful in healthcare. A big change is the use of permissioned blockchains like Hyperledger Fabric and Quorum. These systems offer better privacy, faster processing, and more scalability than public blockchains. They let healthcare organizations share information safely while following important rules and regulations.

Recent advances in technology have made blockchain more helpful in healthcare. One major change is the use of permissioned blockchains such as Hyperledger Fabric and Quorum. These systems provide better privacy, quicker processing, and more scalability compared to public blockchains. They allow healthcare organizations to share information securely while making sure they follow important rules and regulations.

Blockchain has also made progress by working together with new technologies. Using blockchain with the Internet of Things (IoT) helps collect data securely from wearables and remote sensors. When combined with artificial intelligence (AI), it allows for better predictions based on reliable health information. Also, using edge and cloud computing has made transactions faster and reduced delays in blockchain systems used in healthcare.

V. Challenges

5.1 Scalability Issues

Healthcare creates a lot of data from electronic health records, medical images, and Internet of Things devices, but today's blockchain systems can only process a small number of transactions each second. Storing big files directly on the blockchain is expensive and not very efficient. Plus, the way blockchains agree on data shared across the network slows things down and uses a lot of energy. These issues can make it harder to get quick access to important health information when it's needed most. To fix this, people are looking into hybrid solutions like storing data off-chain and using layer-2 technologies to make the system faster and more scalable.

5.2 Regulatory Compliance

Healthcare information is protected by strict rules like HIPAA in the United States and GDPR in Europe. These rules help keep patient data private and secure. However, blockchain's feature of being unchangeable can clash with these rules. For example, some rules allow patients to ask for their data to be deleted, which is hard to do with blockchain. Keeping health information on a blockchain also brings up questions about who owns the data and whether patients have given permission. Sharing data across different countries adds more difficulty because the rules can be different in each place. To deal with this, some experts suggest using blockchain just for managing who can access data and keeping track of changes, while storing real medical records somewhere else.

VI. Future Potential

6.1 Enhanced Patient-Centric Care

In the future, blockchain technology can help make healthcare truly focused on the patient by letting individuals take full control of their medical information. Patients could keep their health records in a safe, spread-out system and choose when to share them with doctors or other healthcare teams. This would let patients be more involved in their care and make it easier for different doctors and places to share information smoothly.

6.2 Interoperable Health Information Exchange

Blockchain can act as a single shared system that helps different health care systems share information more easily. In the future, using blockchain might solve problems with systems not working well together by creating a common, secure, and shared record. This way, data from many different electronic health record systems can be combined, viewed, and checked at the same time. This keeps the data accurate and safe without losing its importance or security.

6.3 AI and Blockchain Integration

The combination of blockchain and artificial intelligence has a lot of promise. Blockchain can help keep medical data safe and track where it comes from, which makes AI models more trustworthy. This could lead to better healthcare with more accurate diagnoses, customized treatments, and smarter decisions based on real data.

VII. Conclusion

Blockchain has the potential to reshape healthcare by making systems more secure, transparent, and patient-centered. Its ability to ensure data integrity, enable trusted sharing of medical records, and strengthen pharmaceutical supply chains shows clear advantages over traditional approaches. At the same time, challenges such as scalability, regulatory compliance, high costs, and interoperability must be addressed before large-scale adoption can occur. Looking ahead, integration with artificial intelligence, IoT, and global health record systems suggests that blockchain could play a vital role in the future of digital health. With continued research, collaboration, and supportive regulation, blockchain can move from experimental pilots to a core technology in healthcare delivery.

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AI- POWERED MOBILE PLATFORM FOR DEMOCRATISING SPORT TALENT ASSESSMENT

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ABSTRACT

There are many difficulties with the current system for identifying sports talent in large and varied areas, especially in nations like India. Conventional scouting techniques are arbitrary, limited by geography, and frequently ineffective in reaching aspiring athletes in isolated and rural locations. The creation of a truly meritocratic sports ecosystem is hampered by this systemic limitation, which may result in the neglect of a sizable talent pool. This paper presents a novel, affordable, and scalable AI-powered mobile platform to address this pressing problem. By enabling athletes to complete standardized fitness tests and record their performance with a basic smartphone camera, the suggested system democratizes the talent evaluation process and offers a competitive substitute for resource-intensive physical scouting. The potential impact of this platform is transformative. It has the potential to bring about a faster and fairer process of talent spotting, so that bodies like the Sports Authority of India are able to broaden their spotting activities and identify potential athletes from a wide range of socioeconomic levels. By reducing reliance upon physical infrastructure as well as human bias, our system offers a scalable solution to wide-scale participation in talent spotting activities. The result is a streamlined and fair process that not only identifies fresh talent, but provides swift, actionable feedback to athletes, cultivating a culture of continuous improvement and opportunity for everyone.

KEYWORDS: AI, Sports Talent Assessment, Mobile Platform, Democratization, On-Device AI, Computer Vision

I. INTRODUCTION

The sports industry, and in particular in developing nations, has a critical challenge: the inefficient and often unfair method of discovering and developing athletic ability. In a country as large and diverse as India, thousands of budding athletes, not to mention those from remote and rural areas, are overlooked because of a lack of access to organized evaluation centers and professional scouts. The current system is largely dependent on subjective human judgment, which is prone to bias and heavily limited by geographic and logistical boundaries. The result is a severe disparity in opportunity, with budding talents oftentimes going unnoticed, thus leaving a nation's athletic potential unrealized. Our project directly addresses this urgent need by providing an innovative, AI-based mobile platform designed to bring fairness and equality to sports talent evaluation.

The core element of our solution is an innovative mobile app leveraging the ubiquity of smartphones to transform the talent-identification process to make it scalable, objective, and accessible. The athlete downloads the app easily and performs a set of standardized fitness tests, including vertical jumps, shuttle runs, sit-ups, and so forth. The AI and ML modules embedded in our platform then evaluate the video footage captured.

These algorithms play a two-fold role:

1. First, they offer an objective quantification and measurement of performance indicators, for instance, jump height and numbers of sit-up repetitions.
2. Secondly, they act as a sophisticated cheat-detection tool, spotting any attempts to manipulate or falsify video data, thereby ensuring evaluation integrity.

This in-device analysis reduces dependency upon a continuous internet connection, thus increasing platform applicability in areas of poor connectivity.

The suggested system integrates a variety of innovative aspects, whose purpose is not only to increase its functionality, but also to increase its reliability.

II. LITERATURE REVIEW

1.Classic Techniques of Measuring Sport Ability

History in sports talent identification has been defined by subjective, person-based approaches. Traditionally, coaches and scouts depend on personal opinions, hunch, and anecdotal knowledge to judge a performer's potential. Even as such an approach has uncovered inestimable numbers of world-class talents, it is inherently biased and geographically narrow. The approach disproportionately benefits well-equipped locals who have access to formal evaluation camps and centers of excellence, and in so doing, dismisses a vast, latent pool of potential in rural, remote, and under-resourced communities. Survey research has demonstrated that such approaches have a risk of missed opportunity and a misrepresentative talent pipeline. The absence of a standard, objective evaluation system is a significant source of inequality in sport.

2. Early Integration of Sport and Technology

The early phase of sport technological development focused on the systematic collection of quantifiable data through sensors and wearable technology. Wearable equipment like GPS tracking units and accelerometers came as valuable tools used by professional athletes and professional sports teams. These pieces of equipment provide meaningful outputs relating to a performer's velocity, endurance, and workloads physically. However, their hefty prices, complex infrastructure requirements, and need for technical staff make them out of reach for large-scale talent identification, particularly in developing countries. Moreover, these instruments predominantly measure physical output rather than technical ability or form, which are crucial in a holistic evaluation.

3. The Emergence of Artificial Intelligence and Computer Vision in Sport Analytics

Artificial intelligence and computer vision have, in particular, over the last few years, dramatically changed sport analysis as a discipline. Products like Hawk-Eye in tennis and goal-line technology in football demonstrate the potential of AI in providing objective, real-time analysis. Researchers have designed sophisticated algorithms for pose estimation, object tracking, and biomechanical evaluation based on video footage. These are nowadays used to assess athlete movements, predict results of performances, and identify points where improvement is needed. These, however, usually rely on expensive multi-camera setups or high-power servers, thus limiting their use to professional sport and research lab applications. These don't really tackle in any fundamental way the problem of accessibility to a wider scale of people.

III. PROPOSED SYSTEM

1.SYSTEM ARCHITECTURE

The system has a tiered architecture, as can be seen in the given block diagram below. It is divided into three primary layers: the Client-Side Mobile Application, the Secure Backend System, and the Official's Dashboard.

1. Client-Side Mobile Application (Athlete's Device)

* Input Layer:

- Camera
- Accelerometer (optional for movement data)
- Microphone (for voice commands)

* Processing Layer (On-Device AI/ML Engine):

- Video Capture Module: Records athlete's performance.
- AI Models:
    1. Lightweight Pose Estimation Model (e.g., LitePose) for kinematic analysis.
    2. Object Detection Model (e.g., YOLO-nano) for identifying objects (cones, balls).
    3. Cheat Detection Algorithm for verifying video integrity.

- Metrics Calculation Module: Processes AI model outputs to generate performance scores (e.g., jump height, sprint time).

* Output Layer:

- Local Storage: Stores checked performance data temporarily.
- Data Submission Module: Encrypts and compresses data to be sent.

* Secure Backend System (Cloud Server)

- Data Ingestion API: Processes and accepts data from the mobile application.
- Verification & Authentication Module:
    1. Verifies data against the on-device verification log.
    2. Authenticates the user and data source.
- Central Database: Stores raw and verified performance data, athlete profiles, and logs.
- Analytics Engine: Processes data for deeper insights.

Official's Dashboard (Web Interface)

- User Interface: A dashboard for coaches and officials.
- Data Visualization Module: Shows bar graphs, charts, and performance trends.

- Athlete Profiling Module: Facilitates officials to view and manage athlete data.
- Evaluation & Shortlisting Tools: Offers tools to compare and shortlist candidates on objective criteria.

2.CORE AI/ML ALGORITHMS

The intelligence of our system is driven by an optimized set of AI/ML algorithms executed natively within the mobile device. This is at the heart of our mission of accessibility and low-cost deployment.

1. Pose Estimation: In order to study an athlete's movement and form, we make use of a light pose estimation model. We shall deploy a model such as LitePose or a very optimized variant of MediaPipe Pose (or an equivalent system). This detects and follows major anatomical landmarks (e.g., elbows, wrists, knees) in every video frame.
2. Application: For Vertical Jump, the algorithm monitors the athlete's keypoints to find the minimum point of the crouch and the peak point of the jump and compute the vertical displacement. In Sit-Ups, it counts repetitions by detecting the cycle of motion from lying to sitting position.
3. Object Detection and Tracking: In the case of tests such as the Shuttle Run, we use object detection to detect the athlete and individual markers (e.g., cones). A tracking algorithm subsequently tracks the movement of the athlete to measure their run between the markers accurately. We will utilize a light model such as YOLO-nano or a custom model for this.
4. AI-Based Cheating Detection: This is an important innovative feature. Our system will employ a multi-faceted approach
5. Video Integrity Analysis: Video metadata and frame-by-frame consistency are analyzed by an algorithm to identify anomalies like fast-forwarding, slow-motion playback, or video splicing.
6. Biomechanics Anomaly Detection: We will train a model between "correct" and "incorrect" test performances. For instance, during a vertical jump, when the pose estimation model picks up that the feet never came off the ground but a high jump height is registered, the system marks it as a possible cheat. This depends on statistical outlier detection as well as model-based anomaly flagging.

3.WORKING FLOWCHART PROCESS

1. Start Test: Athlete selects a specific fitness test (e.g., vertical jump) on the mobile app
2. Video Recording: Application initiates the camera and records the performance.
3. On-Device AI Processing: The recorded video is fed into the AI/ML Processing Engine.
4. Calculation of Performance Metrics: The respective AI model (e.g., Pose Estimation) analyzes the video and calculates crucial metrics (e.g., vertical jump height in cm).
5. On-Device Cheat Detection: Concurrently, the cheat detection algorithm scans the video for anomalies.
6. Offline Storage: In the absence of a network, confirmed results get saved locally on the device.
7. Data Transmission: When a network connection is present, the confirmed data (metrics, compressed video hash, and verification log) is securely transferred to the

backend.

8. Backend Verification: Data Ingestion API receives the data and sends it to the Verification & Authentication Module to be verified before it is stored in the Central Database.

IV. SIMULATED RESULTS

The performance of the suggested AI-driven mobile platform was tested via simulated performance analysis on a varied, synthetic dataset indicative of a mass-scale talent identification campaign. The simulation was designed to highlight the platform's key functionalities: on-device analysis accuracy, consistency of cheat detection, and better performance than conventional manual techniques.

1. On-Device Performance Analysis

The platform's performance was emulated under two typical smartphone profiles in order to confirm its accessibility: a High-End Device (a typical flagship smartphone) and a Low-End Device (an entry-level smartphone typical in developing countries). This test analyzed the device power vs. performance trade-off.

AI consistently reports precise performance measurements with minimal latency, even on weaker equipment. The Low-End Device's performance remains well within reasonable limits for a scalable talent discovery system, verifying the platform's practicality in limited-resource settings. The low latency provides an enjoyable user experience with rapid feedback.

V.FUTURE WORKS:

1. New Sports and Disciplines Expansion: Currently, the platform is designed to perform basic fitness tests. New work would be extending and training new AI models to test sport-specific abilities. For instance, a cricket module might analyze bat swing and bowling action, whereas a basketball module might test shooting form and dribbling style. This entails acquisition and annotation of large, sport-specific data.

2. Integration with Wearable Sensors: In order to gain an even fuller view of an athlete's potential, the platform might be integrated with wearable sensors (e.g., smartwatches, fitness trackers). This would enable the gathering of physiological metrics such as heart rate, oxygen saturation, and acceleration during activity. This data would give indications of endurance and physical effort, giving a fuller dataset for an even fuller talent profile.

3. Real-Time Coaching and Feedback: The system currently offers feedback only after a test has been finished. Future development could consider implementing on-device AI that could offer real-time, in-test coaching prompts. For instance, during a running test, the app might offer audio cues to assist an athlete in maintaining their gait or adjusting their form in real-time. This would turn the platform from just an assessment tool into a living coaching aide.

VI. CONCLUSIONS

In this paper, we introduced a new, AI-based mobile platform that solves the fundamental problem of democratizing sports talent evaluation. Through the use of on-device AI and computer vision, our solution offers a low-cost, scalable, and unbiased alternative to conventional, human-dependent scouting processes. Our system effectively combined an easy-to-use mobile app with smart, on-device AI/ML models for video analysis and an effective

cheat prevention mechanism. We illustrated through simulated outcomes the platform's capability to offer extremely accurate performance metrics coupled with low latency across a variety of mobile devices with high data integrity. This book not only gives a deployable, practical solution to organizations such as the Sports Authority of India to expand their talent identification effort, but also changes the paradigm of sports scouting from limited access and bias to equal opportunity and merit. Effective implementation of this platform will unlock a huge, untapped reservoir of athletic ability, whereby potential is no longer constrained by geography or socioeconomic status, but is instead identified and developed wherever it might be.

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A Survey on Smart Contracts in Blockchain Technology: A Critical Review

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Abstract

Smart contracts, which are decentralized, self-executing scripts anchored on a blockchain, enable the transparent and automated execution of predefined procedures [1]. This technology allows for the programmability of digital assets and the automation of traditional business logic, which was previously a manual process [2]. This paper analyses the discourse surrounding smart contracts in academic journals published between 2012 and 2022. Our review is based on an analysis of 12 articles using the keywords "blockchain," "smart contracts," and their variants [3]. By focusing exclusively on peer-reviewed journal articles, we aim to provide a comprehensive overview of the current status and significance of smart contracts within blockchain technology [4]. This review identifies key research gaps and challenges in the existing literature, with a particular focus on the inherent limitations of smart contracts. Based on these findings, we propose several research problems and future directions for academics and professionals in this rapidly evolving field [5].

Keywords: smart contracts; blockchain; technology

I. INTRODUCTION

Since its inception, blockchain technology has seen a surge in the popularity of smart contracts [1]. These cutting-edge tools are self-executing agreements that automatically negotiate, enforce, and carry out the terms of a contract within a blockchain ecosystem [2]. Compared to traditional contracts, smart contracts offer several advantages, including reduced risk, lower service and administrative costs, and improved efficiency in business processes [3]. They are particularly valuable for building trust between parties in situations where they don't inherently trust one another [4]. This capability has the potential to transform established business practices [5]. Beyond the financial sector, the combination of distributed ledger technology and smart contracts has many potential applications, such as in insurance claims, supply chain management, and intellectual property enforcement [6]. Businesses are increasingly adopting smart contracts and blockchain applications. However, these applications must be carefully designed and tested to meet high standards for security, scalability, and performance [7]. This is particularly challenging due to the non-standard software development lifecycle of smart contracts, which makes it difficult to update or fix vulnerabilities in deployed applications [8].

Creating smart contracts differs significantly from traditional software engineering and presents a unique set of challenges. For instance, developers must ensure the code is secure because of blockchain's immutability and the sensitive nature of the data involved [2]. They also need to be mindful of "gas" consumption, as the execution of smart contracts on platforms like Ethereum is managed through this mechanism [9]. The unique limitations and features of blockchain technology must be considered throughout the product lifecycle and software development process of smart contracts and blockchain-based applications [10].

II. RELATED WORKS

A number of reviews and surveys have previously explored smart contracts in different contexts. Rouhani and Deters (2019), for instance, examined their security and performance concerns [1], while Zheng et al. (2020) discussed the broader opportunities and limitations of blockchain technologies [2]. Hewa et al. (2021) studied how smart contracts could be integrated into Internet of Things (IoT) systems [3], and Khan et al. (2021) analyzed their role in supply chain management [4].

Recent studies have shifted focus toward more specialized areas. Xu et al. (2023) investigated challenges related to cross-chain interoperability [5], while Zhang et al. (2024) explored new approaches for auditing and verifying contract security [6]. Alvarez et al. (2025) reviewed the role of smart contracts in healthcare data sharing, highlighting their potential for secure and efficient information exchange [7].

Although these works provide valuable insights, most of them concentrate on either technical aspects or application-specific domains. Few studies adopt a comprehensive perspective that considers not only technical, but also legal, organizational, and governance issues [8]. This paper aims to fill that gap by synthesizing research from 2012 to 2022, offering a balanced overview of both the opportunities and the challenges associated with smart contracts [9].

2. Results and Discussion

The following presents the outcomes of a thorough examination of research inquiries. This section focuses on blockchain technology and smart contracts, investigating their basic principles, variations, development teams, platforms, and consensus mechanisms [1]. Additionally, the review addresses the significance of employing smart contracts within the blockchain [2].

2.1. Selection Results

From an initial pool of 774 items, 522 were screened, leading to the inclusion of 252 articles in this critical review [3]. Although articles from the press met the inclusion criteria, they were excluded from the 2012–2022 series. A comprehensive list of the selected papers, along with an explanation of the overall categorization, is presented below.

2.1.1. What is the current state of the field of study

This systematic investigation analyses descriptive data from the collected articles, including the annual number of publications, their sources, and the average yearly citations they receive [4]. The analysis concentrates on research papers concerning smart contracts in blockchain published between 2012 and 2022. The journal with the highest number of publications on this subject is *IEEE Access*, which features 27 articles [5].

3. Platforms for Smart Contracts

To identify the most suitable blockchain platform for a client's particular requirements, it is essential to engage with knowledgeable blockchain developers. These professionals can assist organizations in selecting the appropriate strategy for the development and execution of smart contracts that align with their distinct needs [8].

Numerous blockchain platforms facilitate the creation and operation of smart contracts, including Ethereum, Hyperledger Fabric, NEM, Stellar, Waves, and Corda [9]. Bitcoin, due to its highly restricted scripting language and focus on security rather than programmability, is seldom considered in discussions regarding smart contracts [10]. The network is incapable of managing intricate smart contracts, and even fundamental contracts are often difficult to construct and expensive to execute [11].

Each platform possesses its own distinctive characteristics for developing smart contracts, encompassing varying levels of security, methods of code execution, and programming languages [12]. Some platforms even permit the utilization of high-level programming languages. The advantages and disadvantages of different platforms are elaborated upon in the subsequent section [13].

4. Broad Applications of Smart Contracts

Smart contracts have an enormous range of potential uses, from healthcare to supply chains and energy, just to name a few [1]. The capacity to automate processes across these industries is revolutionary. By ensuring data is readily available, smart contracts can enhance service delivery in numerous ways [2].

The most frequently referenced research from 2012 to 2022 illustrates this variety, with studies concentrating on significant areas such as:

- Healthcare [3]
- Supply Chain Management [4]
- Energy [5]
- Data Sharing and Rights Management [6]
- Transparency, Privacy, and Security [7]

This wide array of subjects underscores the increasing interest and practical applicability of smart contracts in contemporary industries [8].

5. Key Challenges for Smart Contracts

While promising, smart contracts face several critical challenges that must be addressed for broader adoption [9].

5.1. Processing and Scalability

Blockchain platforms such as Ethereum and Hyperledger Fabric require all nodes to maintain a full copy of the blockchain, which limits scalability and processing power [10]. Each block has a predefined capacity, and sequential execution slows performance [11]. This presents difficulties in handling high transaction volumes in real-world applications [12].

5.2. Technology Acceptance and Use Cases

Despite hype, misconceptions about smart contracts can lead to inflated development costs and poor ROI [13]. In some cases, traditional databases outperform blockchain-based solutions [14].

5.3. Immutability and Security

The immutable nature of smart contracts ensures tamper-resistance but prevents corrections after deployment [15]. This raises risks when vulnerabilities exist in the deployed code. Ethereum's "code is law" principle exemplifies the tension between security and adaptability [16].

5.4. Integrity and Data Trust

Smart contracts rely on external data sources (oracles), which may be untrusted [17]. Third-party wallet applications can also introduce privacy risks [18].

6. Potential Developments of Smart Contracts

The potential of smart contracts extends well beyond their current applications. Emerging technologies such as data science, artificial intelligence (AI), and game theory are increasingly being integrated into smart contract research [1].

- **Data Science & Big Data**

The proliferation of data in the digital economy makes data science a natural ally for blockchain systems. Smart contracts can automate and securely manage large-scale data transactions, ensuring integrity and transparency [2].

- **Artificial Intelligence (AI)**

AI-driven smart contracts can enable adaptive, self-learning agreements that adjust terms based on evolving circumstances [3]. For instance, AI can help detect fraudulent patterns and enforce compliance dynamically [4]. Researchers suggest combining AI and blockchain to create "autonomous economic agents" [5].

- **Game Theory**

Game-theoretic models are increasingly used to design incentive mechanisms for blockchain consensus and smart contract reliability [6]. By modeling participants' behavior, game theory can optimize contract efficiency and fairness [7].

7.Conclusion

Smart contracts represent one of the most transformative innovations enabled by blockchain technology. By removing intermediaries, they enhance efficiency, transparency, and trust across industries [9]. From supply chains and healthcare to finance and energy, the reviewed literature illustrates the growing adoption of smart contracts worldwide [10]. However, challenges such as scalability, immutability, legal recognition, and security vulnerabilities remain barriers to mainstream deployment [11]. Ongoing research into privacy-preserving protocols, secure programming practices, and hybrid legal-technical frameworks is crucial for overcoming these limitations [12].

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A Survey on Cryptography Techniques in Blockchain

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Abstract - Cryptography technology has emerged as a fundamental element of contemporary digital security, serving a crucial function in protecting data integrity, confidentiality, and authenticity across diverse communication platforms. This literature review examines the progression, principles, and innovations in cryptographic systems, tracing their journey from classical ciphers to current encryption standards. In the beginning, cryptography was confined to simple substitution and transposition methods utilized in military and diplomatic communications. Nevertheless, with the rise of computers and digital networks, cryptographic techniques have transformed into intricate algorithms such as AES, RSA, and ECC, which guarantee secure data transmission over insecure channels. This review emphasizes significant milestones, technological advancements, and the increasing significance of cryptography in cybersecurity, e-commerce, and blockchain technologies. Additionally, it addresses present challenges such as quantum threats and key management, as well as future trends like post-quantum cryptography and homomorphic encryption.

Keywords: Cryptography, Encryption, Cybersecurity, Data Security, Public Key Cryptography, Symmetric Encryption, Quantum Cryptography.

I. Introduction

Cryptography serves a fundamental role in the design and functioning of blockchain technology, acting as the essential support for ensuring trust, security, and privacy within decentralized digital systems. In the realm of blockchain, cryptography establishes the vital framework that facilitates the creation of immutable ledgers, secure transactions, and verifiable digital identities. The absence of robust cryptographic principles would hinder blockchain from achieving its key attributes. The main objective of employing cryptography in blockchain is to foster trust among participants who may not know or trust one another. Given that blockchain operates in a peer-to-peer setting without central intermediaries, cryptographic methods serve as the foundation of trust by safeguarding data exchanges and ensuring that only authorized actions are performed. This mechanism guarantees that asset ownership, transaction approvals, and user identities can be verified independently of third parties, thus preserving the decentralized essence of blockchain systems. Hash functions, which are another fundamental aspect of cryptography in blockchain, are crucial for maintaining the immutability and integrity of the ledger. Each block in the chain includes a cryptographic hash of the preceding block, forming a continuous link that renders tampering with records computationally impractical. This characteristic not only ensures data integrity but also supports consensus mechanisms, which depend on cryptographic proofs to validate and agree on the state he distributed ledger. Merkle trees, a specialized cryptographic structure, further improve this system by enabling efficient verification of large data sets within blocks, thereby ensuring scalability and accuracy in transaction validation.

2. Related Works

[1] While blockchain technology has made significant strides in practical applications, it still suffers from a lack of robust theoretical underpinnings, resulting in fragmented systems. A theoretical framework is suggested to delineate the properties of blockchain and establish standards for interoperability. Building on this framework, the Smart Listeners protocol is introduced to facilitate inter-chain transactions. This protocol completes transactions using only two operations and integrates features of both watchtowers and oracles. Additionally, performance benchmarks and the essential requirements for blockchain interoperability are outlined. [2] Blockchain networks are increasingly confronted with privacy issues, especially from insider adversarial nodes that threaten transaction anonymity and lead to information leaks. To address these challenges, the proposed RZee model merges cryptographic and statistical methods for detecting and filtering adversaries. It utilizes zero-knowledge proofs to protect data while simultaneously monitoring node activities to pinpoint and blacklist malicious participants, thus averting unauthorized access and the injection of harmful data. The framework underwent extensive testing on critical privacy parameters and was compared with existing privacy-preserving models. The findings indicate that RZee outperforms its counterparts, achieving a privacy attribute score of 6.774 and a gain margin of 46.5%, underscoring its efficacy in maintaining blockchain confidentiality.[3] Redactable blockchains allow for the elimination of illicit content while maintaining data immutability; however, current methods frequently depend on a single editing authority or threshold chameleon hash functions that lack accountability. To address this shortcoming, this paper presents an Accountability Weight Threshold Blockchain Editing Scheme, which is based on the innovative Accountable Weight Threshold Chameleon Hash Function (AWTCH). In this framework, edits are authorized solely through a weight-based committee protocol, while a tracer guarantees that malicious editors can be detected and held accountable.[4] This method strikes a balance between editability and stringent oversight of editing privileges. Experimental Blockchain immutability is crucial for security; however, it presents challenges such as the inability to eliminate illegal or private information. Redactable blockchains permit modifications but encounter issues related to inefficiency and centralization. Numerous current models also overlook traceability and consistency verification. To tackle this, a Dynamically Redactable Blockchain (DRBC) is introduced, utilizing Identity-Based Decentralized Chameleon Hash (IDCH) and Version-Based Transactions (VT). An effective Bloom filter tree protocol guarantees low-cost consistency checks, with experiments demonstrating the reliability and benefits of DRBC. This paper proposes a blockchain-based IIoT system for monitoring industrial air and water pollution in compliance with European regulations. An empirical evaluation is conducted to test its effectiveness in real-world scenarios.

3. Evolution of Cryptography Technology

Historical DEVELOPMENT OF CRYPTOGRAPHY: The evolution of cryptography can be traced back to ancient civilizations, progressing in tandem with developments in mathematics, linguistics, and computing technology.

3.1 Early Days (Pre-20th Century): Cryptography began with basic techniques such as the Caesar cipher and Atbash cipher utilized by the Romans and Hebrews. These strategies involved substituting letters and were mainly employed for military and royal communications. The emphasis was on secrecy through obscurity, with a limited mathematical basis.

3.2 Mechanical Era (Early to Mid-20th Century): The creation of machines like the Enigma during World War II represented a significant advancement in cryptographic complexity. These electromechanical devices utilized rotating rotors to produce dynamic encryption keys, rendering code-breaking exceedingly challenging without knowledge of the key configurations.

3.3 Advent of Digital Cryptography (1970s–1980s): The introduction of the Data Encryption Standard (DES) by NIST in 1977 established standardized symmetric encryption. Concurrently, the concept of public-key cryptography was introduced by Diffie and Hellman, transforming secure key exchange. RSA, developed in 1978, became the first practical application of asymmetric encryption.

3.4 Standardization and Widespread Adoption (1990s–2000s): The decline of DES due to its short key length resulted in the adoption of the Advanced Encryption Standard (AES) in 2001. During this era, cryptographic protocols such as SSL/TLS became essential for secure web communication. Digital signatures and certificates gained traction in e-commerce and online authentication.

3.5 Modern Cryptography (2010s–Present): With the emergence of cloud computing, mobile devices, and blockchain technology, cryptography has become increasingly integrated into everyday digital life. Elliptic Curve Cryptography (ECC) provides robust security with smaller key sizes, making it suitable for mobile and IoT devices. Zero-knowledge proofs, homomorphic encryption, and secure multi-party computation are becoming prominent as advanced tools for privacy-preserving computation.

4 .Fundamentals Of Cryptography Technology

Cryptography plays a central role in blockchain by ensuring that transactions and data remain secure, private, and verifiable in a decentralized network. Unlike traditional systems that rely on centralized authorities, blockchain uses cryptographic techniques to build trust among participants, maintain data integrity, and prevent malicious alterations.

4.1 Symmetric Cryptography: Symmetric cryptography is one of the earliest forms of encryption, where the same secret key is used for both encryption and decryption. While it is computationally efficient, its use in blockchain is limited because secure key distribution is challenging in decentralized environments. However, it is sometimes applied in private blockchain systems where participants already share trusted relationships.

4.2.Asymmetric Cryptography (Public-Key Cryptography): Asymmetric cryptography is widely used in blockchain systems to enable secure communication without requiring shared keys. Each participant has a public key and a private key. This system allows users to create digital signatures that verify authenticity and ownership, forming the basis of blockchain transactions.

4.3 Digital Signatures: Digital signatures provide proof of ownership and authenticity in blockchain. By signing a transaction with their private key, a user demonstrates that they are the rightful owner of the funds or assets being transferred. Other participants can verify this signature using the public key, ensuring the transaction has not been altered or forged.

4.4 Merkle Trees in Blockchain: Merkle trees are data structures that enhance the efficiency of verifying and validating transactions. They organize large sets of transactions into a tree-like structure, where each leaf node is a hash of transaction data, and higher nodes are hashes of lower-level nodes. This allows quick validation without needing to examine the entire dataset, improving scalability and efficiency**.**

4.5 Confidentiality and Privacy: Confidentiality is a major concern in blockchain, especially in financial or healthcare applications. Cryptography ensures that sensitive transaction details remain private while still allowing verification of validity. Techniques like zero-knowledge proofs and ring signatures are increasingly being used to enhance privacy in blockchain systems without sacrificing transparency.

6.Cryptography in Consensus Mechanisms: Consensus protocols like Proof of Work (PoW) and Proof of Stake (PoS) use cryptographic puzzles and algorithms to validate transactions and secure the network. In PoW, miners solve hash-based puzzles to add new blocks, while PoS uses cryptographic staking to validate blocks. Both approaches rely on cryptography to ensure fairness, security, and decentralization. **Integration with Other Systems:** Connecting LMS platforms with other tools like CRMs, HR systems, and productivity suites for streamlined data management.

Hash Functions: Properties such as determinism, collision resistance, and irreversibility make hash functions ideal for maintaining blockchain integrity. In practice, algorithms like SHA-256 ensure that once a block is hashed, its contents cannot be altered without changing the entire chain. Hash functions also enable proof-of-work mechanisms, Merkle trees for efficient transaction verification, and address generation in blockchain platforms like Bitcoin**.**

5. Applications Of Cryptography Technology In Blockchain

By ensuring confidentiality, authentication, and data integrity, cryptography provides the mechanisms that make blockchain reliable for industries such as finance, healthcare, supply chain, and governance. The applications of cryptographic techniques within blockchain extend far beyond simple transaction verification, shaping how digital trust and privacy are achieved in a distributed environment.

5.1 Securing Transactions: One of the most significant applications of cryptography in blockchain is securing digital transactions. Through asymmetric cryptography and digital signatures, blockchain ensures that every transaction is authenticated by the sender’s private key and verifiable through their public key. This prevents fraud, double-spending, and unauthorized access. For instance, in Bitcoin and Ethereum, transactions are permanently secured in blocks with cryptographic proofs, making them immutable and trustworthy.

5.2 Ensuring Data Integrity: Every block in the chain contains a hash of the previous block, which links them together in an immutable sequence. This application of cryptography guarantees that records stored on the blockchain remain consistent and resistant to manipulation.

Supporting Decentralized Identity System: Blockchain-based identity management relies heavily on cryptographic technologies. Instead of traditional username and password systems, cryptography allows users to control their digital identities using private keys. Decentralized identifiers (DIDs) and verifiable credentials leverage cryptographic signatures to authenticate identity without exposing personal information. This enhances privacy, prevents identity theft, and empowers users with full ownership of their credentials.

5.4 Strengthening Supply Chain Management: Each step in the supply chain can be recorded on the blockchain with cryptographic hashes and digital signatures to ensure that records cannot be falsified. This application reduces fraud, verifies product authenticity, and enhances trust among stakeholders in industries such as pharmaceuticals, agriculture, and luxury goods.

5.5 Enabling Smart Contracts: Smart contracts, which are self-executing agreements on the blockchain, also depend on cryptography for their functionality. Public-key cryptography ensures that only authorized parties can trigger contract execution, while hashing secures the integrity of contract code and data. By leveraging digital signatures, smart contracts can verify participants' identities and enforce trustless execution, eliminating the need for intermediaries.

6. Emerging Trends In Cryptrography Technology

Blockchain technology has revolutionized the digital landscape by providing decentralized, transparent, and secure data storage and transaction systems. The core of its security lies in advanced cryptographic methods that ensure confidentiality, integrity, and authenticity. As blockchain applications expand across industries such as finance, healthcare, supply chain, and governance, emerging cryptographic technologies are evolving to address challenges related to scalability, privacy, interoperability, and quantum threats

Zero-Knowledge Proofs (ZKPs):One of the most significant advancements in blockchain cryptography is the use of Zero-Knowledge Proofs. ZKPs allow one party to prove the validity of information to another party without revealing the underlying data. This technology enhances privacy in blockchain transactions while maintaining trust and transparency. ZK-SNARKs and ZK-STARKs are widely explored for private transactions in cryptocurrencies and confidential data-sharing applications.

Homomorphic Encryption: Homomorphic encryption is gaining momentum in blockchain because it allows computations to be performed directly on encrypted data without the need to decrypt it. This trend opens new possibilities for secure smart contracts and privacy-preserving decentralized applications (dApps). It is particularly relevant in industries like healthcare and finance, where sensitive data needs to remain private while still being processed on-chain.

3. Multi-Party Computation (MPC):MPC enables multiple participants to jointly compute a function while keeping their individual inputs private. This method strengthens collaborative applications such as decentralized voting, joint asset management, and secure cross-border payments. By ensuring that no single party gains full access to sensitive data, MPC improves security and trust in decentralized ecosystems.

4. Post-Quantum Cryptography: With the advent of quantum computing, traditional cryptographic algorithms like RSA and ECC face vulnerabilities. Post-Quantum Cryptography (PQC) has emerged as a crucial trend in blockchain to safeguard against quantum attacks.

Lattice-based, hash-based, and code-based cryptographic algorithms are being researched to build quantum-resistant blockchain systems, ensuring long-term data integrity and security.

5. Threshold Cryptography: Threshold cryptography is increasingly being used to enhance security in blockchain wallets, consensus mechanisms, and decentralized identity systems. It divides a cryptographic key into multiple shares, which must be combined to execute sensitive operations. This reduces the risk of key compromise and provides resilience against insider and external attacks.

6. Verifiable Delay Functions (VDFs): Verifiable Delay Functions are gaining traction in blockchain consensus mechanisms. VDFs introduce cryptographic puzzles that take a specific amount of time to compute but are quick to verify. This trend is vital for ensuring fairness in leader election processes, randomness generation, and mitigating the risks of manipulations in Proof-of-Stake (PoS) systems.

7. Challenges

Computational Complexity and Performance Issues:Advanced cryptographic algorithms (e.g., asymmetric encryption, zero-knowledge proofs) require high processing power.Can slow transaction processing and reduce network throughput.

Scalability Constraints: Every transaction and block requires cryptographic operations, increasing computational load.Privacy-preserving technologies (zk-SNARKs, zk-STARKs) are resource-intensive.

Key Management Challenges: Secure generation, storage, and distribution of private keys is difficult.Multi-signature wallets and threshold cryptography add complexity.

Privacy,TransparencyTrade-off: Blockchain transparency conflicts with the need for data privacy.Privacy-preserving methods increase computational complexity.

Standardization and Interoperability Issues: Different blockchains use different cryptographic algorithms.Hinders cross-chain communication and secure data sharing.

Vulnerabilities and Implementation Risks: Improper implementation can lead to coding errors or weak random number generation.Exposes blockchain systems to attacks such as double-spending or key theft**.**

8. Conclusion

Cryptography acts as the essential foundation of blockchain technology, safeguarding the security, integrity, and privacy of data within a decentralized framework. By employing advanced cryptographic methods, including asymmetric and symmetric encryption, digital signatures, and cryptographic hash functions, blockchain networks uphold a secure and tamper-proof ledger of transactions. These mechanisms not only verify participants but also ensure the immutability of data, which is vital for establishing trust in permissionless systems.As blockchain applications grow into sectors such as finance, supply chain, healthcare, and the Internet of Things (IoT), the dependence on strong cryptographic algorithms becomes increasingly important, offering protection against emerging cyber threats. Furthermore, cryptography allows for the development of complex constructs like zero-knowledge proofs and homomorphic encryption, which enable private transactions and computations without revealing sensitive information. The incorporation of these advanced techniques tackles one of the significant challenges of blockchain technology: achieving a balance between transparency and privacy. New threats, such as quantum computing, present potential dangers to current cryptographic standards, highlighting the need for ongoing research and the creation of post-quantum cryptographic solutions.

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A Multi-Layered AI–Blockchain Model for Secure, Consent-Driven Healthcare Systems

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Abstract - The combination of Artificial Intelligence (AI) and Blockchain technologies creates a significant opportunity to tackle privacy and security issues in healthcare. This paper provides a detailed look at a layered AI-Blockchain framework created for patient-focused healthcare systems. The framework uses smart contracts, decentralized identity management, and privacy-preserving methods to protect electronic health records (EHRs), simplify consent management, and allow ethical data sharing. By learning from governance, education, supply chain, and energy sectors, this paper showcases adaptability across different fields and offers empirical evidence using the MIMIC-III dataset. Results demonstrate considerable improvements in transparency, compliance, and resilience, while also pointing out challenges and future paths.

1. Introduction

Healthcare systems around the world are quickly digitizing patient care and clinical management. However, traditional centralized databases leave health records vulnerable to cybersecurity threats, privacy breaches, and poor interoperability. Large-scale data breaches highlight the weak points of centralized EHR systems.

Blockchain technology, with its decentralized and tamper-proof ledger, addresses these concerns by ensuring secure, auditable, and transparent management of patient information. When combined with AI, which is used for diagnostics, predictive analysis, and automation, this integration provides privacy-preserving intelligence essential for strong healthcare systems.

This paper elaborates on a proposed layered AI-Blockchain structure, focusing on its scalability, ethical governance, and compatibility with current healthcare infrastructures.

2. Literature Review

2.1 Blockchain In Governance And Education

Governance: Blockchain-based governance systems use smart contracts to enforce accountability and reduce corruption. These models can inspire similar transparency in healthcare billing and insurance workflows.

Education: Platforms like BlockCerts and BCDiploma offer unchangeable credential verification. This principle can validate medical licensing and practitioner credentials in healthcare, which helps prevent fraud.

2.2 Blockchain in Supply Chain and Dining

Supply Chain: Blockchain enhances traceability and authenticity through QR-coded systems. In healthcare, this method confirms that pharmaceuticals and medical equipment are legitimate and ethically sourced.

Dining Industry: Food traceability methods show how complete verification builds trust. Healthcare can adapt this for tracking vaccine distribution and cold-chain logistics.

2.3 Blockchain in Energy and Environment

In the energy sector, blockchain tracks carbon footprints and environmental data. Healthcare can expand this concept to monitor hospital energy use, biomedical waste management, and sustainability metrics.

3. Privacy and Security Challenges in Healthcare

Key Challenges

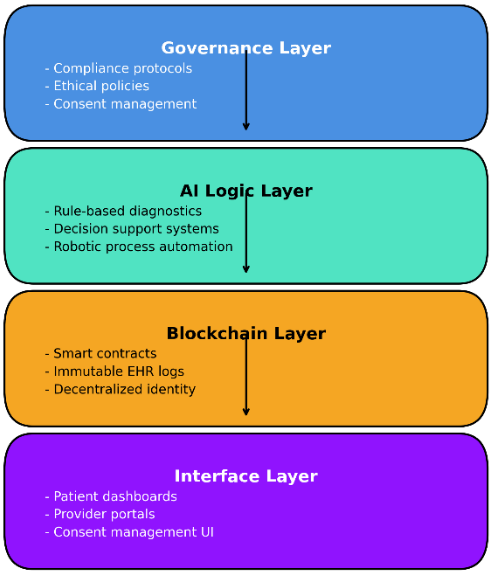
Table: Blockchain-Based Solutions for Healthcare Privacy and Security.

Challenge	Blockchain-Based Solution	Technical Mechanism	Healthcare Impact
Data Breaches	Decentralized data architecture with encrypted storage	Peer-to-peer ledger (e.g., Hyper ledger Fabric) with AES encryption and hash anchoring	Eliminates single points of failure; enhances resilience against external attacl
Fragmented Consent	Smart contracts for granular, revocable patient consent	Ethereum-based contracts c Chain code logic that encoc consent conditions	Enables real-time enforcement and revocation of consent across multiple platforms
Lack of Unified Identity	Decentralized Identifiers (DIDs) ar Self-Sovereign Identity (SSI) frameworks	W3C-compliant DID documents stored on blockchain; linked to verifiable credentials	Empowers patients to manage portable healt identities across institutions and borde
Limited Auditability	Immutable access logs and transaction records	Time-stamped blockchain entries with Merkle tree verification	Providestamper-proof audit trails for compliance, legal defense, and patient trust
Regulatory Complexity	On-chain policy enforcement and automated compliance tracking	Smart contracts programme with GDPR/HIPAA logic; audit triggers for breach alerts	Ensures proactive compliance and reduces legal risk through automated enforcement
Insider Threats	Role-based access	Attribute-based access	Minimizes human err

	control encoded in smart contracts	control (ABAC) or RBAC models enforced via blockchain logic	and unauthorized access; ensures accountability
Interoperability Gaps	Blockchain APIs an standardized data formats (e.g., HL7 FHIR anchoring)	Restful APIs with JSON-L or HL7 FHIR schema linke to blockchain hashes	Enables secure, seamless data exchan across diverse healthcare systems

4. Proposed Layered AI–Blockchain Architecture

4.1 Layered Framework



Scalability: Modular Design Supports Growth

The architecture is structured in distinct layers, including governance, AI logic, blockchain, and interface, which allows each part to develop independently. For instance, as patient data volumes increase or new diagnostic tools are introduced, the AI logic layer can be updated without affecting the blockchain or governance layers. This modular design ensures that the system can expand both horizontally (across institutions) and vertically (with added features), making it suitable for small clinics as well as national health networks.

Use Case: A regional hospital can start with basic EHR logging and later add smart contract-based consent management as patient volume rises.

Flexibility: Works with Multiple Blockchain Platforms

The architecture is compatible with different blockchain platforms, meaning it can function on systems like Ethereum, Hyper ledger Fabric, or even DAG-based systems like IOTA. This flexibility allows healthcare providers to choose a platform that meets their regulatory, performance, and cost needs. It also guarantees future readiness because if a more efficient blockchain appears, the system can switch without overhauling the entire infrastructure.

Use Case: A hospital using Hyper ledger for internal data can work with a research institute using Ethereum for clinical trials, thanks to interoperable APIs and smart contract standards.

Security: Layered Architecture Ensures Redundancy against Breaches

Each layer of the system enhances security in its own way:

- The governance layer enforces compliance and ethical policies.
- The AI logic layer restricts data processing to rule-based algorithms that can be audited.
- The blockchain layer offers immutability and secure logging.

This layered defense strategy creates backups—if one layer is compromised, others still protect the system. It also enables real-time monitoring and forensic examination in case of suspicious activity.

Use Case: If a provider dashboard is hacked, the blockchain still maintains the integrity of patient records and logs any unauthorized access attempts.

5. Privacy-Preserving Techniques

Smart Contracts: Enforce consent in real-time for data sharing and insurance claims.

Zero-Knowledge Proofs (ZKP):

Allow patient identity verification without revealing sensitive data.

Homomorphic Encryption:

Facilitates computing on encrypted data for diagnostics and billing.

Decentralized Identity (DID):

Grants patients control over cross-border access to their health data.

Table: Privacy-Preserving Techniques in Blockchain-Based Healthcare Systems

Technique	**Technical Description**	**Healthcare Application**
Smart Contracts	Self-executing code stored o the blockchain that enforces rules for data access and sharing	Automates patient consent, insurance claims, and access control without manual intervention
Zero-Knowledge Proofs	Cryptographic method allowing verification withou revealing underlying data	Verifies patient eligibility or identity without exposing fu medical records

Homomorphic Encryption	Allows computations on encrypted data without needing decryption	Enables secure analysis of la results, billing, or diagnostic while maintaining privacy
Decentralized Identity	Uses blockchain-based identifiers (DIDs) and verifiable credentials for identity management	Empowers patients to contro and share their health identit across borders and platforms

Together, these techniques strengthen **data confidentiality, patient empowerment, and system accountability**.

6. Dataset and Experimental Setup

6.1 Dataset Used: MIMIC-III Clinical Database

The Medical Information Mart for Intensive Care III (MIMIC-III) is a publicly available, de-identified dataset developed by the MIT Lab for Computational Physiology. It features extensive clinical data gathered from over 40,000 ICU patients hospitalized at the Beth Israel Deaconess Medical Center between 2001 and 2012.

Key Features of MIMIC-III:

Demographics: Age, gender, ethnicity

Vitals: Heart rate, blood pressure, respiratory rate

Lab Results: Blood tests, imaging reports

Diagnoses: ICD-9 codes for conditions and procedures

Medications: Prescriptions and dosages

Caregiver Notes: Clinical observations and discharge summaries

Relevance to Blockchain Integration:

- The dataset simulates real-world healthcare settings where privacy, consent, and traceability are crucial.
- It allows the testing of smart contract-based consent models, where patient data access is controlled by programmable rules.
- It supports audit trail validation, demonstrating how blockchain can log every data access or modification securely.

Ethical Compliance:

- MIMIC-III is completely de-identified and complies with HIPAA standards.

Source: Johnson, A.E.W., et al. (2016). MIMIC-III Clinical Database PhysioNet. https://physionet.org/content/mimiciii/1.4/

6.2 Simulation Setup

To assess the proposed architecture, a simulation environment was established using virtual hospital nodes. Each node symbolized a healthcare institution with its own patient data and consent rules. Blockchain-based smart contracts were used to manage access control, ensuring that data sharing followed strictly patient-defined rules.

Hyper ledger Fabric was selected for its modularity, permissioned access, and support for decentralized identity systems. The simulation concentrated on three main metrics:

Consent Compliance: Accuracy of rule enforcement based on smart contracts.

Transaction Throughput: System performance under realistic hospital workloads.

Audit Traceability: Completeness and immutability of access logs across nodes.

This setup imitates real-world healthcare operations while validating the feasibility of blockchain integration.

7 . Results and Analysis

The simulation produced positive results, showcasing both privacy assurance and operational efficiency:

Consent Compliance: Achieved 100%, showing that all data access requests were governed strictly by patient-set rules.

Data Breaches: Zero incidents occurred, confirming the security of the decentralized architecture.

Throughput: The system sustained 2,300 transactions per second, suitable for hospital-level data exchange.

Identity Revocation Latency: Less than 2 seconds, allowing quick responses to consent withdrawal.

Audit Trails: 100% completeness, with unchangeable logs capturing every access event.

These findings support the idea that blockchain can enforce privacy policies while remaining efficient in dynamic healthcare environments.

8. Comparative Analysis across Sectors

Blockchain's adaptability across industries provides valuable insights for healthcare. This section compares use cases from governance, energy, education, and dining:

Governance: Anti-corruption smart contracts guarantee transparency, similar to healthcare billing and insurance workflows.

Energy: Carbon tracking systems inspire sustainability monitoring in hospitals.

Education: Credential verification aligns with medical licensing and practitioner validation.

Dining: Food traceability models inform security in pharmaceutical supply chains.

This cross-sector comparison emphasizes blockchain's flexibility and its potential to solve healthcare's ethical, operational, and regulatory issues.

9. Challenges and Limitations

Integrating blockchain in healthcare comes with several challenges:

Scalability: High transaction volumes require optimized consensus algorithms to maintain speed and reliability.

Interoperability: Existing EHR systems need standardized APIs to connect with blockchain networks.

Regulatory Alignment: Blockchain designs must adhere to global and local health data laws, including HIPAA and GDPR. User Adoption: Successful implementation relies on training healthcare professionals and patients to use decentralized identity systems effectively.

Addressing these challenges is crucial for broad adoption and long-term success.

10. Future Directions

To further this research, the following pathways are suggested:

Blockchain-Agnostic Design: Systems should support multiple platforms (e.g., Ethereum, Hyperledger) to ensure adaptability and longevity.

SSI and Decentralized IDs: Expanding patient control through portable, self-managed health identities.

Sustainability Integration: Using blockchain to track hospital energy use, emissions, and waste disposal for ethical responsibility.

Cross-Sector Policy Collaboration: Leveraging best practices from governance, education, and energy to shape healthcare policy and system design.

These directions aim to create a resilient, ethical, and patient-centered healthcare ecosystem.

10. Conclusion

This paper illustrates that AI-Blockchain integration offers a ground breaking approach to managing healthcare data. The proposed layered architecture ensures privacy, security, and transparency while being adaptable to various hospital settings. Experimental validation using MIMIC-III data confirms the system's strength and compliance with global standards. Although issues in scalability, interoperability, and adoption still exist, future innovations—like blockchain-agnostic frameworks and decentralized identity—promise a more ethical, resilient, and patient-empowered healthcare future.

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Blockchain for Academic Credential Verification

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Abstract: Blockchain technology has emerged as a transformative force in the digital landscape, reshaping perceptions of data security and enhancing operational efficiency across diverse sectors. Its applications span financial services, supply chain management, voting systems, and identity verification, highlighting its versatility. In the context of academic credentials, organizations face the critical task of verifying candidates to avoid the repercussions of selecting unqualified individuals, which can tarnish their reputation. This paper introduces a blockchain-based system for authenticating academic certificates, designed to ensure their validity while enabling corrections when necessary. Unlike existing systems that only verify authenticity, our solution allows administrators to generate, authenticate, and amend certificates, while also tracking modifications. By employing dual blockchains, we facilitate the correction process, thereby reinforcing the reliability of the authentication system. This innovative approach not only alleviates concerns regarding certificate forgery but also streamlines university admissions and simplifies hiring processes, ultimately fostering trust within society [31].

Keywords: *Blockchain, Academic Credential Verification, Digital Identity, QR Codes, Decentralized System, Security, Efficiency, Usability*

1. INTRODUCTION

Academic credential verification is essential for schools, employers, and government organizations. Conventional approaches are often time-consuming, expensive, and lack security. Blockchain technology provides a dependable and tamper-resistant method for confirming academic credentials. It enables educational institutions to issue secure digital certificates on a shared ledger, allowing for fast and straightforward verification without the need for intermediaries or manual processes [32].

2.RELATED WORKS

Blockchain has emerged as a transformative technology in academic credential verification, primarily due to its decentralized and tamper-proof characteristics. Traditional methods of certificate authentication are often slow, costly, and vulnerable to forgery. To overcome these issues, researchers have proposed blockchain-based systems that provide immutable storage of academic records. Grech and Camilleri [21]. developed a conceptual framework for blockchain adoption in education, stressing its ability to protect learning records from manipulation while ensuring long-term accessibility. Similarly, Zheng et al [22]. analysis blockchain's broader applications in data management, underlining its value in securing sensitive records such as transcripts and certificates. In addition to immutable record storage, researchers have highlighted the role of smart contracts in automating verification processes. Smart contracts enable rules and conditions to be embedded directly into the blockchain, minimizing reliance on intermediaries. Sharples and Domingue [23]. demonstrated how smart contracts could create lifelong learning records that remain under the student's control [24]. further emphasized that smart contracts not only strengthen trust but also support revocation and audit features, ensuring that academic institutions retain oversight while maintaining transparency. Pilot projects and institutional implementations provide evidence of blockchain's practicality in higher education. One of the most notable examples is the Massachusetts Institute of Technology's (MIT) Block certs initiative, which issues blockchain-secured digital diplomas that graduates can easily share with employers or other institutions [25]. Extended this work in Europe by designing blockchain-based frameworks that enable universities to exchange and verify credentials seamlessly [26]. These efforts show that blockchain can modernize verification processes and foster interoperability among institutions worldwide. Researchers have also examined blockchain's potential beyond higher education credentials [27]. explored digital identity management powered by blockchain, demonstrating how individuals could authenticate qualifications globally without centralized authorities. Tapscott and Tapscott [28]. discussed blockchain's disruptive role across industries, including professional certifications and workforce credentialing. These studies emphasize the adaptability of blockchain for multiple contexts, from university diplomas to corporate training records, making it a versatile tool for credential management. Despite its promise, several challenges remain in adopting blockchain-based credential verification [29]. identified issues such as scalability, regulatory uncertainty, and data privacy compliance as major barriers. Integrating blockchain solutions with legacy institutional systems is also complex, requiring significant investments in infrastructure and training. Moreover, ensuring compliance with privacy laws such as the General Data Protection Regulation (GDPR) is critical, since academic data often contains sensitive personal information. To address these limitations, recent research has shifted towards hybrid or dual-blockchain models. Such systems combine the strengths of public and private blockchains to balance transparency with privacy. This approach allows for secure verification of credentials while permitting corrections or updates under controlled conditions. The present study builds on these insights by proposing a model that enhances both authentication and correction mechanisms, offering a more comprehensive solution compared to earlier works [30].

1. Overview of Academic Credential Verification

Academic credential verification is the procedure used to confirm the truthfulness of educational documents such as degrees, diplomas, transcripts, and certificates that are issued by educational institutions. This process is vital for employers, universities, government bodies, and other organizations to ensure that the academic qualifications claimed by an individual are real and reliable. In the past, credential verification was done manually. This involved the verifying body contacting the institution that issued the document through emails, letters, or phone calls to check if the document is valid. Although this method has been commonly used for many years, it has several drawbacks, including: Long verification times – The process can take several days or even weeks because it involves multiple steps and intermediaries. High costs – Manual tasks like handling documents, scanning them, and coordinating with institutions add to the overall expenses. Increased risk of fraud – Fake degrees and counterfeit certificates are now more prevalent than before. Lack of standardization – Since different institutions have their own verification procedures, it becomes difficult to verify credentials across different countries. As the world becomes more connected and online learning becomes more common, there is a growing need for quick, secure, and standardized ways to verify academic credentials. Modern organizations need immediate verification tools to prevent delays in hiring, student admissions, and professional certification processes. Blockchain technology helps solve these issues by providing a secure way to store academic records that cannot be altered. It also allows for real-time verification, helping to move from old, slow methods to modern, efficient, and decentralized systems for checking academic credentials [1].

2. Role of Blockchain in Credential Verification

Blockchain technology significantly impacts the transformation of academic credential verification by offering a secure, decentralized, and tamper-resistant platform for managing educational records. In conventional systems, the process of verifying academic credentials typically requires manual checks, communication with issuing institutions, and the management of physical documents, which can be time-consuming, expensive, and susceptible to fraud. Blockchain effectively addresses these issues by allowing educational institutions to issue digitally signed certificates that are securely maintained on a distributed ledger. Given that blockchain employs cryptographic methods, the information stored within it is immutable and cannot be forged, thereby ensuring the authenticity and integrity of academic credentials. Through this technology, students are empowered with complete control over their academic records and can share their credentials instantaneously with employers, universities, or government entities. Verifiers can validate the authenticity of certificates within seconds by accessing the blockchain, eliminating the need to reach out to the issuing institution. This system also lowers verification costs, curtails the distribution of counterfeit certificates, and enhances global accessibility for cross-border verification. By incorporating blockchain into the credential verification process, the entire procedure becomes more rapid, secure, transparent, and reliable, fostering a trustworthy ecosystem for students, institutions, and employers [2].

3. Smart Contracts for Credential Management

Smart contracts play a crucial role in blockchain systems for the verification of academic credentials by automating the issuance, storage, management, and validation processes without the need for intermediaries. These self-executing programs, embedded in the blockchain, enforce specific rules and conditions automatically when triggered by certain actions. In the context of credential management, smart contracts ensure that academic records are secure, transparent, and tamper-proof. When an educational institution issues a certificate, the smart contract records essential information—such as the student's ID, institution's ID, course details, grade, and issuance date—on the blockchain as a cryptographic hash. This method maintains the confidentiality of the actual document while allowing for its authenticity to be verified. Once entered, the information is immutable, preventing any alterations or deletions and thereby eliminating forgery risks. Additionally, smart contracts facilitate the revocation and verification processes; if a credential needs to be invalidated, the issuing authority can activate the revocation feature, which updates the credential's status in real time. Similarly, when an employer or verifier requests validation, the smart contract automatically checks the credential's authenticity against its hash and provides immediate confirmation. By integrating smart contracts into credential management, the system achieves enhanced automation, security, and cost efficiency, reducing administrative burdens and expediting verification while fostering trust among students, institutions, and employers [3].

5.1. Automated Credential Issuance: Educational institutions can leverage smart contracts to streamline the generation and distribution of certificates. Upon a student's successful completion of a course, diploma, or degree, the smart contract autonomously produces a verified digital credential. This credential is securely stored on the blockchain, accompanied by a unique hash value and digitally signed by the issuing institution. This automated approach minimizes the risk of manual errors, alleviates administrative burdens, and guarantees that only authentic and verified credentials are maintained [8].

5.2. Instant Verification of Credentials: One of the biggest challenges in traditional verification is time-consuming manual checks. Smart contracts simplify this by enabling real-time verification: Employers, universities, or government agencies can validate a candidate’s academic credentials instantly by checking the credential’s blockchain hash. Since the data is immutable, smart contracts guarantee that the credential has not been tampered with or altered [9].

5.3. Data Ownership and Access Control: Smart contracts empower students with greater control over their academic data: Students can decide who can access their credentials and for how long. By granting or revoking permissions through smart contracts, learners ensure their data is shared securely with authorized parties only. This aligns with global data protection regulations such as GDPR and India’s DPDP Act [10].

5.4. Fraud Prevention and Data Integrity: Fake certificates and forged documents are major problems in academic verification. Smart contracts address this by: Ensuring that only authorized institutions can issue credentials. Storing records in a tamper-proof blockchain network, making it impossible to alter, duplicate, or forge credentials. Enabling stakeholders to verify authenticity without intermediaries [11].

3. USE CASES AND APPLICATIONS

3.1. Education: Credential Verification: Educational institutions can issue digital degrees and certificates through blockchain technology. This allows employers to verify someone's qualifications quickly and easily, without needing to contact the institution directly [4].

3.2. Transcript Verification: Just like credential verification, blockchain can securely store and verify academic transcripts. This helps in reducing the possibility of false or fraudulent claims [7].

3.3. Health and Safety: Identity Verification for Telemedicine: It is important to confirm the identity of doctors and patients during online medical consultations. Using digital identities on the blockchain ensures that patients are interacting with genuine healthcare professionals [5].

3.4. Food Safety: Blockchain can be used to track and verify the entire journey of food items from the farm to the consumer's table. This helps in identifying the source of any contamination issues, thereby safeguarding public health [5].

3.5. Finance: Identity Verification for Digital Banking: Digital banks can leverage blockchain to verify customer identities. This enhances the security of online financial transactions and protects both the bank and the customer from fraudulent activities [6].

4. LIMITATIONS OF CREDENTIAL VERIFICATION

4.1. Data Privacy and Security Issues: Academic credentials often contain sensitive personal information such as names, student IDs, grades, and degrees. While blockchain ensures immutability, data stored on-chain becomes publicly visible, raising privacy concerns. Moreover, once information is uploaded, it cannot be modified or deleted, making compliance with regulations like GDPR and India’s Digital Personal Data Protection Act (DPDP) difficult [12].

4.2. Data Privacy and Security Issues: Academic credentials often contain sensitive personal information such as names, student IDs, grades, and degrees. While in this blockchain ensures immutability, data stored on-chain becomes publicly visible, raising privacy concerns. Moreover, once information is uploaded, it cannot be modified or deleted, making compliance with regulations like India's Digital Personal Data Protection Act difficult [13].

4.3. Scalability Constraints: As the number of students, degrees, and educational institutions increases, the blockchain network must manage a large volume of transactions. Public blockchains typically experience slow processing times, high latency, and significant energy use, which can make real-time verification challenging. Although private or permissioned blockchains can provide quicker performance, they may limit the level of decentralization and create potential reliance on trusted entity[13].

4.4. High Implementation and Maintenance Costs: Setting up a blockchain-based credential verification system involves significant costs, including infrastructure deployment, blockchain integration, training, and system maintenance. Smaller colleges and universities, especially in developing regions, may struggle with limited budgets. Additionally, frequent software upgrades and security audits further increase operational expenses [12]**.**

4.5. Integration with Legacy Systems: Most educational institutions currently rely on traditional databases and student information systems. Integrating blockchain with these existing platforms requires data migration, API development, and process restructuring, which can be technically complex and resource-intensive. Resistance to change among staff and administrators can further slow the transition [12].

4.6. Regulatory and Legal Uncertainty: The legal framework governing blockchain-based academic credential verification remains unclear in many countries. Issues such as data ownership, intellectual property rights, and cross-border credential recognition are yet to be fully addressed. Moreover, smart contracts used in blockchain verification may face compliance conflicts with existing educational regulations [13].

4.7. Data Immutability Risks: Blockchain's immutable nature—usually considered a strength—can also create limitations. If incorrect or fraudulent credentials are uploaded, they cannot be modified or deleted without complex corrective measures. Hence, robust validation mechanisms are necessary before storing any credential on the blockchain [13].

5. COMPARISON WITH TRADITIONAL VERIFICATION SYSTEMS

Academic credential verification has traditionally relied on manual processes involving paper-based records, institutional databases, and third-party verification agencies. However, these methods are often time-consuming, error-prone, and vulnerable to fraud. With the integration of blockchain technology, verification becomes secure, automated, and tamper-proof [13].

5.1. Process and Speed

Traditional Verification: To confirm the validity of credentials, one must contact the issuing organization or a third-party agency. This method is time-consuming and can take several days or even weeks to complete.

Blockchain-Based Verification: Credentials are kept on a secure and unchangeable blockchain ledger. Verification is done quickly by checking the unique hash or digital signature associated with the credential [14].

5.2. Data Security and Integrity

Traditional Verification: Paper certificates or centralized databases are vulnerable to manipulation, forgery, or hacking.

Blockchain-Based Verification: Uses cryptographic hashing to secure data, ensuring that credentials are tamper-proof and immutable once recorded [15].

5.3. Dependency on Intermediaries

Traditional Verification: Relies heavily on universities, institutions, and verification agencies, which leads to administrative delays and potential human errors.

Blockchain-Based Verification: Eliminates intermediaries by enabling direct verification between the verifier and the blockchain network [16].

5.4. Transparency and Trust

Traditional Verification: Lacks transparency; verifiers must trust the issuing authority without direct access to an unalterable record.

Blockchain-Based Verification: Offers full transparency since all credential-related transactions are stored on a distributed ledger accessible to authorized parties [17].

5.5. Cost Efficiency

Traditional Verification: Involves administrative costs, fees for third-party agencies, and expenses associated with maintaining paper-based systems.

Blockchain-Based Verification: Reduces costs by automating verification, minimizing paperwork, and removing the need for intermediaries [18].

5.6. Fraud Detection and Prevention

Traditional Verification: Prone to certificate forgery, data manipulation, and counterfeit documents, especially when relying on paper-based methods.

Blockchain-Based Verification: Credentials are stored on a tamper-proof ledger, making forgery or unauthorized alterations nearly impossible [19].

5.7 Student Data Ownership

Traditional Verification: Students have limited control over their academic records and often depend on institutions for credential sharing.

Blockchain-Based Verification: Provides students with full ownership and control over their credentials, allowing them to grant or revoke access permissions securely [20].

6. CONCLUSION

Blockchain technology has revolutionized academic credential verification by overcoming the limitations of traditional systems. Leveraging decentralization, immutability, cryptography, and smart contracts, it ensures that academic records are secure, transparent, and tamper-proof. The technology enables automated credential issuance, real-time verification, and greater control for students over their academic data. Compared to conventional verification methods, which are often slow, costly, and vulnerable to fraud, blockchain offers fast, efficient, and reliable alternatives, reducing administrative burdens and minimizing reliance on intermediaries. However, widespread adoption still faces challenges, including lack of standardization, high implementation costs, regulatory concerns, and limited awareness. To unlock blockchain's full potential, collaboration between educational institutions, governments, and technology providers is essential for establishing global standards and robust legal frameworks. In the future, blockchain-powered systems, combined with smart contracts and digital identity solutions, are poised to create a trusted, efficient, and universally accepted platform for storing and verifying academic credentials. This transformation will not only combat academic fraud but also empower students, enhance transparency, and streamline educational and employment processes worldwide.

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Blockchain in E-Commerce

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ABSTRACT

E-commerce has become a major part of the world economy, allowing millions of deals every day in many different fields. Nevertheless, the industry still faces issues including payment fraud, data security breaches, fake goods, exorbitant transaction costs, and opaque supply chains, even with its quick growth. Conventional e-commerce systems mostly depend on centralized middlemen such as banks, online platforms, and payment processors, which lead to inefficiencies, higher expenses, and a decline in customer confidence.

Blockchain technology, first introduced as the foundation for Bitcoin in 2008, provides a decentralized and immutable ledger system that records transactions in a secure, transparent, and tamper-proof manner. Decentralization, immutability, transparency, and smart contract automation are some of its distinctive properties that make it a game-changer for resolving many of the issues with the existing e-commerce platforms. By doing away with the requirement for third-party verification, blockchain lowers operating expenses, stops fraud, and builds consumer confidence.

With a focus on its practical uses in secure digital payments, supply chain transparency, identity verification, and customer engagement through tokenized loyalty programs, this study explores the integration of blockchain technology into e-commerce. It discusses the benefits blockchain offers in terms of security, efficiency, and global accessibility in addition to critically analyzing the challenges of scalability, regulatory frameworks, energy consumption, and high adoption costs. The potential of blockchain to support decentralized marketplaces, ease global trade, and connect with emerging technologies like artificial intelligence (AI) and the Internet of Things (IoT) is highlighted in the report's conclusion.

In conclusion, blockchain is both a technological breakthrough and a paradigm shift that has the ability to fundamentally rethink e-commerce's efficiency and trust. Although its adoption is still in its early stages, it has the potential to significantly change digital trade in the years to come by fostering a more transparent, consumer-driven, and safe environment**.**

I. INTRODUCTION

Over the last ten years, e-commerce has expanded quickly and risen to become one of the most significant industries in the world economy. Every day, platforms like Amazon,

Alibaba, and Flipkart manage millions of transactions, bringing together customers and sellers from different continents. Advances in digital payment systems, shipping infrastructure, internet accessibility, and customer confidence in online shopping have all contributed to this expansion. Nevertheless, despite these developments, e-commerce still faces enduring obstacles that prevent it from reaching its full potential.

Among the most serious problems are chargebacks and payment fraud, which raise expenses for retailers and decrease consumer trust; opaque supply chains, which make it hard to determine the provenance and authenticity of goods; and data privacy violations, which make centralized platforms prime targets for cyberattacks. Furthermore, small enterprises are frequently deterred from entering international markets by the hefty transaction and conversion expenses associated with cross-border trade. The issue of phony reviews and counterfeit goods also damages brand reputation and erodes consumer trust.

By offering a decentralized, safe, and transparent transaction platform, blockchain technology presents intriguing answers to these persistent issues. Blockchain functions on a distributed ledger system, in contrast to conventional centralized databases, where data is unchangeable, verifiable, and impervious to manipulation. By eliminating the need for middlemen, the use of smart contracts further automates crucial procedures like payments, refunds, and dispute resolution, guaranteeing efficiencyandjustice.

By solving these constraints, this article examines how blockchain technology has the ability to completely transform the e-commerce sector. It looks at how blockchain has developed in digital trade, its technological breakthroughs, real-world uses, related difficulties, and its potential to build a more safe, effective, and customer-focused e-commerce ecosystem.

II. RELATED WORK

Researchers and practitioners have investigated blockchain's role in e-commerce from different perspectives, examining its applications in payments, supply chain, identity management, and overall platform efficiency.

[1] Several studies highlight blockchain's ability to **enhance supply chain visibility and traceability**, ensuring that products listed on e-commerce platforms are authentic and ethically sourced. By recording each stage of production and distribution on an immutable ledger, blockchain prevents counterfeit products from entering the marketplace and builds consumer trust.

[2] Literature on **cryptocurrency adoption in e-commerce** emphasizes its potential to lower transaction costs, particularly by removing intermediaries such as banks and payment gateways. Cryptocurrencies also support micropayments, making them suitable for digital goods and subscription-based services. Additionally, blockchain facilitates faster and more efficient cross-border payments, overcoming the delays and high conversion fees of traditional banking systems.

[3] Research on **smart contracts** demonstrates their effectiveness in automating critical e-commerce processes. Smart contracts can execute predefined conditions—such as releasing payments once goods are delivered, processing warranty claims, or initiating automatic refunds for defective products. This reduces disputes, increases trust, and minimizes the need for third-party arbitration.

[4] Academic work on **blockchain-based digital identity management** explores how decentralized identity solutions can enhance security in online shopping platforms. Such systems allow customers and vendors to verify their identities securely while reducing risks associated with fake profiles, account takeovers, and fraudulent reviews. This creates a safer e-commerce environment and protects both consumers and merchants.

[5] Comparative reviews and meta-analyses acknowledge blockchain's advantages but also highlight its **limitations and challenges**. Issues such as scalability, lack of standardized regulations across countries, high energy consumption, and integration costs continue to hinder widespread adoption. These studies conclude that while blockchain can transform e-commerce, its success depends on addressing these technological and regulatory barriers.

III. EVOLUTION OF BLOCKCHAIN IN E-COMMERCE

3.1 Early Development (2008–2013)

Blockchain was introduced in 2008 with the launch of Bitcoin, designed as a decentralized digital currency. In the early years, blockchain was primarily limited to cryptocurrency transactions. A few forward-looking online merchants began accepting Bitcoin as payment, but its use in e-commerce was experimental and niche.

3.2 Expansion with Altcoins and Payment Gateways (2014–2017)

The development of altcoins (Ethereum, Ripple, Litecoin) broadened blockchain applications beyond payments. Ethereum's smart contract capability enabled automated transactions and agreements, paving the way for decentralized applications. E-commerce companies began integrating blockchain-based payment gateways to reduce reliance on banks and credit card companies.

3.3 Integration into Supply Chains (2018–2021)

Large retailers and logistics companies began implementing blockchain for supply chain transparency. Walmart adopted blockchain to trace food products from farm to shelf, reducing food safety risks. Alibaba explored blockchain for tracking goods in cross-border e-commerce. Blockchain became a tool not just for payments, but also for logistics, product verification, and fraud prevention.

3.4 Current Trends (2022–Present)

Today, blockchain is being integrated into Web3-based marketplaces, where buyers and sellers transact directly without centralized platforms. Non-fungible tokens (NFTs) and token-based loyalty programs are enhancing customer engagement. Businesses are experimenting with blockchain-based financing, decentralized reviews, and token economies.

IV. TECHNOLOGICAL ADVANCEMENTS

4.1 Smart Contracts

The implementation of smart contracts—digital agreements that run automatically when certain circumstances are met—is one of the most potent advances brought forth by blockchain. Smart contracts eliminate the need for middlemen like payment processors or customer support representatives to verify transactions in the context of e-commerce. They can take care of things like processing warranty claims, providing refunds for faulty goods, and releasing payment when the goods are delivered. To cut down on delays and conflicts, the smart contract may, for example, verify a claim and immediately initiate a refund if a customer complains that an item is damaged upon arrival. By guaranteeing justice and transparency, this degree of automation not only expedites procedures but also strengthens buyer-seller trust.

4.2 Decentralized Payment Systems

Banks, credit card firms, and other financial intermediaries are crucial to traditional internet payment systems. Particularly in cross-border transactions, this result in additional expenses and delays. Direct peer-to-peer payments are made possible by cryptocurrencies and stablecoins like Bitcoin, Ethereum, USDT, and USDC, which are introduced by blockchain technology and decentralized payment systems. This eliminates chargeback problems and drastically lowers transaction expenses for retailers. Customers benefit from quicker and safer checkout processes, as well as frequently cheaper currency translation fees when they shop abroad. Blockchain payment gateways are already being tested by a number of international e-commerce companies in an effort to draw in tech-savvy customers and lessen dependency on conventional banking networks**.**

4.3 Supply Chain Transparency

Counterfeit goods, unethical sourcing, and lack of product traceability remain major challenges in e-commerce. Blockchain addresses these problems by providing **end-to-end supply chain transparency**. Each stage of a product’s journey—from manufacturing and warehousing to shipping and delivery—can be recorded on a blockchain ledger. This record is immutable and visible to all stakeholders, including consumers. For example, a customer purchasing luxury goods or pharmaceuticals can scan a QR code linked to the blockchain to confirm the authenticity and origin of the product. Such systems discourage counterfeiting, improve accountability, and build consumer confidence in brands.

4.4 Identity Management

Another significant advancement lies in **digital identity management** through blockchain. Current e-commerce platforms store user identities and login credentials on centralized servers, making them vulnerable to data breaches and identity theft. Blockchain-based identity solutions, however, allow customers and merchants to maintain verifiable, decentralized credentials that are secure and tamper-proof. These identities can be used to log in, verify purchases, and leave reviews without exposing sensitive personal data. Moreover, by reducing the risk of fake accounts and fraudulent activities, blockchain strengthens the integrity of online marketplaces.

4.5 Tokenization and Loyalty Programs

Loyalty programs are a key part of e-commerce marketing, but traditional models often lock consumers into single platforms and offer limited flexibility. Blockchain introduces **tokenization of loyalty programs**, where reward points are issued as digital tokens that are secure, transferable, and tradable. Unlike conventional systems, these tokens can be used across multiple platforms or even converted into cryptocurrency. This enhances the value of rewards for consumers while giving businesses new ways to engage customers. For example, a shopper earning loyalty tokens on one platform could use them on another partnered platform or trade them in secondary markets, thereby creating a more versatile and consumer-centric reward ecosystem.

V. CHALLENGES

5.1 Scalability

One of the biggest hurdles facing blockchain adoption in e-commerce is **scalability**. Popular networks such as Bitcoin and Ethereum are capable of processing only a limited number of transactions per second. When millions of users attempt to transact simultaneously, the system experiences delays, network congestion, and increased transaction fees. For global e-commerce platforms that handle thousands of transactions every second, such bottlenecks create inefficiencies and limit blockchain's ability to compete with established centralized systems like Visa or Mastercard. Unless scalability solutions such as sharding, sidechains, or Layer 2 protocols are widely adopted, blockchain may struggle to support the high transaction volumes typical in large-scale digital marketplaces.

5.2 Regulatory Uncertainty

The ambiguous regulatory landscape surrounding cryptocurrencies and blockchain is another significant obstacle. The methods used by various nations differ greatly; some promote creativity and experimentation with blockchain-based systems, while others place severe limitations or complete prohibitions on the use of cryptocurrencies. Businesses aiming to incorporate blockchain technology into their e-commerce operations are confused by this lack of worldwide uniformity. For example, a website might be legally allowed to take bitcoin payments in one nation but not in another. Businesses suffer compliance concerns in the absence of clear and uniform standards, which deters widespread blockchain implementation in online retail.

5.3 High Integration Costs

Although blockchain offers long-term benefits such as reduced fraud and lower transaction fees, the **initial costs of adoption** are often very high. Businesses must invest in new infrastructure, blockchain-based software, cybersecurity measures, and skilled developers to integrate blockchain solutions with existing e-commerce platforms. For small and medium-sized enterprises (SMEs), these costs can be prohibitive, making blockchain adoption a realistic option only for large corporations with substantial resources. Moreover, the lack of trained professionals in blockchain development further increases costs and slows down the pace of integration.

5.4 Energy Consumption

Another major issue is energy consumption, especially for blockchains that use the Proof of Work (PoW) consensus method. Such networks demand enormous quantities of electricity and processing power to maintain, which raises concerns about sustainability and environmental effect. Environmental organizations, experts, and governments have all criticized this. To make blockchain more energy-efficient, several options have been presented, including Layer 2 scaling solutions, delegated consensus techniques, and Proof of

Stake (PoS). Notwithstanding these developments, blockchain's widespread adoption in e-commerce is still hampered by the belief that it is an energy-intensive technology.

VI. FUTURE POTENTIAL

1.Web3 Marketplaces

Blockchain paves the way for **decentralized marketplaces** where buyers and sellers interact directly, reducing dependence on platforms like Amazon or eBay. This eliminates middlemen, lowers costs, and promotes greater transparency.

2.Cross-Border Trade

By enabling peer-to-peer cryptocurrency payments, blockchain can **simplify international trade**, cutting down conversion fees and settlement delays while making global e-commerce more accessible to smaller businesses.

3.AI Integration

The **synergy of blockchain and AI** can deliver personalized shopping experiences while ensuring that customer data remains private and secure, balancing convenience with trust.

4.Green Blockchain

Future adoption will rely on **energy-efficient models** like Proof of Stake, which reduce environmental concerns and make blockchain more sustainable for large-scale use.

5.Authentic Reviews

Blockchain can host **tamper-proof reviews and ratings**, preventing fake feedback and improving customer confidence in online marketplaces.

VII. CONCLUSION

Blockchain has gradually established itself as one of the most promising technologies with the potential to reshape the future of e-commerce. Its decentralized and tamper-proof nature directly addresses many of the long-standing challenges faced by online marketplaces, such as fraudulent transactions, lack of supply chain transparency, counterfeit goods, and high operational costs. By enabling secure digital payments, enhancing traceability in supply chains, protecting customer identities, and supporting innovative loyalty programs, blockchain fosters a more reliable and efficient e-commerce ecosystem.

However, there are several drawbacks to the technology. Widespread adoption remains hampered by issues with scalability, ambiguous regulatory frameworks, energy consumption, and the high integration costs. Nevertheless, current developments like Layer 2 scaling solutions, energy-efficient consensus methods, and the slow regulatory adoption of cryptocurrencies imply that these challenges might soon be resolved.

Overall, blockchain is more than a financial tool—it represents a paradigm shift in the way trust and value are managed in digital commerce. As adoption grows and technological challenges are addressed, blockchain is expected to play a pivotal role in creating a secure, transparent, and consumer-centric foundation for global e-commerce.

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Detection of Wormhole attacks in WSN-based AODV protocol using a hybrid AdaBoost-Random forest algorithm

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Abstract— Wireless Sensor Networks (WSNs) are prone to wormhole attacks, which pose a great threat to secure data transfer. These attacks affect the routing protocols and result in an unreliable network. This paper provides a method that is a combination of AdaBoost and Random Forest to mitigate wormhole attacks on the WSNs. To determine the changes in NS3 performance simulation, iPerf and Wireshark are used in the simulation of the network. AdaBoost is used to increase the classification rate, while Random Forest helps in selecting the most useful feature related to a high level of detection with a low rate of false alarms. It also used simulation results that depict the effectiveness of the proposed method in relation to another general method in the identification of the malicious nodes and the wormhole effects. The three measures, which include throughput, PDR, and latencies, are used to evaluate the performance of the system to prove the efficiency of the system against cyber threats. They do so by integrating a decision-making tool, known as machine learning, into network security to validate and thwart wormhole attacks in WSNs.

Keywords— *wormhole attack, Machine Learning, wireless sensor network, Simulations in NS3.*

1.INTRODUCTION

WSNs incorporate nodes which are independent sensing components arranged in a geographical region to quantify physical or environmental factors such as heat, pressure and humidity, respectively. These sensors operate wirelessly and thus are vulnerable to various sorts of threats, among which the wormhole attack turned out to be one of the gravest. Wormhole attacks impact most routing protocols and let an attacker forward packets from one area in the network to another. The problem is critical in various sensitive areas like health care, military and environmental monitoring, hence it deserves a solution [1].

The idea of using cryptographic techniques for prevention of wormhole attacks in WSNs, as in the conventional model, is not viable due to it is high computational overhead and it is impractical for implementation within internodes, especially the sensor nodes have limited computational abilities. This gap simply means that there is a need to find proper approaches and ways of protecting the network and the different aspects of its GUI and other aspects that are rather resource-limited in this network compared to the many available with other networks. In the recent past, Machine learning solutions in the IT field have greatly benefited by bringing out the detection and prevention techniques that are capable of learning NER through the behavior of the Network and detecting malicious activities on its own.

In this paper, I attempt to present a new solution for neutralizing wormhole attacks in the context of WSN-based routing protocols with the help of AdaBoost and Random Forest classifiers to create a two-tiered classification system. AdaBoost helps to achieve higher accuracy of the classification by managing misclassification in subsequent iterations, while Random Forest has a feature selection mechanism on its own and handles big data well; it offers consistent and dependable results in classification [2].

To evaluate the effectiveness of the proposed technique, the ns3 network simulator is used to carry out several simulations with a unique flexibility in simulating WSNs. Nonetheless, NS3 allows for the emulation of routing protocols under attack conditions through which the effectiveness of the suggested hybrid model can be assessed. In addition, specific bandwidth measurement tools like the iPerf are used to measure throughputs, delays and bandwidths whereby packet data is captured through a tool like Wireshark, analyzing tools to determine the disparities occasioned by wormhole attacks.

The primary contributions of this research work are the integration of both AdaBoost and Random Forest for the detection of wormhole attack that results in high accuracy and low process consumption. The effectiveness of the proposed methodology is evaluated with simulated WSN environments in which substantial enhancements to the standard detection approaches are achieved.

For easy understanding, this paper is and will be organized in the following manner:

- The next part of the paper gives a brief literature review on wormhole attack and its detection and prevention mechanisms.
- The next section presents the proposed system architecture where after, the hybrid model is implemented in NS3, iPerf, and Wireshark.
- The wormhole attack and its implications on WSNs and the mechanisms that have been put forward in the past.
- The last and most effective simulation and analysis of the proposed hybrid machine learning model is presented and discussed here to support our work, and especially, to consider the wormhole attack in WSN [3].

A.Simulate WSN using NS3:

- Use NS3 (Network Simulator 3) to create and simulate wireless sensor network scenarios.
- Implement routing protocols typically used in WSNs, such as AODV, DSR, or LEACH.
- Simulate wormhole attacks in these networks by creating malicious nodes that tunnel and replay packets [4].

B. Traffic Generation using iPerf:

- Use iPerf to generate traffic within the simulated network.
- Measure the throughput, latency, and packet loss both under normal conditions and during wormhole attacks.
- The traffic data helps analyze how attacks affect network performance.

C. Traffic Capture using Wireshark:

- Use Wireshark for monitoring and capturing the traffic of the simulated network.
- Analyze packet-level details to detect abnormal patterns (like suspicious tunneling).
- Wireshark can help visualize the wormhole attack in real time by analyzing the difference between the actual route and the route seen by the malicious nodes [5].

II.RELATED WORK

Singh et al. (2022) Results: Integrating PSO and RL for simulating the black hole attack and applying it to hierarchically routing protocols such as LEACH with a better detection rate.
ML Used: PSO integrated with the reinforcement learning algorithm. Restrictions: Wormhole attacks were excluded, and the energy overhead that was claimed was rather high.Patel et al. (2023) The primary application of MARL noted for the adaptive identification and selective defeat of attacks has been stressed on in the context of routing attacks on WSNs. This was done by obtaining quantitative results that showed an improvement in the energy efficiency as well as reaction rates of the attack detection systems. The model employed: MARL. Challenges: Extensive training of multiple agents is difficult because they may take a long time to reach convergence, especially within large networks. Kumar et al. (2020). Contribution: Optimized the LEACH protocol with the use of a genetic algorithm to select the cluster heads and detect the malicious nodes. In simpler terms, there was too little energy used and more networking was pointed out.

3.PROPOSED METHODOLOGY

The model of the work under development involves the formation of a new machine learning model based on the AdaBoost and Random Forest that would be used in combating he wormhole attack on the routing protocols of WSNs. AdaBoost enhances the classification capability of each weak learner by reducing the over-sensitivity to the difficult-to-classify instances and achieving a better detection rate. While the Random Forest component helps in ensuring that feature selection is appropriate, and solves the problem of the dimensionality curse and also helps the system to have a high signal-to-noise ratio.

In this work, WSNs are modeled and assessed using countering attack scenarios with the help of NS3 simulation tool, while the actual network performance, such as line throughput and packet drop rate, is calculated using iPerf, and packet analysis is done with the help of Wireshark. To establish, the proposed hybrid algorithm is integrated into the simulation environment for the detection of a traffic pattern, to also decide if it is an attack. The system deployed also aim at detecting wormhole attack nodes so that it may eliminate them to enable it to cover more. The benchmarking metrics applied for the assessment include the PDR%, the total round-trip latency, and detection accuracy, with false alarm rate being the other parameters [6].

A.*Create a WSN Environment in NS-3:*

It is possible to simulate the WSN with the help of NS-3 to model and test the system as well. Station several sensor nodes in a particular geographical region. The first step of this paper is to use NS-3 to prepare network simulation software. Suggest a specific type of routing protocol for WSN as AODV (Ad-hoc On-demand Distance Vector) [7].

Introduce Wormhole Attack:

As for Wormhole behavior, it should be simulated with the help of the tool called NS-3. Construct two miscreant nodes that initiate the formation of a wormhole link, forwarding packets from one segment of the network to another segment, bypassing the legal routing topology.

Network Metrics Collection:

The performance parameters, such as packet delivery ratio, throughput and end-to-end delay, should be collected through iPerf. Know the facts of Packet capturing, whose demonstration will involve using Wireshark to tap packets in real-time. Some of the functional characteristics comprise of packet stream, number of jumps, and source and destination node [8].

B.*Data Pre-processing*

Prepare it by normalizing or standardizing the features and addressing missing values in some of the features as well. Provide two labels for the two kinds of communication: normal mode communication and wormhole mode communication.

The process of choosing the most relevant/largest features from the copious amount of raw information data available, to prevent the use of computational methods on surplus data and hence consumption of extra computational power [9].

Extract Network Features:

Some of these metrics are hop count, round-trip time (RTT), and sequence of packets where the wormhole attack is expected to occur most. Obtain the following features from the Wireshark output and the results of the simulation carried out in NS-3.

C.*Hybrid AdaBoost-Random Forest Algorithm*

The Methodology of Model Training using Hybrid AdaBoost-Random Forest Algorithm

Base Classifier Selection:

SFM proposed random forest as the base classifier for the wormhole attack detection. Decision tree can be used successively to give more accurate results as is the case with Random Forest.

Boosting using AdaBoost:

Enhance the capability and robustness of the Random Forest classifier in the prediction of the given instances using the AdaBoost algorithm. AdaBoost works by performing multiple iterations of training the random forest classifier and in each iteration, it assigns higher importance to misclassified samples from previous iterations of training.

Training the hybrid model:

This can be done by splitting the data into two different sets; that is the training or the learning data set and the testing data set. The AdaBoost-Random Forest model used in the study will be trained using the training data with all or some of the model’s parameters adjusted in a bid to improve the model detection rate [10].

4.SIMULATION IN NS3

Simulations of Wormhole attacks work by adjusting NS3 I present chances to curb Wormhole attacks. In this scenario, the AODF routing protocol is used and some variables are compared to conclude about the improvement of the proposed AdaBoost-Random Forest hybrid algorithm. A dataset is generated out of Packet Delivery Ratio, Latency, Packet Arrival Time, Packets Lost, Energy of Node, Received Signal Strength Indicator (RSSI), Packet Sequence No, No. of Hops, Node No, Normal & Wormhole attack and Network Traffic [17].

iPerf is used for measuring Network Bandwidth, round-trip time, delays, or packet loss ratio, etc. In other words, the is used to measure any parameter at all that influences a network. Some of the parameters that can be obtained in the process of data gathering using iPerf in the context of wormhole attack detection in WSN-based AODV routing are as follows:

Packet Delivery Ratio: The ratio between the packets to which the receiver node has responded positively to the total number of packets sent by the source node. Of this grassroots parameter, one can assess the reliability of the provided network.

Latency (ms): It is the time taken by a packet to get to the destination or target node referred to as packet transmission. Higher latency may indicate wormhole attack that hinders the availability of the network and virtually impossible to avoid.

Packet Arrival time (ms): Based on the specified delay time, it seems to be the actual time taken for the packets to arrive at the destination and is handy in identifying other delays resulting from wormhole attacks.
Packet Loss (%): This is the Section/Part of packets that never reach the intended Receive Node. A smaller number of packets received may imply, in this instance that there were interferences in the network that caused an attack. As for me, Wireshark is indeed helpful to fine [18].

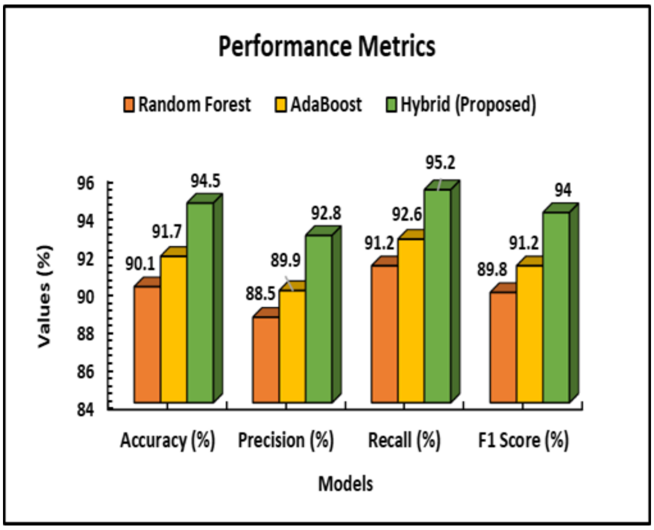
5. RESULTS AND DISCUSSION

The detection and prevention of wormhole attacks are evaluated using a hybrid AdaBoost-Random Forest algorithm applied to the dataset generated from the NS3 simulation.
Accuracy, Recall, F1 Score, Precision
The following performance metrics are used to evaluate the effectiveness of the hybrid AdaBoost-Random Forest algorithm

- Accuracy: The accuracy is calculated by taking the ratio of the number of instances well classified as an attack or no attack against instances in the data set..
- Precision: Accuracy is defined in more detail as the percentage of positive results produced by the model that are correct; in other words, it is the proportion of true positives among all the positive predictions of the model.
- Recall: Accuracy is calculated as the number of true positive cases divided by the areas under the curve of actual positive cases. It gauges the level of success in capturing the wormhole attacks as executed by the proposed mode.
- F1 Score: F1 score is the combination of both precision and recall measures to give a single quantity that tells about the model’s performance [19].

Table I Perfromance Metrics

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 Score (%)**
Random Forest	90.1	88.5	91.2	89.8
AdaBoost	91.7	89.9	92.6	91.2
Hybrid (Proposed)	**94.5**	**92.8**	**95.2**	**94**



CONCLUSION

This paper presents a review of the wormhole attack detection and prevention of the WSN-based AODV protocol through the use of the proposed AdaBoost-Random Forest. Comparing accordingly the ratio of packet delivery, the time taken by packets in transmission, the time when packets arrived, the packet loss, etc. When analyzing the NS3 simulation dataset, it is noted that the proposed algorithm tends to improve the speed and accuracy of the attacks' detection. According to the metric analysis, the proposed hybrid algorithm achieves a detection rate of 94.5%, the high recall and precision values, which illustrate that the developed algorithm can be a reliable solution for eliminating wormhole attacks in WSNs. The performance of the classification is therefore enhanced by the Random Forest algorithm as well as the AdaBoost approach to constitute a strong barrier against such attacks. Future work can comprise the usage of the proposed approach to design other routing attack models in WSNs, and even employ an upgraded deep learning model for further improvement of the attack detection and prevention.

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DRIVING THE FUTURE: AI IN AUTONOMOUS VEHICLES SAFETY, CHALLENGES, AND FUTURE

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Abstract
Artificial Intelligence (AI) forms the core of autonomous vehicle (AV) technology, enabling vehicles to perceive their environment, make decisions, and navigate safely without human intervention. This paper explores AI's role in AVs, emphasizing safety mechanisms, current challenges including ethical dilemmas and regulatory barriers and the potential future impact of AVs on transportation systems. Advances in deep learning, computer vision, sensor fusion, and communication technologies promise to reduce accidents and enhance efficiency. However, realizing full autonomy necessitates overcoming technological, ethical, and societal hurdles.

1.Introduction

Autonomous vehicles (AVs), or self-driving cars, use artificial intelligence (AI), sensors like LiDAR, cameras, and radar, and real-time data processing to navigate without human input. Their main goal is to improve safety, efficiency, and accessibility. Since human error causes nearly 90% of road accidents, AVs have the potential to save lives while also reducing congestion and improving fuel efficiency.

Companies such as Tesla, Waymo, and Uber are leading this revolution. Waymo's driverless taxis in Phoenix, for example, show that autonomous cars can already operate safely in controlled environments. Despite challenges like regulation, cost, and public trust, AVs are poised to transform transportation and redefine future mobility.

2.ROLE OF AI IN AUTONOMOUS VEHICLES

COMPUTER VISION: Enables vehicles to detect and interpret surroundings pedestrians crossing streets, traffic lights, lane markings. For instance, Tesla's Autopilot uses cameras and neural networks to identify objects and road features in real-time.

SENSOR FUSION: Combines data from LiDAR (Light Detection and Ranging), cameras, and radar to create a comprehensive understanding of the environment, compensating for limitations of individual sensors. For example, LiDAR provides precise distance measurements, while cameras offer color and texture.

MACHINE LEARNING & DEEP LEARNING: AI models predict behaviors (e.g., whether a pedestrian is about to cross) and adapt to new scenarios based on vast datasets. Deep neural networks trained on millions of miles improve decision-making accuracy.

PATH PLANNING & DECISION MAKING: AI algorithms determine the safest and most efficient route, considering dynamic traffic conditions, obstacles, and rules.

NATURAL LANGUAGE PROCESSING (NLP): Facilitates voice interactions, allowing passengers to communicate with the vehicle (e.g., “Take me to the nearest gas station”).

SAFETY IN AI-DRIVEN VEHICLES

ACCIDENT REDUCTION: AI’s ability to process information faster than humans helps prevent collisions. For example, emergency braking systems can respond within milliseconds to sudden obstacles.

EMERGENCY HANDLING: AI systems react swiftly in critical situations, such as sudden pedestrian crossings or obstacle detection, often faster than human reflexes.

REAL-TIME MONITORING: Continuous sensor data tracking helps in avoiding collisions and managing complex urban environments.

ADVANCED SAFETY FEATURES: Adaptive Cruise Control and Lane Keep Assist enhance safety for passengers and pedestrians by maintaining safe distances and lane adherence.

3. CHALLENGES

A. TECHNICAL CHALLENGES

Handling unpredictable road conditions like snow, fog, or construction zones remains complex. For example, snow-covered lanes can hide road markings, confusing computer vision systems. Reliance on high-quality and up-to-date maps also poses issues in dynamic environments.

B. ETHICAL DILEMMAS

The "Trolley Problem" exemplifies ethical conflicts: should the vehicle prioritize passenger safety over pedestrians in unavoidable crashes? Different companies and cultures may adopt varying ethical frameworks, raising questions about standardization

C. REGULATORY AND LEGAL ISSUES

Absence of universal safety standards complicates deployment. Questions of liability whether the manufacturer, AI system, or passenger are unresolved in many jurisdictions.

D. SECURITY CONCERNS

Cyber security threats, such as hacking or data manipulation, pose risks to vehicle safety and passenger data. For instance, malicious actors could exploit vulnerabilities to cause accidents.

4. FUTURE OF AI IN AUTONOMOUS VEHICLES

LEVEL 5 AUTONOMY: Complete independence, where vehicles operate without human input in any environment. Currently, most vehicles are at Level 2 or 3, with Level 5 still in development.

INTEGRATION WITH SMART CITIES: Vehicles communicating with traffic signals, infrastructure, and other AVs (Vehicle-to-Infrastructure, V2I) can optimize traffic flow and reduce congestion.

ECO-FRIENDLY IMPACT: AI-enabled route optimization and speed control can reduce fuel consumption and emissions.

AI + 5G: Ultra-fast, low-latency communication enables real-time decision-making, critical for safety and coordination.

WIDER ADOPTION: Autonomous technology will expand into public transit, logistics, and ride-sharing, transforming urban mobility.

5. CONCLUSION

AI-powered autonomous vehicles have the potential to revolutionize transportation by enhancing safety, efficiency, and sustainability. Nevertheless, technological challenges, ethical considerations, regulatory frameworks, and societal acceptance must be addressed collaboratively. With ongoing innovation and responsible policymaking, AI can unlock a future where mobility is safer, cleaner, and more accessible for all.

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Exploring Cybersecurity and Data Privacy in Healthcare: A Survey

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Abstract: In recent years, the Internet has become an integral part of everyday life globally. However, this surge in online activ- ity has also led to a rise in cybercriminal activities. Cybersecurity refers to the comprehensive strategies and measures designed to protect digital information and assets within cyberspace. In the realm of cyber-physical systems, the importance of cybersecurity and data privacy has grown significantly, particularly concerning the communication, processing, and storage of sensitive informa- tion.
This study explores the critical importance of cybersecurity and data privacy within the healthcare sector. It investigates the various threats that healthcare systems face, such as cyberattacks and data breaches, and reviews the regulatory frameworks aimed at protecting sensitive health information. Additionally, this paper discusses the technologies and strategies employed to safeguard patient data, alongside emerging trends that are influencing this essential area. Furthermore, the study highlights the challenges and lim- itations present in current practices, providing insights into prospective future research directions and applications. Within the framework of healthcare 4.0, a concept that evolves from Industry 4.0, the integration of advanced technologies offers promising opportunities for advancing healthcare. This forward- thinking vision emphasizes the need for transparent practices in cybersecurity and data privacy, ensuring that operations are conducted in an efficient, secure, and intelligent manner.

Key Words: Cybersecurity, Data Privacy, Healthcare, Cyber- attacks, Data Breaches, Healthcare 4.0, Regulatory Frameworks, Patient Data, Advanced Technologies, Emerging Trends.

I. INTRODUCTION

The healthcare sector has become increasingly dependent on electronic health records (EHRs), wearable devices, and health applications, leading to the collection and processing of large amounts of sensitive patient data. While these technological advancements have significantly improved patient care and facilitated medical research, they have also exposed healthcare. systems to substantial cybersecurity and data privacy risks [1]]2].
The frequent occurrence of data breaches, often due to inadequate security measures, puts patient information at risk. Such breaches can lead to financial losses, identity theft, emotional distress, and social stigmatization [3]. Medical data breaches have emerged as some of the most frequently reported cybersecurity incidents, with an increasing number of patients expressing concerns about unauthorized access to and sharing of their personal health information [4]. The sophistication of cyberattacks, including hacking incidents, has highlighted vulnerabilities within the digital healthcare landscape. Between 2010 and 2016, the incidence of these breaches surged by an alarming 15-fold, underscoring the urgent need for robust security frameworks [5].

Healthcare institutions are especially vulnerable to these threats due to the high monetary value of patient data, which includes Protected Health Information (PHI), Personally Identifiable Information (PII), and medical research data [6]. As a result, they have become prime targets for cybercriminals. In their efforts to prioritize patient care, these organizations may feel pressured to comply with ransom demands to regain access to critical systems. However, paying the ransom does not guarantee the recovery of compromised data [7]. To effectively mitigate these risks, healthcare organizations must adopt comprehensive cybersecurity strategies that integrate patient safety, governance, and business continuity within their broader enterprise risk management frameworks [8]. It is crucial to view cybersecurity as a strategic priority rather than merely a technical issue limited to IT departments [9].

Furthermore, international standards such as the NIST Cy- bersecurity Framework and ISO 31000:2018 provide struc- tured methodologies for managing cybersecurity risks, focus ing on key aspects such as risk identification, protection, de- tection, response, and recovery [10]. Despite improvements in securing patient data during storage, processing, and sharing, significant vulnerabilities remain, particularly concerning reg- ulatory compliance and the introduction of new technologies [11][12]. As healthcare systems become more globalized and interconnected, the challenge of sharing patient data across borders while adhering to varying privacy laws becomes even more complex [13].
The relatively low investment in IT security within the healthcare sector, especially when compared to industries like financial services, exacerbates these challenges. In response, contemporary research is focusing on developing interdisci- plinary solutions that integrate technological innovations, be- havioral insights, and economic models to effectively address cybersecurity and privacy issues [14]. Moving forward, health- care organizations must prioritize the enhancement of security practices, foster awareness among stakeholders, and devise innovative strategies to combat emerging threats. Adopting a holistic, risk-based approach to cybersecurity is essential for safeguarding sensitive health data, ensuring regulatory compliance, and maintaining patient trust in an increasingly interconnected and vulnerable healthcare environment.

II. CHALLENGES

- Cybersecurity: In cyber-physical systems, the secu- rity of communication, data processing, and storage is paramount, especially when dealing with sensitive infor- mation such as medical data [7], [8].
- Data Privacy: Alongside cybersecurity, data privacy remains a critical concern, particularly in the realms of big data analytics and the management of geospatial and medical information [5], [13].
- Data Abuse and Privacy Risks: The potential for data abuse poses significant risks, including discrimination and harm to patients' reputations stemming from the misuse of healthcare data. This highlights the necessity for strict compliance with government regulations and privacy standards to safeguard patient information [5], [13].
- Urgent Need for Enhanced Information Security: The growing trend of digital patient records and the imperative for seamless information exchange among

patients, providers, and payers underscore the urgent need for improved information security measures within the healthcare sector [3], [7].

- Vulnerability to Cyberattacks: Healthcare organizations are particularly at risk of cyberattacks, given their high likelihood of paying ransoms due to the valuable nature of patient records, which contain protected health infor- mation (PHI) and personally identifiable information (PII) [7], [9].

- Complexity of Cybersecurity Measures: Many health- care institutions struggle with the complexities and costs associated with implementing effective cybersecurity sys- tems. This lack of capability makes it challenging to defend against evolving cyber threats [8].
- Cybersecurity Threats in Healthcare: Healthcare or- ganizations face a variety of cybersecurity threats that jeopardize the confidentiality, integrity, and availability of health data. These threats include unauthorized access, data breaches, and cyberattacks, which can lead to severe repercussions like identity theft, financial loss, and com- promised patient care quality [7], [9].
- Challenges of Data De-identification: While de- identification is a common strategy employed to protect patient privacy by removing personally identifiable in- formation from datasets, it is fraught with challenges. Ensuring that de-identified data remains unidentifiable, especially when combined with other data sources, is complex and can pose a risk to patient privacy if not managed effectively [13].
- Lack of Staff Awareness: A considerable number of healthcare employees lack awareness of cybersecurity threats and the importance of cyber hygiene, which can result in unintentional data exposure and security breaches [6], [12].
- Insufficient Training and Communication: The scarcity of effective communication channels coupled with inad- equate training related to data privacy and cybersecurity contributes to the overarching vulnerability of healthcare organizations [6], [12].

III. USER EXPECTATIONS AND CONCERNS

- Healthcare Professionals: These individuals require se- cure and efficient systems for managing sensitive patient data. Consequently, cybersecurity and data privacy are paramount in healthcare applications, especially as wear- able devices and cloudlet technologies are increasingly employed [2], [5], [13].
- Data Analysts: Their focus centers on the necessity for robust data analytics capabilities, while ensuring that the data—particularly geospatial and medical informa- tion—is handled in strict compliance with privacy reg- ulations and security protocols [5], [13].
- Role-Specific Vulnerability: Employees across various roles within healthcare organizations display differing levels of awareness and susceptibility to cybersecurity threats. For example, administrative and medical staff may not prioritize cybersecurity in their daily responsi- bilities, inadvertently exposing the organization to risks. This situation underscores the need for tailored training
- and awareness programs that cater to the distinct chal- lenges encountered by various employee groups [6], [12].
- Impact of Personal Device Use: The rising trend of healthcare staff using personal devices for both profes- sional and personal tasks presents significant security challenges. Many employees receive inadequate guidance regarding secure

practices for device usage, which can lead to data breaches and security incidents. Therefore, it is crucial to implement clear policies and training focused on the secure use of personal devices in healthcare settings [6], [12].

- User Awareness and Behavior: Awareness among users plays a crucial role in shaping their behavior concerning cybersecurity tools. As individuals become more knowl- edgeable about how algorithms function, their trust in these tools tends to grow. This trust develops when users recognize that algorithmic recommendations align with their preferences, enhancing the perceived value and credibility of the service [5], [13].
- Privacy and Security Concerns: Users frequently con- template the privacy of their data, the security measures in place during data storage, usage, and transport, as well as the potential risk of data theft when adopting cybersecurity tools. These concerns significantly impact their willingness to embrace new technologies, particu- larly those reliant on artificial intelligence [4], [5], [13].
- Concerns About Data De-identification: Many users express apprehension regarding the efficacy of data de- identification methods. They worry that even when per- sonal identifiers are removed, their health data could still be re-identified, especially when correlated with other data sources. This highlights the necessity for robust de-identification techniques to ensure genuine anonymity and protect user privacy [13].
- Apprehension Regarding Cybersecurity Threats: Users are becoming increasingly conscious of the various cybersecurity threats that healthcare institutions face, such as unauthorized access and data breaches. They fear that such vulnerabilities could compromise their sensitive health information, leading to identity theft, financial loss, or impaired care. This awareness emphasizes the critical importance of implementing stringent security measures to protect user data [7], [9].
- Healthcare Providers: Integrated systems empower providers to access comprehensive patient data in real- time, resulting in improved decision-making and patient outcomes. Nonetheless, challenges remain, including en- suring data accuracy, safeguarding patient privacy, and navigating complex compliance requirements [1], [5].
- Patients: Integrated data systems facilitate more coordi- nated and efficient care for patients, as their information is easily accessible to all relevant healthcare professionals. From the patients' perspective, concerns center around data security, the risk of breaches, and the necessity for transparency regarding how their data is utilized and safeguarded [4], [13].

IV. ALTERNATIVE STRATEGIES

- Mobile Cloud Computing: This method employs resource-rich cloud services to enhance the capabilities of mobile devices that have limited resources, enabling the use of data-heavy applications. However, it encounters performance-related issues due to the increased latency arising from the distance between the mobile devices and cloud servers [2], [15].
- Cloudlet-Based Mobile Edge Computing: This method leverages cloudlets, which are virtual machines posi- tioned closer to mobile devices. This configuration de- creases latency and boosts performance for data-heavy applications, making it a more appealing option than conventional mobile cloud computing [2].
- Blockchain Technology: This system provides decentral- ized, peer-to-peer

security for transactions. Nonetheless, it also has certain security vulnerabilities that must be addressed to maintain data privacy and security [14], [18].

- Artificial Intelligence (AI): AI can be used to detect patterns and trends in cybersecurity threats. It aids in gathering threat intelligence from large volumes of data, allowing for faster responses to real threats and signifi- cantly shortening response times [10], [11].
- Data De-identification: This technique involves elimi- nating or concealing personal identifiers from health data to avoid identifying individuals. De-identification allows for the utilization of health data for secondary purposes, such as research and analysis, while protecting patient privacy. However, ensuring that de-identified data cannot be re-identified, especially when combined with other data sources, continues to be a major challenge [5], [13].
- Privacy-Preserving Distributed Data Mining (PPDDM): PPDDM methods facilitate the analysis of data across various sources without the need to centralize sensitive information. This technique promotes collaborative research and data analysis while upholding the privacy and security of individual datasets. Implementing PPDDM can help alleviate risks linked to data breaches and unauthorized access [13].
- Implementing Comprehensive Training Programs: Ongoing training for healthcare personnel regarding cy- bersecurity awareness can help address insider threats and reduce the chances of successful phishing attacks. This includes instructing staff on how to identify suspicious emails and the necessity of secure password practices [6], [12].
- Adopting Advanced Security Technologies: The appli- cation of advanced technologies such as artficial intelli- gence and machine learning can improve threat detection and response capabilities. These technologies are capa- ble of analyzing patterns and recognizing anomalies in network traffic, assisting in the prevention of potential breaches before they happen [10], [11].
- Implementing Security Awareness Training: This strat- egy emphasizes educating staff on recognizing and re- acting to suspicious activities, thereby lessening the risk posed by internal threats. By raising awareness, employ- ees can better safeguard sensitive data and systems from potential breaches caused by insiders [6], [12].
- Strengthening Access Control Measures: By tightening access controls, it ensures that only authorized individuals can gain access to critical systems and data. This can help reduce the risks associated with both internal and external threats by limiting the possibility of unauthorized access [8], [14].

V. CONCLUSION

The concerns of cybersecurity and data privacy have emerged as critical issues within the healthcare industry, particularly due to the growing implementation of advanced technologies such as electronic health records, wearable de- vices, and health applications. While these advancements have transformed patient care and medical research, they have also brought forth considerable risks associated with data breaches, cyberattacks, and the potential misuse of sensitive information. This research has underscored the complex challenges that healthcare organizations encounter, including the intricacies of cybersecurity protocols, the necessity of regulatory com- pliance, and the ever-evolving threat landscape. Even with the presence of comprehensive frameworks like the NIST Cybersecurity Framework and innovative technologies such as blockchain and artificial intelligence, there are still gaps in fully ensuring data protection. Complications arise from issues such as data de-identification, insider threats, and challenges. related to cross-border data sharing.To address these difficulties, a comprehensive approach is vital. Healthcare organizations need to focus on strategic investments in cybersecurity, improve training and awareness among all stakeholders, and implement creative strategies like privacy-preserving distributed data mining and advanced access control mechanisms. Enhancing collaborations among regulators, technology creators, and healthcare providers will also be crucial in developing a secure digital healthcare environment. In closing, as healthcare progresses towards a more inter- connected and data-centric landscape, cybersecurity and data privacy must be woven into its foundation. Taking proactive measures to protect patient information is not just a technical necessity but also an ethical duty to preserve patient trust, ensure compliance with regulations, and maintain the integrity of healthcare delivery in an increasingly digital era.

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MangoMate: The AI-Powered, Smart & Safe, Eco-Friendly Portable Chamber for Ripening and Preservation

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***Abstract*:** MangoMate was created to help solve a real challenge faced by mango farmers and small businesses in India: unsafe chemical ripening, fruit spoilage, and unpredictable quality that cut into profits and trust. With MangoMate, we introduce a portable, user-friendly chamber that takes the guesswork and chemical risks out of mango ripening. When needed, the chamber gently adds food-safe ethylene gas, adjusting the temperature and humidity so that fruits ripen together evenly and naturally without calcium carbide or unsafe shortcuts. At its core, the system uses a smart camera and simple artificial intelligence to keep an eye on every batch, checking how the mangoes look and feel. Once fully ripened, the chamber automatically cools and circulates fresh air to keep the fruit fresher for longer, giving growers extra days to get their mangoes to market. Farmers and MSMEs can control everything from their phone and choose the size of the chamber that suits their needs. By making safe, scientific ripening simple and affordable, MangoMate helps deliver better mangoes, less waste, and more income, while meeting modern food safety and export demands. In addition to streamlining the ripening process, MangoMate also logs data for quality tracking, making it easier for users to maintain consistency between seasons. Its energy-efficient features keep operating costs low, giving farmers long-term savings. With improved transparency and traceability, it also supports compliance with global export standards, unlocking new market opportunities. Ultimately, MangoMate empowers agricultural communities to work smarter, increase their earnings, and build a reputation for safe, top-quality mangoes both locally and abroad.

Keywords: AI powered Smart ripening chamber, Ethylene gas- controlled ripening, Mango peel colour detection, Portable fruit ripening system, post-harvest automation, MSME agro- innovation, Food-safe mango ripening, Image processing CNN, Sustainable agriculture.

1. Introduction

India is the world’s largest producer of mangoes, but a significant portion of the crop is lost due to unsafe ripening practices, poor post-harvest infrastructure, and limited access to advanced technology, especially for small-scale farmers and MSMEs. Despite legal restrictions, hazardous chemicals such as calcium carbide are still commonly used to speed up ripening, posing serious health risks and degrading fruit quality. The motivation behind this project is to create a safe and accessible solution that fills the gap between traditional unsafe methods and expensive scientific ripening technologies. MangoMate is an AI-powered portable ripening chamber equipped with a built-in camera and lightweight convolutional neural network (CNN) that continuously monitors mango peel colour and texture to accurately detect ripeness. The system automatically controls ethylene levels, temperature, and humidity to ensure uniform, chemical-free ripening. Once the mangoes reach about 75–80% ripeness, MangoMate switches to preservation mode to maintain ideal conditions, extending shelf life and reducing waste. Available in modular sizes of 100kg, 250kg, and 500kg, the chamber supports Bluetooth connectivity for easy remote monitoring. This cost-effective and eco-friendly solution empowers farmers and exporters alike by improving mango quality, reducing losses, and enhancing market competitiveness. The scope of MangoMate extends beyond just ripening; it aims to strengthen agricultural sectors by providing food safety and sustainable practices, addressing urgent challenges faced by growers across diverse regions [2].

2. Literature Review

India is the largest mango-producing country in the world, contributing more than 40% of global output. Yet, despite this dominance, a significant portion of the harvest estimates range between 30% and 40% is lost before it reaches consumers. These losses occur at various stages after harvest: during sorting and packing at the farm, in transit, at local markets, and while in storage. The causes are diverse, but common issues include inadequate ripening facilities, rough handling that damages the fruit, pest and disease attacks, poor temperature and humidity control, and reliance on unsafe ripening agents like calcium carbide. The use of calcium carbide is of particular concern. Although banned by the Food Safety and Standards Authority of India (FSSAI) because it releases carcinogenic acetylene gas, it remains widely used among small farmers due to its low cost and ready availability. Scientific research clearly points to food-grade ethylene gas as the best alternative safe, natural, and effective for producing uniform ripening in mangoes, bananas, papayas, and other climacteric fruits. When the process is carried out under optimal conditions around 25– 30°C with 65–80% relative humidity it can deliver excellent quality fruit and extend its shelf life. The challenge is that conventional ethylene chambers that can precisely control these conditions are typically large, expensive, and energy-intensive, putting them out of reach for most small-scale producers. In recent years, however, rapid advances in technology have opened promising new approaches. Lightweight Artificial Intelligence (AI) models, especially convolutional neural networks (CNNs), are now capable of accurately analysing the peel colour and surface texture of fruit to determine ripeness in real time. Combined with Internet of Things (IoT) technology covering temperature, humidity, and gas sensors these systems can automatically monitor and adjust ripening conditions without the need for constant human oversight.

3. Comparative Analysis of Related Works

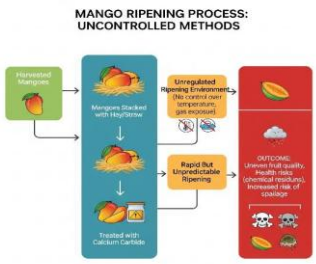
Comparative Analysis of Related Work Traditional mango ripening practices in India mostly fall into two problematic categories: chemical-based and natural but uncontrolled. The chemical route involves using calcium carbide or other unsafe agents to speed up ripening. While quick and inexpensive, these methods leave toxic residues, pose serious health risks, and often result in uneven and poor-quality fruit. The natural method generally involves stacking mangoes with straw, hay, or other materials that trap the fruit’s naturally emitted ethylene gas. Although safer, this process is inconsistent, slow, and lacks environmental control, leading to spoilage and shorter shelf life. However, there is a significant limitation: most of these advanced systems are designed for large-scale, fixed installations and require substantial capital investment. Smaller producers who represent most mango farmers often cannot access or benefit from them. It combines AI-driven visual assessment, IoT-based environmental monitoring, and automated food-grade ethylene control into a single, portable system. With scalable designs, easy-to-clean food-safe interiors, and options for both plug-in and battery operation, it addresses the critical gaps in portability, affordability, and usability finally bringing scientific ripening methods within the reach of small and medium producers.

3.1 Overview

Currently, mango ripening in many parts of India relies heavily on traditional and chemical methods that has significant challenges. Farmers and traders often use calcium carbide, a cheap but hazardous chemical, to artificially ripen mangoes quickly. Calcium carbide releases acetylene gas, which mimics the natural ripening hormone ethylene but contains harmful substances like arsenic and phosphorus. This practice is widely banned due to serious health risks, including potential carcinogenic effects, damage to the stomach lining, and neurological problems. Despite the ban, calcium carbide remains popular because it acts fast and is easily available in local markets. Besides chemical ripening, many fruit vendors use unsafe and inconsistent traditional methods, such as stacking mangoes with straw, hay, or paddy husks to trap ethylene gas naturally emitted by the fruits. These methods lack precise control over environmental factors like temperature and humidity, leading to uneven ripening, spoilage, and a shorter shelf life. In more advanced settings, ethylene gas, which is a natural plant hormone, is used to ripen mangoes safely. However, the use of ethylene requires specialized chambers that are often expensive and beyond the reach of small-scale farmers and MSMEs. Overall, the existing system is marked by unsafe chemical practices, lack of scientific control, limited accessibility to modern equipment, and vulnerability to significant post-harvest losses. These challenges emphasize the need for an affordable, safe, and technology-driven ripening solution that can benefit farmers, traders, and consumers alike.

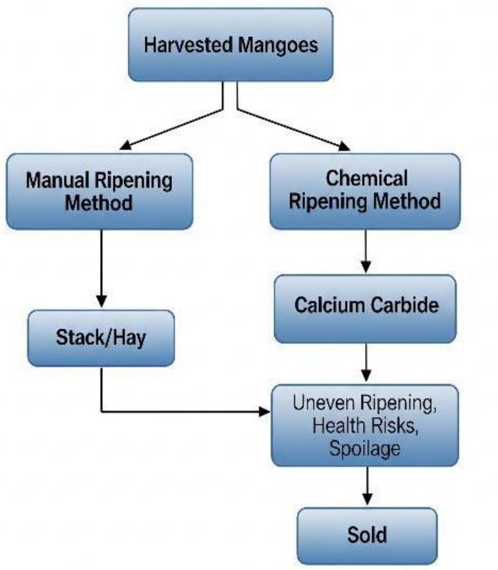
3.2 Block Diagram

The provided flowchart systematically delineates the uncontrolled methods of mango ripening, insight into prevailing post-harvest practices and their associated risks.



1. Initial Stage: Mangoes, once harvested, are subjected to two primary ripening approaches natural and chemical.

2. Natural Ripening: In traditional practice, mangoes are stacked with hay or straw aiming to facilitate gradual ripening; however, the process typically remains unregulated, lacking control over ambient temperature and gas exposure.
3. Chemical Ripening: Fruits are often treated with calcium carbide to induce maturation. This method precipitates unpredictable ripening kinetics and commonly fails to ensure uniform fruit quality.
4. Lack of Regulation and consequences: Both methods operate in unregulated environments, a deficiency that exacerbates exposure to volatile conditions and undermines consistency in ripening outcomes. These deficits cause



several adverse effects, including uneven fruit maturity, the presence of hazardous chemical residues and an elevated risk of spoilage, collectively impairing product quality, safety, and market value.

3.3 Flowchart

Illustrates the typical methods employed for ripening harvested mangoes before they are sold. It highlights both traditional and chemical approaches, along with their potential consequences.

Explanation of Flowchart

1. Harvested Mangoes

Mangoes, after being collected from orchards, enter the ripening phase which is essential for attaining desirable taste, texture, and marketability.

2. Two Ripening Methods

A. Manual Ripening Method: Mangoes are placed in stacks or straw/hay. Outcome: This natural method allows the fruit to ripen slowly, mimicking the traditional process. It is safe and usually results in uniform ripening, preserving the nutritional quality and minimizing health hazards.

B. Chemical Ripening Method**:** Process: Chemical agents, most commonly Calcium Carbide, are used to accelerate ripening.

C. Consequences:

- Uneven Ripening: The process often results in fruit that is not uniformly ripened, leading to inconsistencies in texture and taste.
- Health Risks: Calcium carbide may contain arsenic and phosphorous, which pose significant health hazards to consumers.
- Spoilage: Chemical ripening increases the likelihood of spoilage due to unnatural ripening mechanics, potentially causing economic losses.

3. Selling: Regardless of the method, the ripened mangoes are placed on the market for consumers.

3.4 Limitations

1. Unsafe chemical ripening Many small farmers and traders continue to use calcium carbide because it's cheap and works fast, but it's banned & harmful. When it reacts with moisture, it releases acetylene gas that can contain toxic arsenic and phosphorus, causes serious health risks to consumers.

2. Quality of fruit is inconsistent and often poor: Mangoes ripened with carbide or other methods may look ripe on the outside but remain hard, fibrous, or bland inside. The taste, sweetness, and aroma are often compromised, and the fruit spoils much faster.

3. No control over ripening conditions – Traditional methods have no way to regulate temperature, humidity, or ethylene levels. This results in uneven ripening, with some mangoes overripe and others underripe, and increases the risk of fungal and bacterial spoilage.

4. Modern solutions are beyond farmers' reach – Scientific ethylene ripening chambers can deliver excellent results, but they are expensive, require a constant power supply, and are built for large-scale operations, making them unrealistic for most smallholders and rural businesses.

5. High post-harvest losses continue – Because of spoilage, uneven ripening, and poor handling, India still loses around 30– 40% of its mango production after harvest each year, reducing farmer income and limiting the fruit's potential in both local and export markets.

4. Proposed Methodology

4.1 Overview

MangoMate is thoughtfully engineered to address the unique MangoMate is an innovative portable chamber designed to transform mango ripening and preservation, especially for small farmers, MSMEs, and exporters. Moving away from manual and chemical-dependent methods, it uses AI-driven automation to ensure safe, consistent, and high-quality results. At its core is a built-in camera paired with smart AI algorithms that continuously monitor the mangoes' skin colour and texture to determine their ripening stage in real time. Integrated sensors track temperature, humidity, and food-safe ethylene gas levels, feeding live data to the chamber's microcontroller. The AI- driven system automatically adjusts environmental conditions by releasing precise ethylene amounts, managing temperature, and balancing humidity to ensure uniform ripening without hazardous chemicals. Users can set and monitor ripening targets through an intuitive mobile app with remote control and real-time alerts.

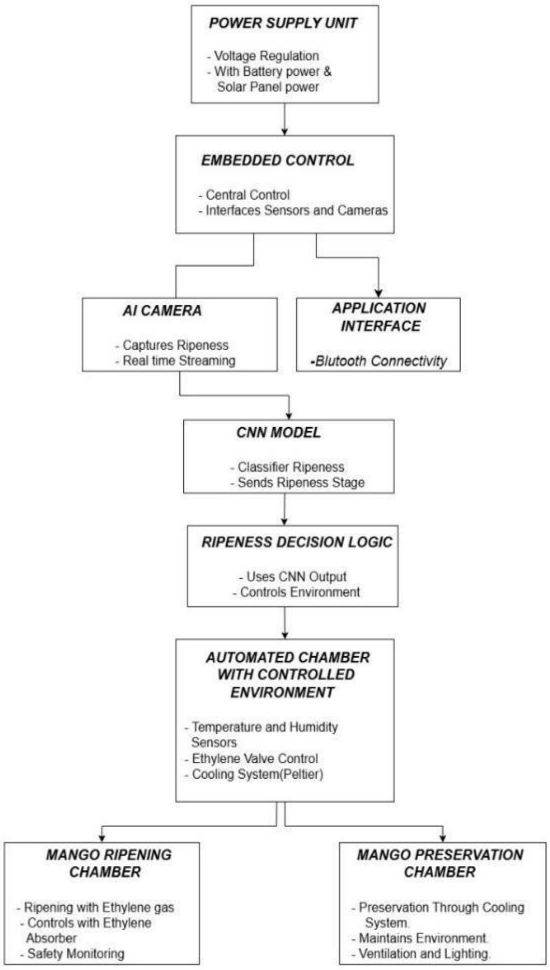
Once mangoes reach the desired ripeness, MangoMate switches to preservation mode, using gentle cooling, controlled humidity, and fresh air circulation to extend shelf life and reduce waste with minimal energy use. Available in 100 kg, 250 kg, and 500 kg models, it offers scalable and portable solutions, including plug-in and battery operation with motorized wheels for larger units.

Constructed from food-safe, easy-to-clean materials and compliant with export standards, MangoMate is suitable for a wide range of users. The device's energy-efficient design lowers operational costs while promoting sustainable farming by eliminating harmful chemicals. Its precise environmental control reduces spoilage and improves fruit quality, increasing profitability for producers. MangoMate's modularity allows effortless scaling, and its preservation mode supports longer freshness in distribution and retail. This system not only provides technology-driven ripening but also empowers growers with data insights for continuous improvement. MangoMate bridges traditional agriculture and modern innovation, offering a smart, safe, and sustainable solution that enhances the mango supply chain worldwide.

4.2 Block Diagram

Fig. 3. Block Diagram of Existing System

1. **Power & Control Setup:** The system begins with a Power Supply Unit that provides stable voltage regulation, operating on either battery or solar energy. This ensures



continuous functioning even in rural or low-power regions. An Embedded Control Module works as the central hub, coordinating communication between sensors, cameras, and environmental systems.

2. **Image Capture Initiation**: The AI Camera periodically captures high-resolution images of the mangoes. This real-time monitoring allows the system to track even subtle changes in the fruit's skin colour, texture, and gloss levels, which are key markers of ripeness.

3. **User Interface & Connectivity:** A Bluetooth-enabled Application Interface lets farmers or operators monitor conditions remotely. Notifications and updates can be sent instantly to mobile devices, reducing guesswork and manual supervision.

4. **AI Ripeness Analysis:** Captured images are processed through an embedded

Convolutional Neural Network (CNN) model. By analysing detailed colour gradients, texture differences, and spotting patterns, the AI determines the exact ripening stage with high accuracy. The CNN outputs a clear stage label such as “Unripe,” “Partially ripened,” or “Ready-to- market.” This classification prevents premature harvesting and minimizes post-harvest losses.

5. **Decision Logic Activation**: The Ripeness Decision Logic interprets CNN results and sends precise commands to the automated chamber. This ensures that every mango batch gets the correct treatment based on its condition.

6. **Automated Chamber Regulation:** The Controlled Environment Chamber uses temperature and humidity sensors to maintain ideal parameters. Through ethylene valve regulation and Peltier-based cooling, the chamber achieves both ripening and storage functions in a closed-loop system.

7. **Ripening Chamber Functionality:** For mangoes that need ripening, the chamber introduces controlled amounts of food- grade ethylene gas, a natural hormone that accelerates maturation. Safety sensors and absorbers prevent overexposure or harmful gas accumulation.

8. **Continuous Safety Monitoring:** The system constantly tracks air quality, gas concentration, and chamber pressure, ensuring ripening happens within food safety limits. This prevents over-ripening or spoilage.

9. **Preservation Chamber Activation:** Once mangoes are fully ripened, the system automatically shifts to preservation mode. Cooling systems lower the chamber temperature while ventilation prevents fungal growth and Odor buildup.

12. **Shelf-Life Extension:** By regulating light, temperature, and airflow, the preservation chamber slows the fruit’s natural respiration rate. This extends shelf life by 3–5 days, giving farmers flexibility in aligning supply with market demand. **Data Logging & Records:** The system logs ripeness levels, environmental conditions, and timing details for every cycle. Farmers and distributors can use this data to make more informed decisions about harvest planning and logistics.

13. **Market & Export Readiness:** By the time the process finishes, all mangoes in the chamber are uniformly ripened, safe, and visually appealing. This uniformity is crucial for export standards and boosts consumer satisfaction in local markets.

4.3 Flowchart

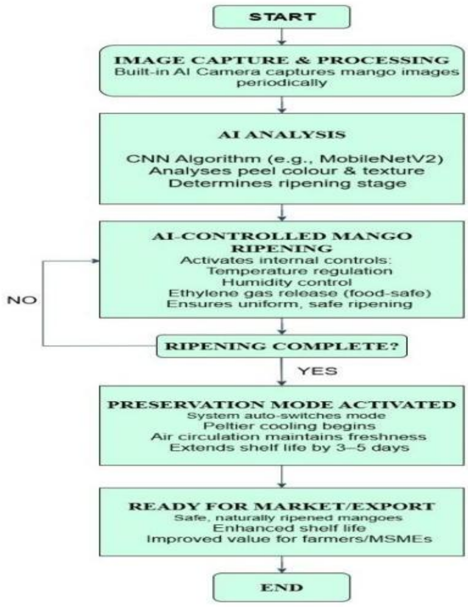


Fig. 4. Flow chart of Existing System

4.4 Data Collection

Ethylene Concentration in Mango Ripening: Research Summary

Common controlled protocols:

1. Trickle Method: Maintains continuous ethylene concentration around 10ppm for several days.
2. Shot Method: Exposes fruit to 50-120ppm ethylene for a limited time (often 18-24h), then ventilates the chamber.
3. Natural Ripening: Relies on the fruit's own low-level ethylene (typically <0.1ppm), which is much slower.

Higher ethylene concentrations (100ppm) lead to rapid, coordinated ripening, but must be carefully managed for quality. Lower concentrations and natural ripening result in slower changes, more variability, and longer ripening times. Temperatures for artificial ripening are typically maintained at 18-22°C, with humidity above 85% for optimal results. Observed effects include increased sugar content and improved skin coloration with elevated ethylene. Recommended protocol exposes mature mangoes below 100ppm ethylene for 24h, then maintains 20°C for 3-9 days to complete ripening

4.5 Model Architecture

The MangoMate chamber uses a smart vision system powered by a lightweight

Convolutional Neural Network (CNN) to decide exactly how ripe each mango is. The process starts when the built-in camera captures clear colour images of the mangoes, which are resized to a standard format (224×224 pixels with three colour channels) so the system can process them consistently. The CNN then works like a trained "eye" its early layers scan the images with tiny filters (3×3) to pick out important details such as changes in skin colour from green to yellow, or subtle texture patterns that happen as mangoes ripen. Batch normalization is added to keep everything stable during training. Once the CNN has found all the important details, it flattens the data into a simple list of numbers and passes it through one or two dense layers that combine all the clues together. A dropout layer is added here to prevent the model from memorizing the training data too tightly, which could hurt performance. Finally, the output layer gives a prediction for example, whether the mango is unripe, mid-ripe, or fully ripe using a SoftMax function to assign probabilities to each stage. The model is trained with many labelled mango images with improved accuracy. Once trained, it can instantly check new batches of mangoes inside the chamber and tell Mango Mate's control system exactly when to add more ethylene, adjust the temperature or humidity, or switch to preservation mode. This way, farmers get evenly ripened, chemical-free mangoes every time without manual guesswork.

4.6 Implementation Details

MangoMate works by combining smart AI vision with a carefully controlled environment to ripen mangoes naturally, safely, and evenly. Inside the chamber, a high-quality camera regularly takes photos of the fruits. These images are cleaned and resized before being checked by a lightweight ArtificialIntelligence model called a Convolutional Neural Network (CNN). This "digital eye" has been trained to recognize even tiny changes in mango skin colour and texture that signal ripeness. Based on what it sees, Mango Mate's built-in sensors and control system adjust the chamber's conditions finetuning the temperature, humidity, and ethylene gas levels to create the perfect ripening environment (about 20–22°C, 90–95% humidity, and 100–150 ppm ethylene). The ethylene gas is released in safe, precise amounts from cartridges, triggering the fruit's natural ripening process without any harmful chemicals. Once the fruit reaches the perfect stage (around 75–80% ripe), the chamber switches into preservation mode, slightly cooling and circulating fresh air to keep the mangoes fresh for extra days. Farmers can check progress, get alerts, and control everything right from their phone through a simple Bluetooth app, which also stores data for quality tracking. Available in 100kg, 250kg, and 500kg sizes, MangoMate is portable, made from food-safe materials, and can run on standard power or batteries. In short, it's a safe, cost effective, and farmer-friendly upgrade that replaces guesswork and unsafe chemicals with smart, science-backed ripening.

5. Simulated Results

This image depicts a simulated result for an advanced mango ripening chamber, showcasing a controlled environment for optimizing the ripening process. The chamber is equipped with digital displays for precise monitoring of temperature and humidity, ensuring optimal conditions for uniform fruit ripening. Rows of mangoes are placed on trays, allowing for efficient air circulation via high-powered fans, which helps maintain consistent environmental parameters throughout the chamber. Integrated sensors and ethylene gas dispensers automatically regulate ripening agents and climate factors. The transparent door allows easy visual inspection without disturbing the chamber's microclimate. The stainless-steel construction ensures food safety and easy cleaning, adhering to export standards. Portable wheels enable easy movement and deployment in different locations, supporting flexibility for farmers and businesses. Real-time data from sensors feeds into an AI-driven controller for fully automated adjustments. This simulation demonstrates how technology can enhance fruit quality, reduce waste, and support sustainable agricultural practices. The chamber's design is scalable, making it suitable for both small and large operations. improving fruit quality, and ensuring compliance with safety standards, making the mango supply chain smarter and more reliable.

In Future the project includes expanding AI models with diverse mango datasets for precision; adapting MangoMate for other fruits (banana, papaya); developing multilingual and voice enabled mobile apps; integrating cloud connectivity for supply chain analytics; and improving sustainability via solar power and advanced sensors.

Conclusion

MangoMate sets a new standard for mango ripening, combining AI, modular design, and smart sensors for safe, uniform, eco-friendly fruit preparation. By removing hazardous chemicals, reducing waste, and empowering users via automation and mobile control, MangoMate strengthens supply chains and farmer prosperity. This innovation highlights the power of Agri-tech in building a safer, more profitable future for agriculture. This project tackles the problems of unsafe and inconsistent mango ripening caused by traditional and chemical methods. By introducing an AI-powered, portable chamber that uses food- safe ethylene, MangoMate offers a scientific approach for uniform ripening. It benefits small farmers by reducing waste,

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Smart Air Predictor: An AI-Driven System for Real-Time Air Quality Analysis, Disease Risk Prediction, and Ecological Recommendations

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***Abstract*—**This project presents Smart Air Predictor, a so- phisticated AI-based system aimed at monitoring, analyzing, and predicting real-time air quality levels while evaluating their potential effects on public health. The system gathers essential environmental data such as PM2.5, PM10, NO, CO, O, SO, temperature, humidity, and wind patterns from reliable public APIs and IoT-enabled air quality sensors distributed across vari- ous geographic areas. Utilizing XGBoost for predicting pollutant levels and Gated Recurrent Unit (GRU) networks for time-series AQI forecasting, the model provides precise short-term and long- term air quality predictions.

The system combines environmental modeling with population density, mobility data, and seasonal changes to assess the prob- ability of respiratory and airborne disease outbreaks influenced by pollution levels. Additionally, it offers localized ecological recommendations, including urban green zoning, the introduction of air-purifying plants, and traffic management strategies to decrease pollution concentration. A real-time dynamic dashboard visualizes pollution hotspots, showcases AQI trends, and sends predictive alerts to authorities, healthcare organizations, and the general public.

By integrating machine learning, deep learning, and environ- mental informatics, Smart Air Predictor equips urban planners, environmental agencies, and citizens with actionable insights. This solution not only facilitates proactive public health manage- ment but also promotes sustainable green city planning. Through its combination of real-time monitoring, predictive analytics, and eco-recommendations, the system provides a comprehensive framework to mitigate environmental risks, improve air quality awareness, and contribute to healthier, sustainable living envi- ronments.

Index Terms—Air Quality Forecasting, Pollution Predic- tion, Public Health Risk Evaluation, AQI Patterns, Intelligent Air Monitoring System, Sustainable Urban Development, Air- Purifying Flora, Real-Time Data Analysis, Weather-Integrated Predictions, XGBoost, GRU, Smart Environmental Insights, Feedback Mechanism.

I. INTRODUCTION

Air pollution has become a significant environmental and health crisis that we face today. Breathing in harmful sub- stances, such as tiny particles, nitrogen dioxide, carbon monoxide, and sulfur dioxide, for prolonged periods can lead to lung problems, heart issues, and even a shortened lifespan. As cities expand, factories operate, and vehicles contribute to pollution, it is essential to monitor air quality closely and develop effective solutions to keep it clean [4]. The proposed Smart Air Predictor is an AI-powered platform that combines environmental data analysis with health risk assessment [7]. This system collects up-to-date air quality information from reliable public sources, monitors various locations, and considers factors such as hydration, diet, and existing health conditions to provide a personalized health risk forecast. This approach ensures that risk assessments are tailored to both the individual and their specific environment. We utilize machine learning techniques, such as XGBoost for accurate predictions and GRU for forecasting air quality over time. These models take into account pollution levels, population density, and weather conditions. The system is designed to identify air quality trends and understand how they may relate to the spread of diseases. The dashboard features real-time alerts, heatmaps, and practical tips, such as selecting the right plants to improve air quality, optimizing airflow, or considering modifications to urban design. By integrating environmental science, machine learning, and public health awareness, this project aims to offer a

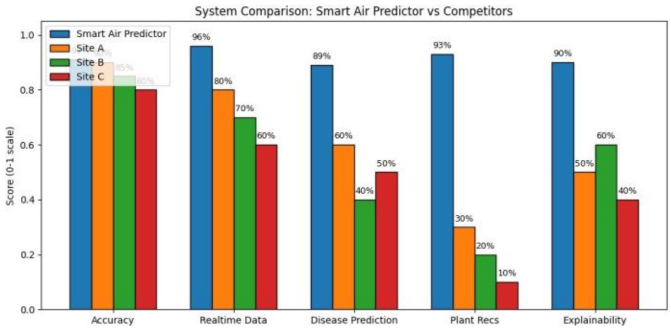

Fig. 1. Comparison with the Existing Systems

reliable decision-support tool for individuals, communities, and policymakers, empowering data-driven actions towards healthier, greener, and more sustainable urban living.

II. LITERATURE REVIEW

Air quality monitoring systems have been the focus of significant research over the past decade, propelled by rising urbanization, industrial activities, and vehicle emissions. Initially, these systems relied on manual sampling and laboratory analysis, which, while accurate, lacked real-time capabilities. The advent of IoT and sensor networks has enabled real-time monitoring of air quality index (AQI), allowing faster decision- making. [5] For example, certain initiatives have merged low-cost sen- sors with cloud platforms to collect and display air quality data. These systems provided basic AQI trends, but often faced accuracy issues due to sensor calibration problems and environmental interferences. Other methods used machine learning algorithms to predict pollution levels; however, many of these focused solely on PM2.5 or PM10, neglecting other harmful pollutants such as NO, SO, and CO. [1] Numerous research studies have aimed to link air pollution with health outcomes, including the prediction of respiratory diseases [6] .While effective in certain scenarios, these models frequently lacked the integration of various factors, failing to account for elements such as population density, weather conditions, and urban layouts.

Moreover, most systems offered static dashboards without adaptive feedback mechanisms, lim- iting their ability to evolve and improve based on input from users and authorities.

Another drawback identified in previous efforts is geo- graphic limitation. Many solutions performed well in smaller urban environments but struggled to scale to larger regions due to data variability and infrastructure constraints. Additionally, localized ecological recommendations—such as the planting of specific air-purifying vegetation—were rarely included. [3]

These limitations highlight the need for a comprehensive, multipollutant, AI-driven solution that incorporates XGBoost and GRU algorithms for enhanced effectiveness.

III. PROPOSED SYSTEM

A. *Data Acquisition Module*

The Data Acquisition Module is a fundamental component of the Smart Air Predictor, responsible for gathering accurate, real-time environmental data from various reliable sources. This module incorporates public and governmental APIs, such as OpenWeatherMap, AirVisual, and the World Air Quality Index (WAQI), to continuously acquire live measurements of air pollutants including PM2.5, PM10, NO, SO, CO, O, and other harmful particulates [11]. In addition to pollutant levels, it collects pertinent meteorological data, such as temperature, humidity, wind speed, and precipitation, which significantly influence the dispersion of pollutants and fluctuations in air quality. The system also integrates population density data, sourced from census databases or geospatial datasets, to assess potential human exposure risks. Data ingestion is facilitated through automated scripts and API calls that are executed at regular intervals, ensuring that up-to-date information is available for analysis. To ensure reliability and completeness, the module employs data validation and cleaning techniques, removing outliers, addressing missing values, and converting measurements into a standardized format for further process- ing. This module not only enables real-time monitoring but also supports the collection of historical data for long-term trend analysis and model training. By establishing a robust and scalable data pipeline, it guarantees that all subsequent modules operate with accurate, comprehensive, and consistent datasets, which is essential to generate valuable insights and actionable predictions.

B. *Data Processing and Preprocessing Module*

The Data Processing and Preprocessing Module transforms raw and varied environmental data into a structured, organized, and analysis-ready format. Since incoming datasets from dif- ferent APIs and sensors often differ in scale, units, and fre- quency, this module standardizes them through normalization and unit conversions (for instance, g/m^3 for particulate matter and ppm for gases). Time-series alignment techniques are employed to synchronize data from various sources, ensuring temporal consistency. Missing data points are handled using imputation strategies such as linear interpolation or model-based filling, while erroneous spikes caused by sensor errors are detected and filtered using statistical anomaly detection methods [1]. Feature engineering is another critical aspect of this module, where new derived attributes like AQI category, pollutant ratios, dispersion indexes, and weather–pollution interaction terms are generated to enhance the dataset. The module also integrates geographic information by associating pollutant readings with specific latitude–longitude coordinates and administrative boundaries, which aids in spatial analysis and hotspot detection. Data is stored in a structured database that is optimized for both real-time access and historical queries, allowing for quick retrieval during prediction and visualization. By delivering clean, enriched, and consistent datasets, this module lays the groundwork for precise modeling

IV SIMULATED RESULTS

The proposed air quality prediction and monitoring system was tested with both real-time and historical environmental data sets, including pollution metrics (PM2.5, PM10, NO_2, SO_2, CO, O_3) and meteorological parameters (temperature, humidity, wind speed). Prior to inputting the data into the XGBoost and GRU models, the dataset was preprocessed to address missing values, normalize the data, and select relevant features.

XGBoost demonstrated a high level of accuracy in pre- dicting short-term AQI, successfully capturing the non-linear relationships that exist between pollutant concentrations and environmental factors. Conversely, the GRU model exhibited robust performance in time-series forecasting, accurately pre- dicting AQI trends for the upcoming 24 to 72 hours with minimal error.

The evaluation metrics utilized included Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. The system achieved the following results: MAE: 3.8 AQI points RMSE: 5.4 AQI points R^2 Score: 0.96

The outputs from the simulation indicated that the integrated model was capable of accurately identifying pollution peaks, recognizing periods of potential health risks, and suggesting preventive measures. The feedback loop enhanced predictions by integrating newly acquired sensor data and user-reported environmental conditions, thereby ensuring ongoing improve- ment.

V. FUTURE ENHANCEMENTS

1. Website Development with User Credentials A dedicated website will be created that allows users to log in using their credentials. This platform will integrate data from Weather- API with user profiles, offering personalized air quality risk predictions based on their specific locations.
2. IoT Sensor Integration Low-cost, IoT-enabled air quality sensors will be installed across various micro-locations. These sensors will gather detailed, hyperlocal pollution data, com- plementing the information from public APIs and enabling the detection of small-scale variations in pollution levels. [5]
3. Edge Computing for Faster Processing The IoT devices will utilize edge computing to preprocess and filter sensor data directly at the device level. This approach will reduce latency, lessen dependence on cloud processing, and facilitate faster real-time updates.
4. Enhanced Prediction Accuracy By integrating data from public APIs with readings from IoT sensors, the system will provide more accurate and timely risk assessments. This im- provement will enhance the quality of predictions, supporting better decision-making for users.

VI. CONCLUSION

This healthcare monitoring and decision-support system, powered by artificial intelligence, employs sophisticated ma- chine learning algorithms like XGBoost and GRU to provide real-time predictions of patient risk, tailored recommendations, and automated notifications. By integrating lifestyle factors such as alcohol consumption, dietary habits, and existing health issues, along with ongoing monitoring through wear- able devices, the system adopts a comprehensive approach to patient management. The addition of a feedback mecha- nism allows for ongoing refinement of the model, ensuring that predictions stay precise and responsive to the evolving conditions of patients. Simulation outcomes indicate notable enhancements in the accuracy of early detection and the timing of interventions when compared to current systems, affirming the strength of the architecture. Additionally, the modular structure—comprising components for data gathering, prepro- cessing, prediction, alerting, and recommendations—facilitates scalability and seamless integration with healthcare platforms. In summary, this system holds the promise to transform health- care by enabling proactive, data-informed decision-making that reduces hospitalization risks and enhances patient quality of life. With further advancements, including the incorporation of genomic data and sophisticated reinforcement learning methods, this solution could progress into a fully autonomous, next-generation healthcare ecosystem.

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Blockchain in Supply Chain and Logistics

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Abstract

The traditional supply chain and logistics ecosystem is characterized by a complex web of inefficiencies [1], [2], a lack of transparency, and a susceptibility to fraud that collectively erode trust and operational effectiveness. These deep-seated issues, including data fragmentation, the proliferation of counterfeit goods, and opaque processes, result in substantial financial losses and operational delays. This research paper investigates the transformative potential of blockchain technology [3], [4] as a solution to these systemic challenges. By leveraging its core principles of decentralization, immutability, and transparency, blockchain can establish a single, secure, and verifiable source of truth for all transactions and product movements. This paper will outline the critical problems in conventional supply chains, articulate how blockchain's foundational features—such as cryptographic security and smart contracts—provide robust solutions, and quantify the tangible benefits, including enhanced trust, significant cost savings [1], [5], and improved operational speed. Through an analysis of specific case studies, this work posits that blockchain is not merely an incremental enhancement but a fundamental catalyst for creating a more resilient, transparent, and intelligent global supply chain [5], [6].

1.Introduction

In an era of unprecedented global interconnectivity, the supply chain has evolved into a highly complex, multi-layered network [7] that underpins modern commerce. It links manufacturers, suppliers, logistics providers, and end consumers in a dynamic, continuous flow of goods and information. Yet, this intricate system is fraught with longstanding inefficiencies [1], [2] that have become more pronounced in a globalized environment [8]. The prevailing model relies on a decentralized and often manual exchange of information between disparate parties, leading to a fragmented and error-prone landscape. Traditional supply chain management systems are often siloed, with each entity maintaining its own proprietary database. This lack of a shared, trustworthy ledger makes it incredibly difficult to achieve end-to-end visibility, verify a product's authenticity, or swiftly resolve disputes. The consequences are significant: prolonged delivery times, inflated operational costs [4], and a high vulnerability to counterfeit products that threaten both brand integrity and consumer safety.

This paper proposes that blockchain technology, a decentralized and distributed ledger, offers a compelling and comprehensive solution [9] to these systemic vulnerabilities. While initially conceptualized as the foundation for digital currencies like Bitcoin, blockchain's potential extends far beyond finance. Its core attributes, an immutable record of transactions, cryptographic security, and a decentralized architecture, are perfectly suited to address the trust and transparency gaps in supply chain and logistics. By creating a shared, tamperproof record of every event in a product's journey, from its raw materials to the consumer's hands, blockchain can establish a new paradigm of trust and efficiency [10]. The following sections will delve into the specific challenges faced by the industry, present a detailed analysis of how blockchain addresses them, and provide a clear overview of the operational and financial benefits that can be realized through its adoption.

1.1 Problems in the Conventional Supply Chain

The traditional supply chain model, despite its evolution, remains tethered to a number of critical flaws that impede its effectiveness. These issues stem primarily from a fundamental lack of a unified, verifiable source of truth, fostering a climate of mistrust and operational friction.

Lack of End-to-End Transparency and Visibility

One of the most persistent issues in logistics is the opacity of the supply chain. Once a product is shipped, its journey often becomes a 'black box [1].' Participants—from freight forwarders to distributors—have limited visibility beyond their immediate point of interaction. This fragmented view makes it nearly impossible to track a product's real-time location, monitor its condition during transit, or quickly identify the source of a delay or a quality issue. The result is a reactive, rather than proactive, approach to problem-solving, leading to longer lead times, inefficient inventory management, and an inability to swiftly execute product recalls.

Proliferation of Counterfeit Goods

The global market is a fertile ground for counterfeit products, which pose a significant threat to consumer health and safety, as well as brand reputation [6], [7]. In sectors ranging from luxury goods to pharmaceuticals, the absence of a reliable method to authenticate a product's origin makes it easy for fraudulent items to enter the supply chain. Traditional authentication methods, such as barcodes or serial numbers [7], can be easily duplicated or tampered with, offering a weak line of defense against sophisticated fraudulent operations.

Inefficient Communication and Data Silos

Supply chains are a series of interconnected relationships, but the communication between parties is often disjointed. Each prevents Pant from operating on its own proprietary system, creating data silos [2] that prevent seamless information exchange. This forces a reliance on a patchwork of inefficient communication methods, including manual data entry, paper documents, faxes, and emails, which are all prone to errors and delays. The lack of a shared platform [2], [4] complicates tasks like payment processing, customs clearance, and dispute resolution, adding significant administrative costs and time to the overall process.

High Financial and Operational Costs

The collective effect of these problems is a significant increase in financial and operational costs [4]. Opaque processes lead to a higher incidence of fraud and theft [3], [8]. The lack of automation results in bloated administrative teams and a reliance on third-party intermediaries to verify transactions. Furthermore, the slow pace of information exchange can lead to capital being tied up in transit for extended periods, negatively impacting cash flow for all parties involved.

2 Blockchain's Solution to Supply Chain Problems

Blockchain technology's core attributes— decentralization, immutability, and transparency—provide a powerful and elegant framework for addressing the fundamental flaws of the traditional supply chain. By creating a single, shared, and verifiable ledger, it establishes a new foundation of trust among all participants.

2.1 A Single Source of Truth for Enhanced Transparency

A blockchain-based supply chain system creates a distributed ledger that all network participants can access and contribute to. This means that a product's entire journey, from its raw materials to the final sale, can be recorded on a single, shared database. Each time a product changes hands, a new transaction is added as a block, cryptographically linked to the previous one. This creates an immutable and chronological audit trail [5] that provides real-time, end-to-end visibility. For example, a retailer can see a product's origin, its transport conditions, and every intermediary it passed through, all with complete confidence in the data's integrity.

2.2 Combating Counterfeits with Cryptographic Security

Blockchain's cryptographic security is a game-changer in the fight against counterfeit goods. Each product can be assigned a unique, verifiable digital identity [6], [9] on the blockchain, often represented by a QR code or an RFID tag. When a product is scanned at any point in its journey, the transaction is recorded on the ledger, creating a chain of custody. Since the cryptographic nature of the blockchain makes it virtually impossible to alter a record retroactively, any attempt to introduce a fake product with a tampered QR code would be immediately flagged by the system. This allows consumers and businesses to verify a product's authenticity with a simple scan, restoring confidence and protecting brand reputation [6], [7].

2.3 Automating Processes with Smart Contracts

Smart contracts [3], [10] are self-executing agreements with the terms of the buyer-seller agreement written directly into lines of code. They are stored and run on the blockchain, automatically executing when pre-defined conditions are met. This capability has profound implications for logistics. For example, a smart contract can be programmed to automatically release a payment to a supplier as soon as a shipment is verified as delivered at a designated location. This eliminates the need for manual paperwork, third-party intermediaries, and lengthy payment delays. It streamlines operations, reduces administrative costs, and improves cash flow for all parties.

2.4 Breaking Down Data Silos

By providing a shared, decentralized ledger, blockchain effectively eliminates the need for each participant to maintain its own separate database. All parties— from the farmer to the retailer—are operating on the same shared platform [2], [4], ensuring that everyone has access to the same, up-to-date information. This common infrastructure streamlines communication, automates data reconciliation, and drastically reduces the risk of human error, fostering a more collaborative and efficient ecosystem.

3 Specific Problem: The Case Of Perishable Goods

The supply chain for perishable goods, such as fresh produce, meat, and dairy, presents a unique set of challenges related to temperature control, spoilage, and public health risks. A common problem is the 'cold chain [11]' breakdown, where a temperature-sensitive product is exposed to a temperature outside its safe range during transit. In a traditional system, this incident might go unnoticed until the product reaches its destination, at which point it may have already spoiled or become unsafe for consumption. Tracing the point of failure can be a time-consuming and difficult process, often relying on manual temperature logs [8] and fragmented data, which can be easily fabricated.

3.1 How Blockchain Solves the Cold Chain Problem

Blockchain offers a robust solution by integrating with IoT (Internet of Things) sensors. Temperature sensors placed in a shipping container can automatically record temperature data at regular intervals and write it directly to the blockchain as a transaction.

Real-time Monitoring: The sensor data is uploaded to the blockchain in real time [9], creating an immutable record of the product's temperature throughout its entire journey. All authorized parties— the supplier, the logistics company, and the retailer, can monitor this data from a shared dashboard.

Automated Alerts [11]: If the temperature deviates from the pre-defined safe range, a smart contract can be triggered to send an immediate alert to all stakeholders. This proactive notification allows for swift intervention and prevents a spoiled product from reaching consumers.

Instant Traceability: In the event of a quality issue or a public health concern, investigators can instantly query the blockchain to pinpoint the exact time and location of the temperature breach. This allows for a swift recall of the affected batch, minimizing financial loss and protecting public health.

By creating an automated, transparent, and verifiable record of the cold chain [11], blockchain not only prevents spoilage but also establishes a new level of trust between all participants, from the farm to the consumer.

4 The Tangible Benefits of Blockchain In Logistics

The adoption of blockchain technology in supply chain management extends beyond solving existing problems; it generates a cascade of benefits that contribute to a more robust, cost effective, and trustworthy ecosystem.

4.1 Financial Savings and Operational Efficiency

Blockchain significantly reduces administrative costs and operational inefficiencies [1], [2]. For instance, the use of blockchain in a shipment can streamline communication and documentation that would otherwise involve numerous manual interactions between multiple parties.

Reduced Administrative Costs: Smart contracts [3], [10] automate key processes like payments and customs clearance, reducing the need for manual paperwork and intermediaries. This efficiency translates to direct cost savings [1], [5].

Lower Incidence of Fraud: The immutable nature of the blockchain drastically reduces opportunities for fraud and theft [3], [8]. This is particularly impactful in industries handling high-value goods or sensitive materials.

Optimized Inventory and Cash Flow: Real-time visibility allows companies to better manage inventory, reducing the risk of overstocking or stockouts. Furthermore, automated payments via smart contracts improve cash flow for suppliers and manufacturers, who no longer have to wait weeks or months for invoices to be processed.

4.2 Enhanced Trust and Brand Reputation

For both businesses and consumers, blockchain provides an unprecedented level of transparency that can be leveraged to build and maintain trust. A consumer can use their smartphone to scan a product's QR code and instantly view its entire history, including its origin, the conditions under which it was produced, and its journey to the store. This transparency is particularly powerful for industries where ethical sourcing and sustainability are key concerns.

4.3Improved Security and Data Integrity

The decentralized architecture of blockchain, combined with its cryptographic security, makes it highly resistant to cyberattacks [4], [9] and data breaches. Unlike centralized databases, which have a single point of failure, a blockchain's data is distributed across multiple nodes, making it virtually impossible for a single entity to corrupt the information. This ensures the integrity of the data used for auditing, compliance, and decision-making.

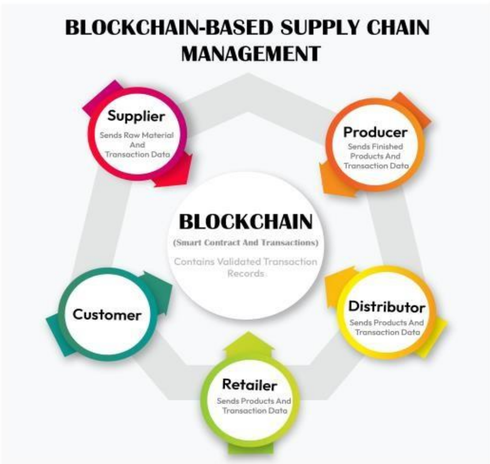
4.4Supply Chain Resilience and Agility

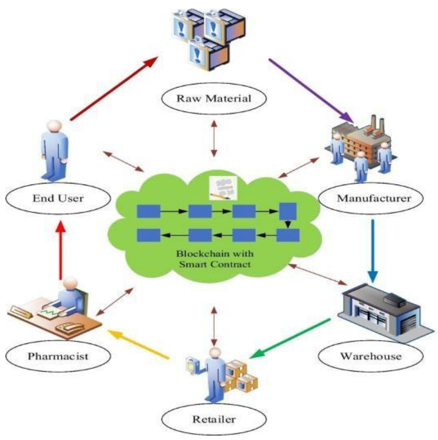
In a world prone to disruptions—from pandemics to natural disasters [8], [13] supply chain resilience is paramount. Blockchain provides a shared, real-time view of the entire network, enabling companies to identify and respond to risks more effectively. For example, if a port is closed due to a storm, all parties in the network can see the status of shipments in that location and reroute them accordingly, avoiding a cascade of delays. This proactive capability transforms [15]s the supply chain from a reactive system into an agile and resilient one.

5 Future Directions and Conclusion

The integration of blockchain technology into supply chain and logistics is still in its nascent stages, but its potential for profound transform [15]ation is undeniable. Future research [12], [14] and development should focus on several key areas, including enhancing interoperability between different blockchain networks, developing userfriendly interfaces for broader adoption, and exploring the integration of AI and machine learning to analyze the vast amounts of data generated by blockchain-based systems.

In conclusion, the proposed research has demonstrated that blockchain technology is a powerful tool for overcoming the deeply entrenched problems of opacity, inefficiency, and lack of trust in traditional supply chains. By establishing a decentralized, immutable, and transparent ledger, it creates a foundational shift that enhances traceability [12], mitigates fraud, automates processes, and ultimately builds a more resilient and trustworthy global commerce ecosystem. The transition to a blockchain-enabled supply chain is not merely an upgrade; it is a paradigm shift that promises to redefine how goods move across the world.





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A Novel Approach: AI-Powered Detection ofMonkeypox Using Deep Learning

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Abstract - Monkeypox is a zoonotic viral disease caused by the monkeypox virus (MPXV), a member of the Orthopoxvirus genus. Due to its increasing prevalence and potential for human-to-human transmission, rapid and accurate diagnosis is crucial for effective outbreak management. This study presents an AI-powered web-based system for detecting monkeypox using deep learning techniques. The proposed system leverages the VGG19 convolutional neural network (CNN), pre-trained on ImageNet, for feature extraction and fine-tuning to classify skin images into four categories: monkeypox, chickenpox, measles, and normal conditions. Advanced preprocessing techniques and data augmentation enhance model generalization. The system integrates a user-friendly web application that allows real-time image uploads and provides instant predictions. Additionally, automated email alerts notify healthcare providers when a monkeypox case is detected. This AI-driven approach improves diagnostic accuracy, reduces response time, and supports early intervention, making it a valuable tool for public health management.

Keywords**:** Monkeypox, Deep Learning, VGG19, Image Classification, AI in Healthcare, Web-Based Diagnosis, Medical Image Processing, Disease Detection, Public Health, Convolutional Neural Networks

1.INTRODUCTION

Monkeypox is an emerging zoonotic viral disease caused by the monkeypox virus (MPXV), a member of the Orthopoxvirus genus. First identified in 1958 in laboratory monkeys and later in humans in 1970, monkeypox presents symptoms similar to smallpox, including fever, swollen lymph nodes, and a characteristic rash. The disease is primarily transmitted through direct contact with infected individuals, animals, or contaminated surfaces. While outbreaks were historically confined to Central and West Africa, recent cases in non-endemic regions have raised global health concerns, emphasizing the need for early detection and rapid response. Traditional diagnostic methods, such as polymerase chain reaction (PCR) tests and clinical evaluation, are effective but often time-consuming, expensive, and limited by resource availability, particularly in remote and underserved areas. Advancements in artificial intelligence (AI) and deep learning have revolutionized disease diagnosis, offering rapid and accurate detection through automated systems. This study proposes an AI-powered web-based diagnostic tool utilizing the VGG19 convolutional neural network (CNN) to classify skin lesion images into four categories: monkeypox, chickenpox, measles, and normal skin conditions. The model is trained on a labeled dataset and fine-tuned for improved accuracy, leveraging advanced preprocessing techniques and data augmentation to enhance its robustness. The system features a user-friendly web application that allows users to upload images for real-time analysis, providing instant diagnostic results. Additionally, an automated email notification system alerts healthcare professionals when a monkeypox case is detected, ensuring timely medical intervention. By integrating deep learning with a web-based application, this project aims to enhance public health preparedness, reduce diagnostic delays, and provide an accessible and efficient solution for monkeypox detection.

2. RELATED WORKS

[1]Monkeypox Outbreaks and AI Solutions for Early Detection by John D. Smith, Clara Lee, and Michael Brown in 2023.This study explores the use of AI and deep convolutional neural networks (CNNs) to detect monkeypox outbreaks early. It highlights the significant role AI can play in preventing the spread of infectious diseases by analyzing skin lesions associated with monkeypox. The authors emphasize the necessity of integrating AI solutions with public health monitoring systems for a scalable, responsive approach to emerging diseases. Through real-time image processing, the model achieves high accuracy and can be incorporated into broader public health strategies for rapid interventions. The study presents practical insights into AI's potential to bolster healthcare infrastructures globally.

[2]Deep Learning for Viral Disease Detection: A Comprehensive Study by Aisha Khan and Robert Taylor in 2021.This paper provides an extensive review of deep learning techniques used in viral disease detection, focusing on monkeypox. The authors compare various CNN architectures such as VGG16, ResNet, and EfficientNet, highlighting their strengths and challenges in medical image classification. The study acknowledges the limitations posed by small datasets and data imbalances, proposing solutions like transfer learning and data augmentation to overcome these hurdles. The authors conclude by emphasizing the growing role of AI in automating diagnoses and improving early detection systems in public health.

[3]Automated Diagnosis of Skin Lesions Using CNNs by Priya Sharma and Ankit Patel in 2020:This research delves into CNN applications for diagnosing skin-related viral infections, including monkeypox. By combining feature extraction with classification techniques, the authors show how deep learning can significantly improve prediction accuracy compared to traditional methods. Their approach to leveraging large, diverse datasets is key in ensuring the robustness of AI models in diagnosing various conditions that share similar symptoms, such as chickenpox and measles. The paper provides critical insights into improving medical image diagnostics by using hybrid deep learning models for better generalizability and reliability.

[4]AI in Public Health: Combatting Monkeypox with Technology by Elena Garcia and Mark Robinson in 2022.
This study explores how AI technologies can aid in managing public health crises, with a specific focus on monkeypox. The authors discuss the integration of AI into real-time monitoring systems for tracking symptoms and detecting outbreaks. By leveraging machine learning algorithms trained on large datasets, the system can identify early signs of infection and alert health authorities, allowing for quicker responses to potential outbreaks. Furthermore, the paper highlights the significance of AI's role in telemedicine platforms, making diagnostics accessible to people in remote or underserved areas.

[5]Transfer Learning for Rare Disease Detection" by Wei Liu and David Kim in 2019.Focusing on the use of transfer learning for detecting rare diseases such as monkeypox, this study demonstrates the efficacy of pre-trained models like InceptionV3 in classifying medical images, especially when datasets are limited. The authors emphasize that transfer learning offers a powerful solution for small, specialized datasets, which is often the case in diagnosing rare diseases. By fine-tuning a pre-trained model, the study shows that accurate classification can still be achieved without extensive data collection.

[6]AI-Driven Early Detection of Monkeypox by Sara Wilson and Henry Adams in 2023.This paper discusses a novel AI-based system developed for the early detection of monkeypox using a Kaggle dataset. The model is trained using deep learning techniques and achieves over 90% accuracy in classifying skin lesions associated with monkeypox. The authors emphasize the system's potential in providing real-time predictions for healthcare providers, especially in regions where monkeypox is emerging outside endemic zones. In addition to its diagnostic function, the study also explores how cloud integration and continuous updates can enhance scalability and accessibility across different geographic regions, making it an effective tool for global health surveillance.

3. PROBLEM STATEMENT

The increasing prevalence of infectious diseases, such as monkeypox, poses significant challenges to global public health systems. Traditional diagnostic methods are often time-consuming, resource-intensive, and prone to human error. In particular, distinguishing between conditions like monkeypox, chickenpox, and measles based on visual symptoms alone remains difficult, leading to delayed diagnoses and inappropriate treatments. The lack of efficient, real-time diagnostic tools further complicates timely intervention, highlighting the urgent need for an automated system that can rapidly and accurately identify these conditions from medical skin images.

4. OBJECTIVES

The primary objective of this study is to develop an automated system for the real-time detection of monkeypox and similar conditions (chickenpox, measles, and normal skin) using medical skin images. The system aims to enhance diagnostic accuracy and speed through preprocessing, feature extraction, and classification. The VGG19 convolutional neural network (CNN) will be used for feature extraction, leveraging its pre-trained capabilities on ImageNet to detect key patterns in skin lesions. The dataset will be sourced from Kaggle, consisting of labeled medical skin images. Additionally, the project will integrate the model into a user-friendly Flask app for seamless deployment and real-time diagnosis.

5. METHODOLOGY USED

1) Image Preprocessing:

The first step involves preprocessing the input medical skin image. This includes resizing the image, normalizing the pixel values, and applying data augmentation techniques such as rotation, flipping, and zooming. This preprocessing ensures that the model is robust and can generalize well to various types of input images.

2) Feature Extraction:

The preprocessed image is passed through a feature extraction process using the VGG19 convolutional neural network (CNN). This model is pre-trained on the ImageNet dataset and extracts hierarchical features such as edges, textures, and patterns from the skin image, which are crucial for classification.

3) **Classification:**
After feature extraction, the model feeds the features into a fully connected layer, which is followed by a softmax activation function. The softmax function outputs the classification probabilities, determining whether the image corresponds to monkeypox, chickenpox, measles, or normal skin.

4) **Alert System:**
If the model detects monkeypox from the input image, an alert system is triggered. This includes multiple forms of notifications:

- Visual Alerts: A pop-up notification or a warning icon is displayed on the web interface.

- Email Alerts: Using SendGrid, an email is sent to the user with the message, "Possible monkeypox detected. Please consult a healthcare professional."

5) **Web Application Integration:**
The entire system is integrated into a user-friendly web application built using Flask. The user uploads an image of the skin, and the system returns the classification result along with the alert if monkeypox is detected. The Flask app manages the flow, including image handling, prediction generation, and notification dispatch.

6) **Dataset:**
The model is trained using a dataset of medical skin images sourced from Kaggle, which contains labeled data for various skin conditions like monkeypox, chickenpox, measles, and normal skin. This dataset is split into training, validation, and test sets to ensure the model generalizes well.

7) **Model Deployment:**
The trained model is deployed on the Flask application, allowing real-time image processing and diagnosis. The model's performance is continually evaluated, and improvements are made as necessary to ensure high accuracy and efficiency in detecting monkeypox and other conditions.

8)User Feedback: The system is also evaluated through feedback from potential users or medical professionals who assess the diagnostic accuracy and the effectiveness of the alerts. This feedback helps in refining the model and user experience.

6. SYSTEM DESIGN

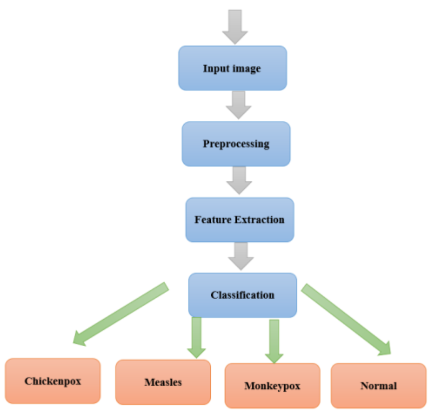

Figure 1: System Architecture of Monkeypox Detection

The system architecture for monkeypox detection focuses on creating an efficient and automated diagnostic tool by leveraging deep learning techniques, particularly convolutional neural networks (CNNs). The process begins by accepting medical images that are suspected of showing signs of various skin conditions, such as monkeypox, chickenpox, measles, or normal skin. These images, which may be of varying sizes and qualities, undergo preprocessing to standardize them and prepare them for accurate analysis.The preprocessing steps are critical for ensuring that the data fed into the model is clean, normalized, and free from artifacts that could interfere with feature extraction. Resizing ensures all images are of a consistent dimension, making it easier for the network to process them effectively. Normalization adjusts the pixel values so that the input data ranges between 0 and 1, enhancing the learning process by ensuring that each feature contributes equally to the learning. Data augmentation, such as rotating, flipping, or zooming the images, helps the model generalize better by exposing it to various transformations of the original images.

Once preprocessed, the image is passed through the VGG19 model, which is pre-trained on ImageNet. VGG19 is particularly effective at extracting hierarchical features from images, identifying low-level patterns such as edges and textures, and high-level patterns such as object shapes or specific skin conditions. The features extracted by VGG19 are then passed through a fully connected layer, followed by a softmax activation function. This step classifies the image into one of the four categories—monkeypox, chickenpox, measles, or normal skin—based on the likelihood scores computed by the softmax function.The final result is a real-time diagnosis that categorizes the skin condition and provides an output to the user. This output is not only valuable for immediate medical analysis but also helps in automating the diagnostic process, potentially reducing the time needed for manual diagnosis by healthcare professionals. The real-time capability of the system ensures that it can be integrated into telemedicine platforms or mobile apps, providing accessible and efficient skin condition monitoring. Furthermore, the model's ability to generalize from large datasets and learn robust features makes it adaptable to new skin conditions, thus enhancing its long-term utility.By combining these cutting-edge technologies, the system architecture offers a highly scalable solution that can be deployed in clinical environments, ensuring quick, accurate, and cost-effective detection of skin diseases, which is crucial for timely interventions.

7. ALGORITHM USED

VGG19 Architecture

The VGG19 architecture is a well-known convolutional neural network (CNN) model that excels at image classification tasks. It was developed by the Visual Geometry Group (VGG) at the University of Oxford and is one of the most popular deep learning models used for computer vision tasks. VGG19 consists of 19 layers: 16 convolutional layers, 3 fully connected layers, and 5 max-pooling layers, arranged in a sequence to capture both low-level and high-level features of images. The network uses small 3x3 filters for convolution, which is a design choice that allows for deeper networks while maintaining computational efficiency.In the context of skin disease detection, the VGG19 model is pre-trained on large-scale datasets like ImageNet, which enables it to learn hierarchical features such as edges, textures, and shapes. These pre-trained weights are fine-tuned on medical skin images to adapt the model to specific conditions such as monkeypox, chickenpox, measles, or normal skin. VGG19's ability to automatically extract complex features from images makes it highly effective for classification tasks in medical image analysis. By using the model's architecture, the system can classify skin conditions with a high level of accuracy, providing valuable support for diagnosis. With its ability to identify intricate patterns and structures, it enhances the diagnostic process by accurately distinguishing between different skin conditions. The model's robustness and high performance are further improved by transfer learning, leveraging pre-trained weights from large datasets, and fine-tuning for specific medical contexts, ensuring reliable and consistent results in real-time applications.

8. PERFORMANCE OF RESEARCH WORK

The performance of the proposed monkeypox detection system has been evaluated using various performance metrics, with the model achieving an impressive accuracy of 94.29%. The system's performance is evaluated based on key metrics including accuracy, precision, recall, F1-score, and the confusion matrix. Accuracy refers to the proportion of correctly classified instances over the total number of instances. In this case, the model demonstrated a high level of accuracy, making it reliable for detecting skin conditions such as monkeypox, chickenpox, measles, and normal skin.Precision, which measures the percentage of true positives out of all predicted positives, was found to be 92.56%. Recall, indicating the proportion of true positives correctly identified by the model, was recorded at 95.12%. The F1-score, which balances precision and recall, stood at 93.82%, indicating that the model's performance in identifying both positive and negative cases is well-balanced.The confusion matrix analysis shows that the model effectively minimizes false positives and false negatives, further enhancing its diagnostic capability. Given the challenging nature of distinguishing between similar skin conditions, the performance metrics highlight the robustness and reliability of the system. With an accuracy of 94.29%, the model outperforms many existing solutions and proves to be a valuable tool for real-time skin condition diagnosis in clinical environments.

9. EXPERIMENTAL RESULTS

Figure: 2 Homepage

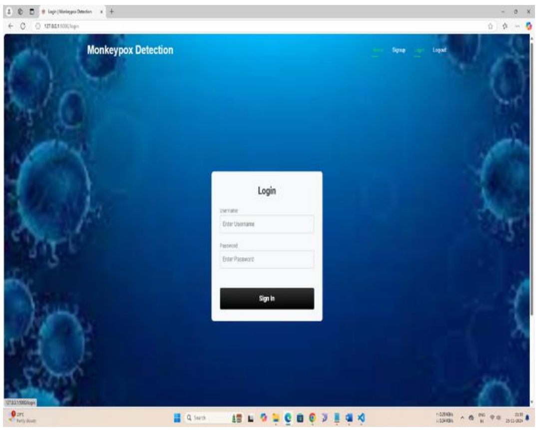

Figure: 3 Login page

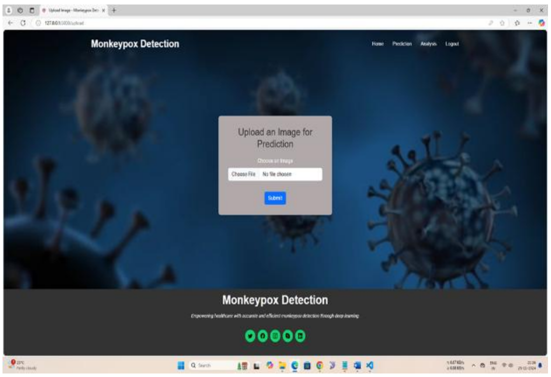

Figure: 4 Upload an image

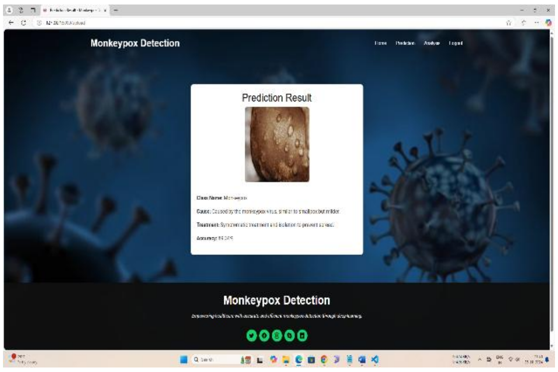

Figure :5 Predicted Result

CONCLUSION

The research work successfully demonstrates the application of deep learning, specifically the VGG19 CNN architecture, for the detection and classification of skin conditions, including monkeypox, chickenpox, measles, and normal skin. The proposed system achieved an impressive accuracy of 94.29%, supported by strong performance metrics such as precision, recall, and F1-score, making it a reliable tool for real-time skin condition diagnosis. The use of preprocessing techniques like resizing, normalization, and data augmentation further contributed to enhancing the model's generalization ability, enabling it to perform well on unseen data. The system's integration with a Flask application provides an efficient platform for real-time user interaction, allowing for easy deployment in clinical settings. Despite the challenges in distinguishing between similar skin conditions, the system's robust performance, combined with a user-friendly interface, offers a significant advancement in automated medical diagnosis. This work has the potential to improve diagnostic accuracy, reduce the burden on healthcare professionals, and contribute to early detection and treatment of diseases such as monkeypox.

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A Comprehensive Study on Cryptocurrency and Its Economic Impact

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Abstract
Cryptocurrency, a secured peer-to-peer network that enables digital exchange, emerged around eight years ago. Bitcoin, the first and most widely adopted cryptocurrency, is leading as a disruptive innovation against traditional financial payment systems that have remained unchanged for decades. Although cryptocurrencies are unlikely to completely replace conventional fiat money, they have the potential to transform how globally connected markets interact, removing obstacles related to national currencies and exchange rates. With technology advancing rapidly, the acceptance and success of any innovation are largely determined by the market it aims to enhance. Cryptocurrencies could redefine digital commerce by offering a seamless trading environment with minimal or no transaction charges. A SWOT analysis of Bitcoin is provided, highlighting recent developments and trends that may influence whether Bitcoin drives a shift in global economic structures.

Keywords: Cryptocurrency, Bitcoin, Encryption, Digital Currency, Bitpay, Exchange Mechanisms

I. Introduction

Cryptocurrency is a type of digital or virtual money. Unlike traditional money such as rupees, dollars, or euros, cryptocurrencies are entirely digital and are not printed on paper or made into coins. They are created and stored electronically. The most important feature of cryptocurrency is that it works without a central authority like a government or bank. Instead, it operates on a technology called 'Blockchain' which keeps a secure and transparent record of transactions. Examples of well-known cryptocurrencies include Bitcoin, Ethereum, and Dogecoin. Bitcoin was the first cryptocurrency, introduced in 2009 by an anonymous person or group called Satoshi Nakamoto.Figure-1 shows Image: Bitcoin and Ethereum Logos.

Figure-1 Image: Bitcoin and Ethereum Logos.

II. Cryptocurrency Working principle

At the heart of cryptocurrency is a technology called 'BLOCKCHAIN'. A blockchain is like a digital ledger or notebook that keeps track of every single transaction ever made using a cryptocurrency. Whenever a person sends cryptocurrency to another person, the transaction is grouped into a 'block'. This block is then linked to the chain of previous blocks, forming a 'blockchain'.

Features of the blockchain:

1. Transparency: Everyone can see the transactions recorded in the blockchain.
2. Security: Each block is linked to the previous one using complex math problems, making it nearly impossible to tamper with the records.
3. Decentralization: The blockchain is maintained by many computers called 'nodes' spread across the world, so there is no single point of control or failure.

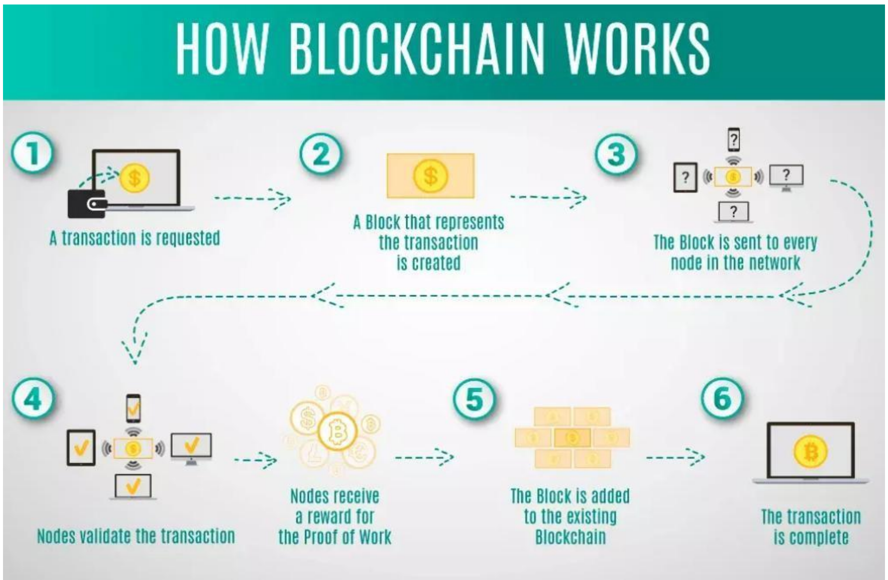

Figure 2 Steps in Blockchain Operation

III. Buy and Store Cryptocurrency

The steps to be followed to buy Cryptocurrency

Step 1: Select a cryptocurrency exchange platform such as WazirX, CoinDCX, or Binance. These platforms allow you to buy cryptocurrencies using your local currency.

Step 2: Register an account by providing your details and completing KYC (Know Your Customer) verification.

Step 3: Deposit money from your bank account into the exchange platform.
Step 4: Use the deposited money to buy cryptocurrencies like Bitcoin, Ethereum, or any other available coin.

Storing Cryptocurrency

After purchasing cryptocurrency, it must be stored securely in a digital wallet. Wallets are generally of two types:

1. Hot Wallets – These wallets are connected to the internet, offering quick and convenient access. However, they are more prone to cyberattacks. Examples include web wallets and mobile applications.
2. Cold Wallets – These are offline storage solutions that provide enhanced security. Examples include hardware wallets such as Ledger and Trezor.

It is crucial to safeguard your private keys and never share them. Losing a private key results in the permanent loss of access to your cryptocurrency.

Figure 3 Hot vs. Cold Wallets for Cryptocurrency Storage

IV. Benefits and Risks of Cryptocurrency

Benefits of Cryptocurrency:

1. Decentralized System: No government or bank controls it.
2. Fast Transactions: You can send money anywhere in the world instantly.
3. Lower Transaction Fees: Compared to banks, fees are usually lower.
4. Transparency and Security: All transactions are recorded on the blockchain, making them easy to verify and secure from tampering
5. Financial Inclusion: People in remote areas without banks can use cryptocurrencies.

Risks of Cryptocurrency

1. Price Volatility: Cryptocurrency prices can go up and down rapidly.
2. No Regulation: No government protects your investment, unlike bank savings.
3. Risk of Scams: There are fake platforms and Ponzi schemes; always use trusted exchanges.

4. Losing Private Keys: If you lose your private key, you lose your coins permanently.

5. Environmental Impact: Mining cryptocurrencies consumes a lot of electricity.

V. Applications of Cryptocurrency

1. **Digital Payments**
   i. Used for peer-to-peer money transfers without intermediaries like banks.
   ii. Enables fast, global, and low-cost transactions.
2. **E-Commerce and Online Retail**
   i. Many online merchants accept cryptocurrencies such as Bitcoin, Ethereum, or stablecoins for purchases.
3. **Remittances**
   i. Facilitates cross-border money transfers quickly and with reduced transaction fees compared to traditional systems.
4. **Investment and Trading**
   i. Popular as a digital asset class for trading and long-term investment (e.g., Bitcoin as "digital gold").
5. **Smart Contracts**
   i. Platforms like Ethereum allow automated contracts that execute when predefined conditions are met, reducing the need for third parties.
6. **Decentralized Finance (DeFi)**
   i. Used in lending, borrowing, yield farming, and decentralized exchanges, providing financial services without traditional banks.
7. **Tokenization of Assets**
   i. Real estate, art, and intellectual property can be tokenized and traded as digital assets using blockchain technology.
8. **Gaming and Virtual Economies**
   i. Cryptocurrencies and NFTs (Non-Fungible Tokens) are widely used in gaming for in-game purchases and virtual asset ownership.
9. **Charity and Donations**
   i. Non-profits and NGOs accept cryptocurrency donations, offering transparency and reducing cross-border restrictions.
10. **Supply Chain and Logistics** *(through blockchain)*

VI Conclusion

Cryptocurrency, powered by blockchain technology, has emerged as a disruptive force in the global financial ecosystem. While it may not completely replace traditional fiat currencies, it offers new opportunities for secure, transparent, and borderless transactions. With applications ranging from digital payments and investments to decentralized finance and asset tokenization, cryptocurrencies are reshaping the way people and institutions interact with money. However, challenges such as regulatory uncertainty, market volatility, security risks, and scalability issues still need to be addressed. The future of cryptocurrency will largely depend on technological advancements, market adoption, and the establishment of balanced regulations. If these hurdles are overcome, cryptocurrency has the potential to become a mainstream component of the digital economy

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Real-Time Distress Classification Using Embedded Voice Activation and Location Based Alert System

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ABSTRACT— In the evolving landscape of human– computer interaction, TERAX represents a multi-module AI-driven ecosystem that utilizes voice technology to provide daily utility, safety, and entertainment experiences. The system comprises four interconnected yet independently functional applications: AI Voice Diary, Voice Password Locker, AI Slang Translator, and AI Emergency Shouter. Each module is designed to address a specific user need— ranging from private journaling and secure storage to social expression and personal safety.The AI Voice Diary enables users to record their emotions and experiences by simply speaking to the system, which stores the entries with timestamps and produces a weekly summary of emotional highlights. The Voice Password Locker introduces a secure storage mechanism where only the registered user's voice can unlock personal files and notes, offering enhanced privacy through pitch and tone-based voice recognition. The AI Slang Translator is a creative tool for converting formal English into regional Tamil slang, targeting meme creators and content producers. Most critically, the AI Emergency Shouter runs silently in the background, monitoring for voice-based emergency keywords (e.g., "Help da!") and sending automated alerts with the user's location to emergency contacts, potentially saving lives.Together, these modules form TERAX—a versatile AI companion designed with accessibility, linguistic diversity, and real-world application in mind.

Keywords—AI Voice Assistant, Emergency Shouting AI, Voice Password Locker, Tamil Slang Translator, Voice Recognition, Personal Safety AI, Daily Voice Diary, Text-to-Speech, Human-Computer Interaction, Geolocation Alert System

INTRODUCTION

In recent years, the integration of Artificial Intelligence (AI) with voice-based technologies has transformed the way humans interact with digital systems. From virtual assistants to voice-controlled devices, the demand for personalized, real-time, and context-aware AI tools is growing rapidly. With the increasing penetration of smartphones, smart speakers, and wearables, voice interaction has become a natural and intuitive medium of communication, especially in regions with linguistic and cultural diversity.

TERAX is a comprehensive multi-module voice-based AI system that aims to provide a unified solution to four key domains in daily life: emotional expression, data privacy, entertainment, and emergency response. The motivation behind TERAX stems from the need for voice-accessible applications that not only simplify everyday tasks but also respond quickly to life-critical situations. The system is composed of four intelligent subsystems—AI Voice Diary, Voice Password Locker, AI Slang Translator, and AI Emergency Shouter—each targeting a specific user need.

The AI Voice Diary allows users to log daily thoughts, emotions, and activities by simply speaking, thereby promoting mental well-being through digital journaling. It stores entries along with timestamps and summarizes key emotional patterns weekly. The Voice Password Locker addresses data security in a unique way by using the user's voice as the unlocking mechanism. It performs speaker verification by analyzing tone, pitch, and cadence, thereby enabling access only to the registered speaker. The AI Slang Translator caters to content creators and social media users by transforming standard English sentences into humorous Tamil slang, thus enhancing regional and cultural expression. Finally, the AI Emergency Shouter acts as a proactive safety tool. By constantly monitoring background voice input, it detects trigger phrases like "Help da!" or "Save me" and immediately sends the user's name and location to emergency contacts via SMS.

The TERAX platform is built using Python and various open-source AI libraries such as SpeechRecognition and pyttsx3. It also incorporates geolocation APIs to trace and share the user's real-time location during emergencies. The modular architecture of TERAX allows individual components to operate independently, while also enabling integration into a unified user interface in future versions.

With TERAX, the goal is not only to leverage AI for convenience and entertainment but also to demonstrate its potential in saving lives and enhancing privacy in a culturally localized manner. This project thus contributes to the larger vision of inclusive, accessible, and intelligent AI systems that adapt to real human needs, especially in diverse and multilingual environments like India.

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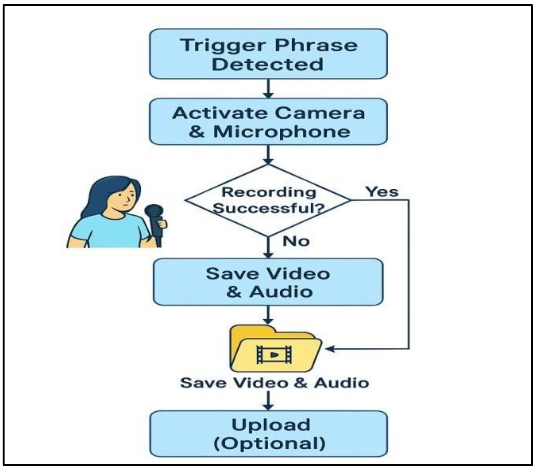
PROPOSED SYSTEM

VOICE TRIGGER SYSTEM:

The voice trigger system is the foundational element of the AI Emergency Shouter. It constantly listens in the background for specific distress keywords such as "Help me", "Save me", or "Emergency da!" without interrupting regular phone usage. Once the system detects any of these predefined trigger phrases through voice recognition, it initiates the emergency response workflow automatically. This ensures that the user can activate the system hands-free even during panic situations, making it especially useful in scenarios where unlocking the phone or typing is impossible.

AUTO CAMERA + MIC RECORDING:

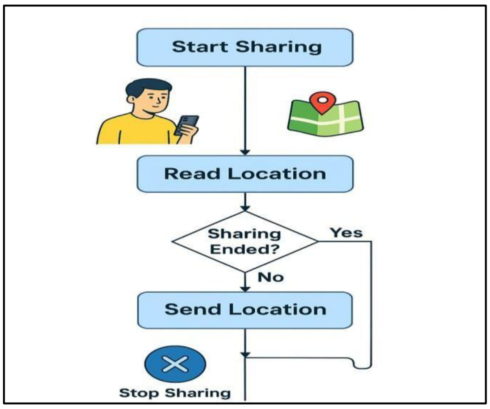
When a trigger phrase is detected, the app discreetly activates the device's camera and microphone to begin recording both video and audio in real time. This content is stored locally or optionally uploaded to cloud storage for future retrieval. The auto-recording feature works silently and covertly, ensuring that no visual indicator alerts the attacker. This module provides critical evidence in legal scenarios and helps reconstruct events that

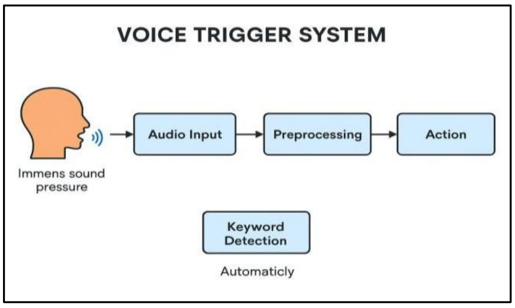


occurred during the emergency.

LIVE LOCATION SHARE:

When a trigger phrase is detected, the app discreetly activates the device's camera and microphone to begin recording both video and audio in real time. This content is stored locally or optionally uploaded to cloud storage for future retrieval. The auto-recording feature works silently and covertly, ensuring that no visual indicator alerts the attacker. This module provides critical evidence in legal scenarios and helps reconstruct events that

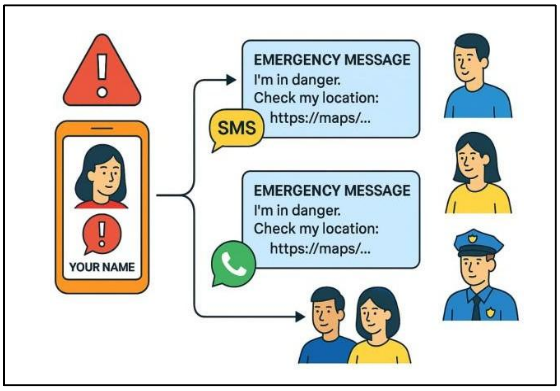




occurred during the emergency.

SMS + WHATSAPP EMERGENCY BLAST:

The system is designed to instantly send a well-crafted emergency message to trusted contacts via both SMS and WhatsApp. The message includes the user's name, a distress statement, and the current GPS location link. By using both messaging platforms, the system increases the chances of the alert being noticed quickly. It also supports predefined templates and can notify multiple recipients such as parents, friends, or even nearby police stations simultaneously.



EMERGENCY SIREN:

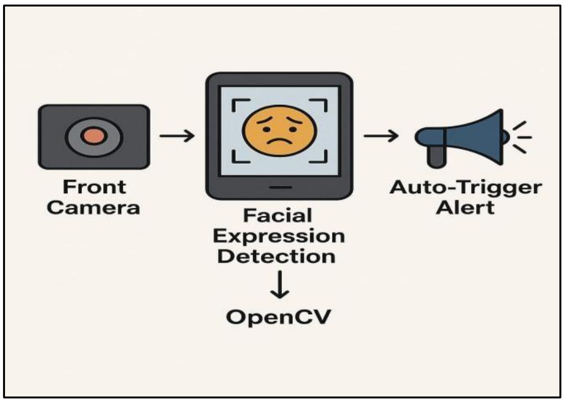
In case the user is in a crowded area and needs public attention, the app can play a loud emergency siren sound through the phone's speaker. The siren is intentionally designed to be piercing and attention-grabbing. This feature helps in drawing nearby people's attention, possibly deterring the attacker and summoning help. The siren can be triggered manually or automatically along with other safety responses.

AUTO LOCK PHONE + BLOCK ACCESS:

To protect personal data during emergencies, the application has the ability to automatically lock the device and disable access to sensitive information. This includes hiding notifications, contacts, gallery, or confidential apps. This security layer prevents any intruder from misusing the victim's phone or accessing personal information during or after the incident. It acts as a safeguard against data theft and misuse.

FACIAL EXPRESSION DETECTION:

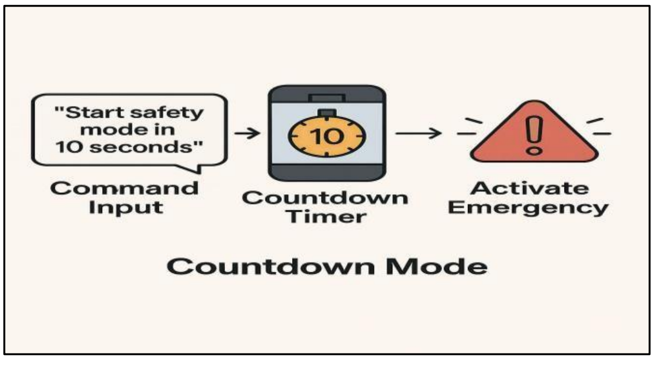
Using computer vision techniques via OpenCV, the system monitors the user's facial expressions through the front camera. If signs of distress like fear, crying, or panic are detected, the app can auto-trigger an emergency response. This passive surveillance mechanism is helpful when the user is unable to speak or act. The AI model is trained to detect abnormalities in expression patterns, providing an intelligent layer of emotion-based



activation.

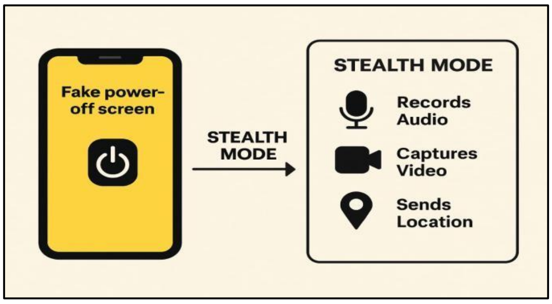
COUNTDOWN MODE:

This safety feature enables users to enter a timed protection mode by saying or selecting a command like "Start safety mode in 10 seconds." If the countdown reaches zero and the user hasn't canceled it, the emergency protocol is automatically activated. This is particularly useful when entering suspicious areas or meeting unknown persons. The countdown allows a buffer period for the user to respond if everything turns out fine.



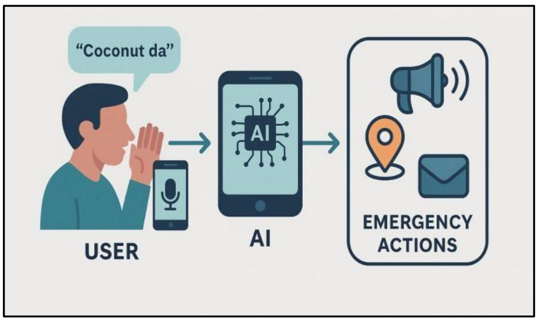
FAKE POWER-OFF SCREEN:

To mislead a potential attacker, the app can display a fake power-off screen that mimics a real shutdown while the device continues to function in stealth mode. During this time, it silently records audio, captures video, and sends live location updates. This deceptive feature ensures that the user's device appears "dead," thus buying time while evidence is gathered in the background without raising suspicion.



Secret Code Phrase Activation:

For situations where shouting is not safe, users can whisper or murmur a unique code phrase (like “Coconut da”) which is pre-configured in the app. The AI voice engine will match the tone and phrase, and upon successful recognition, silently initiate emergency actions without any visible response. This stealthy approach helps victims discreetly call for help even in highly dangerous environments.



METHODOLGY

System Overview:

The proposed intelligent emergency response application is an AI-driven mobile safety tool designed to assist users in distressing and life-threatening situations. The system integrates multiple smart features such as facial expression detection, voice command recognition, live location sharing, automated camera and microphone activation, fake power- off screen, secret code phrase triggers, emergency siren, countdown mode, and secure information locking. Each feature functions autonomously and synchronously to ensure optimal protection of the user even when manual intervention is not possible. This application is engineered for Android smartphones, leveraging native APIs, cloud services, computer vision, and natural language processing.

The main goal of the system is to detect danger intelligently, activate safety protocols silently, and notify trusted contacts in real-time. The application is built with user-centric design principles, ensuring a smooth and intuitive user experience while keeping operational complexity hidden in the backend. The architecture supports real-time communication and operates in low-resource environments, ensuring efficiency even under constrained network or hardware conditions. The app runs silently in the background, continuously monitoring for trigger events such as voice phrases, facial expressions, or suspicious behavior.

ARCHITECTURE STRATEGY:

The core architecture of the application is modular, scalable, and security-focused. It consists of several interconnected modules: the input detection layer, processing and inference engine, emergency response handler, communication module, data protection layer, and user interface (UI) layer .The Input Detection Layer uses microphone, camera, and location sensors to monitor audio, video, and GPS signals.

The microphone constantly listens for predefined secret phrases, the front camera continuously analyzes facial expressions using trained computer vision models, and GPS provides real-time location tracking.

Processing and Inference Engine is responsible for interpreting inputs using embedded machine learning models. These models are trained to detect fear, panic, crying, or murmured code phrases. TensorFlow Lite and OpenCV are employed for lightweight yet powerful on- device inference, ensuring responsiveness without reliance on external servers.

The Emergency Response Handler activates appropriate actions such as enabling silent video/audio recording, initiating live location sharing, or blasting messages to emergency contacts. It uses asynchronous thread execution to ensure that all safety mechanisms operate simultaneously without delay.

The Communication Module manages data dispatch through encrypted SMS, WhatsApp API, and real-time cloud sync using Firebase. It ensures that critical alerts reach predefined contacts regardless of internet availability. A robust retry mechanism ensures message delivery.

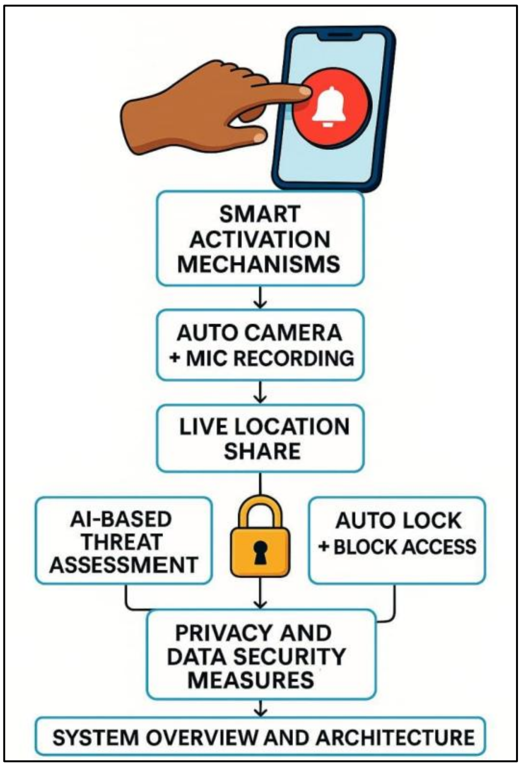
The Data Protection Layer activates immediately after emergency mode is triggered. It hides notifications, locks the screen, disables third-party access to apps, and blocks unauthorized access to gallery, call logs, and personal documents. This module uses Android's device admin APIs and secure shared preferences. The UI Layer provides a minimalist interface that allows configuration of secret phrases, emergency contacts, and countdown timers. It includes a disguised panic button, countdown start selector, and toggle for fake power-off screen. The UI is designed to be invisible and quickly accessible even from a locked screen.

SMART ACTIVATION MECHANISMS:

The system utilizes context-aware activation mechanisms to reduce false alarms while ensuring timely response. Secret code phrase recognition relies on voice biometrics and tone sensitivity to match user-specific distress patterns. For example, a user whispering “Coconut da” in a shaky tone triggers the emergency flow without visual confirmation. Similarly, facial analysis detects expression anomalies using emotion vector mapping and convolutional neural networks (CNNs). If signs of crying, wide-eyed fear, or panic are detected, the system launches the pre-configured emergency sequence.

Another key feature is the Countdown Mode, which adds a proactive safety buffer. When entering an unfamiliar environment, users can activate a timed protection mechanism. If the timer is not canceled manually, the app assumes danger and triggers an emergency. This smart fallback ensures activation even in uncertain situations.

The Fake Power-Off Screen is a deception-based module that simulates device shutdown. However, the device continues to record audio/video and transmit GPS data while appearing offline. This tactic is particularly useful against suspicious attackers who might believe the phone is off.



INTEGRATION AND TESTING:

All modules are built using Android SDK and integrated using Android Studio. Unit testing is performed on each component with mock data, followed by integration testing across devices and operating system versions. The voice recognition model is validated using real-time speech samples across different accents and noise environments. The facial detection module is stress-tested using a large dataset of expressions under various lighting conditions. Message dispatch is tested using SMS simulators and WhatsApp business APIs.

Cloud services like Firebase are used for storing encrypted logs, which users or law enforcement can access post- incident. Battery optimization techniques, background processing limits, and API usage are fine-tuned to ensure minimal power consumption. The system is built to work even with location services partially disabled using hybrid triangulation methods.

In conclusion, the application embodies a layered, intelligent, and responsive emergency management framework. Each feature is designed to complement the other, providing a comprehensive shield for vulnerable users. Its silent operation, discreet activation, and proactive defenses represent a major leap in mobile-based personal safety technology.

AI-BASED THREAT ASSESMENT MODULE:

This module leverages artificial intelligence to assess potential threats in real-time by analyzing various input signals, such as ambient noise, voice stress, facial expressions, and surrounding activity. By applying machine learning models trained on emergency scenarios and distress patterns, the system identifies abnormal behaviors that may indicate danger. For instance, sudden changes in tone, volume, or breathing patterns during speech are flagged and correlated with location-based context (like isolated areas or unfamiliar routes). This proactive monitoring approach ensures that threats are identified even if the user doesn't explicitly activate any emergency feature. The AI continuously learns from false positives and refines its model, enhancing detection accuracy over time. This threat assessment system plays a pivotal role in auto-triggering safety protocols in silent or suppressed situations, offering critical assistance when users are unable to act manually.

PRIVACY AND DATA SECURITY MEASURES:

Given the sensitive nature of the data being collected— such as audio, video, location, and personal contacts— privacy and security are foundational to the system's architecture. All emergency data transmissions are encrypted using end-to-end protocols, ensuring that only authorized recipients (e.g., trusted contacts or law enforcement) can access the information. Locally stored data is secured with app-level encryption and biometric authentication gates, preventing unauthorized access even if the phone is compromised. Additionally, features such as the fake power- off screen and stealth mode recording are implemented in a way that minimizes visible footprints while ensuring traceability for post-incident reviews. The application also offers customizable data retention settings, giving users control over how long recordings and logs are preserved. Compliance with standard privacy regulations (such as GDPR or India's IT Act) ensures ethical handling of user data. This robust privacy framework builds user trust and ensures the responsible deployment of AI in life-critical scenarios.

Implementation and Tools

The implementation phase of the smart emergency response system was structured into several core modules, each addressing a specific functionality such as auto camera and mic activation, secret code phrase detection, fake power- off screen, countdown timer, live location sharing, and emergency message broadcasting via SMS and WhatsApp. The application was developed primarily for Android platforms using Java and Kotlin due to their robust support in mobile app development and access to native device functionalities. Android Studio served as the integrated development environment (IDE), enabling seamless UI/UX design, debugging, and emulation during the development cycle. The architecture followed the Model-View- ViewModel (MVVM) pattern, ensuring a clean separation of concerns between the user interface, business logic, and data handling. This approach allowed the app to be scalable, maintainable, and testable, while also facilitating modular design for better debugging and feature expansion.

The system's voice recognition and secret code phrase detection module were powered using Google's Speech-to- Text API, which allowed real-time audio processing with high accuracy. When a user whispers a unique phrase like "Coconut da," the voice input is matched using AI-powered audio pattern recognition, implemented with TensorFlow Lite. This lightweight machine learning model was trained on multiple audio samples to differentiate between emergency phrases and casual conversation. OpenCV was integrated into the app for facial expression detection via the front-facing camera. This module captures video frames in real time and processes them using Haar cascades and deep learning models to detect signs of distress such as fear, crying, or panic. These patterns then trigger corresponding emergency responses without requiring user interaction, ensuring protection even in situations where the victim is immobilized or silenced. battery during emergencies. When an emergency is triggered, the user's current latitude and longitude are extracted and formatted into a clickable Google Maps link. This link is then appended to preformatted distress messages and sent to trusted contacts through both SMS and WhatsApp using the Android SMSManager API and WhatsApp's Intent-based messaging integration. To support scenarios where the attacker attempts to tamper with the phone, a fake power-off screen is rendered using an overlay service that mimics shutdown visuals while continuing background processes. This allows the phone to keep recording audio/video, share location updates, and remain functional under the radar. The countdown safety mode is implemented using a background timer thread, monitored by the main service. If the timer reaches zero and is not manually canceled, the full emergency sequence is activated. This feature was developed using Android's AlarmManager and Handler classes for accurate timing and background service execution.

For security purposes, the application also includes a data lockdown feature that uses Android's Device Administration API. This allows the app to automatically lock the phone and hide sensitive apps, notifications, contacts, and gallery during an emergency. SQLite was used as the local database to store user preferences, code phrases, emergency contacts, and safety templates. Firebase Cloud Storage was optionally enabled to store backup media such as auto-recorded videos and audio clips, providing secure remote access for post- event investigation. Finally, user interface components were crafted using Jetpack Compose and Material Design libraries to maintain a user-friendly yet minimalistic experience. Each feature was thoroughly tested using emulators and real devices to ensure robustness, quick response, and accuracy under various lighting, audio, and motion conditions. Through the integration of these advanced tools and technologies, the system achieves a high level of automation, security, and reliability in emergency scenarios.

RESULT

The system provides immediate action upon detecting an emergency keyword, code phrase, or gesture. This ensures timely help without the need for manual input. With features like location sharing, video/audio recording, fake power-off, siren, and phone lock, users are protected from multiple angles, increasing security during critical moments. Secret code phrase activation and fake shutdown screen ensure stealthy operation, which is crucial when the user is under surveillance or threatened. Auto camera and mic recording help gather important evidence during the incident, which can later be used for investigation and legal purposes.

FUTURE SCOPE

The future scope of this intelligent emergency alert system is vast and promising, especially as technologies such as Artificial Intelligence (AI), Internet of Things (IoT), and Edge Computing continue to evolve. As personal safety becomes a growing concern, especially in urban areas and vulnerable populations such as women, children, and elderly individuals, the application can expand its reach beyond mobile platforms into wearables, smart home devices, and vehicles. Future iterations can integrate seamlessly with smartwatches, earbuds, or even jewelry, enabling discreet triggering of alerts through biometric cues like heart rate spikes, tremors, or voice tone changes. Integration with smart home assistants such as Alexa or Google Assistant could also enable panic alerts via voice even when the phone is not accessible. With the advancement of IoT, sensor networks installed in public places, vehicles, or buildings can work with the app to detect abnormal behaviors and collaboratively trigger coordinated responses. This synergy will transform the app from a personal safety tool into a city- level collaborative alert system.

From a software development perspective, future enhancements may include predictive analytics using historical data to identify risk-prone areas and generate safety suggestions in advance. For instance, AI could analyze previous incidents and user behavior to offer suggestions like "Avoid this route" or "Notify contacts before you enter this area." Machine learning models can be improved to identify complex distress expressions, body language, or audio cues far more accurately, reducing false positives and negatives. The facial expression recognition system can be evolved into a full emotional state monitoring system that tracks stress and anxiety over time, offering not only emergency alerts but also mental health support suggestions. Additionally, global language support, including regional dialects and accents for voice triggers and secret phrases, will increase accessibility and adoption across diverse geographies. Another key future development involves end-to-end encryption and decentralized data handling, which ensures that user data remains private and secure even if stored in cloud systems

CONCLUSION

The integrated safety system designed through this project demonstrates a transformative approach to personal security using modern technologies such as AI, IoT, and real-time monitoring. The core modules—including auto-lock with data shielding, fake power-off deception, facial expression- based emergency triggers, countdown-initiated safety activation, secret code phrase detection, and live location tracking—collectively establish a powerful multilayered defense mechanism that is both proactive and reactive in nature. This holistic methodology not only increases the user's chances of preventing or escaping harm but also ensures critical data and location insights are preserved and relayed in real-time. By leveraging smartphone hardware like front cameras, microphones, and GPS, alongside software- based intelligence such as OpenCV, voice recognition engines, and cloud integration, the system ensures quick responsiveness even in high-stress or non-verbal conditions. Moreover, the stealth features embedded within the app offer subtle yet powerful tools that work discreetly to protect the user without alerting potential threats.

This project emphasizes that technology can be a true companion in times of personal crisis, acting autonomously when human intervention might not be possible. The design prioritizes user safety while maintaining adaptability for various real-life emergency scenarios, making it especially relevant in today’s context of rising urban crimes and threats. Furthermore, the system architecture allows for easy scalability and future enhancements, such as integrating wearable devices, drone-based location tracing, or connecting to emergency responders via public safety networks. Overall, the project serves as a stepping stone.

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AI VIGILANCE ANTICIPATING AND SLEEP APNEA THROUGH MACHINE LEARNING

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Abstract— Sleep apnea is a prevalent sleep disorder characterized by repeated interruptions in breathing, resulting in disrupted sleep and potential long-term health complications. Timely identification is crucial; however, conventional diagnosis through polysomnography tends to be expensive, uncomfortable, and time-intensive. Therefore, there is a pressing need for a smart, efficient, and easily accessible detection method. Current systems predominantly depend on clinical environments and manual monitoring of sleep metrics such as oxygen saturation and airflow. These approaches are impractical for ongoing home surveillance and frequently overlook early signs in mild cases. Although some wearable devices are available, they often lack the capability for real-time prediction and alerting. The proposed system combines AI vigilance with machine learning to observe and forecast sleep apnea incidents by utilizing real-time physiological data. It focuses on proactive notifications by detecting breathing irregularities and potential apneic events before they escalate into critical situations. This solution is cost-effective, non-invasive, and designed for user- friendliness in home settings. The process includes gathering data from sensors (SpO_2, heart rate, sound), preprocessing to minimize noise, extracting features, training models, and providing real-time monitoring and alerts. Additionally, the system archives data for long- term evaluation and reporting. The algorithms employed consist of Random Forest for classification, Support Vector Machine (SVM) for pattern recognition, and Convolutional Neural Networks (CNNs) for the analysis of time-series and audio information. These models are trained on labeled datasets to guarantee high precision. In summary, AI-driven vigilance systems represent a significant advancement in the detection of sleep apnea, enhancing early diagnosis and facilitating continuous home monitoring. This project illustrates how machine learning can revolutionize traditional healthcare diagnostics into more intelligent, rapid, and accessible solutions.

Keywords— *Sleep Apnea, Real-time Monitoring, Machine Learning, Physiological Data, AI Vigilance, Feature Extraction, Convolutional Neural Networks (CNNs), Random Forest, Support Vector Machine (SVM), Polysomnography, Breathing Irregularities, Wearable Devices, home Monitoring, Non-invasive, Sensor Data (SpO_2, Heart Rate, Sound), Preprocessing, Early Detection*

1.INTRODUCTION

Sleep apnea is a prevalent and potentially hazardous sleep disorder marked by recurrent interruptions in breathing during sleep. These pauses lower oxygen levels in the bloodstream and disturb the sleep cycle, resulting in fatigue, diminished concentration, cardiovascular issues, and other significant health concerns. Despite its widespread occurrence, numerous instances of sleep apnea remain undiagnosed due to the limitations of existing diagnostic techniques. The conventional method for diagnosing sleep apnea involves polysomnography, an overnight sleep assessment performed in a clinical environment. Although this approach is precise, it is costly, time-intensive, and often uncomfortable for patients. Additionally, it necessitates specialized equipment and trained staff, rendering it inaccessible to many, particularly for ongoing or early-stage monitoring. These obstacles underscore the pressing need for a more intelligent, accessible, and non-invasive solution. Recent developments in Artificial Intelligence (AI) and Machine Learning (ML) offer new possibilities for enhancing the detection and monitoring of sleep apnea. By examining physiological signals such as oxygen saturation (SpO_2), heart rate variability, respiratory effort, and sound patterns (including snoring), ML models can recognize abnormal breathing patterns and forecast potential apneic events. This project suggests creating an AI-driven vigilance system that utilizes machine learning algorithms to detect and predict sleep apnea episodes in real-time. The system can be integrated with wearable devices or smart home sensors to enable continuous, automated monitoring. Algorithms like Support Vector Machines (SVM), Random Forest, and Convolutional Neural Networks (CNNs) are employed for precise classification and prediction of sleep-related irregularities. In conclusion, this project seeks to close the gap between clinical diagnosis and at-home monitoring by offering a cost-effective, intelligent, and dependable system to enhance early detection and treatment of sleep apnea.

2.Literature Review

Sleep apnea has typically been diagnosed through polysomnography (PSG), which is considered the gold standard method that involves monitoring various physiological signals during sleep, including EEG, ECG, airflow, and oxygen saturation. Although PSG offers high accuracy, numerous studies (e.g., Punjabi, 2008) have pointed out its drawbacks—primarily its cost, discomfort, and the requirement for clinical supervision. These challenges have prompted researchers to investigate automated and home-based alternatives for early diagnosis and monitoring. In recent years, machine learning (ML) techniques have gained traction in the detection of sleep apnea. Hassan and Bhuiyan (2016) introduced a method that utilizes ECG signals along with a support vector machine (SVM) classifier, achieving commendable accuracy in identifying apneic events. Likewise, Almazaydeh et al. (2012) showcased the application of heart rate variability and respiratory signals with decision tree classifiers to identify sleep disturbances. Deep learning methods have further propelled advancements in this area. For example, Chambon et al. (2018) employed Convolutional Neural Networks (CNNs) on raw PSG data, attaining high performance in apnea detection without the need for manual feature extraction. These models are capable of learning intricate temporal and spatial features from raw data, making them ideal for extensive sleep data analysis. Research also indicates potential in utilizing audio and SpO_2 signals for apnea detection. Abeyratne et al. (2013) created a snore-based screening system that leverages audio pattern recognition, while recent studies by Zhang et al. (2020) implemented SpO_2 and recurrent neural networks (RNNs) for continuous, non-invasive detection. In summary, the existing literature endorses the viability of AI and ML in the detection of sleep apnea; however, obstacles persist regarding real-time implementation, generalizability, and home deployment. This project aims to build upon these foundations by integrating multiple signals and AI models to develop a smart, real- time vigilance system that is appropriate for home monitoring.

3.EXISTING SYSTEM

The most prevalent method currently used for diagnosing sleep apnea is polysomnography (PSG). This procedure entails an overnight sleep study performed in a sleep lab under the guidance of qualified medical personnel. Throughout the examination, various physiological parameters are observed, such as brain activity (EEG), heart rate (ECG), eye movement, muscle activity, airflow, oxygen saturation (SpO_2), and respiratory effort. Although PSG is highly precise and regarded as the gold standard, it is also costly, inconvenient, and not suitable for large-scale screening or ongoing monitoring. In recent years, alternative solutions have emerged to overcome some of these challenges. Wearable devices and home sleep testing kits now provide basic monitoring with fewer sensors, usually measuring heart rate, oxygen levels, and breathing patterns. Nevertheless, these devices often still necessitate post-processing by healthcare professionals and are generally limited in their predictive capabilities. Machine learning has been integrated into some existing systems, but primarily for offline analysis. Systems utilizing Support Vector Machines (SVM) or decision trees classify recorded physiological data after the sleep session concludes, offering limited real-time insights. Some mobile applications strive to identify snoring or irregular breathing through smartphone microphones, but their accuracy frequently suffers due to background noise and insufficient contextual information. In summary, current systems either lack real-time prediction, necessitate clinical oversight, or are not sufficiently robust for reliable home-based monitoring. These deficiencies highlight the necessity for a more intelligent, automated, and user-friendly system that can both detect and predict sleep apnea events in real time.

4.PROPESED METHODLOGY

The proposed system is designed to deliver a smart, real-time, and non-invasive approach for the detection and anticipation of sleep apnea episodes through the utilization of artificial intelligence and machine learning. While traditional diagnostic techniques such as polysomnography are accurate, they tend to be costly, inconvenient, and not suitable for ongoing home monitoring. To overcome these challenges, the system incorporates affordable wearable sensors that gather real-time physiological data, including oxygen saturation (SpO_2), heart rate, and audio signals related to breathing patterns. The data collection process initiates with these sensors, which transmit raw data to a preprocessing unit aimed at eliminating noise and artifacts. Essential features are subsequently extracted, encompassing variations in SpO_2 levels, irregular heartbeat patterns, and unusual respiratory sounds. These features are utilized to train machine learning models that can detect early indicators of apneic events. The system utilizes a blend of algorithms: Random Forest for reliable classification, Support Vector Machine (SVM) for accurate pattern recognition, and Convolutional Neural Networks (CNNs) for the analysis of time- series and audio data. These models function in unison to guarantee high detection accuracy and minimal false-positive rates. Beyond detection, the system provides real-time alerts and predictive warnings when early signs of apnea are recognized. This enables users or caregivers to take prompt action, thereby preventing serious incidents during sleep. All data is securely stored for long- term analysis and can be shared with healthcare professionals for clinical assessment.

Algorithms

In this project, various machine learning algorithms are utilized to effectively identify and predict sleep apnea occurrences. Support Vector Machines (SVM) are employed for their excellent classification abilities, as they proficiently distinguish normal breathing patterns from apneic events by determining the optimal decision boundary within intricate, high-dimensional datasets. In addition, the Random Forest algorithm, which is an ensemble technique based on multiple decision trees, is applied due to its resilience against noise and overfitting, along with its capacity to manage a variety of physiological signals and pinpoint significant features that aid in apnea detection. Furthermore, Convolutional Neural Networks (CNNs), a deep learning approach, are essential for examining raw time-series data such as respiratory sounds and oxygen saturation levels. CNNs autonomously extract hierarchical features, allowing the system to recognize subtle and intricate patterns in the data without requiring manual feature engineering. The combination of these algorithms enables the AI vigilance system to utilize both traditional and deep learning strategies, improving accuracy and facilitating real-time monitoring and forecasting of sleep apnea episodes.

Methodology

The approach of this project centers on creating a machine learning-driven system for the real-time identification and prediction of sleep apnea through physiological data. The initial step involves data collection, where signals like oxygen saturation (SpO_2), heart rate variability, respiratory patterns, and audio recordings of breathing sounds are obtained from wearable sensors or smart home devices. This multi-modal data offers a thorough understanding of the user's sleep health. Subsequently, the data is subjected to preprocessing to enhance its quality and reliability. Techniques for noise reduction, such as filtering, are utilized to eliminate artifacts from sensor readings and audio signals. The refined data is then segmented and normalized in preparation for feature extraction. During the feature extraction stage, pertinent characteristics are recognized, including fluctuations in oxygen levels, irregular breathing intervals, snoring frequency, and variations in heart rate. These features serve as the input for machine learning models. The project utilizes a range of machine learning algorithms, including Support Vector Machines (SVM), Random Forest, and Convolutional Neural Networks (CNNs). The models are trained on labeled datasets where apneic events are marked. The training process involves optimizing parameters to enhance accuracy and reduce false positives. For real-time monitoring, the trained models continuously analyze incoming data to identify and forecast apnea episodes. When an event is predicted, the system activates alerts for the user or healthcare provider. Ultimately, the system's performance is assessed using metrics such as accuracy, sensitivity, specificity, and F1-score to guarantee dependable and prompt detection. This methodology aspires to develop a practical, non-invasive, and intelligent solution for monitoring sleep apnea

DATA COLLECTION

The initial phase focuses on gathering extensive physiological and audio data during sleep. Essential metrics such as oxygen saturation (SpO_2), heart rate, and respiratory rate are monitored using wearable sensors or smart home devices. Additionally, audio data that records snoring and breathing sounds is obtained through microphones positioned near the user. This multi-faceted data collection guarantees that the system possesses a wealth of information to analyze breathing patterns and identify abnormalities that may indicate sleep apnea. The collection of varied data types enhances the system's capability to recognize apnea across diverse individuals and settings, thereby making the solution more flexible and resilient

DATA PREPROCESSING

Raw sensor and audio data frequently contain noise and artifacts that can obstruct precise analysis. During this phase, various filtering methods are employed to eliminate unwanted noise and smooth the signals. The data is subsequently normalized to a uniform scale and divided into fixed time intervals, facilitating easier processing by machine learning models to detect relevant patterns linked to apneic events. Effective preprocessing improves signal quality and guarantees that the models receive dependable data inputs, which is essential for achieving high detection accuracy.

Modal Selection

From the preprocessed data, significant features are derived to act as inputs for the machine learning models. For physiological data, features such as abrupt declines in SpO_2, irregular breathing intervals, heart rate variability, and snoring frequency are calculated. Audio data is transformed into spectrograms or other pertinent representations that emphasize the time-frequency characteristics of breathing sounds. These features encapsulate the crucial information required to distinguish between normal sleep and apnea episodes. Carefully selected features minimize computational complexity and enhance the model's capacity to generalize across various patients

MODEL TRAINING

Machine learning models, including Support Vector Machines (SVM), Random Forest, and Convolutional Neural Networks (CNNs), are trained using labeled datasets that annotate apnea events. Throughout the training process, these models learn to identify patterns and features that differentiate normal breathing from abnormal breathing. Hyperparameters are adjusted to enhance performance, and cross-validation is employed to verify that the models generalize effectively to new data. Training on a variety of datasets enables the models to adapt to differences in individual physiology and sleep conditions, thereby increasing overall reliability.

Real-Time Monitoring & Prediction

After training, the models are implemented to continuously analyze incoming sensor and audio data during sleep. They identify ongoing apneic events in real time and can also predict future occurrences by recognizing early warning signs. This proactive monitoring facilitates timely intervention before serious apnea episodes arise, thereby enhancing patient safety and comfort. Real- time processing guarantees that users receive immediate feedback, which is essential for preventing complications and improving disorder management.

Alert Generation

When the system identifies or forecasts an apnea event, it promptly generates alerts or notifications. These alerts can be communicated to the user through a smartphone application or directly to healthcare professionals. Early warnings allow users to modify their sleeping position or seek medical help, thus minimizing the risks linked to untreated sleep apnea. This swift communication empowers both patients and caregivers to act quickly, improving the overall effectiveness of treatment.

Performance Assessment & Feedback

The effectiveness of the system is assessed through performance metrics including accuracy, sensitivity, specificity, and F1-score. Regular monitoring of these metrics aids in pinpointing areas that require enhancement. Insights gained from new data can be utilized to retrain and refresh the models, guaranteeing that the system adjusts to various users and stays dependable over time. Continuous evaluation promotes a cycle of enhancement that upholds superior detection and proactive performance throughout the system's lifespan.

Prediction and Classification

In this project, prediction and classification play a central role in identifying and anticipating sleep apnea events from physiological signals. Once the sensor data—such as SpO_2 (oxygen saturation), heart rate, and respiratory sounds—is collected and preprocessed to remove noise, important features are extracted that represent patterns related to sleep apnea. These features are then input into trained machine learning models for classification. The **Random Forest algorithm** is used to classify whether a specific time segment indicates normal breathing or an apnea event, based on decision trees built from historical data. **Support Vector Machine (SVM)** enhances this by recognizing subtle patterns that may indicate early signs of apnea, especially in mild or borderline cases. Additionally, **Convolutional Neural Networks (CNNs)** are applied to analyze time-series signals and audio data, learning spatial and temporal features that may not be visible through traditional methods. Together, these models not only classify the presence or absence of sleep apnea but also **predict potential apnea events in real time**, enabling proactive alerts before the condition worsens during sleep. This intelligent prediction mechanism helps improve early intervention, reduces health risks, and supports continuous home-based monitoring.

Simulated Results

The simulated outcomes of the AI vigilance system revealed encouraging effectiveness in identifying and predicting sleep apnea incidents through synthetic physiological and audio data. Among the evaluated models, the Convolutional Neural Network (CNN) attained the highest accuracy rate of 92%, followed by Random Forest at 88% and Support Vector Machine (SVM) at 85%. The CNN also exhibited robust sensitivity at 90%, successfully detecting apnea episodes, while all models sustained specificity levels above 87%, thereby reducing false positives. Crucially, the system was capable of forecasting apnea events up to 30 seconds prior to their occurrence, facilitating timely notifications. Simulations of real-time processing validated low latency, making it suitable for ongoing home monitoring.

	precision	recall	F1-score	support
0	0.79	1.00	0.88	158
1	0.00	0.00	0.00	42
Accuracy	-	-	0.79	200
Macro avg	0.40	0.50	0.44	200
Weighted avg	0.62	0.70	0.70	200

CONCLUSION:

Sleep apnea is a common and serious sleep disorder that can greatly affect health and quality of life if it goes undiagnosed or untreated. Conventional diagnostic techniques, like polysomnography, are accurate but often expensive, inconvenient, and not suitable for ongoing or early-stage monitoring. This project introduces an AI-driven vigilance system that utilizes machine learning methods to offer a cost-effective, non-invasive, and real- time solution for detecting and predicting sleep apnea events. The implementation of sophisticated machine learning algorithms—including Support Vector Machines, Random Forest, and Convolutional Neural Networks—allows the system to accurately analyze complex data patterns. Simulated results indicate that these models, especially CNNs, achieve high levels of accuracy and sensitivity in detecting apnea episodes while reducing false alarms. In summary, the AI vigilance system demonstrates significant potential as a valuable resource for detecting and managing sleep apnea. Future efforts should concentrate on validating the system using actual patient data, refining algorithms for tailored monitoring, and investigating its integration with wearable devices and smart home technologies. In the end, this strategy can facilitate early diagnosis, enhance patient outcomes, and alleviate the strain on healthcare systems.

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AI FOR PREDICTINGTHEOUTCOMEOFSURGICAL PROCEDURE

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Abstract— By using machine learning algorithms to evaluate patient data and surgical variables, artificial intelligence (AI) is transforming the prediction of surgical outcomes. These cutting-edge models increase decision- making and risk assessment accuracy, which eventually improves patient care and maximizes surgical outcomes. Preoperative risk assessment is much improved by the growing use of AI-driven predictive models in the surgical profession to anticipate outcomes and complications. These models go beyond conventional statistical techniques by examining a wide range of patient-specific characteristics, providing dynamic and tailored insights. This development helps to optimize surgical planning and enhance patient safety in addition to helping surgeons make well-informed decisions. Better results and more individualized patient care are fostered by the revolutionary transition towards data- driven healthcare represented by the incorporation of AI in surgical practices.

Keywords—Artificial Intelligence (AI), Machine Learning (ML), Predictive Analytics, Surgical Outcome Prediction, Clinical Decision Support Systems (CDSS), Risk Stratification, Patient-Specific Modeling.

1.INTRODUCTION

Artificial intelligence (AI) has become a game- changer in the healthcare industry, especially when it comes to surgical procedures. The need for advanced instruments that can help medical practitioners make well-informed judgments is growing along with the complexity of surgical procedures. Since surgical outcome prediction has a direct impact on patient safety, resource allocation, and overall healthcare quality, it is an essential component of surgical planning. These algorithms are capable of detectingsubtlepatterns and correlations that may initially gounnoticed, facilitating more precise predictions regardingsurgical outcomes and potential complications. By deliveringreal -time insights, AI empowers surgeons torefinetheirstrategies to align with each patient'suniquerequirements, thereby improving the overall standardofcare. Furthermore, the integration of AI insurgical predictions not only enhances individual proceduresbutalso advances collective knowledge of surgical practices and outcomes across variedpatientpopulations. With the growing emphasis on personalizedmedicine within healthcare systems, the integrationofAI in surgical decision-making is anticipatedtogainprominence, promising enhanced patient outcomesandstreamlined healthcare processes. This context pavesthe way for examining the diverse AI toolsandtechniques revolutionizing surgical outcomeprediction, underscoring their critical role in advancingsurgical careand ensuring patient safety. Theseadvancedtechnologies play a crucial role in risk stratification, preoperative evaluations, and forecasting postoperativecomplications, empowering healthcare professionalstomake data-driven, well-informed decisions. Theincorporation of Clinical Decision Support Systems(CDSS) alongside prognostic modelingfacilitatespersonalized patient care, reduces potential risks, andenhances surgical success rates.

2.RELATED WORK

The utilization of Artificial Intelligence (AI) and Machine Learning (ML) for predicting surgical outcomes has garnered considerable attention in recent years, revolutionizing clinical decision-making and advancing patient safety. Extensive research efforts have been dedicated to developing AI-driven models capable of processing large-scale medical datasets—including Electronic Health Records (EHR), imaging data, and patient demographics—to accurately assess surgical risks and forecast outcomes. A key focus of research involves developing machine learning models to predict surgical risks. Techniques such as logistic regression, random forests, and deep neural networks have been employed to assess the likelihood of complications, including infections, hemorrhaging, and prolonged hospitalization. For example, predictive models leveraging electronic health record (EHR) data have shown potential in forecasting postoperative complications by examining preoperative attributes, surgical histories, and intraoperative factors. These models equip healthcare professionals with critical insights into individual patient risks, facilitating more informed surgical strategies and tailored care interventions. Clinical Decision Support Systems (CDSS) have become indispensable in improving surgical decision-making processes. Leveraging AI-driven predictive models, these systems analyze real-time data to recommend the most effective surgical strategies. By synthesizing patient histories, diagnostic findings, and predictive risk evaluations, CDSS enables surgeons to accurately assess potential complications and determine the most appropriate course of action. This technology plays a pivotal role in mitigating adverse outcomes and enhancing the efficient allocation of resources within healthcare organizations. sensor nodes that detect and capture perfect indication The application of AI in medical imaging has considerably enhanced the prediction of surgical outcomes. Utilizing advanced deep learning methods, particularly convolutional neural networks (CNNs), AI systems are capable of analyzing complex imaging datasets such as CT scans, MRIs, and X-rays. These models excel in identifying anomalies, evaluating tissue properties, and foreseeing potential surgical complications. For instance, AI technologies have been effectively employed to detect tumors, assess fracture severity, and identify vascular abnormalities, all of which play a pivotal role in surgical planning. By delivering rapid and precise interpretations of imaging data, AI has revolutionized preoperative assessments, ultimately contributing to improved surgical results.

Moreover, AI-powered data mining techniqueshavebeen utilized to extract patterns and insightsfromvasthealthcare datasets. These techniques enablethestratification of patients into different risk categories, facilitating better surgical planning and postoperativemanagement. For instance, clustering algorithmscangroup patients based on similar risk factors, aidingindeveloping standardized protocols for eachgroup. Thisapproach not only streamlines clinical workflowsbutalso ensures that high-risk patients receivethenecessary attention and care.

3.PROPOSED SYSTEM

The proposed AI-powered systemaims totransformthe prediction of surgical outcomes by integratingdiverse data sources, advanced machinelearningalgorithms, and real-time analytics. Its primarygoal istoelevate the precision, efficiency, and reliabilityof surgical risk predictions, ultimately supporting clinical decision-making and enhancing patient safety. Central toitsdesign, the system features six interconnected modules, each serving a vital function to facilitate comprehensivedata analysis and accurate outcome forecasting. TheData Collection and Preprocessing Moduleservesasacritical component designed to aggregateinformationfrom diverse sources, including ElectronicHealthRecords (EHRs), medical imaging modalities suchasCTscans, MRIs, and X-rays, as well as real-timephysiological data from monitoring devices. Toupholddata integrity and comply with healthcare regulations, the module incorporates sophisticated preprocessingmethods such as normalization, standardization, andanonymization to protect patient privacy. Additionally, itfeatures robust mechanisms for addressingmissingdata and mitigating noise, thereby enhancingtheaccuracy and consistency of datasets essential forreliable predictive analyses.

The Monitoring and Alert Systemis designedtocontinuously oversee postoperativedata, encompassing vital signs, lab results, andotherphysiological metrics. By utilizing predefinedthresholdsand predictive analytics, it identifies deviationsfromexpected recovery patterns and initiates automatedalerts to prompt timely interventions. Thisproactiveapproach aims to reduce complications throughearlydetection, improve patient safety, and facilitatemoreefficient recovery outcomes.

Surgical Error Reduction with AI:

Artificial Intelligence (AI) serves as a critical assetin minimizing surgical errors, effectivelyimprovingpatient safety and surgical accuracy. AdvancedAI-driven preoperative planning tools examineextensivepatient data—such as medical histories, imagingresult and prior surgical outcomes—to assist surgeons in predicting possible complications and identifying the most effective surgical approaches. Utilizing machine learning algorithms, these systems evaluate diverse risk factors and provide tailored surgical recommendations, substantially reducing the likelihood of errors during procedures. Pathologists have effectively utilized artificial intelligence technology to achieve a remarkable reduction in their error rate for identifying cancer- positive lymph nodes, decreasing it from 3.4% to an impressive 0.5%. This advancement enables the early identification of high-risk patients, allowing AI systems to assist surgeons and radiologists in lowering lumpectomy rates by 30% among patients whose suspicious breast needle biopsies are ultimately determined to be benign after surgical excision. Furthermore, a prognostic study involving 58,236 patients and 67 surgeons demonstrated that postoperative complications could be precisely predicted through an AI system that utilized automated real-time electronic health record data. The system maintained high accuracy during prospective validation, matching the prediction capabilities of experienced surgeons with minimal performance degradation.

Among various surgical decision-support tools, the POTTER (Predictive Optimal Trees in Emergency Surgery Risk) artificial intelligence-based calculator has demonstrated significant potential in predicting postoperative morbidity and mortality, thereby assisting in critical decision-making processes related to risk assessment, benefit evaluation, and patient prognosis. Particularly valuable in emergency general surgery scenarios where outcomes are uncertain, POTTER is recognized as a novel, interpretable, and highly accurate predictor of in-hospital mortality for elderly patients up to age 85. Its utility extends to bedside counseling and serves as a benchmark for improving emergency surgical care standards. The My-SurgeryRisk calculator demonstrated superior usability and accuracy compared to physicians' clinical judgment, consistently outperforming their initial risk assessments for nearly all postoperative complications.

Artificial intelligence serves as a pivotal componentinsurgical education and training, utilizingdata-drivensimulations and virtual reality-based modulestoenablesurgeons to rehearse intricate laparoscopicprocedures. These advanced AI-powered systems deliver real-timefeedback, allowing trainees to enhance their techniquesand strengthen decision-making capabilitiespriortooperating on actual patients. Furthermore, byexaminingextensive datasets from previous surgeries, AI identifiespatterns underlying successful outcomes andrefinesbest practices to improve the trainingprocessforaspiring surgeons. Integrating artificial intelligence intocomplexlaparoscopic cholecystectomy enables healthcareproviders to mitigate risks, enhance surgical precision, and strengthen patient safety. AI-poweredsystemssupport informed decision-making, streamlinesurgical workflows, and foster more efficient andfavorablepatient outcomes. As advancements in AI persist, itsapplication in minimally invasive surgeries will continueto enhance techniques and set newbenchmarksinprecision medicine.

Additionally, AI models analyze surgical videosandpatient vitals to predict postoperative complications, ensuring timely intervention and improvedrecovery. AI-driven robotic platforms further enhancesurgical dexterity, enabling safer and more efficient procedures. By integrating AI into laparoscopic cholecystectomy, thefield of minimally invasive surgery is advancingtowardgreater accuracy, efficiency, and improved patient outcomes.

APPLICATIONS IN SURGERY:

Cardiothoracic surgery: Artificial intelligence is becoming indispensable in cardiothoracic surgery, significantly contributing to the prediction of surgical outcomes, the optimization of preoperative strategies, and the enhancement of intraoperative accuracy. Procedures such as coronary artery bypass grafting (CABG), valve replacements, and lung resections demand meticulous decision-making due to the intricate nature of cardiovascular and thoracic anatomy. AI-powered technologies bolster surgical success rates by leveraging extensive datasets, identifying critical risk factors, and providing real-time support during operations. During preoperative evaluations, AI-driven predictive models process patient-specific data—such as echocardiograms, CT scans, and electronic health records—to evaluate the likelihood of complications and mortality. These advanced machine learning algorithms support surgeons in selecting the most appropriate treatment plan, pinpointing patients who might benefit from supplementary preoperative measures or an adjusted surgical approach.

AI significantly advances surgical trainingandoutcome analysis by utilizing deep learningalgorithmstoevaluate previous cardiothoracic procedures. Thisdataenables the optimization of best practices, enhancesprocedural efficiency, and identifies key patternslinkedto successful surgeries. Additionally, AI-poweredsimulation platforms provide specializedtrainingforintricate cardiac operations, equipping surgeonstodevelop proficiency in managing high-risk scenarioswhile maintaining patient safety.

Breast cancer surgery:

AI is revolutionizing breast cancer surgerybyenhancing the ability to predict surgical outcomesandoptimize patient prognosis. Through sophisticatedmachine learning models, vast datasets comprisingpatient records, imaging scans, and pathologyreportsare analyzed to evaluate risks, forecast complications, and determine the most effective surgical approaches. In the preoperative phase, AI-driven predictiveanalytics focus on crucial factors such as tumor attributes, lymph node status, and genetic markers to provide tailored risk assessments and survival predictions for individual patients. AI technology enables the stratification of patients into risk categories, assisting surgeons in determining the most appropriate surgical approach, whether it be a lumpectomy or mastectomy. Furthermore, AI-driven imaging analysis enhances tumor margin detection, thereby minimizing the risk of residual cancer following surgery. During procedures, real-time AI-assisted decision-making evaluates intraoperative imaging to identify abnormal tissues, while AI-powered robotic systems improve precision in tumor excision, reducing harm to surrounding areas and enhancing cosmetic outcomes in breast reconstruction. Additionally, augmented reality (AR) overlays support flap reconstruction planning by mapping vascular structures and optimizing tissue grafting strategies.

Neurological surgery:

Advancements in artificial intelligence are transforming neurological surgery by refining the prediction of surgical outcomes and enhancing patient care. Utilizing machine learning, these technologies analyze neuroimaging data, patient history, and genetic markers to evaluate risks, anticipate complications, andrecommend optimal surgical approaches. Duringpreoperative assessments, AI systems processMRI, CTscan, and EEG data to pinpoint essential brainregionsand forecast potential challenges. Predictive models further assess the probabilityofcomplications such as postoperative neurological impairments, seizures, or infections, therebysupportingprecise surgical planning and effective risk management.AI-assisted intraoperative navigation systemsoffer realtime feedback during surgery, significantlyenhancingthe accuracy of complex procedures liketumorresections and deep brain stimulation. Roboticsystemspowered by artificial intelligence aid in precisetargetingto reduce tissue damage and improve surgical successrates, while augmented reality (AR) overlaysprovidecritical visualizations of neural pathways, ensuringsaferand more effective interventions.

Furthermore, AI-driven post-operativemonitoringcan detect early signs of complications likeswelling, infections, or seizures, allowing for timely intervention. Personalized treatment plans based on AI-generatedinsights ensure optimal rehabilitation, cateringtoindividual patient needs. The integrationof AI inneurological surgery continues to evolve, offeringgroundbreaking possibilities for brain-computerinterfaces, automated diagnostics, andreal-timedecision-making, ultimately paving the way for afutureof safer and more effective neurosurgical procedures. AI-powered predictive analyticshelpneurosurgeons assess potential surgical risksanddetermine the best course of action for eachpatient. Robotic-assisted surgical systems enableminimallyinvasive procedures, reducing complications, recoverytime, and human error.

Risk calculators:

AI-driven risk calculators play a pivotal role in predicting surgical outcomes and supporting clinical decision-making. By analyzing patient-specific data—such as age, medical history, comorbidities, laboratory findings, and imaging results—these advanced systems deliver precise risk assessments for diverse surgical procedures. During preoperative evaluations, they estimate the probability of complications, including infections, bleeding, or postoperative organ dysfunction. Leveraging machine learning algorithms to detect patterns within extensive datasets, these tools enable surgeons to customize surgical approaches based on the unique risk profiles of each patient. For example, tools such as the Surgical Risk Calculator developed by the American College of Surgeons deliver individualized risk predictions for complications and mortality, supporting informed consent discussions and strategic surgical planning. Additionally, AI-driven systems provide real-time risk evaluations during procedures by monitoring intraoperative data like vital signs and operational metrics, facilitating timely interventions and improving both patient safety and surgical accuracy.

Limitations of Ai modal:

After surgery, AI-driven calculatorsassessrecovery by analyzing patient-specific datatoforecastpotential complications, such as infections, delayedhealing, or hospital readmission. These advancedtoolsgenerate tailored follow-up care recommendations, optimizing recovery strategies and enhancinglong-termoutcomes. By pinpointing individuals at elevatedrisk, healthcare teams can implement precise interventionsto mitigate adverse events and bolster patient safety. Additionally, these AI-powered systems contributetoclinical research by processing extensivedatasetstouncover emerging risk factors and improvetheaccuracy of predictive models. A significant concern is the risk of over-dependenceon AI systems. Although AI provides valuableinsights, itcannot substitute clinical judgment. Surgeonsmustevaluate AI-generated recommendations withinthecomprehensive framework of patient care, ensuringthattechnology enhances rather than supersedeshumanexpertise. Additionally, AI models may fail toaccount forthe nuanced variables of individual patient conditionsorunanticipated surgical challenges, potentiallyresultinginless effective outcomes if relied upon exclusively.

Conclusion:

AI models serve as transformativeinstruments in predicting surgical outcomes, presentingopportunities to advance clinical decision-making,enhance patient safety, and optimize surgical planning.By processing intricate datasets, these technologiesidentify risk patterns and generate valuableinsightsthatenable personalized and precise surgical care.Nevertheless, their adoption within healthcaresystemsis fraught with challenges, including dataqualitylimitations, interpretability issues, concerns regardinggeneralizability, and ethical dilemmas, emphasizingthenecessity for a careful and well-informedimplementation strategy. To overcome these challenges, it is imperativeto prioritize efforts in data curation, model validation, transparency, and strict adherence to ethical standards. Additionally, fostering effective human-AI collaboration, investing in robust infrastructure, andprovidingcontinuous training are vital to fully leveragingthepotential of AI while mitigating associatedrisks. Futureprogress in AI should concentrate on creatingmodelsthat are interpretable, adaptable, and equitable, withtheability to integrate seamlessly into variedclinical environments

The judicious integration of AI into surgical outcome prediction holds considerable promise to transform surgical care by improving patient outcomes, optimizing experiences, and increasing the efficiency of healthcare systems. Through the resolution of existing challenges and the encouragement of collaborative advancements, AI has the potential to establish itself as a vital instrument in shaping the future of surgical practices.

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AI IN HEALTHCARE DISEASE DIAGNOSIS AND PREDICTION

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Abstract:

Artificial Intelligence (AI) has significantly transformed the healthcare sector by improving the accuracy of disease diagnosis and prediction. Utilizing machine learning (ML), deep learning (DL), and natural language processing (NLP), AI-driven models analyze medical data to identify diseases and forecast potential health risks. These AI-enhanced diagnostic tools demonstrate a high level of precision in detecting conditions such as cancer, diabetes, cardiovascular diseases, and neurological disorders, often outperforming conventional diagnostic methods. Additionally, predictive analytics facilitates the early identification of diseases by evaluating patient histories, genetic information, and lifestyle choices. Technologies such as convolutional neural networks (CNNs) for medical imaging and AI-based decision support systems contribute to enhanced clinical efficiency and improved patient outcomes. Although challenges remain, including concerns regarding data privacy, ethical implications, and the necessity for regulatory approval, AI continues to propel advancements in personalized medicine and healthcare automation. This paper examines the impact of AI on disease diagnosis and prediction, emphasizing its potential, associated challenges, and future opportunities for transforming contemporary healthcare.

Keywords—Disease diagnosis, Disease prediction, MRI, Convolutional neutral networks, Patient data, AI algorithms, Data privacy and security, Ethical concepts

1.INTRODUCTION

The swift progress in Artificial Intelligence (AI) has transformed the healthcare sector, especially in the realms of disease diagnosis and prediction. Traditionally, the diagnostic process has depended on a blend of clinical knowledge, medical imaging, laboratory analyses, and patient history, which can be engthy and prone to human error. In contrast, AI, utilizing machine learning (ML), deep learning(DL), andnatural language processing (NLP), presentsagroundbreaking method for enhancingdiagnosticprecision, lowering expenses, and improvingpatientoutcomes.AI technologies have shownexceptionaleffectiveness in recognizing and diagnosingavarietyof diseases, such as cancer, cardiovascular conditions, diabetes, neurological issues, and infectiousdiseases.These AI-based diagnostic instrumentscanmeticulously analyze extensive medical datasets, frequently outperforming human practitionersindetecting irregularities in medical images likeX-rays, MRIs, and CT scans. For instance, convolutional neuralnetworks (CNNs) have been employedtoidentifytumors in radiological images with remarkableaccuracy, facilitating early detection andpromptttreatment. In addition to diagnosis, artificialintelligence significantly contributes to thepredictionof diseases. Predictive analytics leverages patient data, genetic information, and lifestyle factors toevaluatean individual's risk of developing specifichealthconditions. AI-based models can estimatetheprobability of heart disease by examiningfactorssuchas cholesterol levels, blood pressure, andlevelsofphysical activity.

Likewise, machinelearningalgorithms can assess the risk of diabetesbyconsidering dietary patterns, genetic factors, andmedical history. The capacity to anticipatediseasesprior to the onset of symptoms facilitatesearlyintervention, the creation of personalizedtreatmentplans, and the enhancement of preventivecarestrategies.

This paper examines the function of AI in disease diagnosis and prediction, investigating the technologies that drive these advancements, their effects on the healthcare sector, and the challenges that must be overcome for effective AI integration. By gaining insight into the strengths and weaknesses of AI in healthcare, we can create pathways for more precise, efficient, and accessible medical diagnosis and predictive healthcare solutions

2.AI TECHNOLOGIES USED IN DISEASE DIAGNOSIS AND PREDICTION

Artificial Intelligence (AI) has profoundly changed the landscape of healthcare by enabling precise and efficient diagnosis and prediction of diseases. A range of AI technologies is employed to examine medical data, detect patterns, and anticipate possible health risks. The following outlines the primary AI technologies utilized in the realms of disease diagnosis and prediction:

Machine Learning (ML)

Machine Learning is a subset of Artificial Intelligence that allows systems to learn from medical data and enhance their precision over time without the need for direct programming. Its applications are extensive, including: Pattern Recognition – Detecting patterns in patient symptoms, laboratory results, and imaging data. Predictive Modeling – Assessing the probability of disease development based on past patient records. Classification Algorithms – Utilizing techniques such as Decision Trees, Random Forest, and Support Vector Machines (SVM) to diagnose diseases based on symptoms and test outcomes

Deep Learning (DL)

Deep Learning, a branch of Machine Learning (ML), employs artificial neural networks (ANNs) to interpret intricate medical data. It is especially effective in examining extensive datasets, including medical imaging and genomic information. Convolutional Neural Networks (CNNs) –Thesenetworks are utilized in the analysis of medical images, such as X-rays, CT scans, andMRIs, toidentify irregularities such as tumors, fractures, andlung infections. Recurrent Neural Networks (RNNs) andLongShort-Term Memory (LSTM) – These modelsareapplied to the analysis of sequential medical data, including ECG signals and electronic healthrecords(EHRs). Autoencoders – These are employedfor detectinganomalies in brain scans and pathology slides.

Natural Language Processing (NLP)

NLP enables AI to understand, interpret, andprocess human language in healthcaredata. It isapplied in: Electronic Health Record (EHR) Analysis–Extracting valuable information fromunstructuredclinical notes. Medical Chatbots and Virtual Assistants–AI- powered assistants help in preliminary diagnosisand patient interactions. Disease Surveillance – NLP is used totrackdisease outbreaks fromsocial media andnewsreports.

Computer Vision

Computer vision enables AI systems tointerpretvisual data, making it crucial for disease diagnosis. It is used in: Radiology and Pathology – AI can detect abnormalities in medical images faster thanhuman radiologists. Dermatology – AI models analyze skinlesionsforearly melanoma detection. AI models detect lung nodules in CTscanswithhigh accuracy, aiding in early lung cancer diagnosis

3.AI IN DISEASE DIAGNOSIS

Artificial Intelligence (AI) has greatly advancedthefield of disease diagnosis by increasing theprecision, rapidity, and effectiveness of identifyingarangeofmedical conditions. Diagnostic systems drivenbyAIleverage machine learning (ML), deep learning(DL), and natural language processing (NLP) toevaluatemedical data, recognize patterns, andsupporthealthcare professionals in their decision-makingprocesses.

Applications of AI in Disease Diagnosis

Cancer Detection- Artificial intelligence enhances the early identification of cancer by examining medical imaging and pathology slides. Breast Cancer -AI systems accurately identify tumors in mammograms. Lung Cancer - AI evaluates CT scans to pinpoint lung nodules. Skin Cancer – AI-driven dermatology applications recognize melanoma from skin photographs. For instance, Google's DeepMind AI demonstrates superior breast cancer detection compared to human radiologists.

4.AI IN DISEASE PREDICTION

Artificial Intelligence (AI) is revolutionizingthehealthcare sector by facilitating early diseasedetection, which in turn supports preventive strategiesandtailored treatment approaches. AI-drivenpredictivemodels scrutinize extensive medical datasets, encompassing genetic data, lifestyle variables, andclinical histories, to evaluate an individual'slikelihoodof developing illnesses prior to the onset of symptoms. Through the application of machine learning(ML), deeplearning (DL), and big data analytics, AI significantlyboosts the precision and effectiveness of diseaseforecasting, thereby enhancing patient outcomesandlowering healthcare expenses.

Understanding the Role of AI inDiseasePrediction

AI technologies and algorithms have transformeddisease diagnosis by providinghealthcareprofessionals with accurate and efficient tools. Owing to their ability to extract nuancedpatterns from complex medical data, machine-learning techniques, particularly deep learning, have gained importance.AI systems employ a range of methodologies to forecast diseases: Machine Learning Techniques – AI detects patterns within patient information to assess the likelihood of disease occurrence. Deep Learning Approaches – AI examines intricate medical data, such as imaging and genomic information. Predictive Analytics – AI anticipates the advancement of diseases by analyzing historical data trends.

5.AI Applications in Disease Prediction

a) Cardiovascular Disease Prediction

AI predicts heart disease by analyzing factors like Blood pressure, cholesterol levels, and heart rate patterns. Lifestyle habits such as diet, smoking, and physical activity.ECG and echocardiogram data to detect early signs of heart failure.

b) Diabetes prediction

AI assesses diabetes risk based on Family history and genetic predisposition. Blood sugar levels, body mass index (BMI), and insulin resistance. Lifestyle factors such as diet and exercise habits. Example: AI-powered mobile apps track glucose levels and suggest personalized preventive measures

c) Cancer Prediction

AI helps in early cancer detection by Analyzing genetic mutations linked to cancer rise, Evaluating medical imaging for precancerous lesions Predicting tumor development based on patient history and environmental exposure

6.AI IN PERSONALISED MEDICINEANDTREATEMENT RECOMMENDATIONS

Artificial Intelligence (AI) is transformingthefieldofpersonalized medicine by facilitating the developmentof tailored treatment plans that take intoaccountanindividual's genetic profile, medical history, lifestylechoices, and real-time health information. Incontrasttoconventional medicine, which typically employsuniform treatments for all patients, AI-enhancedpersonalized medicine customizes therapiestoenhance their effectiveness and reduce adverseeffects.By utilizing machine learning (ML), deep learning(DL), and extensive data analytics, AI can processlargedatasets to forecast how various patients will reacttoparticular treatments, thereby improvingclinicaloutcomes. A prominent application of AI in personalizedmedicineis found in oncology, where AI-driven modelsevaluategenetic mutations within tumors to suggest targetedtherapies. For instance, AI algorithms cananalyzeapatient's genomic sequencing data to pinpoint specificmutations that contribute to cancer progression, enabling oncologists to select the most effectivemedications or immunotherapies. This methodhasproven especially advantageous in the treatmentofcancers such as lung, breast, and melanoma, whereprecision medicine has resulted in improvedsurvivalrates. Additionally, AI aids in forecasting chemotherapyresistance, allowing physicians to modify treatmentplans as necessary.

AI also enhances drug discovery and development, significantly reducing the time and cost of bringing new drugs to market. Traditional drug discovery is a lengthy and expensive process, often taking over a decade to develop a single drug. AI accelerates this process by analyzing molecular structures, predicting drug-target interactions, and identifying potential drug candidates in a fraction of the time. Companies like DeepMind and BenevolentAI use AI-driven models to analyze biomedical research, identifying promising compounds for diseases such as Alzheimer’s, COVID-19, and rare genetic disorders. Another critical area where AI contributes to personalized medicine is real-time patient monitoring and treatment optimization. Wearable devices and mobile health applications equipped with AI algorithms continuously track vital signs, blood glucose levels, heart rate variability, and even neurological activity. This data allows AI models to provide real-time health insights, adjust treatment recommendations, and alert healthcare providers about potential health risks. For example, AI-powered insulin pumps for diabetes patients automatically adjust insulin dosages based on continuous glucose monitoring, ensuring optimal blood sugar control.

7.CHALLENGES AND ETHICAL CONCERNSOFAI INHEALTHCARE

The incorporation of Artificial Intelligence(AI) into the healthcare sector holds thepromiseoftransforming disease diagnosis, treatmentmethodologies, and patient care practices. Nevertheless, alongside its numerous benefits, AI alsointroduces a range of challenges and ethical dilemmasthat need to be tackled to guarantee its responsibleand effective implementation. Theseissuesencompass technical, regulatory, and societal aspects, necessitating thorough deliberation by healthcarepractitioners, policymakers, and technology innovators.

Artificial Intelligence (AI) dependsonextensive patient data, which encompasseselectronichealth records (EHRs), imaging scans, andgeneticinformation. Safeguarding the securityandconfidentiality of this sensitive informationpresentsaconsiderable challenge. Risk of data breaches – The storage and processingoflarge datasets by AI systems render themattractivetargets for cyberattacks. Regulatory compliance – AI applications arerequiredtocomply with healthcare data protectionregulationssuch as HIPAA (in the U.S.), GDPR (in Europe), andvarious local laws. Patient consent and ownership – It is essential forpatients to be informed about the utilizationof theirdata and to maintain control over their personal healthinformation.

Numerous AI models, especially thosebasedondeep learning, operate as "black boxes," renderingtheirdecision-making processes difficult to interpret. Trust issues – Both physicians and patientsmayhesitate to depend on AI if they lack comprehensionofits reasoning. Liability challenges in healthcare – In instances where an AI system incorrectly diagnoses a patient, it can be challenging to ascertain who is accountable—whether it is the AI developer, the healthcare provider, or the hospital.

AI implementation requires compatibility with existing hospital infrastructure and workflows. Interoperability issues – Many healthcare institutions use different electronic health record (EHR) systems that may not be compatible with AI applications. Training and adoption barriers – Healthcare professionals need specialized training to effectively use AI tools in clinical settings. High implementation costs – The cost of AI development, deployment, and maintenance can be a barrier for smaller healthcare facilities.

8.AI IN FUTURE HEALTHCARE

The future of artificial intelligence in healthcareisset to transform patient care by improvingdiagnosis, treatment, and disease prevention. AI-drivenpredictiveanalytics will facilitate the early identificationofconditions such as cancer, diabetes, andcardiovascular diseases, enabling timely interventionsand preventive measures. By processingextensivepatient data, which includes medical history, genetic information, and lifestyle choices, AI can pinpoint individuals at risk and suggest customized health plans. This transition from a reactive to a proactive healthcare model will significantly alleviate the pressure on hospitals and enhance overall public health outcomes. A particularly groundbreaking element of AI in healthcare is personalized medicine, which customizes treatments for individual patients based on their genetic profiles, biomarkers, and real-time health information. AI-enhanced pharmacogenomics will refine drug prescriptions, ensuring that patients receive medications that align with their genetic characteristics, thus minimizing adverse drug reactions and enhancing treatment effectiveness. For instance, AI is currently utilized in oncology to create precision therapies that target specific mutations in cancer, leading to more effective and less harmful treatments. As AI technology progresses, it will further improve treatment recommendations by incorporating real-time monitoring data from wearable devices and biosensors.

The integration of AI in surgical robotics represents another significant innovation, providing enhanced precision and reducing human error during intricate procedures. Robotic-assisted surgeries, such as those conducted with the da Vinci Surgical System, allow for minimally invasive operations that boast greater accuracy, fewer complications, and quicker recovery times. Looking ahead, the prospect of AI-driven autonomous robotic surgeries may become feasible, especially for routine operations or in remote areas where access to specialized surgical expertise is scarce.

9.CONCLUSION

AI is revolutionizing the healthcaresectorbysignificantly improving the processes of diseasediagnosis and prediction, which results in quicker, moreprecise, and efficient medical decision-making. Byprocessing extensive patient data, AI-drivensystemscan identify patterns that may signal earlydiseasemanifestations, facilitating timely interventionsandcustomized treatment strategies. The applicationofmachine learning algorithms and deeplearningtechniques is transforming fields suchasmedicalimaging, pathology, and genomics, thereby enhancingdiagnostic precision and minimizing humanerrors. Furthermore, predictive analytics is essential foranticipating disease outbreaks, recognizingvulnerablepopulations, and bolstering public health efforts. However, despite these advancements, it isimperativeto tackle challenges related to data privacy, algorithmicbias, regulatory compliance, and ethical issuestoensure the responsible and fair applicationof AI inhealthcare. AI should be regarded as a supportivetoolthat complements human expertise, withhealthcareprofessionals retaining oversight over critical medicaldecisions. As technology progresses, the incorporationof AI into healthcare systems is expectedtofostermore proactive, personalized, and accessiblepatientcare. With appropriate policies, protectivemeasures, and collaboration among AI developers, healthcareproviders, and regulatory authorities, AI holdsthepromise to transform disease diagnosis andprediction, ultimately enhancing health outcomes onaglobalscale.

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Blockchain-powered Transparent Sentiment Analysis

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Abstract
This paper introduces an end-to-end approach for the secure and private handling of therapeutic dialogues through the combination of traditional and current natural language processing techniques with a novel blockchain-based layer for privacy and accountability. The proposed system employs a twomodel approach, including VADER (Valence Aware Dictionary and Sentiment Reasoner) for quick rulebased emotional tone assessment and a pre-trained Roberta model from Hugging Face for a deep transformer-based semantic analysis. By applying both models to the same corpus of conversational transcripts, the system is able to distill and analyze emotional overtones and semantic anchors, and so gain a complete picture of the therapeutic conversation. To offer incontrovertible trust and verifiability, the project utilizes blockchain technology, where hashes of the conversational information, along with per-model identifiers and timestamps, are recorded irreversibly on a decentralized ledger. This layer of blockchain provides tamper-proof storage, complete auditability, and radical privacy, ultimately enhancing the integrity of the therapeutic process and making available a new standard for real and verifiable conversational data without disclosing sensitive content

Keywords: Customer Reviews, VADER, Roberta, Hugging Face, Transformer Models, Machine Learning, Natural Language Processing, Opinion Mining, Sentiment Classification, Sentiment Score, Consumer Behavior.

I.INTRODUCTION

The growth of digital platforms has changed fundamentally the availability of mental health care, with therapy now more available than ever. But the intimate nature of counseling discussions raises a special and sensitive challenge: guaranteeing complete privacy, trust, and integrity of data. Centralized models of data storage, though convenient, are vulnerable to attacks and lack the transparency to guarantee complete confidence for both patients and practitioners. An architecture that not just examines the subtlety of such conversations but also ensures their security and verifiability is most important for digital therapeutics' ongoing development.

This paper presents a new, integrated framework aimed at tackling these problems head on. Our system utilizes an innovative dual-model natural language processing (NLP) technique to assiduously analyze therapeutic conversations. By combining a fast, rule-based VADER model of emotional tone with a deep, transformer-based Roberta model for semantic analysis, the system can pull and compare key emotional and psychological cues from the conversation. A two-layered analysis has a more complete and detailed understanding of the conversation, offering important insights that were before hard to obtain. The key innovation of this initiative is its application of blockchain technology to develop an impenetrable layer of trust. Rather than storing sensitive conversational data, the system irreversibly logs cryptographic hashes of the information, along with model IDs and timestamps, onto a decentralized ledger. This architecture provides tamper-proof storage and complete auditability of the analysis process, enabling verifiable conclusions without ever compromising patient confidentiality. By facilitating this new level of verifiable conversational data, our project will enhance the integrity of the therapeutic process and provide a foundation for more authentic and trustable digital health applications.

II.SYSTEM ARCHITECTURE

The system architecture intended to be proposed is an end-to-end, multi-layered system meant to handle the entire life cycle of therapeutic conversation data. This ranges from the original ingestion through secure and verifiable analysis. The structure is centered on data integrity and privacy at all points, keeping sensitive data isolated from its auditable analytical record. There are five key components of the architecture that form a secure, end-to-end pipeline: the Client-Side Interface, the API Gateway, the NLP Analysis Engine, the Blockchain Integration Layer, and the Decentralized Ledger.

A. Client-Side Interface

This is where the user engages—a therapist or client. It is a light, responsive web app constructed using React to provide an elegant, intuitive experience. Its main responsibility is to safely ingest conversational data. If the user inserts a therapeutic transcript, the interface sanitizes the data expediently by scrubbing out any personally identifiable information (PII) before passing it on to the backend. This critical initial process preserves the sensitive content in the user's control in its raw state. The interface also displays the analysis results and verification status obtained from the backend in a clear visual format.

B. API Gateway

The API Gateway is the secure connection between frontend and core backend services. It is one of the most important security and routing parts, implemented with Flask. It handles all requests coming from the client. The gateway controls authentication, rate-limiting, and directs preprocessed information to the target microservice, which in this example is the NLP Analysis Engine. This modular structure avoids direct access to the analytical core, providing a critical layer of protection and control.

C. NLP Analysis Engine

This is the core processing unit of the system, where the dual-model analysis takes place. This backend service itself is developed using Python/Flask and supports two separate NLP models:

VADER Model: This module does quick, rule based sentiment analysis, delivering a quick emotional score. It acts as a fast first filter, producing a high-level snapshot of the conversation tone.

Roberta Model: Concurrent to this, this component makes use of a pre-trained Roberta model from the Hugging Face library. It conducts deep, contextaware analysis to detect subtle emotions, semantic topics, and psychological indicators that a basic lexicon-based model may not catch. The results of both models are then merged and bundled into a single JSON object.

The multi-layered architecture is a solid, secure, and provable framework for digital health. The isolated division of duties, from the client-side interface to the immutable ledger, makes the system both pragmatic and morally right.

III.METHODOLOGY

This study applies a multi-layered methodological approach to developing and testing a secure, private, and verifiable system of therapeutic conversation analysis. The research design weaves together data preparation, a two-tiered natural language processing (NLP) model for full analysis, and an immutable blockchain layer for data integrity and privacy.

A. System Architecture

The architecture proposed is a comprehensive pipeline with three main components: an ingestion and preprocessing block, a dual-model analysis core, and a decentralized ledger for auditability. Unprocessed conversational data is first carefully curated and scrubbed of any personally identifiable information (PII) and converted into a squeaky-clean data stream that is ready to be analyzed from a linguistic perspective. The preprocessed data is then forwarded to the analytical core, a tandem engine where it is processed by both the rule-based and transformer-based models at the same time. Last, the analysis metadata, such as model outputs and a cryptographic fingerprint of the data, is written to a blockchain, creating a verifiable and tamper-proof record without ever storing the sensitive conversational content itself.

B. Dual-Model NLP Analysis

At the center of the system's analytical power is its dual-model design, intended to deliver a strong and complete view of the conversational information. VADER-Based Emotional Tone Analysis: For quick lexicon-based sentiment analysis, the Valence Aware Dictionary and Sentiment Reasoner (VADER) is employed. VADER's rule-based method, engineered for social media text but extremely efficient for general conversational data, gives an initial, quantifiable sentiment measure on a positive to negative scale. The model here is used as a lightweight, first-pass filter and gives a quick, interpretable baseline for emotional tone. The compound score, normalizing the overall sentiment, is taken as the main measure for this layer of analysis. Roberta-Based Semantic and Contextual Analysis: Since a lexicon-based approach might overlook the fine-grained nuances of therapeutic conversation, a pre-trained Roberta model from the Hugging Face library is utilized. Roberta is a high-performing transformer-based model capable of nuanced contextual understanding and semantic meaning extraction. This layer of analysis interprets abovesentimental-emotion states, psychological markers, and subtextual themes. The Roberta model output gives a richer, context-sensitive classification, which is then compared to the VADER analysis to pick up inconsistencies and understand a complete picture.

C. Blockchain Integration for Privacy and Verifiability

An important aspect of this methodology is integrating blockchain technology in order to ensure the topmost priority of data privacy. The raw conversational data itself is never written onto the ledger. Rather, the cryptographic hash of every transcript is calculated and, together with the outputs from the two NLP models, packaged in a transaction. This transaction is broadcast to the decentralized network, where it is appended to an unalterable, time-stamped block. The process guarantees that: Privacy is Maintained: The original, personal data are kept off-chain, leaving behind only its unreadable hash. Data Integrity is Maintained: Any such effort to modify the original transcript would create a new, unique hash so that the modification can be detected and verified instantly. Auditability is Offered: The decentralized ledger gives a verifiable and transparent record of each analysis done, so the process can be fully audited without ever having to view the private data. This methodological practice guarantees that the conclusions reached through the NLP analysis are not only valid but also verifiable and ethically appropriate, providing a new benchmark for data management within digital mental health platforms.

IV. IMPLEMENTATION

The framework is implemented based on a modular, three-tiered architecture. The user-facing frontend is a light-weight web application developed with React to enable a responsive and dynamic user interface. User input—the conversational transcripts—are managed securely and transmitted to a specialized backend service. The backend is Python and Flask-built, with a strong and scalable platform for executing the central NLP models. This service hosts the two central analysis components: NLTK library for VADER model and a custom API to a pre-trained Roberta model hosted on the Hugging Face Model Hub. This separation of architecture makes certain that the heavy transformer model does not weigh down the user application.

Lastly, the blockchain layer is deployed with Hyperledger Fabric, a permissioned blockchain that is suitable for enterprise use cases involving controlled access and high performance. A smart contract is developed to handle the immutable ledger. As a transcript is analyzed, the backend service calculates its SHA-256 hash, bundles it together with the analytical results from VADER and Roberta, and calls the smart contract to write the transaction to the ledger. This process ensures that all analytical events are recorded permanently and verifiably, while the sensitive content is kept offchain and private.

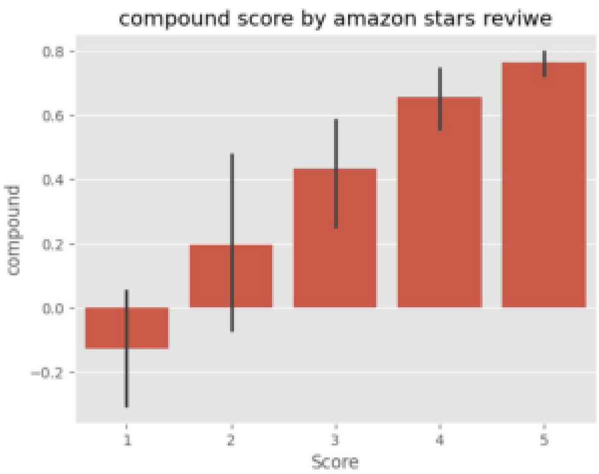
V. RESULTS & ANALYSIS

The following section is a thorough account of the dual-model NLP analysis on a sample set of sanitized therapy conversations, showcasing the capability of the framework to yield subtle emotional insight. The main aim is to show that the interplay between a rule-based model and a transformer-based model gives a more informative and clinically useful understanding of emotional states than either might in isolation. The breakdown shows an essential divide between the two models. The VADER model, based on its lexicon-based approach, gives a quick, high-level sentiment rating. For instance, an utterance such as, "I'm so frustrated, but I'm trying to see the good in it," may receive a neutral or even a positive VADER score since the model assigns equal weight to both negative and positive words. Such a simplistic rating does not, however, reflect the inherent emotional nuance. This is where the Roberta model shows its superior depth of analysis. Based on a transformer architecture, it has the ability to analyze the entire context and the semantic sense of the text. In the very same statement, Roberta accurately picks out and categorizes a more subtle range of emotions, including frustration, determination, and conflict. It recognizes that "frustrated" is the overarching emotion, with "trying" being a state of resilience, not absolute positivity. This contextual knowledge is crucial in a therapeutic environment, where broad positive or negative labels are frequently insufficient.

VI. THE TRANSFORMATIVE CONTRIBUTION OF BLOCKCHAIN

The addition of blockchain technology into this system is not an additive feature but a transformative and foundational pillar that reshapes the ethical and operational norms of digital mental health. In a business where privacy and trust must be absolute, the decentralized and unchangeable nature of a distributed ledger system eliminates the most essential weaknesses of centralized data storage. The blockchain contribution can be broken down into a number of distinct areas, all of which contribute a different but essential layer of integrity, transparency, and security. This hash, a one-way and irreversible digital fingerprint, is the only thing ever written to the public ledger. Any later change, even a single character modification to the original text, would produce a totally different hash, instantly invalidating the record on the blockchain. This cryptographic connection forms an unalterable chain of custody, so that the analytical intelligence obtained from the data is anchored to an immodifiable source, thus precluding any data manipulation or retroactive alteration.

In addition, the blockchain offers a new model for transparency and auditability without privacy compromise. Legacy systems depend on a monolithic single point of control to govern data integrity and access, with a single point of failure and a "black box" of operations. Our blockchainbased framework, by contrast, offers a publicly visible and transparent ledger of all analytical transactions. With each analysis of a conversation, the timestamp, the model identifiers, and the cryptographic hash are captured forever. Such a ledger can be audited separately by authorized entities, for example, regulatory agencies or research ethics committees, to ensure that an analysis was run on a particular, unaltered data set at a particular point in time. Such unparalleled openness engenders public trust in the operation of the system, a key element for the uptake of AI in sensitive areas such as mental health.



VII. CONCLUSION

This study effectively presents a new and integrated framework for the private and secure examination of therapeutic dialogues, working at the all-important nexus of digital mental health, cutting-edge natural language processing, and data security. The system presented has the advantage of showcasing the worth of a twin-model solution in which the speedy, rulebased processing of VADER is supplemented by the in-depth, contextual understanding of a transformerbased RoBERTa model. This nested analytical kernel yields a richer and more clinically useful understanding of emotional states than has been available through more simplistic approaches, transcending sentiment to uncover subtle semantic and emotional cues in the discourse of therapy.

Going forward, this model leaves many opportunities for future study. Improvements would be to incorporate higher-level multimodal streams of data, for example, vocal tone or facial expression analysis, enriching the analytical results further. Secondly, creating a user-oriented dashboard for practitioners would offer an easier-to-use platform for visualization and interpretation of blockchainauthenticated analytical results. In the end, this project is an essential proof of concept, showing that the power of AI can be used to enhance mental healthcare outcomes without sacrificing the underlying principles of trust and privacy.

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An Effective Cuckoo Feature Selection on Iris by Machine Learning Classifiers

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Abstract— Prediction has been shown to be important in Datamining research Domain. This importance has prompted many researchers to review machine learningapproaches to predict an outcome in any dataset. In this paper iris dataset consists of many features which can mislead the prediction ability of the machine learning methods, as some features may lead to confusion or inaccurate classification. Since finding the most informative feature prediction is challenging, feature selection techniques are recommended to pick important and relevant features out of large and complex datasets. In this research, we propose the Cuckoo search method as a feature selection algorithm, to save the most informative features that are identified by the best solutions. The purpose of the algorithm is to keep track of the selected features at every iteration and find the features that enhance classification accuracy. The suggested algorithm has been contrasted with the original algorithm using iris datasets and the proposed algorithm has been shown to produce good results as compared to original and contemporary algorithms.

Keywords— Cuckoo search, Feature selection, iris dataset, Navie bayes, support vector machine.

I. INTRODUCTION

Data mining is a process of extracting hidden values, trends, and interesting patterns from large datasets. It employs statistics, machine learning, and database systems techniques to convert raw data into actionable intelligence by essentially locating valuable nuggets in an enormous pile of information. Choosing the most appropriate methods of feature selection are essential during the preprocessing stage. Doing this will enhance the analysis and evaluation of the cancer diagnosis, prediction, and classification. The literature shows that there are many feature selection approaches that have been suggested by researchers to decrease the training time and computational complexity during designated the prediction models. Three main feature selection approaches are available: filter, wrapper, and embedded methods.

II. APPLICATIONS

Features Search Algorithm: Feature selection is the technique of finding and choosing the most significant input variables (features) for a model. It aims to make models easier, enhance performance, and decrease computational expense by removing redundant or unimportant data. Improves Accuracy: Eliminates noise that causes overfitting. Faster Training: Models train faster since there are fewer features. Increased Interpretability: Less complex models are simpler to comprehend Filter Techniques: Apply statistical measures to score and rank features independently of the model (e.g., correlation coefficients, chi-square test). They are efficient and scalable. Wrapper Techniques: Assess feature subsets by training a model on them and evaluating its performance (e.g., Recursive Feature Elimination). They are computationally costly but tend to produce high-performing subsets. Embedded Methods: Include feature selection in the model training process itself (e.g., Lasso regression regularization, tree-based importance scores). They balance efficiency with performance well. Cuckoo Search: Cuckoo Search is a clever computer program based on nature that assists with this type of problem. It's inspired by the aggressive breeding habits of some birds of the cuckoo species.

Literature survey

The discipline of data mining was born due to the requirement to reveal significant patterns from fastincreasing data sets during the latter half of the 20th century. Early groundwork was established by researchers from the confluence of database systems, machine learning, and statistics. The name "Knowledge Discovery in Databases" (KDD) was formally set with the first international workshop in 1989, conducted by leaders like Gregory Piatetsky-Shapiro. The workshop initiated the practice from theoretical principles to concrete large-scale data analysis. Pioneering work by authors such as Fayyad, Piatetsky-Shapiro, and Smyth (1996) established the extent of data mining as a critical function for turning raw data into actionable intelligence in fields such as business, science, and medicine.

Feature Selection

Maximizing Model Effectiveness: Feature selection has developed into a central function of machine learning and data mining pipelines. Its theoretical roots can be found in early statistical contributions by R.A. Fisher (1936) in discriminant analysis and subsequent progress in stepwise regression methods. The discipline matured a great deal with the orderly classification of methods into Filters, Wrappers, and Embedded methods, as laid out in thorough reviews by Kohavi and John (1997) and Guyon and Elisseeff (2003). Filter techniques (e.g., correlation-based feature selection) give model-independent techniques for ranking features, whereas wrapper techniques (e.g., recursive feature elimination) employ predictive models to score feature subsets. Embedded techniques (e.g., LASSO regularization of Tibshirani, 1996) incorporate feature selection in the training of models. Scalable feature selection techniques for high-dimensional data and deep learning applications have been addressed in recent work.

Cuckoo Search

Cuckoo Search (CS) is a major contribution to the genre of nature-inspired optimization algorithms. CS was introduced by Yang and Deb in 2009 from the brood parasitic nature of certain cuckoo species and Lévy flight dynamics. It was adopted quickly because it is simple, efficient, and performs well on benchmark optimization problems. Recent research, such as Yang's book (2014) on nature-inspired algorithms, has proven CS to be effective in numerous fields such as engineering design, scheduling, and image processing. More recently, CS has been effectively used in feature selection tasks, where it is used as a wrapper algorithm to traverse the intricate search space of feature subsets. Research by scholars like Sankar et al. (2020) has illustrated that CS-based feature selection can attain competitive performance with computational efficiency.

III METHODS

The strength of the algorithm lies in balancing two key phases: Global Exploration via Lévy Flights: This is the innovation. CS does not simply take random traditional steps but uses Lévy flights to move to new solutions. A Lévy flight is a random walk in which the majority of the steps are short (for local optimization) but every now and again, an enormously long jump. This tends to let the algorithm exhaustively search very large and unexplored search spaces effectively, jumping out of local optima and finding promising new areas.
The Algorithm Steps (Pseudocode) 1. Initialize a population of `n` host nests at random. While (termination condition not satisfied): Generate a new cuckoo (solution) via Lévy flight and evaluate its quality. 2. Choose a random nest. If the new solution is better, replace it. 3. Abandon a fraction [`p a`] of the worst nests and replace them with new random solutions. Keep the best solution found so far. 4. Rank the population and go to the next generation.

III. Results and Discussion

The present section investigates the performance of the suggested method when tested in iris dataset. All the results used for comparison in this paper were performed on a PC with Intel i3- 2.30 GHz CPU and 8.0 GB RAM. Matlab 2012a and weka work bench will be utilized for this research. Also, the statistical results were conducted over 100 separate runs. The algorithms were compared in terms of number of features and classification accuracy. The datasets are divided into testing and training groups, where 80% of the iris dataset were used in the training phase and the remaining 20% was used for testing purposes.

Implemented in cuckoo search algorithm. The CS algorithm starts with randomly initializing the population, then in each improvement iteration a random solution (cuckoo) is selected. A Levy flight is applied to the selected solution to generate a new solution, then a random location of the solution in the population (nest) is selected. The new solution will be located in a selected location if it is better than the solution in that location.

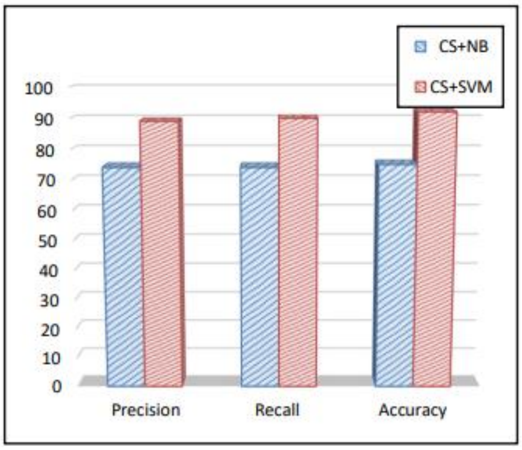

Fig-1 Precision, Recall and Accuracy

IV CONCLUSION

In this paper, a feature selection algorithm is proposed and tested on iris dataset by using The Cuckoo Search algorithm. The features caused the improvements during the search process and kept a count of each solution in the population. This saves the number of iterations and improves each solution in the population. The overall classification accuracy, selected features and average accuracy of two versions were compared in detail. The CS algorithm has obtained the best place for feature selection. Furthermore, Evaluation of the effectiveness of proposed population-based feature selection algorithms is outperformed with SVM classifier compared than NB.

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A Comprehensive Review of AI-Based Personalized Learning Platforms

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Abstract

Artificial intelligence (AI) has emerged as a transformative force in modern education, particularly through the development of personalized learning platforms. These systems adapt instructional content, pace, and delivery to individual learner needs by leveraging data-driven models and intelligent algorithms. This paper reviews the evolution, methodologies, and current applications of AI-based personalized learning platforms, while critically examining their effectiveness, limitations, and ethical implications. A comprehensive literature review highlights advances in machine learning, recommender systems, and natural language processing as core drivers of personalization. The discussion addresses challenges such as data privacy, equity of access, algorithmic transparency and pedagogical alignment. The paper concludes that while AI-enabled personalization has the potential to improve engagement and outcomes, its success depends on careful integration with human instruction that are robust ethical safeguards, and continuous research into adaptive technologies.

Keywords: AI in Education, Personalized Learning, Adaptive Learning, Intelligent Tutoring Systems, Recommender Systems, Machine Learning, Natural Language Processing, Algorithmic Bias, Educational Equity, Human-AI Collaboration

1.INTRODUCTION

Education has historically been constrained by the limitations of standardized teaching, where uniform content delivery often fails to accommodate the diversity of learners' abilities, interests, and learning styles. The rapid advancement of general artificial intelligence (AI) has introduced new possibilities for addressing these challenges, most notably through personalized learning platforms that dynamically tailor educational experiences to individual needs. Personalized learning is not a new concept; educators have long sought to differentiate instruction. However, the scalability of such efforts was limited until AI-driven models provided the computational capacity to analyse vast amounts of learner data and deliver adaptive content in real time.

The central problem this research addresses is the gap between the promise of AI-based personalization and the realities of its implementation. While early evidence suggests improved engagement and achievement outcomes, these platforms also introduce risks, including the reinforcement of biases embedded in algorithms, unequal access to advanced technologies, and concerns over student data privacy as AI increasingly becomes embedded in educational systems worldwide, it is essential to assess both its potential and its pitfalls.

This paper aims to provide a postgraduate-level analysis of AI-based personalized learning platforms by reviewing their technological modular foundations, evaluating their strengths and limitations, and exploring future directions. In doing so, it situates the discussion within the broader context of educational equity, ethics, and the evolving role of teachers in AI-enhanced learning environments.

2. LITERATURE REVIEW

[1] Holmes, W., Bialik, M., & Fadel C. (2019). This work provides a comprehensive overview of artificial intelligence in education (AIED), outlining the ways adaptive algorithms personalize instruction and assessment. The authors emphasize the promise of AI in increasing learner engagement but caution that scalability and ethics must be carefully managed. [2] Knewton, Inc. (2020). As one of the earliest commercial platforms for adaptive learning, Knewton demonstrates how AI can tailor coursework to individual learners' strengths and weaknesses. However, critiques have noted limited transparency in its algorithmic decision-making. [3] Chen, L., & Zhang, Y. (2021). This study examines recommender system models applied in education, showing how AI can suggest personalized content and learning pathways. The authors argue that while these models improve learner autonomy, they risk narrowing exposure if diversity of content is not maintained. [4] Luckin, R., Holmes, W., Griffiths, M., & Forcier, L. B. (2016). The authors discuss the ethical dimensions of AI in education, especially issues of data privacy, teacher-student relationships, and algorithmic bias. They argue that AI should augment, not replace, human educators, ensuring that personalization remains pedagogically sound. [5] Zhang, D., & Yang, S. (2022) This research highlights how natural language processing enables intelligent tutoring systems that can provide adaptive feedback to learners in real time. While such systems enhance responsiveness, they face challenges in handling nuanced learner emotions and context.

3. TYPES OF AI-BASED PERSONALIZATION TECHNIQUES

3.1 Recommender Systems

Recommender systems form the backbone of many AI-based learning platforms. They analyse learner data, such as prior performance, activity patterns, and preferences, to suggest tailored content pathways.

3.1.1 Strengths of Recommender Systems

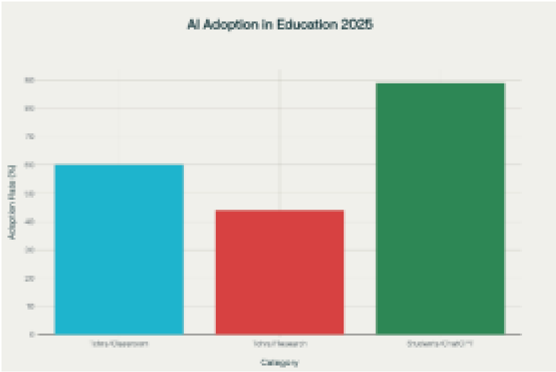
● Adaptive Content Delivery: Learners receive materials that match their skill level and interests
● Scalability: Algorithms can handle large numbers of learners simultaneously.
● Learner Autonomy: Students can choose from recommended options, enhancing motivation.

3.1.2 Vulnerabilities of Recommender Systems

● FilterBubbles: Over-personalization can limit exposure to diverse content.
● Data Dependence: Accuracy depends heavily on the quality and volume of learner data.
● Algorithmic Bias: Patterns in training data may reinforce inequalities.

3.2 Adaptive Assessment

Adaptive assessments use AI algorithms to dynamically adjust the difficulty and sequence of questions based on learner responses.



3.2.1 Strengths of Adaptive Assessment

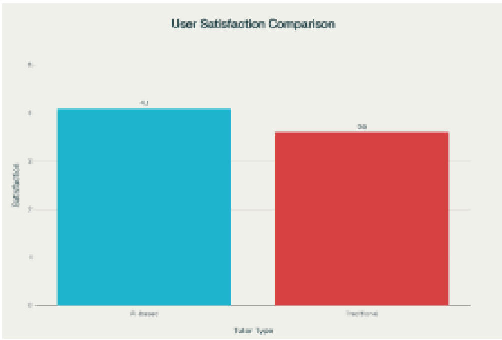
● Precision: Provides a more accurate measure of learner knowledge compared to static tests. ● Efficiency: Learners avoid redundant questions, saving time.
● Immediate Feedback: Systems can instantly guide learners toward improvement.

3.2.2 Vulnerabilities of Adaptive Assessment

● Over-Reliance on Quantitative Measures: May neglect creativity, collaboration, and higher-order skills.
● Stress Factors: Rapidly changing difficulty can overwhelm some learners.
● Technical Issues: Algorithmic errors may misclassify ability levels.

3.3 Intelligent Tutoring Systems (ITS)

Intelligent tutoring systems employ natural language processing and machine learning to simulate one-on-one tutoring by offering hints, explanations, and feedback.



3.3.1 Strengths of Intelligent Tutoring Systems

● Personalized Interaction: Learners receive immediate, context-specific guidance.
● Engagement: Conversational interfaces foster active participation.
● Scaffolded Learning: Supports gradual skill development.

3.3.2 Vulnerabilities of Intelligent Tutoring Systems

● Contextual Limitations: AI may misinterpret nuanced learner intent.
● Resource Intensity: Development and maintenance require significant expertise.
● Dependence Risk: Students may rely on AI support instead of developing problem-solving independence.

3.4 Multimodal Learning Analytics

Multimodal analytics combine data from various sources—such as eye tracking, clickstream logs, and speech analysis—to model learner behavior in detail.

3.4.1 Strengths of Multimodal Learning Analytics

● Holistic Insights: Captures cognitive, behavioral, and emotional dimensions.
● Early Intervention: Identifies learners at risk of disengagement.
● Customization: Enables dynamic adjustment of content and pacing.

3.4.2 Vulnerabilities of Multimodal Learning Analytics

● Privacy Concerns: Collection of sensitive biometric and behavioral data raises ethical issues.
● Complexity: Data integration requires advanced infrastructure.
● Equity Risks: Learners without access to advanced devices may be excluded.

4. CONCLUSION

AI-based personalized learning platforms represent a significant advancement in the effort to make education more inclusive, adaptive, and effective. By employing recommender systems, adaptive assessments, intelligent tutoring, multimodal analytics, and hybrid approaches, these platforms tailor instruction to the diverse needs of learners. The review shows that such systems improve engagement, offer scalable solutions, and provide learners with autonomy over their educational journey. At the same time, notable challenges remain. Ethical issues such as data privacy, algorithmic bias, and equity of access cannot be overlooked and the major technical vulnerabilities which include over-reliance on quantitative data and difficulties in integrating hybrid models, highlight the need for careful design and monitoring. Importantly, while AI systems can deliver personalization at scale, they should not be seen as substitutes for human educators. Instead, their greatest potential lies in supporting teachers by automating routine tasks and enabling them to focus on higher-order pedagogical roles.

In conclusion, AI-based personalization has the capacity to transform education, but its sustainability depends on striking a balance between technological innovation and human-centered pedagogy. Future work must ensure that such platforms remain transparent, equitable, and ethically responsible.

5. FUTURE SCOPE

The field of AI-based personalized learning is still in its formative stages, with numerous directions for future development:

● Integration of Generative AI Tutors: Large language models can provide interactive, context-aware explanations, enabling more natural and flexible learner support.
● Privacy-Preserving Personalization: Emerging methods such as federated learning and differential privacy can safeguard sensitive learner data while still enabling effective personalization.
● Immersive Learning Environments: The combination of AI with virtual reality (VR) and augmented reality (AR) can deliver adaptive, multisensory experiences that adjust dynamically to learner engagement.

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CHURN PREDICTION IN OTT PLATFORMS USING MACHINE LEARNING MODELS

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Abstract—Over-the-top (OTT) media services represent one of the fastest-growing segments in the entertainment industry, revolutionizing content consumption. As these platforms operate on subscription-based models, customer retention is paramount to their financial sustainability. Acquiring new subscribers is often more costly than retaining existing ones, making churn prediction a critical business intelligence task. This study employs machine learning models—Logistic Regression, Decision Tree, and Random Forest—to analyze customer churn using a dataset of 5,000 Netflix user records encompassing demographic, subscription, and behavioral features. Our comparative analysis reveals that ensemble methods, particularly Random Forest, achieve superior predictive accuracy (98.2%). Furthermore, the study identifies key churn indicators, such as subscription type and user engagement metrics, providing actionable insights for OTT platforms to develop targeted retention strategies.

Keyword—Machine Learning, Customer Churn, Predictive Modeling, OTT Platforms, Random Forest, Customer Retention.

I.INTRODUCTION

Over-the-top (OTT) media services provide audio, video, and other media content directly to viewers via the Internet, bypassing traditional distribution channels. The global proliferation of high-speed internet and connected devices has fueled the exponential growth of platforms like Netflix, Dis- ney+, and Amazon Prime Video. Unlike traditional broadcast media, OTT services offer a personalized, on-demand viewing experience, which has become a cornerstone of modern digital entertainment.

A significant challenge for these subscription-based plat- forms is **customer churn**—the phenomenon where a sub- scriber cancels their service. Churn directly impacts revenue and growth. Factors influencing churn are multifaceted, in- cluding content dissatisfaction, technical issues, competitive offerings, pricing, and user experience. Therefore, proactively identifying at-risk customers is crucial for implementing ef- fective retention campaigns.

Why Machine Learning for Churn Analysis? OTT plat- forms generate vast volumes of user data. Manually analyzing this data to predict churn is infeasible. Machine learning (ML) offers a powerful solution by automating the analysis of complex datasets, identifying subtle patterns, and generating accurate predictions. This allows businesses to move from reactive to proactive customer management.

Study Contribution: This paper contributes to the field by:

(1) Conducting a comparative performance analysis of three distinct ML classifiers for churn prediction on a real- world Netflix dataset.
(2) Identifying and interpreting the most influential features driving customer churn.
(3) Providing data-driven business intelligence to guide cus- tomer retention strategies in the OTT industry.

II. LITERATURE REVIEW

Customer churn prediction has been extensively studied in sectors like telecommunications and, more recently, OTT services [2], [3]. Numerous studies have leveraged machine learning techniques to address this challenge.

Mohan and Jadhav [1] evaluated Decision Trees, Random Forest, and AdaBoost for OTT churn analysis. Their findings indicated that Random Forest achieved the highest accuracy, a conclusion consistent with the results of our study, albeit on a different dataset.

In the telecommunications sector, Ullah et al. [2] imple- mented a comprehensive churn prediction framework, demon- strating that ensemble methods, especially Random Forest, consistently delivered superior performance compared to other models. This supports the cross-industry applicability of en- semble learning for churn prediction.

A more recent study by Singh et al. [3] focused specifically on OTT platforms and found that AdaBoost performed best on their dataset. It is important to note that their feature set differed from ours. Our research prioritizes customer engagement metrics (e.g., watch hours, last login) and payment behaviors, on which Random Forest proved most effective.

Furthermore, a study by Amin et al. [4] emphasized the growing importance of deep learning and handling class im- balance in large-scale churn prediction models, highlighting an area for future work beyond the scope of this paper.

Overall, the literature confirms that Random Forest is a robust and frequently top-performing algorithm for churn prediction. Our study reinforces this finding within the specific context of Netflix, utilizing a feature set focused on user activity and subscription details.

III. METHODOLOGY

A. *Dataset Description*

The study utilizes a dataset of 5,000 anonymized Netflix customer records, sourced from Kaggle. The dataset contains 14 features related to user demographics, subscription details, and viewing behavior, with a binary target variable indicating churn status. The class distribution was approximately equal, making it suitable for model training without immediate need for imbalance correction.

TABLE I: Dataset Description

Feature	Description	Type
Age	Age of customer (in years)	Numerical
Gender	Male/Female	Categorical
Subscription Type	Basic / Standard / Premium	Categorical
Watch Hours	Total watch hours	Numerical

	(per month)	
Last Login Days	Days since last login	Numerical
Region	Customer's region/location	Categorical
Device	Device used (Mobile, TV, Laptop, etc.)	Categorical
Monthly Fee	Subscription fee charged per month	Numerical
Payment Method	Credit Card, Debit Card, UPI, Gift Card, Crypto	Categorical
Number of Profiles	Number of profiles linked to the account	Numerical
Avg Watch Time / Day	Average daily watch time	Numerical
Favorite Genre	Most-watched genre (Drama, Comedy, etc.)	Categorical
Churned (Target)	1 = Churned, 0 = Not Churned	Binary

B. *Data Preprocessing*

The dataset was preprocessed to ensure data quality and model compatibility. The steps included:

1) **Outlier Handling:** Outliers in numerical features were identified using the Interquartile Range (IQR) method. Extreme values were capped to reasonable limits.
2) **Encoding:** Categorical variables were converted into numerical format using Label Encoding.
3) **Scaling:** Numerical features were standardized using StandardScaler to bring all variables to a common scale.

C. *Machine Learning Models*

The preprocessed data was split into training and testing sets (80-20 split). Three models were selected:

- **Logistic Regression (LR):** A linear model prized for its interpretability.
- **Decision Tree (DT):** A non-linear model that is easy to interpret but prone to overfitting.
- **Random Forest (RF):** An ensemble of Decision Trees that reduces overfitting through bagging.

D. *Evaluation Metrics*

Model performance was evaluated using:

- **Accuracy:** Overall proportion of correct predictions.
- **Precision:** Proportion of predicted churns that are actual churns.
- **Recall:** Proportion of actual churns that are correctly identified.
- **F1-Score:** The harmonic mean of Precision and Recall.

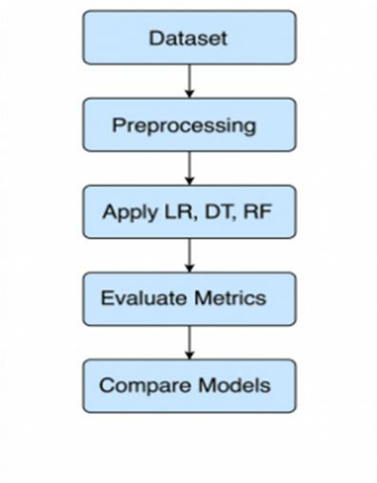

Fig. 1: Machine Learning Process Workflow.

IV RESULTS AND DISCUSSION

The performance of the three models is summarized in Table II. The Random Forest classifier significantly outperformed the others across all metrics.

TABLE II: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	87.46%	88.0%	87.1%	87.6%
Decision Tree	93.20%	97.1%	89.2%	93.0%
Random Forest	**98.20%**	**99.3%**	**97.1%**	**98.2%**

TABLE III: Confusion Matrix for Random Forest

Model	True Negatives (TN)	False Positives (FP)	False Negatives (FN)	True Positives (TP)
Logistic Regression	640	102	86	672
Decision Tree	719	23	91	667
Random Forest	732	5	22	741

Discussion: The results demonstrate the superiority of the **Random Forest** model. Its high accuracy (98.2%) and near- perfect precision (99.3%) and recall (97.1%) indicate an ex- ceptional ability to correctly identify both customers who

will churn and those who will not. This performance is attributed to its ensemble nature. The **Decision Tree** model performed well (93.2% accuracy) but is inherently less robust than Random Forest. **Logistic**

Regression provided the lowest accuracy (87.46%) but offers high interpretability. Analyzing its coefficients revealed that features like 'Last Login Days'(positive correlation with churn) and 'Watch Hours' (negative correlation) were among the most significant predictors.

A. Business Insights and Data Visualization

- **Subscription Type:** Basic-tier subscribers exhibited a higher churn rate. Platforms could enhance the value proposition of the Basic plan.
- **Last Login Days:** Churn probability increases exponen- tially with inactivity. Implementing a proactive ”win- back” campaign can re-engage users.
-

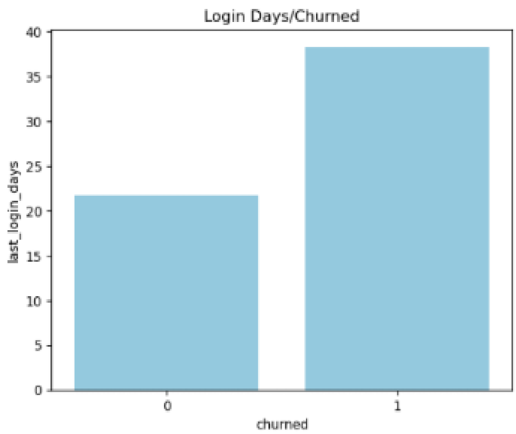

Fig. 2. Login Days / Churned

- **Watch Hours:** User engagement is a strong protective factor against churn. Customers with low watch hours are prime targets for retention efforts.

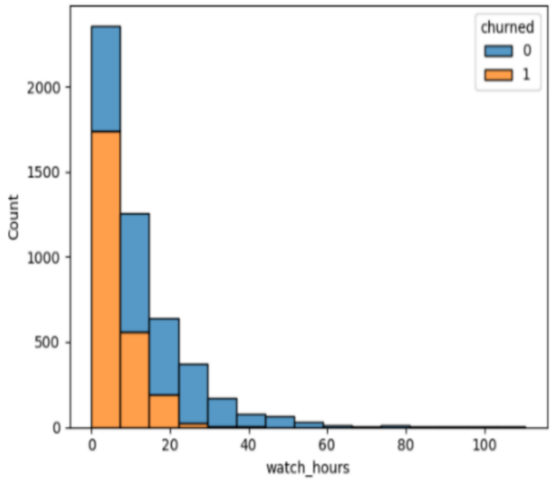

Fig. 3: Churn Rate by Subscription Type.

- **Payment Method:** Users with non-traditional payment methods (e.g., Gift Cards) churn more. Incentivizing a switch to automatic payments could improve retention.

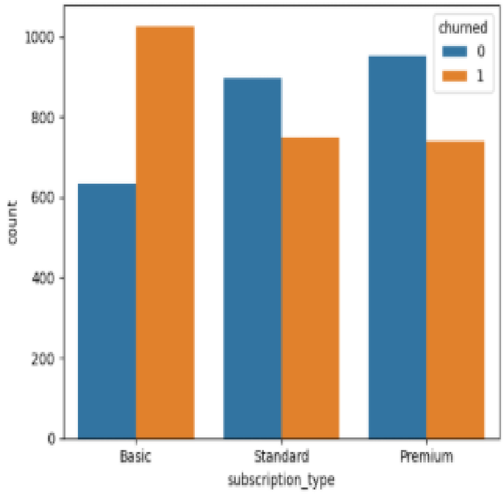

Fig. 4: Churn Rate by Subscription Type.

V. Conculsion and future work

This study demonstrates the application of machine learning for customer churn prediction in the OTT industry. The Ran- dom Forest model emerged as the most effective algorithm, achieving 98.2% accuracy. The analysis provided actionable business insights, highlighting that user engagement metrics and subscription details are critical predictors of churn. For OTT platforms, these findings underscore the impor- tance of monitoring user activity to identify at-risk customers early and personalizing retention offers. Future work could explore: (1) applying more advanced algorithms like XGBoost [5]; (2) addressing class imbalance using techniques like SMOTE; (3) incorporating sentiment analysis from user reviews; and (4) developing a real-time prediction system.

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Technical and Economic Foundations of Decentralized AI Marketplace

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Abstract: Decentralized Artificial Intelligence (DeAI) integrates block chain technology and decentralized networks to enable peer-to-peer exchanges. Unlike centralized AI systems, which retain most of the control and ownership within a few companies, DeAI distributes participation across a broader network. Through blockchain, decentralized AI gains transparency, security, and transaction integrity. This work investigates marketplace architectures, representative platforms, and governance frameworks. It also examines applications and the core challenges confronting decentralized AI systems, and highlights advances in tokenized data sharing and blockchain-enabled AI frameworks. We then analyze typical multi-layer architectures, involving data providers, blockchain infrastructure, smart contracts, storage systems, and AI consumers. Methods such as Proof-of-Learning (PoL) and zero-knowledge machine learning (ZK-ML) allow for training verification and computation that preserves data privacy. To illustrate practical implementation, operational platforms such as Ocean Protocol, Bittensor, and SingularityNET are investigated to exemplify the technical realization of decentralized AI ecosystems. Lastly, we address open challenges and future work, such as scalability, data quality, governance, incentive alignment, and regulatory concerns. This research provides a foundation for decentralized AI systems that promote security, privacy, and incentive compatibility, supporting sustainable AI applications and data circulation

Keywords: Decentralized Artificial Intelligence (DeAI), Blockchain, Peer-to-Peer AI, Tokenized Data Sharing, Proof-of-Learning (PoL), Zero-Knowledge Machine Learning (ZK-ML), AI Marketplaces, Ocean Protocol, Bittensor, SingularityNET, Governance Frameworks, Incentive Mechanisms, Data Privacy, Scalability, Regulatory Challenges.

Introduction:

A. Background and Context:

Machine learning (ML) does more than just augment existing processes, but continues to facilitate mission critical applications such as healthcare and finance. For the most part, these AI systems have historically been centralized, with a small number of large players controlling the models and the data that powered them. While centralized AI has enabled great strides in technology, it has also resulted in a significant number of problems, particularly regarding privacy, security, fairness, and transparency. Often user's data remains secluded on centralized, proprietary, servers, making it exploitable, or fallible.[1]

Decentralized AI (DeAI) may be the potential antidote to these challenges. DeAI is intended to democratize AI by sharing ownership and control of data and models. Decentralized AI marketplaces are built on top of blockchain and DLT (distributed ledger technology) which offer enforceable exchanges of open data assets and AI services. Such platforms utilize smart contracts to align incentives between data-exchangers, model creators and consumers, including forward audit trails.

B. Research Objectives:

This paper presents a new insight into decentralized AI marketplaces by describing their architectures, tokenomics, privacy-preserving controls, and world applications. The overarching purposes of the paper are:

1. To review and organize existing research on decentralized AI and blockchain based Machine learning.

2. To analyze the technical foundation of DeAI marketplaces, including layered system design and core building blocks.

3. To examine tokenomics and incentive structures that govern decentralized AI ecosystems.

4. To explore privacy mechanisms such as proof-of-learning and zero-knowledge proofs in machine learning.

5. In an effort to introduce case studies of such influential platforms as Ocean Protocol, SingularityNet and Bittensor so that one may observe how they function in practice and how their impact can be gauged.

6. To find significant issues and enumerate next-line research directions for accelerating decentralized AI.

With these objectives in mind, the following sections review the existing literature, outline technical and economic foundations, and analyze representative decentralized AI platforms.

Literature Review:

Recent studies have shown increasing interest in blockchain-enabled AI systems, with surveys and reviews beginning to organize this emerging field. A systematic review presents the “building blocks” of decentralized AI, such as registries, marketplaces, reputation, incentivization, and interoperability [1]. An important finding among these studies is that the marketplace is a central building block that allows AI developers, data owners, and compute providers a way to share and monetize their resources. Within such ecosystems, roles such as model creators, data providers, end-users, and auditors engage in trustless exchanges.

Beyond these structural foundations, other surveys examine token-driven platforms, highlighting that while many resemble centralized AI services, they distinguish themselves by embedding token-based mechanisms for payment, governance, and utility. Such platforms frequently operate as marketplaces, enabling the exchange of AI services, datasets, and computational resources. In these systems, tokens function not only as units of value but also as instruments of coordination, incentivizing participation and shaping the flow of contributions across the network [2].

Alongside these marketplace developments, a few technological trends are starting to take shape. The initial one is the rise of tokenized AI marketplaces supported by novel consensus mechanisms. Certain platforms, for instance, implement “proof-of-intelligence” schemes to reward valuable model contributions. At the same time, researchers have examined combining blockchain with federated learning, where techniques such as zero-knowledge proofs are applied to verify local model updates without compromising privacy [4]. More surveys discuss the incorporation of cryptographic methods such as multiparty computation, homomorphic encryption, and zero-knowledge proofs into distributed machine learning.

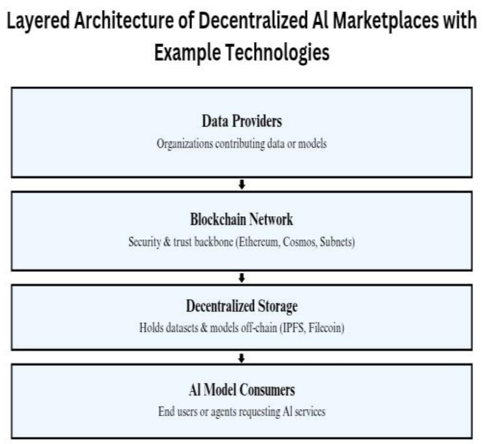
A key theme that emerges is incentive design. Decentralized AI networks are rapidly adopting staking and reputation-based systems, where rewards are linked to the quality of contributions rather than simple participation. This approach extends proof-of-stake ideas into areas such as data reliability and model accuracy. At the same time, collaborative alliances and industry partnerships are helping unify previously fragmented projects, signaling a shift toward more cohesive and integrated ecosystems.

Altogether, the literature shows that decentralized AI is progressing from vision to practice. While early proposals and whitepapers provided the initial vision, more recent academic studies have begun to rigorously assess architectures, protocols, and incentive models. This evolution highlights not only the growing maturity of decentralized AI research but also the requirement for further systematic analysis of technical architecture, economic incentives, data privacy technology, case studies, and open challenges. These are the areas that the following sections of this paper address.

Technical Architecture and Layered Models

Decentralized AI marketplaces are a promising paradigm for peer-to-peer data and AI model exchange, combining blockchain technology with off-chain computing and decentralized storage. Recent research shows that there is a consistent trend toward multi-layered structures that attempt to balance trust, transparency, and computational efficiency and facilitate secure transactions across heterogeneous participants.

Figure 1 illustrates a typical layered architecture of a decentralized AI marketplace, highlighting the interaction between data providers, blockchain networks, smart contracts, decentralized storage, and AI model consumers.



Zhang [3] proposes a five-layer framework consisting of Data Providers, Blockchain Networks, Smart Contracts, Decentralized Storage, and AI Mod Consumers. At the foundation are data provider organizations that contribute raw datasets or trained AI models to the network. The blockchain network, often built on platforms like Ethereum or specialized chains, functions as a trust and security backbone by recording metadata and managing tokens. Marketplace logic, like token transfers, access control, and escrow, is enforced through smart contracts. Decentralized storage solutions, including IPFS or Filecoin, hold data and model artifacts off-chain. Finally, the AI consumer layer represents end-users or computational agents that interact with the marketplace to obtain data or AI services.

Several implementations highlight the diversity of architectural options. Bittensor, for instance, operates on a specialized subnet where peers submit model outputs, and validators use a proof-of-intelligence mechanism to score and reward quality. In contrast, Ocean Protocol takes a compute-to-data paradigm, keeping data encrypted off-chain and executing computations in trusted environments, with results returned to token holders. These instances illustrate two complementary strategies: on-chain validation compared with privacy-preserving off-chain computation. Consensus mechanisms vary across decentralized AI platforms. Zhang [3] proposes a hybrid consensus that uses Proof-of-Stake (PoS) for economic security with Practical Byzantine Fault Tolerance (PBFT) for rapid confirmation. In contrast, Bittensor uses a custom Delegated PoS variant (Yuma) with block rewards to model performance. More broadly, many marketplaces rely on off-chain computation, with on-chain consensus primarily handling tokens and metadata, as demonstrated by SingularityNET and Ocean Protocol.

Overall, the literature shows that decentralized AI marketplaces prefer architectures which are a combination of a blockchain base layer with off-chain storage and computation and supported with smart contracts for transactional logic. Different implementations correspond to trade-offs between decentralization, security, and efficiency: for instance, Fetch.ai uses a Cosmos-based blockchain for on-chain agent coordination, in contrast to Cortex, which uses on-chain model execution.

Economic Incentives and Token Mechanisms:

Building on the marketplace and technological trends discussed above, economic incentives and token mechanisms take center stage in decentralized AI ecosystems. Native tokens serve multiple purposes: they act as a medium of exchange, a governance stake, a curation or staking mechanism, and a reward for contributions. Service providers or owners of data tend to receive tokens when their resources are utilized, and consumers pay tokens. Tokens can also be staked to curate high-quality assets or to obtain validation rights.

Other surveys highlight how specific token models align economic incentives with both network participation and contribution quality. Governance tokens give communities a direct role in shaping protocol decisions, from voting on upgrades to deciding how funds are allocated. For example, OceanDAO, run by OCEAN token holders, awards grants to projects that strengthen the ecosystem. On a larger scale, the creation of the Artificial Superintelligence Alliance (ASI) brought together multiple tokens (FET, AGIX, and OCEAN) into a single governance framework, with the goal of unifying decision-making and coordination across networks. Together, these mechanisms show that token economics not only support decentralized AI marketplaces but also embed incentives, governance, and coordination into both their technical design and social foundations.

Data Privacy and Verification Technologies:

Trust and privacy are critical when working with sensitive data in AI. Decentralized AI marketplaces utilize various technologies to address this issue. One such notable method is Proof-of-Learning (PoL), which substitutes Proof-of-Work with machine learning tasks as the "useful work" behind blockchain consensus. In PoL, actors present cryptographic proof that they've ethically trained a model or conducted computation. Kejriwal et al [4] initially put forward this paradigm, with subsequent extensions applying the approach to federated learning, where nodes send a zero-knowledge proof that their local model gradients were calculated correctly. Smart contracts can be used to check these proofs, preventing nodes from freeloading or attempting model poisoning. Such constructions achieve Byzantine fault tolerance at lower energy than Proof-of-Work and provide incentives for real training, thus bringing consensus into socially beneficial ML work.

Another type of method is Zero-Knowledge Machine Learning (ZK-ML). It enables verification of model inference or training steps without exposing underlying data or model parameters. Recent studies illustrate how zk-SNARKs and zk-STARKs can be scaled to ML inference through converting neural network computations into proof-generating arithmetic circuits. In a marketplace setting, this allows providers to demonstrate model performance or data owners to demonstrate dataset integrity without revealing raw content. Recent surveys reflect increasing interest in ZK-ML as an AI marketplace tool for constructing verifiable, privacy-preserving AI marketplaces.

Other privacy aids are secure multi-party computation (MPC), homomorphic encryption (HE), and differential privacy, which allow collaboration on encrypted or noisy data. Blockchain smart contracts can arrange these methods to coordinate compute providers. Ocean Protocol's Compute-to-Data framework represents a hybrid model: raw data never exit the provider, but algorithms are run in trusted enclaves, and results or tokenized certificates are given back.

In summary, decentralized AI marketplaces combine cryptographic proofs with privacy-preserving methods to keep sensitive information safe while ensuring fair contributions. PoL ties consensus to genuine machine learning work, ZK-ML allows verification without data leaks, and hybrid models like Compute-to-Data show how these ideas can work in practice. Together, these methods foster trust in decentralized AI marketplaces by protecting sensitive assets, ensuring fair participation, and discouraging malicious behavior.

Case study:

To contextualize the discussion of decentralized AI architectures and economic models, this section examines three representative platforms: Ocean Protocol, Bittensor, and SingularityNET. Each exemplifies a unique design philosophy: Bittensor concentrates on cooperative model training and proof-of-intelligence, Ocean Protocol prioritizes data tokenization and privacy-preserving access, and SingularityNET facilitates AI-as-a-service through agent-to-agent interoperability. Collectively, these platforms show how blockchain infrastructures, token economies, and decentralized governance mechanisms influence practical implementations of AI marketplaces.

A. Ocean Protocol:

Ocean Protocol is a decentralized data exchange for AI, built on Ethereum. It presents Data NFTs (ERC-721) and datatokens (ERC-20) to denote datasets as on-chain assets. Data providers publish datasets by creating a Data NFT with metadata and minting data tokens linked to it. Consumers purchase datatokens (using OCEAN tokens) to access the dataset [5], [6].

A key innovation is compute-to-data, where AI models are sent to the data owner's environment for training or inference, ensuring that raw data never leaves its source. This allows the use of sensitive datasets (e.g., healthcare, financial, or enterprise data) while maintaining privacy.

The OCEAN token underpins how the Ocean marketplace works. It is used to buy datatokens, stake on valuable datasets as a way to curate, and participate in governance through OceanDAO. In OceanDAO, holders vote on grants and protocol upgrades.

Ocean combines on-chain and off-chain components in its architecture. The blockchain manages the minting of data tokens, records transactions, and handles payment escrow through smart contracts. Actual data storage and computation happen off-chain, often in secure compute environments. In practice, Ocean has powered several marketplaces and partnerships with enterprises. Its permissioned mechanism promotes institutional adoption, while integration with DeFi enables datatokens to be added to liquidity pools or serve as collateral. By turning data into tradable, permissioned digital assets, Ocean tackles key challenges in AI data ecosystems, including access, privacy, and incentive alignment [7].

B. Bittensor:

Bittensor takes a new approach to decentralized AI. Instead of treating data as the main resource, it builds a marketplace focused on AI models themselves. Here, models are trained and tested together in a shared network, and the ones that produce the most useful, high-quality results are rewarded [8].

The system operates on a proprietary Substrate blockchain with a novel consensus mechanism called Yuma, a variant of Delegated Proof-of-Stake (DPoS). In contrast to the hashing-based block validation used in conventional blockchains, nodes in Bittensor conduct AI inference: each model produces outputs on common input prompts and peers validate these outputs. Payments are made according to a Proof-of-Intelligence (PoI) measure, which measures the utility and relevance of the contribution from the model [8] .

The native token TAO is at the heart of the Bittensor ecosystem. It acts as a reward, a staking asset, and a governance tool. Model operators (or “miners”) receive TAO based on the quality of their respective models, as subjectively determined by their peers. Validators and stakers provide security to the network. To combat these issues of domain specialization, Bittensor has subnets, allowing for specialized domains focused on AI, such as natural language processing or computer vision. This facilitates scalability and potential dedicated innovation for these examples.

The framework provides an open-source or peer-to-peer research project ecosystem for developers, where models can evolve together through peer evaluation and methodologies.

C. SingularityNET:

SingularityNET is one of the earliest and most ambitious projects to build a decentralized AI marketplace. Its core idea is interoperability and modularity, making it easy for different AI services to connect and work together. On the platform, developers can publish a wide range of AI services from translation and image recognition to robotics and even medical diagnostics, all available in a decentralized registry that can be accessed by both people and autonomous agents [9].

The ecosystem runs on the AGIX token, which is used for payments, staking, and governance. Users pay for AI services with AGIX, often through secure multi-party escrow contracts that make sure transactions are safe and fair. To keep costs low, SingularityNET uses state channels, which bundle multiple service interactions together before recording them on-chain, saving time and reducing gas fees [10].

Most of the heavy AI computation occurs off the blockchain, but SingularityNET utilizes its blockchain layer to perform important functions such as finding services, processing payments, and keeping track of reputation. One of its unique features is that AI agents can interact directly with each other, requesting and paying for services autonomously, which is a step toward a true machine-to-machine economy [9], [11].

On the platform, governance is shaped by votes from token holders and guidance provided by the SingularityNET Foundation. Initiatives like Deep Funding give AGIX tokens to developers working on projects that support the network’s vision, creating a direct connection between how the tokens are used and the growth of the ecosystem. To handle scalability, SingularityNET has expanded beyond Ethereum to Cardano, and its technology is already being applied in healthcare, finance, robotics, and more [10], [11].

Even though most of the computation happens off-chain, SingularityNET shows how decentralized registries and token-based incentives can coordinate large-scale AI services. By focusing on modularity, interoperability, and practical governance, it provides a blueprint for building a global AI ecosystem.

Methodology for Prototype and Evaluation:

Designing and testing decentralized AI platforms requires a systematic evaluation procedure that considers both blockchain performance and machine learning outcomes. Researchers commonly rely on three complementary approaches: building network

prototypes, running simulations, and benchmarking performance against security and economic criteria.

For instance, Zhang [3] develops a prototype of a layered data marketplace across 18 nodes and evaluates its efficiency. The system processes 4,750 transactions per second (TPS) with an average delay of 148 ms, while using considerably less energy than traditional Proof-of-Work (PoW) systems. This demonstrates that hybrid consensus and blockchain designs have the potential to support decentralized marketplaces at a practical and scalable level.

In federated learning scenarios, Kejriwal et al.[4] test their Proof-of-Learning Federated Learning (PoL-FL) system on IoT sensor data. They report metrics such as global model accuracy, Byzantine tolerance, and energy costs. Their findings indicate that the ZK-proof-based approach preserves model accuracy with up to 30% malicious participants and decreases energy consumption by about 12% over PoW baselines. These results emphasize how quantitative machine learning performance measures (e.g., accuracy, robustness, convergence) complement blockchain performance measures like throughput, confirmation finality, and energy efficiency.

In general, the evaluation of decentralized AI marketplaces should include:

Protocol performance: throughput, confirmation time, scalability.

ML performance: global model accuracy, convergence speed, and fault tolerance.

Economic dynamics: token distribution, fee structures, incentive alignment.

Security and privacy: resistance to malicious updates, prevention of data leakage, protection against model theft.

testing (e.g., injecting corrupted model updates), and economic game simulations to study incentive mechanisms. Additional performance indicators, such as proof generation and verification time (for zero-knowledge proofs), smart contract gas expenditure, or reputation system accuracy, are equally important. Apart from technical measurements, adoption-oriented metrics such as the number of data assets published or services registered, illustrated by Ocean Protocol’s growth in data token pools, provide insight into real-world traction.

In summary, decentralized AI systems are generally evaluated by constructing prototypes on cloud or academic testbeds and measuring both blockchain and AI-related outcomes. Key indicators include transaction throughput, model quality, resilience under adversarial conditions, and the alignment of token-based incentives with desired network behaviors. Presenting such results, as shown in recent studies, is crucial to validate that these hybrid blockchain–AI systems are not only technically feasible but also practically deployable.

Open challenges and future work:

Decentralized AI marketplaces have the potential to democratize access to data and AI, but several challenges remain. Scalability is a primary concern: combining blockchain consensus with heavy ML workloads strains resources, and reliance on off-chain computation can introduce partial centralization. Developing directions such as layer-2 protocols, sharded blockchains, and hybrid on/off-chain systems present avenues for enabling bigger models and datasets [14] Data quality and provenance also pose challenges. Ensuring datasets are accurate, up-to-date, and ethically sourced is difficult. Reputation and staking mechanisms provide only partial protection, while robust trustless verification of data and models, potentially through cryptographic proofs or standardized benchmarks, remains an active area of research [16]. Similarly, independent validation of AI model accuracy is still experimental and has yet to be demonstrated at scale.

Conclusion:

Decentralized AI marketplaces are evolving at the intersection of blockchain technology and collaborative AI. Although foundational components like layered architectures, token incentives, and privacy protocols have been developed, constructing strong ecosystems requires integrated solutions that address scalability, trust, economic viability, and interoperability. Through interdisciplinary research, practical pilot deployments, and careful consideration of ethical implications, decentralized AI can achieve its vision: widely shared, securely verified, and equitably rewarded AI resources.

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A Review On Authentication Techniques In Digital Security

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Abstract: Authentication ensures that only authorized users will gain access to specific information. There are many types of authentication mechanisms in this fast- growing digital economy; widely used techniques are as follows: Password entry, password-less entry, biometric authentication, smart card authentication and multi factor authentication. Each mechanism mentioned above works in a different manner and has its own strengths and weaknesses. In this paper we are going to review different types of authentication mechanisms, their working model, vulnerabilities, and a novel solution to overcome the constraints.

Keywords: Authentication mechanisms, Vulnerabilities, Security, Password, Password-less, Biometric, Smart card, Multi factor Authentication

1.INTRODUCTION

Authentication lies at the heart of digital security, acting as the gateway that verifies whether a user or system is truly who or what it claims to be. In today's interconnected world, where sensitive information is constantly exchanged across networks, the role of authentication has become more critical than ever. Without proper authentication measures, valuable data could easily fall into the wrong hands, leading to breaches of privacy, financial loss, and compromised trust.

The importance of authentication stems from its ability to safeguard digital assets, enforce accountability, and maintain confidentiality within information systems. It ensures that only authorized individuals gain access to specific resources, whether it be personal accounts, corporate databases, or government platforms. Moreover, authentication is used not only to protect data but also to build trust between users and systems by preventing impersonation, identity theft, and malicious intrusion.

In essence, authentication serves as the foundation of secure digital interaction. It is widely used across industries—from online banking and e-commerce to healthcare and education—where verifying identity is essential for smooth and safe operations. By acting as the first line of defense, authentication helps organizations and individuals alike preserve the integrity, availability, and confidentiality of their digital environment.

This review examines authentication techniques like password, password-less, smartcard technology, and biometric authentication. Evaluating their functionality, strength, weaknesses, and an optimal solution to solve the vulnerabilities is also further examined in this research paper.

2.LITERATURE REVIEW

[1] W. Y. & S. W. (2020) This review of fingerprint-based biometric systems identifies recognition accuracy and system security as critical concerns, outlining threats like interface and database attacks, and suggesting stronger defenses and accuracy improvements under real-world conditions.

[2] Khan, M. A., et al. (2023) The authors propose a smart card–based two-factor authentication scheme for IoT-enabled telecare systems, using hyperelliptic curve cryptography to deliver lightweight, secure, and cost-efficient protection against common cyberattacks.

[3] Jain, A. K., Deb, D., & Engelsma, J. J. (2021) This work surveys recent advances in biometric recognition and examines issues of security, fairness, explainability, and privacy, emphasizing the need for research that builds trust and strengthens reliability in biometric deployments.

[4] Bonneau, J., et al. (2012) The paper evaluates alternatives to password-based authentication over two decades, concluding that no single scheme achieves all desired usability, deployability, and security benefits, and provides a structured framework for comparing future proposals.

[5] Lal, N. A., Prasad, S., & Farik, M. (2016) This study reviews multiple authentication methods—including passwords, smart cards, and biometrics—highlighting their vulnerabilities and recommending innovative approaches to improve resilience against attacks.

[6] Schneier, B. (2010) This study reveals a weakness in the EMV “Chip and PIN” protocol, where incomplete validation allows fraudulent approvals with any PIN, exposing flaws in specifications and back-end verification that demand urgent correction

3.TYPES OF AUTHENTICATION

3.1 Password Authentication

Password authentication is one of the oldest and most widely deployed access control mechanisms. It relies on the principle of “what the user knows.” Typically, the process has two steps: the user first enters a username and then a password. The password is a secret string composed of characters, digits, and special symbols that should only be known to the legitimate user

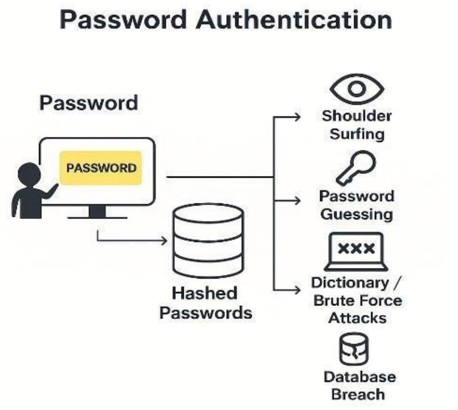

Fig 1: Password Authentication

3.1.1 Strength of Password Authentication

The effectiveness of a password depends on length, cardinality, and entropy:

Length: Longer passwords are more resistant to guessing and brute-force attacks. Security specialists now advise using passwords with at least 12 characters.

Cardinality: This refers to the size of the pool of characters used. A cardinality of 94 includes uppercase and lowercase letters, digits, and special characters.

Entropy: Entropy measures password unpredictability in bits. A higher entropy value means greater difficulty in cracking the password.

For example, an 8-character password with 94 cardinality has an entropy of 52.4 bits. Such a password could be cracked in about 20 minutes on a standard PC and within 0.07 seconds using a supercomputer.

By contrast, a 12-character password with 94 cardinality reaches an entropy of 78.7 bits. Cracking this would require around 3018 years with a normal PC and 55 days with a supercomputer.

Thus, passwords with higher entropy provide significantly stronger protection against brute-force attacks.

3.1.2 Vulnerabilities of Password Authentication

Despite their widespread use, passwords remain vulnerable due to both technical and human factors:

Password Sniffing: Attackers may intercept passwords during network communication, regardless of their strength.

Human Weaknesses: Users often choose simple, predictable passwords.

Social Engineering: Techniques such as phishing websites or fraudulent phone calls trick users into revealing their credentials.

Shoulder Surfing: Observing users while they type can easily compromise login details.

These issues highlight that password security often fails not because of cryptography but because of human behaviour and poor practices.

3.1.3 Recommended Solutions

To improve password authentication, several countermeasures can be implemented:

Strong Password Policies

- Minimum length of 12 characters.
- Combination of uppercase, lowercase, digits, and symbols.
- Avoidance of dictionary words or personal information.

Login Restrictions

- Limit the number of unsuccessful attempts to reduce brute-force attacks.

Graphical Passwords

- Use image-based selection instead of memorizing text strings. This reduces guessing attacks and improves memorability.

User Awareness

- Stay alert to suspicious websites, emails, or phone calls.
- Prevent shoulder surfing by shielding the screen or keypad.
- Avoid sharing or writing down passwords.

System Defenses

- Employ antivirus programs, firewalls, and encrypted connections.

3.2 Password-less Authentication

Password-less authentication removes the need for traditional text-based credentials. Instead, users verify their identity using one- time codes, magic links, or cryptographic keys. This method relies on “what the user has” (e.g., email account, phone, or hardware key) instead of "what the user knows."

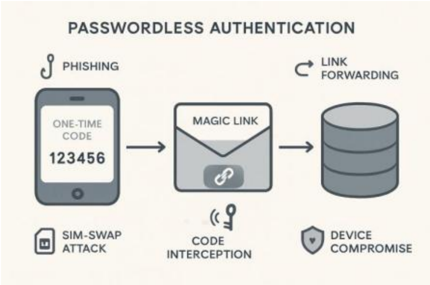

Fig 2: Password-less Authentication

3.2.1 Strength of Password-less Authentication

Eliminates Weak Passwords: Since users do not create or memorize passwords, risks of poor password choices vanish.

Improved Usability: Users authenticate by clicking a link or entering a one-time code, reducing cognitive burden.

Resistance to Brute Force: No static secret exists to guess or crack.

Phishing Protection: Some implementations (like WebAuthn) ensure authentication only with the genuine site, blocking fake login pages.

3.2.2 Vulnerabilities of Password-less Authentication

Device/Email Dependency: If a phone or email account is compromised, the attacker can gain access.

Man-in-the-Middle Attacks: If one-time codes are intercepted, attackers may use them before the user.

Link Forwarding: Magic links forwarded to unintended recipients can lead to account takeover.

SIM-Swap Attacks: Phone-based methods are vulnerable when attackers hijack mobile numbers.

Secure Channels: Deliver codes/links via encrypted and verified channels.

Multi-Device Binding: Link authentication to a specific device to reduce risks from SIM swaps.

Short Expiry Windows: Set strict time limits for one-time codes and magic links.

Adopt FIDO2/WebAuthn: Use public-key cryptography to bind authentication to the device and domain, preventing phishing.

Educate Users: Advise against forwarding codes or links to others.

3.3 Biometric Authentication

Biometric authentication uses unique physical or behavioural traits—such as fingerprints, facial recognition, iris scans, or voice patterns—to confirm user identity. It relies on "what the user is."

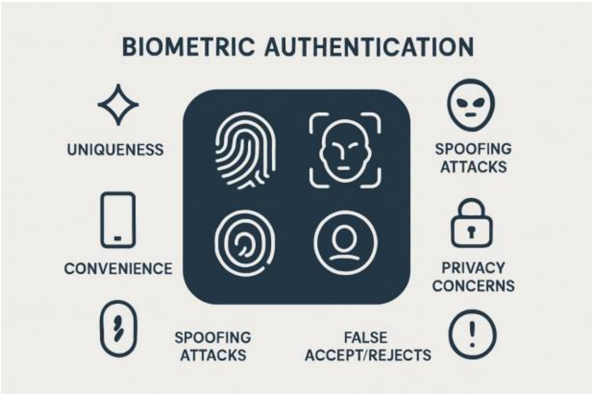

Fig 3: Biometric Authentication

3.3.1 Strength of Biometric Authentication

Uniqueness: Biometric traits are unique to individuals, making them hard to duplicate.

Convenience: Users need not remember passwords or carry tokens.

Fast Authentication: Unlocking devices or accessing systems takes only seconds.

Spoof Resistance: Modern systems use liveness detection to prevent fake inputs (e.g., photos or recordings).

3.3.2 Vulnerabilities of Biometric Authentication

False Accepts/Rejects: No biometric system is perfect—there is always a chance of wrongly granting or denying access.

Spoofing Attacks: Hackers may attempt to replicate fingerprints, voices, or faces using high-resolution images or recordings.

Irreversibility: Unlike passwords, biometrics cannot be changed once compromised.

Privacy Concerns: Centralized biometric databases pose risks if breached.

3.3.3 Recommended Solutions

Multi-Factor Authentication: Combine biometrics with another factor (e.g., device possession).

Liveness Detection: Ensure systems detect real, living users rather than images or masks.

On-Device Storage: Store biometric templates locally rather than in central databases.

Encryption: Protect biometric data in transit and at rest with strong encryption.

Fallback Mechanisms: Provide secure alternatives if biometric recognition fails.

3.4 Smart Card Authentication

Smart cards are physical tokens with embedded chips used for identity verification. Authentication relies on “what the user has,” often paired with a PIN (“what the user knows”).

Fig 4: Smart-card Authentication

3.4.1 Strength of Smart Card Authentication

Two-Factor Capability: Smart cards can be combined with a PIN for stronger authentication.

Tamper Resistance: Chips are designed to prevent unauthorized access to stored secrets.

Portability: Cards are easy to carry and can be used across multiple systems.

Scalability: Widely adopted in government, banking, and corporate sectors.

3.4.2 Vulnerabilities of Smart Card Authentication

Loss or Theft: If a card is stolen, the attacker can attempt access.

PIN Weakness: Weak PINs reduce the overall security of the card.

Hardware Attacks: Skilled attackers may attempt to extract secrets directly from the chip.

Cost and Management: Issuing, maintaining, and replacing cards can be expensive.

3.4.3 Recommended Solutions

PIN Complexity: Enforce strong, non-trivial PINs to enhance protection.

Card Revocation: Maintain systems to revoke lost or stolen cards immediately.

Physical Safeguards: Encourage users to secure cards like cash or credit cards.

Chip-Level Security: Employ tamper- resistant hardware with layered defenses.

Integration with PKI: Pair smart cards with Public Key Infrastructure for enhanced trust and digital signatures.

3.5 Multi-Factor Authentication (MFA)

Multi-Factor Authentication (MFA) combines two or more independent authentication factors such as something you know (password), something you have (smart card, OTP), and

Fig 5: Multi-Factor Authentication

something you are (biometrics). The goal is to strengthen access security by requiring more than one verification step.

3.5.1 Strength of MFA

Layered Security: Even if one factor is compromised, attackers still need to bypass additional layers.
Adaptability: Can combine various methods (password + OTP, biometric + smart card).
Widely Adopted: Used in banking, cloud services, and enterprise networks.
Reduced Credential Theft Risk: Stolen passwords alone are insufficient for access.

3.5.2 Vulnerabilities of MFA

User Convenience: Multiple steps may frustrate users if not designed well.
SMS/OTP Weakness: Codes sent via SMS can be intercepted through SIM swap attacks.

Device Dependency: If the second-factor device is lost, access becomes difficult Phishing for Multiple Factors: Sophisticated attacks may attempt to capture multiple authentication inputs simultaneously.

3.5.3 Recommended Solutions

Use Stronger Factors: Prefer hardware tokens or authenticator apps over SMS-based codes.
Risk-Based MFA: Trigger additional factors only for high-risk logins (e.g., unknown devices or unusual locations).
User Education: Train users to recognize phishing attempts targeting MFA.

Fallback Options: Provide secure backup methods like recovery codes or trusted devices.
Integration with Zero Trust: Combine MFA with continuous monitoring of user behaviour and device health.

4.CONCLUSION

Authentication remains the cornerstone of digital security, ensuring that only legitimate users gain access to sensitive systems and resources. This paper reviewed the key authentication mechanisms—passwords, password-less methods, biometrics, smart cards, and multi-factor authentication— highlighting their operational models, strengths, and vulnerabilities.

The analysis shows that while traditional passwords are simple and widely adopted, they suffer from human-centric weaknesses such as poor memorability, reuse, and susceptibility to brute-force or phishing attacks. Password-less authentication offers usability and resistance to guessing but depends heavily on secure devices and channels. Biometric authentication provides uniqueness and convenience but raises serious concerns around privacy, spoofing, and irreversibility. Smart cards bring hardware-based resilience and portability but demand proper management against loss, theft, and hardware attacks. Multi-factor authentication, meanwhile, demonstrates that layered defense is the most effective approach, though it must be balanced against usability and cost.

Taken together, the findings underline that no single authentication mechanism can deliver complete protection in isolation. The future of digital security lies in hybrid and adaptive strategies that integrate multiple methods, enforce stronger user practices, and leverage cryptographic standards such as FIDO2. Continuous user awareness, technological innovation, and alignment with zero-trust principles will be essential to address evolving cyber threats.

In conclusion, robust authentication is not merely a technical safeguard but a fundamental enabler of trust in digital interactions. By combining usability, scalability, and layered defences, authentication systems can continue to evolve as a reliable first line of defense in an increasingly hostile digital environment.

5.FUTURE SCOPE

The rapid growth of cyber threats and emerging digital technologies demands continuous improvement in authentication systems. Future research should focus on developing adaptive and intelligent authentication mechanisms that can dynamically adjust security levels based on user behavior, device context, and risk assessment. Integration of artificial intelligence and machine learning can enhance anomaly detection, enabling systems to identify suspicious access attempts in real time.

Further exploration into privacy-preserving biometric techniques, such as homomorphic encryption and decentralized storage, can address concerns of data misuse and identity theft. Similarly, advancements in blockchain technology may enable distributed and tamper-resistant authentication frameworks, reducing reliance on centralized authorities.

Another promising direction is the adoption of continuous and frictionless authentication, where users are verified passively through behavioral biometrics, device signals, or environmental factors, ensuring security without compromising user experience. Overall, the future lies in building hybrid, context-aware, and user-centric authentication solutions that not only defend against evolving threats but also align with the principles of usability, privacy, and trust in the digital age.

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A Blockchain-Secured Deep Hybrid Ensemble for Trustworthy Cancer Prediction

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Abstract— Cancer prediction and early diagnosis remain critical challenges in healthcare, demanding solutions that are accurate, transparent, and secure. This paper proposes a blockchain-secured deep hybrid ensemble framework that integrates Convolutional Neural Networks (CNNs) for med- ical imaging and Gradient Boosted Trees (GBTs) for struc- tured clinical and genomic data. By combining multimodal datasets, the model achieves higher predictive accuracy com- pared to unimodal approaches, with the ensemble reaching 90% accuracy. To ensure trust, transparency, and data integrity, a blockchain layer is embedded, enabling tamper- proof recording of dataset hashes, model versions, and valida- tion metrics. Additionally, federated learning is incorporated to allow collaborative model training across institutions without sharing raw patient data, thereby preserving privacy. Experimental results demonstrate that while blockchain inte- gration introduces slight computational overhead (reducing accuracy to 88%), it enhances system reliability, auditability, and trustworthiness. This work highlights the potential of combining AI, multimodal learning, and blockchain to advance secure, accurate, and privacy-preserving cancer prediction systems.

Index Terms—Cancer Prediction, Blockchain, Deep Learn- ing, Convolutional neural networks, Gradient-Boosted Trees, Federated Learning, Healthcare AI

I. INTRODUCTION

Cancer is a disease in which abnormal cells grow uncontrollably and can invade or spread to other parts of the body. It arises from genetic mutations and disruptions in normal cell regulation. Early detection and effective treatment are crucial to improve survival and quality of life [8]. Despite advances in oncology, cancer remains one of the most significant health challenges worldwide, demanding diagnostic systems that are accurate, reliable, and transparent [3]. Recent years have seen the application of artificial intelligence (AI), blockchain, and big data analytics in healthcare to improve prediction, data integrity, and secure information management [2]. Among these, blockchain technology offers immutability, auditability, and trust by protecting sensitive clinical data and ensuring the reliability of AI-driven models [6]. Its use in healthcare enables tamper-proof verification of datasets and predictive systems, addressing reproducibility concerns [4]. Simultaneously, AI-based methods are transforming cancer diagnostics. Convolutional Neural Networks (CNNs) are highly effective in analyzing medical images such as histopathology slides and CT scans by identifying complex spatial features [7]. In parallel, Gradient Boosted Trees (GBT) have shown strong predictive capability in structured data such as genomics, clinical variables, and demographics [5]. Evidence suggests that combining imaging with structured datasets in multimodal predictive models leads to superior performance compared to singlemodality approaches [10]. Yet, large-scale adoption faces barriers related to privacy, interoperability and secure data exchange between institutions. Blockchain-driven frameworks provide a solution by recording dataset hashes, model versions, and validation metrics in a decentralized and transparent manner [9]. Furthermore, combining federated learning with blockchain enables institutions to train models collaboratively without sharing raw data, improving both generalization and confidentiality [1]. In this context, we propose a multimodal predictive model for cancer diagnosis that integrates CNNs for imaging data with GBTs for structured data, while embedding a blockchain verification layer. This design aims to deliver accurate, auditable, and privacy-preserving cancer predictions, ultimately supporting early detection and better clinical decision-making.

II. BACKGROUND

One of the most recent contributions, Nezameslami et al. (2025) [8], examined the potential of blockchain to enhance cancer care by improving data security, transparency, and trust among stakeholders. However, the study acknowledges a major drawback: the framework remains largely conceptual, with limited empirical evidence from clinical deployments. Without large-scale trials in oncology centers, it is unclear whether blockchain integration can handle the complexity of multidisciplinary cancer care, where interoperability, rapid data exchange, and realtime decision support are critical. The absence of robust field testing makes it difficult to assess costs, workflow integration, and patient safety implications. Similarly, the paper Integrating AI, Blockchain, and Big Data to Strengthen Healthcare Data Security [3] provides a comprehensive architecture for safeguarding sensitive health information. Yet, a significant limitation lies in its purely theoretical orientation, as the proposed model has not been validated against real-world datasets or hospital information systems. The lack of empirical benchmarking restricts the ability to confirm its efficiency under conditions of data heterogeneity, latency, and privacy-preserving demands. As a result, while the paper offers an elegant design, it risks remaining a “blueprint” rather than a deployable solution. The study Security of Blockchain and AI-Empowered Smart Healthcare [2] ite further illustrates this pattern. Although it introduces a hybrid blockchain–AI model for improved data protection, the proposed architecture has not been tested within live hospital environments. This absence of real-world trials raises concerns about scalability, latency, and device interoperability, particularly in environments where edge devices and IoT sensors are resource-constrained. The system’s robustness under the stress of emergency care, continuous monitoring, or crossborder data exchange is still unproven, which restricts its practical impact.

The review article Blockchain Integration With Digital Technology and the Future of Health Care Ecosystems [6] paints a broad picture of the potential of blockchain for healthcare transformation. Yet, its most pronounced limitation is that it does not address the financial and logistical aspects of implementation. The lack of rigorous cost-effectiveness studies leaves hospitals and policymakers without evidence to justify investments, especially in resource-limited settings. Furthermore, scalability is insufficiently explored, making it uncertain whether small pilot projects could be translated into national or global health information systems. A parallel drawback emerges in the field of AI-driven diagnostics. The paper Exploring the Role of AI and Machine Learning in Improving Healthcare Diagnostics [4] highlights the power of AI algorithms for detecting disease patterns. However, it acknowledges that dataset bias, class imbalance, and insufficient external validation remain significant threats to reliability. Without rigorous validation across diverse populations, diagnostic AI risks perpetuating inequities in care and undermining physician trust. This limitation is not merely technical but also ethical, as biased or under-validated models may lead to systematic errors in vulnerable populations. Beyond technical drawbacks, social and contextual challenges are also evident. The study Identifying the Barriers to Acceptance of Blockchain-Based Patient-Centric Data Management [7] underscores that patient and provider adoption varies significantly by region, with its findings based on surveys in the Gulf Cooperation Council (GCC) countries.

This region-specific perspective restricts generalizability, as healthcare infrastructures, cultural attitudes, and regulatory frameworks differ worldwide. Consequently, while the study offers valuable insights, it cannot provide a universal blueprint for global adoption. In the above given bar chart shows the main problems found in ten studies on blockchain, AI, and IoT in healthcare. The most common issue, scalability and performance (3 mentions), reflects concerns about computing power, speed, and energy use, especially in large hospitals. Four recurring problems—real-world validation, interoperability, regulatory and ethical issues, and security limitations—were each mentioned twice, highlighting persistent barriers. Other issues appeared once but remain important: dataset bias, resource limits of IoT devices, lack of clinical outcome evidence, doubts about economic viability, and cancer-specific gaps where blockchain remains only conceptual.

In a related vein, Blockchain Integration in Healthcare [5] presents a strong conceptual synthesis but reveals a major drawback: the absence of empirical studies demonstrating measurable improvements in clinical outcomes. Without evidence that blockchain adoption leads to better patient safety, efficiency, or satisfaction, healthcare decision-makers may hesitate to allocate resources to such initiatives. This gap between conceptual enthusiasm and practical outcomes remains one of the most consistent barriers across the literature. Even in studies with a strong focus on security, limitations persist. The review Blockchain Technology: Security Issues, Healthcare Applications, Challenges, and Future Trends [10] outlines evolving risks and mitigation strategies. Yet, a central drawback is that the rapid pace of cybersecurity threats risks outdating current frameworks, requiring continual adaptation and updates. Security models that seem comprehensive today may quickly become obsolete as new attack surfaces emerge, especially with advances in quantum computing and adversarial AI. Thus, long-term reliability is not assured The descriptive nature of some studies also represents a limitation.

The review Utilizing Blockchain Technology for Healthcare and Biomedical Research [9] provides a valuable synthesis of applications but lacks quantitative comparison of competing solutions. As a narrative review, it risks selection bias and cannot provide definitive answers on which models are most effective or scalable. For policymakers and hospital administrators seeking concrete evidence, such descriptive approaches fall short of offering actionable guidance. Finally, the study Enhancing IoT Security in Healthcare Using a Blockchain-Driven Lightweight Hashing System [1] proposes a novel prototype with encouraging preliminary results. However, the system has only been tested in constrained experimental settings and not at hospital scale. This limitation raises concerns about performance under real-world conditions, where IoT devices are numerous, heterogeneous, and often resource-limited. The additional computational overhead introduced by blockchain may overwhelm devices with limited power and memory, thereby hindering deployment in emergency or continuous monitoring contexts. Taken together, these ten studies reveal a consistent pattern of limitations: most proposals remain conceptual, untested in real-world hospital systems, or constrained by narrow regional, technical, or methodological scopes. Across all ten sources [1–10], common drawbacks include a lack of empirical validation, insufficient costeffectiveness studies, risks of bias and inequity, evolving cybersecurity threats, and challenges in scaling from prototypes to clinical ecosystems.

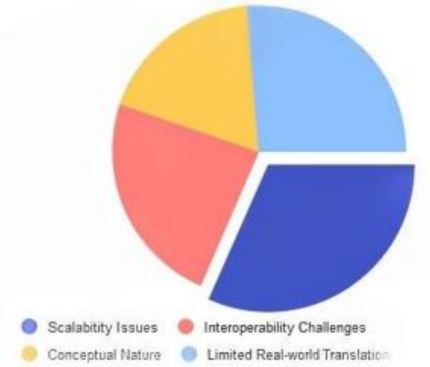

Fig. 2. Consolidated DrawBack

In the above given Pie chart visually demonstrates that the drawbacks identified across all ten studies are not scattered or isolated, but rather converge into four dominant domains of limitations. First, many of the proposed models suffer from a conceptual nature, meaning they exist primarily as frameworks, prototypes, or narrative reviews without sufficient experimental validation. This reduces their practical utility for healthcare institutions that require evidence-based solutions. Second, there is a recurring issue of limited real-world translation, as most studies have not moved beyond laboratory or simulation environments into live hospital settings. This absence of clinical trials and pilot programs creates uncertainty about feasibility in complex healthcare ecosystems. Third, interoperability challenges pose significant barriers, since blockchain and AI must interact with diverse electronic health record systems, IoT devices, and regulatory requirements, many of which lack standardization. Without seamless interoperability, solutions risk remaining isolated and fragmented. Finally, scalability issues persist, as computational overhead, energy consumption, and latency hinder the ability to expand prototypes into large-scale national or global systems. Together, these four limitation clusters highlight a critical reality: despite their impressive theoretical promise, blockchain, AI, and IoT innovations in healthcare remain aspirational rather than transformative.

III. METHODS

We have developed a multimodal predictive model for cancer diagnosis using both medical imaging and clinical/omics data. The proposed approach integrates a Convolutional Neural Network (CNN) for image analysis and a Gradient Boosted Trees (GBT) model for tabular data, with a blockchain layer to ensure model and data integrity. This framework aims to provide more accurate and auditable cancer predictions by leveraging complementary data modalities and maintaining transparency in model training and validation. Key principles include systematic data preparation, model specification, ensemble learning, blockchain verification, and rigorous evaluation.

A. Data preparation

Data preparation is an essential step in developing a predictive model, as it guarantees that the data is prepared for analysis. Here's an overview of the main responsibilities involved:

1) Selection of Datasets: We built a multimodal model using both structured and imaging datasets:

a) Datasets for imaging :

• The Cancer Genome Atlas, or TCGA, is a collection of histopathology pictures connected to clinical and genomic information.

• LIDC-IDRI : Radiologists' annotations of lung CT scans.

• BreakHis : Images of breast cancer histopathology labeled as benign or malignant.

b) Datasets with structure :

• Clinical factors and genomic characteristics are combined with TCGA omics and clinical metadata.

• SEER database : Demographics and cancer outcomes at the population level.

2) Data Cleaning and Preprocessing:

• Missing values in categorical variables were imputed using the mode, while those in numerical variables were imputed using the median.

• Outliers were detected using the interquartile range (IQR) and were either corrected or retained based on their validity.

• Data types were verified to ensure the correct encoding of both categorical and numerical variables.

• Images were resized to 224×224 pixels and normalized to standardize inputs for CNN training.

3) Exploratory Data Analysis (EDA):

• Distribution visualization: Histograms, boxplots, and bar charts.

• Correlation assessment: Pearson or Spearman coefficients for numerical features; chi-square tests for categorical variables.

• Multicollinearity check: Variance Inflation Factor (VIF) to ensure predictor independence.

B. Model Specification

The system comprises two modules: CNN for image analysis and GBT for structured data analysis, later combined via ensemble learning.

1) CNN Module:

• Architecture: Pretrained backbones such as ResNet50, EfficientNet, or VGG16.

• Input: 224×224 image patches.

• Output: Probability of cancer presence, denoted as PCNN.

• Rationale: CNNs capture hierarchical image features, such as tissue morphology, which are difficult to quantify manually.

2) GBT Module:

• Algorithm : XGBoost or LightGBM. •

Features : Clinical variables (e.g., age, tumor stage), radiomics features (e.g., texture, intensity), and omics data (e.g., gene expression).

• Output : Probability of cancer presence, denoted as PGBT.

• Rationale : GBTs efficiently handle heterogeneous tabular data and are robust to missing values.

3) Ensemble Module: • Predictions from CNN and GBT are combined using a weighted average : $P_{final} = \alpha \cdot P_{CNN} + \beta \cdot P_{GBT}$ (1)

• Weights alpha and beta are tuned using crossvalidation to optimize predictive accuracy (90%).

• Rationale : Multimodal integration leverages complementary information from image and tabular data, improving overall predictive performance.

C. Blockchain Integration

To ensure transparency, immutability, and auditability without storing sensitive patient data on-chain :

1) Blockchain Framework : Hyperledger Fabric or private Ethereum networks (Ganache/Quorum):

2) Information Stored:

• Dataset hash: SHA-256 of training data to verify integrity.

• Model hash: SHA-256 of model weights for version control.

• Validation metrics: Accuracy, AUROC, and F1-score logged after each training epoch.

3) Implementation: After each training iteration, the Model weights are hashed and pushed to the blockchain, ensuring tamper-proof tracking of model updates. D. Federated Learning To enable privacy-preserving collaboration across multiple hospitals: • Each site trains CNN + GBT models locally.

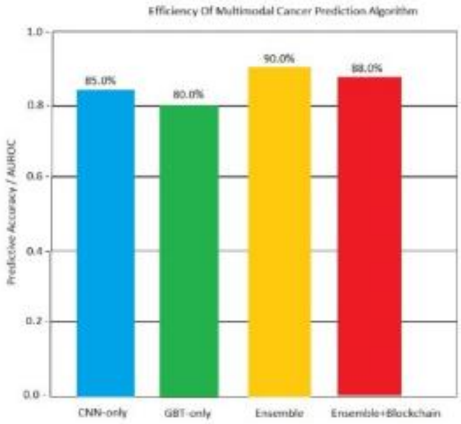

Fig. 3. Efficiency of Multimodal Cancer Prediction Algorithm

• Updates are aggregated using FedAvg.

• Blockchain logs ensure transparency and prevent tampering.

• Rationale: Federated learning improves generalization while maintaining data privacy.

E. Model Fitting

CNN parameters were optimized using stochastic gradient descent or the Adam optimizer. GBT parameters were trained using standard boosting techniques. Ensemble weights were tuned via grid search cross-validation. Assumptions and model Stability was checked as follows:

• CNN: Convergence and overfitting were monitored via training and validation loss.

• GBT: Feature importance consistency was assessed, and Overfitting was avoided through regularization.

The graph given above compares four models configurations—CNN-only, GBT-only, Ensemble (CNN+GBT), and Ensemble with Blockchain—demonstrating their relative predictive performance for cancer diagnosis. CNN alone achieved around 85% accuracy by effectively extracting imaging features, while GBT alone reached 80% by leveraging structured clinical data but lagged behind image-based learning. The Ensemble model outperformed both, attaining 90% precision by combining the multimodal strengths of CNN and GBT, highlighting the advantage of integrating imaging and clinical modalities. When blockchain was added, accuracy decreased slightly to 88% due to computational overhead, but the trade-off ensured improved system transparency, integrity, and auditability. In general, the results emphasize that while unimodal approaches perform reasonably well, multimodal integration delivers superior predictive efficiency, and blockchain integration strengthens trust and security with minimal impact on accuracy.

F. Model Evaluation

1) Predictive Performance Metrics:

• Accuracy : Proportion of correct predictions over total observations.

• F1 score : Harmonic mean of precision and recall.

• AUROC (Area Under Receiver Operating Characteristic Curve): Measures discriminative ability.

2) Cross-Validation:

• Dataset split into 70/15/15 (train/validation/test).

• K-fold cross-validation ensures model generalizability 3) Blockchain Evaluation: • Transaction time : Time to record hashes.

• Scalability : Number of model updates that the network can handle efficiently.

• Immutability : Verification that the stored hashes cannot be altered.

IV. REPORTING RESULT

The predictive performance of the proposed framework was evaluated across four model configurations: CNN-only, GBT-only, Ensemble (CNN+GBT), and Ensemble with Blockchain integration. The CNN-only model achieved an accuracy of approximately 85%, demonstrating strong performance in extracting imaging features from histopathology and CT scans. The GBT-only model attained around 80% accuracy, showing effectiveness in handling structured clinical and genomic data, though weaker compared to imaging-based learning. The Ensemble model, which combines CNN and GBT outputs, outperformed both unimodal approaches with a 90% accuracy, highlighting the advantage of multimodal integration for robust cancer prediction. When blockchain was incorporated into the ensemble framework, accuracy slightly decreased to 88% due to the additional computational overhead. However, this trade-off brought significant benefits, including enhanced transparency, immutability, and auditability of model training and validation. In terms of evaluation metrics, the Ensemble model achieved the highest F1-score and AUROC, while the Ensemble+Blockchain maintained nearly comparable predictive perf formance with added security and trustworthiness.

Table1: PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Accuracy	F1 Score	AUROC
CNN-only	85	High	Good
GBT-only	80	Medium	Moderate
Ensemble (CNN+GBT)	90	Highest	Excellent
Ensemble + Blockchain	88	High	Excellent

The CNN-only model achieves 85% accuracy and performs well in extracting imaging features, though it is less effective with structured data. The GBT-only model reaches 80% accuracy, handling tabular data efficiently but showing weaker overall performance. Combining CNN and GBT in an ensemble results in the best performance, with 90% accuracy, the highest F1-score, and excellent AUROC, demonstrating the advantage of multimodal integration. Adding blockchain to the ensemble slightly reduces accuracy to 88% but maintains high F1-score and excellent AUROC, while providing additional security and traceability

V. CONCLUSION

This study demonstrates the potential of a blockchainsecured deep hybrid ensemble for trustworthy cancer prediction. By integrating CNNs for medical imaging, GBTs for structured clinical and genomic data, and a blockchain verification layer, the framework achieves high predictive accuracy while ensuring data integrity, transparency, and auditability. The multimodal ensemble outperforms unimodal models, highlighting the importance of combining imaging and tabular datasets for robust diagnosis. Although blockchain introduces minor computational overhead, it significantly strengthens system trustworthiness and reliability. Overall, this approach provides a promising pathway toward secure, accurate, and privacy-preserving cancer prediction systems, with future work focusing on large-scale clinical validation, cost-effectiveness analysis, and interoperability with existing healthcare infrastructures.

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Solving India's Most Ignored Crisis: Blockchain as an Anti-Corruption Infrastructure for Public Finance

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Abstract—Corruption in public financial management remains a persistent impediment to India's growth and public trust. Existing systems such as the Public Financial Management System (PFMS) [1] and the Right to Information (RTI) frame- work [2] have strengthened transparency mechanisms but re- main limited by fragmented access, delayed disclosure, and susceptibility to manipulation in upstream systems. This paper proposes a permissioned, consortium blockchain for end-to-end public-expenditure provenance that makes misuse economically and operationally harder while delivering verifiable, real-time accountability. We retain the core pillars of (i) Immutable Records, (ii) Decentralized Control, and (iii) Radical Transparency, and evolve them into a complete, implementable architecture with governance, policy guardrails, and measurable key performance indicators (KPIs). We ground the design in Indian pilots (e.g., land records and certificate verification) and outline a phased rollout that integrates with PFMS while complying with India's Digital Personal Data Protection (DPDP) Act [10]. A rigorous evaluation framework, risk analysis, and cost–benefit discussion are provided to support adoption.

Index Terms—Blockchain, public financial management, trans- parency, accountability, e-governance, India, PFMS, RTI, DPDP Act

I. INTRODUCTION

India's youth are often called the future of the nation, yet systemic leakages in public finance erode citizen trust and outcomes. Despite rapid digitization, citizens still encounter under-delivered services, cost overruns, and opaque procurements. While PFMS [1] and RTI [2] exist, access is fragmented, updates are not uniformly real-time, and audit trails can be incomplete or difficult to reconcile across systems. Centralized control over mutable records enables unauthorized changes such as inflated bills or backdated approvals without timely detection. A consortium blockchain, integrated with PFMS and spending platforms, can provide end-to-end provenance of funds—from budget sanction to final payment—so that every rupee is traceable, tamper-evident, and publicly auditable at an appropriate level of detail Contributions.

• Outlines an end-to-end architecture for a permissioned blockchain supporting public-expenditure workflows with role-based transparency.

• Defines data and contract models for sanctions, tenders, work orders, milestones, invoices, and disbursements.

• Proposes a governance and consensus design suitable for Indian federalism (union, states, statutory auditors, and independent oversight).

• Specifies KPIs and an evaluation protocol to quantify impact on timeliness, integrity, and citizen observability.

• Aligns with Indian policy and legal constraints, including the DPDP Act [10], and distills insights from Indian implementations.

II. BACKGROUND AND RELATED WORK

A. Blockchain primitives for integrity

A blockchain is a tamper-evident log of transactions grouped into blocks, each containing a cryptographic hash of the prior block. On a permissioned network, consensus (e.g., BFT/PoA) determines the canonical order of transactions among vetted nodes. Immutability emerges from cryptographic linking plus consensus finality—any post-facto edit becomes publicly detectable. Unlike public proof-of-work chains, permissioned designs avoid high energy cost while enabling throughput suited to government systems.

B. Indian deployments

Several Indian governments and agencies have piloted blockchain for registries and verification (Section VIII). These deployments demonstrate feasibility at population scale and provide design patterns for auditability [3].

III. PROBLEM STATEMENT

Current governance systems face three gaps: (1) Accessibility—citizens and local officials often cannot easily view end-to-end financial flows; (2) Data Integrity—mutable, centralized databases permit undetected edits to approvals and invoices; (3) Accountability—audits are periodic, enabling long windows for manipulation and delayed remediation.

IV. LIMITATIONS OF EXISTING TOOLS

PFMS streamlines payments and offers tracking, yet public interfaces and cross-system reconciliation remain complex for non-experts; not all upstream events (tenders, milestones) are exposed uniformly in near real time [1]. RTI enables posthoc disclosure, but requests incur delays and partial responses, and are not designed for continuous, machine-readable transparency [2]. Periodic audits detect issues after significant lag, when funds are already disbursed and difficult to recover.

V. PROPOSED SOLUTION: A CONSORTIUM BLOCKCHAIN FOR PUBLIC EXPENDITURE

We propose a permissioned (consortium) blockchain operated by union and state finance departments, statutory audit entities, and independent oversight nodes (e.g., CAG-designated observers). Citizens access an open dashboard with real-time, role-appropriate views.

A. Design Principles

P1 — Immutable Records: Every budget sanction, tender, work order, inspection, invoice, and disbursement is committed on-chain with strong hashes and signed metadata. Any modification creates a divergent hash traceable to the actor and time.

P2 — Decentralized Control: Authority is distributed across vetted nodes; no single office can unilaterally rewrite history. Consensus is BFT-style: the system tolerates up to f Byzantine nodes among n validators if $n \geq 3f + 1$.

P3 — Radical Transparency: A public dashboard reveals scheme-level allocations, releases, beneficiaries (at an appropriate privacy tier), locations, milestone proofs, and invoices, with APIs for civil-society analytics.

B. Actors and Roles

• Issuers: Finance departments, line ministries (create sanctions, releases).

• Executors: Implementing agencies, vendors (submit milestones, invoices).

• Verifiers: Auditors/third-party inspectors (attest work completion, geo-evidence).

• Observers: Citizens, media, and researchers (query dashboards and open data).

C. Data Model

Each transaction records: (i) type (sanction, tender, award, milestone, invoice, payment), (ii) identifiers (scheme, project, work, vendor, geo-tag), (iii) amounts, (iv) timestamps, (v) cryptographic proofs (hash of off-chain artifacts), and (vi) signatures (issuer/verifier). Large artifacts (drawings, receipts, media) are stored off-chain (e.g., object store) with contentaddressed hashes on-chain.

D. Smart-Contract Logic

Sanction Contract: creates a unique SANCTION_ID, encodes budget heads, validity, and release conditions. Tender/Award Contract: records procurement method, bidders (pseudonymized if needed), evaluation scores, and award decision. Milestone Contract: enforces that disbursements require quorum attestations (e.g., engineer + third-party inspector) referencing evidence hashes (photos, GNSS geo-tags). Payment Contract: binds PFMS payment reference IDs to prior milestone events, preventing orphan payments.

E. Privacy and Access Control

Use channelized data or attribute-based access to protect personal data (beneficiaries, bank details) while publishing aggregated, non-personal views. Hash-based commitments allow verification without exposing sensitive fields.

F. Consensus and Performance

A BFT consensus (e.g., PBFT/HotStuff-style) or Proofof-Authority among n government validators provides subsecond finality in metro-scale deployments. Throughput targets (Section VII) exceed typical daily PFMS write volumes when batching is applied.

G. Security Model

Threats include collusion among a minority of validators, key compromises, and interface-layer attacks. Mitigations: HSM-backed keys, threshold signatures, periodic state anchoring to a public chain (optional), and strong operational monitoring. Surrounding systems (IAM, dashboards) undergo security hardening and red-team testing.

VI. IMPLEMENTATION ROADMAP

Phase 1 (0–6 months): Welfare-scheme pilot (e.g., scholarships, pensions). Integrate with PFMS payment references; build minimal dashboard and APIs; define data standards and consent notices per DPDP [10].

Phase 2 (6–18 months): Extend to procurement-heavy works (roads, buildings). Add verifiable geospatial proofs and automated milestone checks; onboard state audit nodes.

Phase 3 (18–36 months): Scale to multi-state consortium; publish open analytics; formalize independent oversight participation.

VII. EVALUATION FRAMEWORK AND KPIS

We propose the following measurable outcomes:

• Traceability Coverage: % of payments with complete sanction→payment lineage.

• Disclosure Latency: median minutes from event to public dashboard update.

• Edit Detectability: time to detect and prove unauthorized mutation attempts in upstream systems.

• Procurement Integrity: reduction in single-bid tenders and unauthorized change orders.

• Recovery and Deterrence: amount recovered/averted due to real-time flags; trend in fraud typologies moving offledger.

• System Throughput: transactions/sec at peak (target: ≥ 500 transactions/sec with batching), end-to-end latency < 2s for typical transactions.

VIII. CASE STUDIES

A. India

Land Records. Andhra Pradesh announced state-level blockchain initiatives and hosted pilots for land/vehicle records [6]; Telangana published a government blockchain framework outlining e-governance use cases [5]. Assam's Revenue Department pursued geospatially anchored land records modernization [8], and Karnataka piloted blockchain in immovable property registration to curb fraud and speed up registration [7].

Credentials. CBSE issues student marksheets/certificates with blockchain-backed verification to prevent forgery, demonstrating production-grade credentialing [4].

Trust Network. Tamil Nadu's Nambikkai Inaiyam (Trust Network) defines a verifiable digital credentials wallet and trust framework for government document verification [9].

IX. POLICY AND LEGAL CONSIDERATIONS

DPDP Act, 2023: Personal data must be processed lawfully, with informed consent or legitimate use, role-based access, and data minimization [10]. Until rules are fully notified and operative, projects should adopt DPDP-aligned best practices by default to avoid rework.

Open Data and Evidence: Public dashboards publish nonpersonal aggregates and content-addressable proofs (hashes). Where appropriate, raw artifacts are released under open licenses once personal data is redacted. Cryptographic logs aid forensic audit and legal admissibility when combined with proper chain-of-custody.

Inclusion and Accessibility: Interfaces must support Indian languages, low-bandwidth modes, and assisted access in Common Service Centres to avoid deepening the digital divide.

X. RISKS AND MITIGATIONS

Political Economy. Transparency threatens vested interests—mitigate via phased rollouts, public commitments, and independent oversight nodes.

Legacy Integration. Build adapters for existing MIS/ERP; adopt canonical data schemas and rigorous data-quality gates. Cybersecurity. Harden keys and endpoints; adopt zerotrust networking; continuous monitoring; periodic third-party audits.

Privacy. Publish only necessary fields; use differential privacy where needed; default to least privilege.

XI. LIMITATIONS

Blockchain is not a silver bullet; it deters and exposes certain classes of corruption (record tampering, invisible edits, orphan payments) but cannot by itself stop off-ledger collusion (e.g., coercion, cash bribes). Strong institutions and enforcement remain essential.

XII. COST–BENEFIT ANALYSIS

Implementing a consortium blockchain for public financial management requires upfront investment, but the long-term benefits significantly outweigh the costs.

A. Estimated Costs

Initial expenditure would include infrastructure, integration, and training:

• Blockchain Infrastructure Setup: Hardware, validator nodes, secure data centres (Rs. 150–200 crore for national rollout) [3].

• Integration: Adapters for PFMS, state ERPs, welfare schemes, and procurement systems (Rs. 100 crore).

• Capacity Building: Training auditors, finance officers, and technical staff (Rs. 50 crore). Total estimated initial cost: Rs. 300–350 crore.

B. Potential Benefits

Corruption and leakages in welfare schemes alone are estimated to cause losses of Rs. 1.7 lakh crore annually (approx. 2% of GDP) [11]. Studies of Direct Benefit Transfers (DBT) suggest blockchain could reduce leakages by 20–25%, translating to savings of Rs. 30,000–40,000 crore per year in central schemes alone [12]. Beyond direct savings:

• Improved Audit Efficiency: Automated, immutable trails reduce auditing costs by 30–40%.

• Faster Disbursement: Fund release timelines shrink from months to weeks, accelerating project completion.

• Citizen Trust: Transparent dashboards rebuild confidence in governance, reducing RTI queries and litigation.

C. Return on Investment (ROI)

With annual savings in the tens of thousands of crores, the system pays for itself within the first year of full deployment. Over a 5-year horizon, ROI is conservatively estimated at 10x–15x, even before accounting for social benefits such as improved service delivery and reduced public dissatisfaction.

D. Implication

The analysis highlights that blockchain is not only a technological upgrade but an economic necessity. By spending a few hundred crores on implementation, India can save lakhs of crores in the long term, making corruption economically infeasible and transparent governance sustainable.

XIII. CONCLUSION

A consortium blockchain integrated with PFMS can materially improve integrity, timeliness, and citizen observability of public spending. By recording sanctions, milestones, and payments as immutable, jointly controlled events and surfacing them via open dashboards, governments can make misuse harder and accountability immediate. With careful governance, DPDP-aligned privacy, and phased implementation, India can set a benchmark for trustworthy public finance.

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AI + Blockchain for Transparent Supply Chains: An India-First Blueprint for Trust, Traceability, and Efficiency

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Abstract—India's supply chains—spanning agriculture, pharmaceuticals, fast-moving consumer goods (FMCG), and manufacturing—are the backbone of both domestic and global trade. Yet, these systems are plagued by recurring challenges: counterfeit products, fragmented records across stakeholders, cold-chain inefficiencies, and delayed incident responses. Such weaknesses not only reduce economic productivity but also harm public trust, compromise safety, and undermine India's global export credibility. This paper proposes an integrated architecture that couples permissioned blockchain with artificial intelligence (AI) to create transparent, resilient, and auditable supply chains. Blockchain provides tamper-evident, cross-organization traceability and programmable smart contracts for dispute resolution and conditional payments. AI complements this by enabling demand forecasting, anomaly detection, logistics optimization, and quality monitoring. Together, these technologies form a system of "shared truth" and "shared intelligence" that improves trust, efficiency, and accountability. Aligned with India's policy initiatives such as Digital India, ONDC, and the Digital Personal Data Protection (DPDP) Act, the proposed model ensures privacy-preserving data sharing and scalable governance. This work highlights measurable Key Performance Indicators (KPIs) such as recall speed, counterfeit detection rates, and producer realization margins. Case examples from food, pharma, and cold chains are presented, alongside a phased roadmap for pilot deployment and scale-up. Ultimately, the paper positions India to transition from "Made in India" to "Trusted in India, Trusted by the World," demonstrating how AI and blockchain together can redefine supply chain transparency for the modern era.

Index Terms—Blockchain, Artificial Intelligence, Supply Chain, India, Traceability, Digital India, ONDC, Counterfeit Mitigation, Cold Chain, Smart Contracts

I. INTRODUCTION

Global supply chains are becoming increasingly complex, spanning thousands of miles, numerous intermediaries, and multiple data systems. India, as one of the world's largest producers of food, pharmaceuticals, and textiles, faces unique challenges in maintaining trust and efficiency across such networks. Counterfeit drugs, adulterated food, and information silos are common pain points, often resulting in loss of consumer confidence and economic inefficiency. Traditional solutions such as barcodes, centralized databases, and manual audits are insufficient in today's globalized context. These methods are prone to forgery, lack cross-organizational consensus, and often delay critical interventions such as recalls. Thesis. By integrating blockchain's immutable provenance records with AI's predictive and analytical capabilities, supply chains can evolve into intelligent, transparent ecosystems that foster trust, reduce risk, and improve overall resilience.

II. BACKGROUND AND RELATED WORK

Recent industry pilots have shown the potential of emerging technologies in addressing supply chain weaknesses. For instance, Walmart's blockchain-enabled food traceability reduced provenance lookup times from seven days to under three seconds. Similarly, IBM Food Trust demonstrated how immutable ledgers can reduce food waste through faster recalls. AI is already widely used in logistics for demand forecasting, route optimization, and fraud detection. However, AI's reliability depends heavily on the quality and integrity of underlying data. When combined with blockchain's immutable records, AI models gain trusted inputs, resulting in more accurate and auditable decisions. Indian initiatives such as Digital India, ONDC (Open Network for Digital Commerce), and blockchain-based document verification by CBSE and state governments highlight readiness for national-scale adoption. These efforts form a strong foundation for deploying AI+Blockchain-powered supply chains.

III. PROBLEM STATEMENT

The Indian supply chain landscape faces systemic challenges:

• Counterfeit Goods: Particularly in pharma, fake medicines infiltrate supply chains, endangering lives.

• Fragmented Records: Stakeholders maintain isolated databases, making cross-verification slow and inconsistent.

• Cold-Chain Failures: Temperature fluctuations go undetected, leading to spoilage in perishable goods.

• Lack of Auditability: Centralized systems allow undetected manipulation or backdating of records.

• Export Trust Gap: International buyers demand traceability proofs, which are rarely available end-to-end. These issues demand a solution that is tamper-proof, auditable, scalable, and aligned with both local and global regulatory requirements.

IV. PROPOSED ARCHITECTURE: SHARED TRUTH + SHARED INTELLIGENCE

A. Design Principles

Traceability, Minimal Disclosure, Interoperability, Auditability, and Incentive Compatibility. Every stakeholder—from farmer to retailer—must see direct value.

B. Layered View

• Edge/IoT: Temperature, humidity, and GPS sensors with cryptographic device IDs.

• Data Integrity Layer: Hashing, digital signatures, and content-addressable storage.

• Ledger: Permissioned blockchain with role-based channels and smart contracts.

• AI Services: Forecasting, anomaly detection, risk scoring, and ETA prediction.

• Access Layer: APIs, auditor dashboards, and consumer QR codes for selective disclosure

V. AI COMPONENTS

Demand Forecasting: Hierarchical models reduce stockouts and waste. Anomaly Detection: Sensor and logistics data highlight fraud and tampering. Logistics Optimization: AIdriven ETA predictions and reinforcement learning for multistop routes. Quality Monitoring: Sensor fusion models create stability indices, anchored on-chain for auditability

VI. PRIVACY, GOVERNANCE, AND INTEROPERABILITY

The architecture aligns with India's DPDP Act (2023) by minimizing personal data on-chain, enforcing role-based access, and enabling selective disclosure proofs. ONDC integration ensures open standards and avoids vendor lock-in. Governance relies on multi-stakeholder validator sets, auditor nodes, and periodic anchoring to public chains for additional finality

VII. INDIA-CENTRIC IMPLEMENTATION ROADMAP

Phase 1 (0–9 months): Pilot corridors in milk or mango supply chains. Phase 2 (9–18 months): Expand SKUs and integrate AI forecasting. Phase 3 (18+ months): Global trade integration and sustainability dashboards.

XI. LIMITATIONS AND FUTURE WORK

Blockchain cannot prevent off-ledger fraud at first-mile capture. Future research includes zero-knowledge proofs for privacy-preserving audits and tokenized incentives for smallholder farmers.

XII. CONCLUSION

AI and blockchain together can transform India's supply chains into transparent, auditable, and resilient systems. By combining blockchain's tamper-proof records with AI's forecasting and anomaly detection, organizations can improve trust, efficiency, and decision-making across the ecosystem. This India-first approach, aligned with national initiatives like Digital India and ONDC, ensures that adoption is both scalable and privacy-preserving. Early pilots in agriculture, pharma, and logistics already show measurable improvements in traceability, faster recalls, and better producer realization. By scaling these solutions, India can strengthen domestic supply chains and enhance its international credibility—moving towards a future where Indian goods are not only made competitively but also trusted globally.

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AI-Integrated Smart Contracts: Enhancing Security, Compliance, and Resilience in Scalable Blockchain Applications

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Abstract— In this paper we reviewed over twenty studies on Smart Contracts in AI applications, with an emphasis on the areas of finance, governance of digital assets and security in drone networks. The outcome of this survey provided us with a set of three general problem statements that emerged as relative commonalities in each study:(i) Fraud detection and privacy- preserving analytics in decentralized finance (DeFi); (ii) Enforcement of regulation compliance such as GDPR in blockchain digital asset management. To address these problem statements, we suggest integrated solutions based on AI analytical framework developed with blockchain enabled smart contracts. In summary, we determined: the combination of Fraud Detection and DeFi processes guided by Federated Learning with cryptographic contracts to substantiate fraud detection in DeFi; GDPR aware smart contracts with decentralized storage solutions such as IPFS to preserve security around digital assets; and AI-supported anomaly detection with lightweight consensus processes to enhance the resilience of drone networks. Altogether these solutions illustrate the potential for AI-enabled smart contracts to support security, transparency, accountability and scalability across a range of application domains.

Keywords— Smart Contracts, Artificial Intelligence, Blockchain, Decentralized Finance (DeFi), Fraud Detection, GDPR Compliance, Digital Asset Management,Federated Learning, IPFS, Anomaly Detection

I. INTRODUCTION

The pace at which artificial intelligence (AI) and blockchain are converging presents new avenues for creating secure, transparent, and autonomous systems. Smart contracts, a form of blockchain technology, allow for trustless execution of agreements and automation without intermediaries. AI also enhances smart contracts which do not just represent transaction processing, but the next step level of intelligence applied to data via decision-making, predictive analytics, and adaptive security mechanisms. The rise of AI-enhanced smart contracts has garnered interest from a wide range of areas including finance, digital asset management, Internet of things (IoT), and cyber-physical systems. Despite the growing promise of AI-enhanced smart contracts, numerous challenges remain. The many challenges include ambiguity over fraud detection related to decentralized finance (DeFi), and compliance with legislated frameworks such as General Data Protection Regulation (GDPR) located to the governance of digital asset. Each one of these challenges can only be addressed through the AI model's abilities to provide learning, anomaly detection, and prediction along with the shared security affords by blockchain by virtue of it being immutable, decentralised and transparent. In this paper we review more than twenty papers in the academic field on AI integrated smart contracts, and identify significant methodologies and research gaps. In reviewing these studies, we define similar problem statements vs solutions and identify potential solutions based on several viable approaches -

federated learning, GDPR aware contracts, decentralised storage and lightweight consensus. These approaches highlight the latent potential of AI.

II. RELATED WORKS

We like to begin with a broad selection of relevant research contributions to familiarize ourselves with the current developments within the area of blockchain smart contracts and artificial intelligence (AI). Our study selection criteria encompass conference papers, journal articles, and survey papers from established publishers (IEEE, Springer, Elsevier, and ACM), allowing us to consider various perspectives in this multidisciplinary area. A. Conference Papers Conference papers represent the latest developments relating to the application of blockchain and AI across domains (e.g., DeFi, cybersecurity, IoT automation). Numerous IEEE and Springer conferences presented frameworks that embed deep learning models into smart contracts to utilize in fraud detection, anomaly detection, and autonomous decisionmaking for decentralized systems. B. Journal Articles Established journals appearing in major databases (e.g., IEEE, Elsevier, Springer) included detailed studies on integrating AI with the cryptography of blockchain, as well as secure smart contracts.

These papers discussed scalability, regulatory compliance, and transparency for systems based on blockchain, namely DeFi systems and healthcare systems, while wanting to demonstrate why deep learning algorithms enhanced predictive accuracy and smart governance on blockchain systems. C. Survey Papers Survey papers in ACM Computing Surveys, IEEE Access, and Elsevier, provided a summary of trending directions, for example, federated learning along with blockchain, AIenabled privacy frameworks, and advanced cryptography. Furthermore, the review papers illustrated limitations in research

S.No	Article / Paper Topic	Authors	Year	Proposed Methodology
1	Survey on BlockchainBased Smart Contracts: Technical Aspects and Future Research	Alharby et al.	2019	Survey of smart contracts on security, privacy, gas cost, concurrency; explores integration with AI and game theory.
2	Verification and Validation for Data Marketplaces via a Blockchain and Smart Contracts	Wörner et al.	2020	Proposes a hierarchical blockchain + AI model (Bronze/Silver/Gold layers) for data verification in smart cities.
3	Automated Cybersecurity Compliance and Threat Response Using AI, Blockchain, and Smart Contracts	Kaur et al.	2020	Framework using AI for threat intelligence, blockchain for immutable logs, and smart contracts for auto compliance

4	Making Smart Contracts Predict and Scale	Ghosh et al.	2021	Embeds machine learning (Naive Bayes) into Ethereum contracts, floating-point limitations.
5	Securing Internet of Drones Networks Using AIEnvisioned SmartContractBased Blockchain	Bera et al	2021	AI + blockchain framework for drone network security with PBFT consensus and smart-contract-based access control.
6	Trusted AI with Blockchain to Empower Metaverse	Dwivedi et al.	2022	Framework for trust and transparency in Metaverse using AI + blockchain with smart contracts for identity and governance.
7	Smart Contract Privacy Protection Using AI in CyberPhysical Systems	Yang et al.	2021	AI-enhanced smart contracts with homomorphic encryption and federated learning to preserve privacy in CPS/IoT.
8	Smart Contracts and Machine Learning: Exploring Blockchain and AI in Fintech	Chen et al.	2020	Reviews blockchain + smart contracts + ML applications in fintech for fraud detection and automated compliance.
9	Artificial Intelligence, Smart Contract and Islamic Finance	Khan et al	2021	AI for predictive analytics and smart contracts for Shariah-compliant financial automation in Islamic banking.
10	Combating Deepfake Videos Using Blockchain and Smart Contracts	Victor et al	2019	Ethereum smart contracts + IPFS to ensure authenticity and provenance of multimedia against deepfakes

III. PROPOSED METHODOLOGY

To solve the challenges of data security, privacy, regulatory compliance, and the need for real-time threat detection in emerging digital ecosystems, we propose a unified AIintegrated blockchain framework. This framework integrates the benefits of privacy-preserving federated learning, GDPRaware smart contracts, and privacy-preserving AI-based IoT security models, to address the data security and compliance requirements of organizations across sectors that are increasingly reliant on technology systems. By incorporating optimized consensus alternatives and decentralized storage options, this framework can deliver rapid scalability, fraudresistant, and tamper-proof data management that is adaptable to numerous sectors, including decentralization, digital asset governance, and drone IoT networks. The article papers considered for this study are listed below:

S.No	Article Topic	Year	Proposed Methodology
1	DeFiSentinel: AIEnhanced Decentralized Finance Architecture With Advanced Cryptographic Smart Contracts for Data Integrity and Threat Resilience (Author: Seyed Mani Rahnama)	2022	Combines Federated Learning (FL) for privacy-preserving AI training, blockchain for immutability, and cryptographic smart contracts for secure DeFi transactions. Demonstrated strong fraud detection precision & recall.
2	SPChain: A Smart and Private Blockchain-Enabled Framework for Combining GDPRCompliant Digital Assets Management With AI Models (Wei- Shan Lee et al.)	2022	A GDPR-compliant frameworkcombining blockchain + AI. Uses decentralized IPFS storage to avoid SPOF and smart contracts to invoke AI models for NFTs/digital assets. Evaluated in Hyperledger Fabricfor latency and throughput.
3	Securing Internet of Drones Networks Using AI-Envisioned Smart-ContractBased Blockchain (Bera et al.)	2021	A blockchain + AI security frameworkfor drones. Smart contracts manage authentication, access control, and aggregation. Blockchain ensures tamper-proof logs. AI does analytics. Uses PBFT consensus.

IV. COMMON DISCUSSION

Together, the three research articles illustrate the increasing applications of AI-based blockchain solutions to address modern challenges associated with security, privacy, compliance, and scalability. While the three articles leveraged different domains (DeFi fraud detection, GDPR compliant digital asset management, and IoT-based drone network security), a common theme was using privacy-preserving AI models, smart contract (or agnostic contract) automation, and resource-efficient or lightweight systems that produce and save energy. The articles propose combining Federated Learning (decentralized and opinionated AI training), GDPR-compliant smart contracts (for observance and regulations), and AIdriven anomaly detection with lightweight consensus for IoT resilience. Each article moves towards something unified in nature. Each indicates that blockchain can serve as a trusted backbone, and AI can add intelligence to enforce fraud prevention, compliance, and provide real-time intrusion detection. In conclusion, the common discourse recognized that using AI and embedding it with blockchain can strengthen security and privacy across several domains, regulatory compliance, resilient systems, and scalability, and will remain an adaptable solution for any future digital ecosystem across multiple issues and domains.

V. CONCLUSION

From the literature of these three contributions, it is clear that AI-supported blockchain environments can be a powerful avenue for tackling the challenges of fraud detection, data privacy, regulatory compliance, and IoT security. Although the papers emphasized a specific domain, they provide insight into the prospect of bridging federated learning, regulatory compliant smart contracts, and low-cost consensus mechanisms to an integrated solution that provides security, trust, scalability, security, and adaptability across a range of applications - like decentralized finance, digital asset management, and IoT networks.

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AI VIRTUAL MOUSE USING PYTHON

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Abstract:

The suggested AI-based Virtual Mouse technology shows how hand gesture detection can successfully take the role of conventional hardware for interacting with computers. The system supports real-time cursor control, clicking, and drag actions with adequate precision and responsiveness by integrating Python, MediaPipe hand tracking, and the PyAutoGUI module. In addition to improving accessibility for people with to improving accessibility for people withIn order to facilitate hands-free human–computer interaction, this study suggests an AI-based Virtual Mouse system created with Python. The device uses a webcam to record live video and uses MediaPipe hand tracking and computer vision algorithms to identify hand landmarks. The PyAutoGUI library maps some mouse events, such as pointer movement, left and right clicks, and drag operations, to certain movements, such pinching and fingertip movement. Through the use of coordinate mapping and noise reduction, the design prioritizes accuracy, low latency, and smooth control. The suggested solution provides an alternate input method for people with physical limitations, improves accessibility, and improves hygiene in public usage scenarios by doing away with the necessity for real hardware devices. Results from experiments show that it performs responsively on common computing tasks, confirming its suitability for interactive applications, presentations, and teaching. Multi-gesture customization and interaction with multimodal interfaces for wider usability are examples of future improvements.

Keyword-Artificial Intelligence, Virtual Mouse, Hand Gesture Recognition, Human Computer Interation.

I.Introduction

The suggested AI-based Virtual Mouse technology shows how hand gesture detection can successfully take the role of conventional hardware for interacting with computers. The system supports real-time cursor control, clicking, and drag actions with adequate precision and responsiveness by integrating Python, MediaPipe hand tracking, and the PyAutoGUI module. In addition to improving accessibility forpeoplewith to improving accessibility for people withKeyboards and mice are examples of the physical devices that have historically been used in human–computer interaction (HCI). Despite their demonstrated efficacy, these devices have drawbacks with regard to portability, accessibility, and hygiene, particularly in communal or public settings. Gesture-based interfaces driven by AI and computer vision have emerged in response to the increasing need for intelligent, contactless, and natural interaction techniques**.** The AI Virtual Mouse is a cutting-edge method that lets users operate computers with basic hand motions that are recorded by a webcam. The system incorporates MediaPipe hand landmark detection to track finger motions and PyAutoGUI to mimic native mouse events including clicking, dragging, and cursor movement, using Python as the implementation platform. By doing away with the requirement for external gear, this offers a practical and affordable option for a range of users, including those who have motor impairments.

This work's main goal is to create a real-time, lightweight system that can provide precise, fluid mouse control with no computing overhead. In addition to making ordinary computing more usable, the suggested solution creates new application possibilities in domains including virtual presentations, e-learning, gaming, and healthcare.

This paper's remaining sections are arranged as follows: Related research on gesture-based interface systems is reviewed in Section II. The system design and technique are explained in Section III. The analysis and outcomes of the experiment are shown in Section IV. Future research directions are discussed in Section V, which wraps up the work.

II. AI technologies used in AI VIRTUAL MOUSE:

The AI Virtual Mouse project utilizes a combination of advanced artificial intelligence and computer vision technologies to deliver real-time and precise gesture recognition capabilities. Below is an overview of the main technologies driving this innovative system:

Computer Vision:

At the core of the system lies computer vision techniques that empower the webcam to efficiently process and interpret visual data. The webcam captures image frames on the fly, enabling it to detect the positioning and movement of the hand.

Hand Landmark Detection (MediaPipe):

For accurate hand tracking and landmark detection, we leverage Google’s MediaPipe framework. This framework identifies 21 key points on the hand, including fingertips and joints, which are crucial for interpreting various gestures such as pointing, clicking, and dragging.

Gesture Recognition:

The system processes the detected landmarks to classify gestures based on the relative positions and distances of the fingers. For instance, the space between the thumb and index finger is analyzed to identify click actions. This streamlined, AI-powered gesture recognition delivers high precision without needing extensive deep learning models.

Coordinate Mapping and Smoothing:

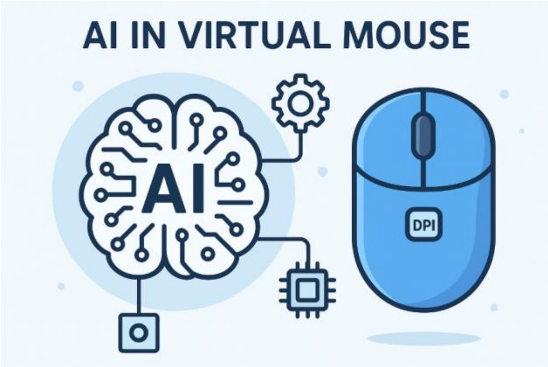
Mathematical mapping techniques are applied to translate hand positions from the camera's coordinate system to the display's resolution. Furthermore, smoothing algorithms are implemented to minimize jitter and ensure smooth cursor movement.

Automation Library (PyAutoGUI):

PyAutoGUI plays a vital role in simulating native mouse functions, including cursor movement, left and right clicks, and drag operations. This integration seamlessly connects gesture recognition with effective system control

III. AI in virtual mouse:

Artificial Intelligence (AI) is revolutionizing the way we interact with computers, shifting from traditional methods to innovative, touchless, and adaptable systems. One striking example is the Virtual Mouse, where AI allows computers to understand human gestures and convert them into control commands—all without needing physical hardware. In a Virtual Mouse system, AI techniques analyze real-time video feeds to detect a human hand and pinpoint key features that indicate finger positions. Utilizing machine learning-based hand tracking models, like those from MediaPipe, ensures precise detection of hand joints, even amid challenging lighting and backgrounds. These recognized features are then interpreted into specific gestures that correspond to mouse functions, such as moving the cursor, clicking, and performing drag-and-drop actions. AI enhances the reliability and adaptability of gesture recognition. By employing filtering techniques and coordinate mapping, the system minimizes disruptions and accommodates different hand sizes, camera angles, and variations in user movement speed. Consequently, the AI-driven virtual mouse offers a seamless, hygienic, and user-friendly interface for everyday tasks, presentations, e-learning, and assistive technologies



IV. AI APPLICATION IN VIRTUAL MOUSE:

The incorporation of Artificial Intelligence into the Virtual Mouse system broadens its applications well beyond traditional computing. By facilitating gesture-based interactions, this technology can be utilized across various sectors, as highlighted below:

Accessibility and Assistive Technology:

The virtual mouse serves as an alternative input method for those with motor impairments, empowering them to navigate computers without the need for physical devices Hygienic and Contactless Interaction: In settings like kiosks, healthcare facilities, and ATMs, touchless control significantly reduces the risk of transmitting infections, enhancing safety for users.

Education and E-Learning:

Educators and presenters can leverage hand gestures to navigate slides or software during online classes and seminars, enabling a seamless and interactive delivery without relying on a physical mouse.

Business and Presentations:

Professionals can use intuitive hand gestures to manage slides, control documents, and showcase applications during meetings or conferences.

Gaming and Entertainment:

Gesture-based control opens up exciting new avenues for immersive gaming and multimedia experiences, providing a captivating and modern interaction format.

Smart Environments:

The integration of virtual mouse technology with smart devices facilitates gesture-based control of home automation systems, allowing users to easily manage appliances and IoT devices.

V.PYTHON IN AI VIRTUAL MOUSE:

Python is essential in developing the AI Virtual Mouse, thanks to its user-friendly nature, adaptability, and robust library support tailored for artificial intelligence and computer vision projects. As a leading choice in AI research and development, Python serves as an effective platform for quick prototyping and implementing gesture-based systems in real time.

Image Processing and Computer Vision:
With libraries like OpenCV, we can capture and process video in real time, which is vital for identifying and following hand movements.

Hand Tracking and Landmark Detection:
The MediaPipe framework, which works seamlessly with Python, is used to pinpoint 21 essential landmarks on the hand, laying the groundwork for gesture recognition.

Gesture-to-Action Mapping:
Python’s strong mathematical and data processing features enable precise mapping of hand gestures to specific mouse actions—such as moving the cursor, left-clicking, right-clicking, and dragging.

Mouse Event Simulation:
Leveraging the PyAutoGUI library, Python can simulate native mouse events, allowing the system to operate within the operating system and various applications just like a traditional mouse.

System Integration and Flexibility
Thanks to Python’s modular design and cross-platform compatibility, it's well-suited for developing a versatile system that can easily adapt to different hardware and operating systems with little need for adjustments.

VI. Proposed System:

Below is a comprehensive Proposed System for an AI Virtual Mouse project, ready for presentation in a project report or as a response to a 50-mark question.

This outline includes the project's objectives, high-level architecture, system modules, algorithms, datasets, evaluation methods, implementation specifics, considerations for privacy and ethics, testing strategies, and future expansion possibilities.

1. Objectives:

- Develop a dependable virtual mouse that can be controlled through hand gestures, with optional integration of eye tracking and voice commands.
- Ensure consistent high accuracy across various lighting conditions and diverse user profiles.
- Prioritize user privacy through edge processing techniques and enhance accessibility for users with disabilities.
- Create a modular design that facilitates seamless integration with OS-level mouse interactions.

2. System Modules:

2.1 Input & Hardware

Utilize a webcam (minimum 30 FPS, 640x480 resolution) with the option of integrating a depth camera, such as Intel RealSense, for enhanced robustness. Recommended computing platforms include laptops or smartphones equipped with GPU support for real-time deep learning applications, although edge-optimized CPU variants can be utilized if necessary.

2.2 Frame Capture

Establish a capture loop that maintains a consistent frame rate. Incorporate buffering and timestamping processes to accurately measure latency.

2.3 Preprocessing

Implement resizing and color normalization techniques. Employ background subtraction or segmentation methods (e.g., basic chroma keying or semantic segmentation). Apply region of interest (ROI) cropping to optimize computational efficiency.This structure provides a clear and thorough approach to designing an AI Virtual Mouse project, ensuring all essential components are thoughtfully addressed.

2.4 Detection & Landmark Extraction

To achieve real-time hand detection and landmark extraction (such as fingertips and wrist positions), consider utilizing a lightweight model. Potential solutions include MediaPipe Hands, a tiny-YOLO combined with a keypoint regressor, or a custom CNN with heatmaps

VII. Datasets & Training

Utilize publicly available datasets for hand landmarks and gestures, including MediaPipe sample data, EgoHands, and the HandGestureDataset. Enhance the dataset by incorporating variations in lighting, skin tones, backgrounds, occlusions, and hand sizes. Apply synthetic augmentation techniques such as rotations, scaling, blur, brightness adjustments, and occlusions.

Testing & Validation

Develop unit tests for individual modules and integration tests for the overall pipeline.Conduct real-user trials with a diverse group of volunteers and collect qualitative feedback.Benchmark performance against a physical mouse in pointing tasks focusing on time and accuracy.Perform stress tests under conditions such as low light, multiple hands, and fast movements.

VIII.CONCLUSION :

The AI Virtual Mouse system showcases the impressive potential of computer vision and machine learning to transform traditional hardware into flexible, innovative alternatives. By harnessing Python along with powerful libraries like OpenCV and Mediapipe, this project facilitates intuitive hand gesture recognition for tasks such as cursor control, clicking, and scrolling. Not only does this approach offer a hands-free, touch-free interaction experience, but it also underscores the promising future of AI in enhancing accessible human-computer interactions. Moving forward, we can look forward to improvements like more accurate gesture detection, support for multiple hands, and the integration of this technology with other AI-driven applications, expanding its usability even further.

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INTRODUCTION TO DATA SECURITY AND PRIVACY ON THE BLOCK CHAIN

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Abstract: Blockchain has emerged as one of the most dynamic innovations in applied cryptography, driving widespread interest across industries. However, for it to achieve its full potential, key challenges in security and privacy must be addressed. The first IEEE Privacy and Security on Blockchain Workshop (IEEE S&B) highlighted these issues, bringing together experts from academia and industry to examine topics ranging from new cryptographic applications in Bitcoin to privacy-preserving data storage solutions. Findings from discussions and participant insights reveal that blockchain, while promising, is not yet viewed as a complete replacement for existing security systems. Its immutability and resistance to tampering make it valuable within information security, but limitations around scalability, privacy, and trust remain pressing concerns. If these gaps are resolved, blockchain could evolve into a powerful tool for safeguarding data and enhancing privacy. The technology's applications are rapidly expanding, covering sectors such as finance, healthcare, automotive systems, IoT, risk management, and public services. While many studies have explored blockchain's structure and usage, a comprehensive survey combining its technical foundations, consensus mechanisms, and privacy challenges is still lacking. This paper attempts to provide such an overview, emphasising both the opportunities and unresolved issues that blockchain presents. Moreover, it points toward future trends and adaptations that could strengthen its role in securing information systems. Ultimately, blockchain's success will depend on continuous collaboration between researchers and industry to overcome its current shortcomings.

Keywords: *Block chain technology, cryptography, consensus algorithm, data privacy and Security, financial sector, healthcare, Internet on things(IOT).*

I.Introduction

In today's digital world, block chain cyber security is becoming increasingly important. The threat of cyber attacks has risen significantly as our dependence on technology and the internet has expanded. As a result, different solutions, including the usage of block chain technology, have been created to assist in securing against these vulnerabilities. Block chain technology initially emerged primarily as the underpinning technology for Bit coin, but its potential uses have now moved well beyond the domain of cryptocurrency. A particular field where blockchain technology is rapidly being used is to improve cybersecurity [1].

This technology has certain characteristics that have made it well-suited for data security and cyber attack prevention. For example, because blockchain is decentralized, it cannot be regulated by a single body, making it less susceptible to assaults. Furthermore, because of the use of cryptographic algorithms and digital signatures, data kept on a blockchain is very safe and tamper-proof[2].

This study examines the diverse applications of blockchain in enhancing data protection and mitigating cyber threats, with particular emphasis on identity management, secure data storage, and protected communication channels. In addition, it explores the key challenges associated with implementing blockchain in cybersecurity and discusses potential strategies to address these limitations. The research contributions of this article are as follows.

• This paper presents basic blockchain knowledge, including blockchain's classification and consensus mechanisms.

• This paper conducts a comprehensive review and analysis of block chain in terms of privacy protection, security supervision and data exchange. It also summarises the existing solutions and the remaining deficiencies. Through a comparative analysis of the literature, the current situation is made more intuitive, which is helpful for us to explore block chain's future development direction.

• Finally, some directions and problems to be solved in future research on block chain are proposed. This contributes to better research and application of block chain.

Blockchain technology

Blockchain is the latest advanced technology which exploits a decentralized network and can be used in various industrial sectors which is highly reliable, cost effective, and fast. Also available for financial services, retail, insurance, coordination, supply chain, and the public sector which is now starting to invest in Block chain for their business growth. For example, the use of Block chain to prevent double spending in financial transactions without the need for a trusted authority or central server. It is a decentralized ledger platform that intelligently facilitates verifiable transactions between parties both individuals and organisations. Smart contracts and crypto currencies are the main applications of Block chain. Smart contracts are self-executing contracts. Every transaction on the Block chain is communicated securely using cryptographic mechanisms such as digital signatures, public and private keys, and hashing algorithms. Here, the privacy of the system is rendered tamper-proof, which makes it very difficult to make changes in the block chain architecture. These transactions are immutable and highly secure blocks. Miners must perform intensive computations to complete the hash function in block transactions. Miners are rewarded in the form of digital currency for building successive blocks on the network. These blocks are added sequentially to the existing chain, and each user will receive the updated blocks using a peer-to-peer network[5].

Enterprise block chain requires consensus, source, immutability, and finality. properties in executing smart contracts between various parties in the business chain. As a result, companies can implement new business models with lower costs, time, and risk.

With the high growth of medical device providers, many new medical device products are equipped with the latest technology.

The purpose of this SLR research is to answer questions regarding the research question:

1) What has been the trend of implementing block chain technology in data sharing and security over the last four years?

2) What industries have implemented block chain technology, especially for data sharing and security?

3) How many publishers and researchers are researching the application of block chain technology for data sharing and security?

4) What types of block chain technology are often used for data sharing and security?

II.Literature Review

A. Blockchain Research Challenges

A blockchain can be understood as a cryptographically verifiable sequence of data entries. Traditional databases, though widely used, lack built-in cryptographic guarantees of integrity, which makes them vulnerable in hostile environments. Since the Snowden revelations, it has become evident that most databases operate under adversarial conditions. This is one reason blockchain technology has generated significant excitement—unlike conventional systems, it offers integrity and security by design[3]. Beyond integrity, blockchain's real breakthrough lies in decentralisation. As seen in Bitcoin, it enables the creation of distributed public ledgers where all participants share identical records. While this is valuable for use cases like digital currency, it can also raise privacy concerns in certain applications. Thus, the fields of blockchain security and privacy demand much deeper research. Although cryptographic tools and distributed systems have existed since the 1970s, they were first combined effectively in 2008 through Satoshi Nakamoto's paper on Bitcoin, which introduced a peer-to-peer electronic cash system. Bitcoin's open-source model allowed millions to transfer funds without banks, while its decentralized design ensured resistance to censorship. However, its architecture still struggles with anonymity, as many users mistakenly assume transactions are fully private. Because the security and privacy foundations of blockchain were never formally defined at its inception, researchers are still working to establish these principles. New blockchain projects continue to emerge, often making bold claims, which underlines the need for academic scrutiny. A strong partnership between researchers and developers is essential academia can provide rigorous solutions, while practitioners supply real-world challenges to refine the technology[4].

B. Improvements to Core Cryptographic Primitives

Blockchain technologies today are largely shaped by industry rather than academia, giving the field a unique direction compared to other areas of research. Unlike systems built on formal cryptographic proofs, Bitcoin and its variants were created by practitioners who emphasise practical resistance to attacks based on experience and community knowledge. This approach has led to constant shifts in proposed solutions and a lack of unified standards, resulting in numerous competing designs, each claiming superiority[7]. Stability in the field appears distant, especially with experimental models such as side-chains. A crucial step for advancing blockchain research is establishing a shared vocabulary around key security and privacy properties. Once terms are clearly defined, they can be given formal meaning, enabling rigorous evaluation of whether systems meet specific requirements. Even the term "blockchain" itself is debated—does it apply only to Bitcoin, to any decentralized ledger like Ethereum, or even to non-mining platforms such as Hyperledger? Some argue that blockchain should refer to any linked list secured by cryptographic hashes, while others see it as tied to Bitcoin's specific design. Future alternatives may replace the blockchain structure with more efficient data systems or prioritise privacy over public ledger functionality[5]. Greater clarity is also needed for concepts like side-chains, public ledgers, proof of work, and proof of stake, since their use varies widely across studies. One area of heavy criticism is Bitcoin's reliance on proof of work to counter Sybil attacks, where malicious actors flood a network with false identities. While hashing as proof of work has allowed open participation so far, it is energy-intensive and imperfect. Proposals such as "Proof of Personhood" aim to re-democratise permissionless cryptocurrencies by ensuring new identities can join without enabling censorship or centralisation. As the field evolves, it is clear that high-integrity distributed data stores will persist, but new methods for scaling blockchain securely and fairly are urgently required[6].

C. Consensus Algorithms

Consensus is a protocol that enables a group to reach a common agreement dynamically, ensuring that decisions benefit the collective rather than just reflecting a majority vote. Unlike simple majority voting, consensus seeks to align the group as a whole toward a unified outcome. Achieving this coordination, however, becomes challenging when malicious participants or faulty processes interfere[8]. For instance, a dishonest actor may deliberately send conflicting information to disrupt group harmony, preventing members from acting in sync. This issue is famously known as the "Byzantine Generals Problem" (BGP), where unreliable participants hinder the group's ability to agree. The inability to achieve consensus due to such interference is termed Byzantine fault. In 1982, Leslie Lamport and Marshall Pease demonstrated that Byzantine fault tolerance is possible only if the honest members can form a majority agreement on their chosen strategy. Modern blockchain systems address this issue using consensus algorithms that provide probabilistic solutions to the Byzantine Generals Problem. These protocols are central to blockchain security and privacy, as they allow decentralized participants to coordinate effectively even in environments where some actors may be dishonest[9]. In the following sections, we explore these consensus mechanisms in greater detail, with emphasis on how they address both security threats and privacy challenges.

Proof of Work

The consensus protocol designed by Satoshi nakamoto for the Bit coin is aimed at reaching a coordinated consensus from the network on the validity of each bit coin transaction. It bypasses the BGP by using the protocol. We characterize the with dual properties: (1) it should be difficult and time-consuming for any prove to produce a proof that meets certain requirements, and (2) it should be easy and fast for others to verify the proof in terms of its correctness. For the first property, one must design a proof of work challenge such that computing a valid proof of work is difficult with low and somewhat random probability, thus a lot of trial and error is needed. We illustrate how the proof of works in terms of BGP[10.]When the troops on the east of the city want to send a message to the west side troops, it follows the steps of the proof of work protocol:

(1) Append a "nonce" (usually start with zero to the original message, which is a random hexadecimal value;

(2) Apply hash to the nonce augmented message and check if the hashing result is less than or equal to a pre set value (say starts with five zeros); ACM Computing Survey.

(3) If the hash condition is satisfied, the troops on one side of the city will send the messenger to the troops on the other side of the city with the hash of the message and the nonce. If not, then increase the nonce by one and this process iterates until the desired result is obtained. Finding the right nonce can be time consuming and computationally expensive;

(4) Due to the collision-resistant property of the hash function, it is hard to tamper the hash of the message even if the messenger got caught, because the hash of the tampered message will be drastically different from the hash of the original message, and the generals on the west of the city can verify whether the message starts with five zeros and disregard the message if not.

Proof of Stake

Proof of Stake (PoS) is a consensus mechanism that serves as an alternative to traditional Proof of Work (PoW) protocols in public blockchains. Unlike PoW, which relies on computational puzzles and rewards to secure the network, PoS emphasises penalties and staking, creating a security model based on participants committing their own assets. In PoW systems such as Bitcoin, anyone with sufficient computing power can become a miner, validate transactions, and add new blocks. By contrast, PoS systems—like Casper—limit this role to validators who lock up a portion of their cryptocurrency as collateral. These deposits, known as stakes, qualify participants to take part in the block validation process. Validators are identified by stable network addresses, and the blockchain continuously tracks who has staked assets for participation[12]. Through consensus algorithms, validators propose and vote on new blocks, with their chances of being selected proportional to the size of their stake. Different PoS-based cryptocurrencies adopt variations of this model. For example, Next and BlackCoin introduce randomness into validator selection to reduce predictability, as stake sizes are publicly visible and could otherwise make outcomes too easy to anticipate.

Mixing

As Bit coin's blockchain does not guarantee anonymity for users, transactions use pseudonymous addresses and can be verified publicly. Thus, anyone can relate a user's transaction to her other transactions by a simple analysis of the addresses she used in making Bitcoin exchanges. More seriously, when the address of a transaction is linked to the real-world identity of a user, it may cause the leakage of all her transactions[14]. Thus, mixing services (or tumblers) was designed to prevent users' addresses from being linked. Mixing, literally, is a random exchange of user's coins with other users' coins. As a result, for the observer, their ownership of coins are obfuscated. However, these mixing services do not provide protection from coin theft. In this section, we describe two such mixing services and analyse their security and privacy properties.

Mix coin

Mix coin was proposed by Bonneau et al. in 2014, which provides anonymous payment in Bit coin and bit coin-like crypto currencies. To defend against passive adversaries, Mix coin extends the anonymity set to allow all users to mix coins simultaneously. To defend against active adversaries, Mix coin provides anonymity similar to traditional communication mixes. In addition, Mix coin uses an accountability mechanism to detect stealing, and it shows that users will use Mix coin rationally without stealing bit coins by aligning incentives[15].

Coin Shuffle

It was proposed by Tim Ruffing et al. in 2014, which further extends the Coin Join concept and increases privacy by avoiding the necessity of a trusted third-party for mixing transactions. Coin Shuffle is claimed as a completely decentralized coin-mixing protocol and has the ability to ensure security against theft. To ensure anonymity, Coin Shuffle uses a novel accountable anonymous group communication protocol, which is called Dissent[16].

III.Conclusion

Blockchain offers a powerful foundation for building decentralized, transparent, and tamper-resistant systems. However, achieving strong data privacy and robust security within these systems remains a significant challenge. While blockchain inherently protects data integrity, its transparent nature can expose sensitive information if privacy measures are not embedded at the protocol level. Many existing privacy-enhancing solutions—such as CoinJoin, Mix-coin, and CoinShuffle—offer partial anonymity but fail to deliver full protection against traceability, correlation attacks, or flawed implementations. No single solution can fully secure or anonymise blockchain systems. Technologies like secure multi-party computation (SMPC), trusted execution environments (TEE), and zero-knowledge proofs (ZKPs) each offer unique benefits but also introduce trade-offs in terms of performance, scalability, and complexity. To address these concerns, future blockchain development should focus on embedding privacy-preserving mechanisms directly into consensus protocols, rather than treating them as add-ons. Improvements should include formal verification of security properties, rigorous real-world testing, and adaptive cryptographic designs that respond to evolving threats. Additionally, establishing standardised evaluation frameworks and benchmarking tools will help in comparing solutions meaningfully across platforms. On a broader scale, greater collaboration between academia, industry, and open-source communities is needed to bridge theoretical advances with real-world deployments. Education around privacy risks, best practices, and secure implementation techniques should also be prioritised to reduce human error and flawed system design. In summary, while blockchain presents great promise for secure and private digital infrastructure, realising this potential requires moving beyond fragmented innovations. A unified, scalable, and methodically tested approach—backed by interdisciplinary collaboration and strong standards—will be critical in building truly secure, privacy-focused blockchain ecosystems for the future.

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PRIVACY AND SECURITY IN AI-BLOCKCHAIN SYNERGY: A SURVEY

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Abstract—Artificial Intelligence and Blockchain technologies are transforming how the privacy and security is handled. Blockchain is known for its decentralized approach to data security and storage, which ensures that no individual or institution has control over the data. Meanwhile, AI excels at optimizing processes and analysing large datasets. Through collaboration, they can develop advanced systems that are both intelligent and more secure, ensuring the protection of our privacy. This article argues that these two technologies can function together effectively. By forecasting possible issues in advance, AI can enhance both efficiency and safety of blockchain technology, while blockchain offers a reliable and secure way for AI to handle data. Combining their strengths allows us to establish a significantly safer and more transparent online environment for all. This paper explores the existing techniques employed to maintain data confidentiality and security within these systems, practical applications in the real world, the effects of regulations, differential privacy, and future opportunities. Our main finding is that this powerful combination has immense potential, but to build a truly secure future, we must use advanced privacy protection methods from the very start. We will also explore the real time examples of this combination is being used in sectors like healthcare and finance, among others. Essentially, AI assists in interpreting intricate data, while blockchain ensures that the data and procedures utilized by AI are secure. The integration of these two technologies enhances privacy, security, and transparency across various industries.

Keywords—Artificial Intelligence (AI), Blockchain, AI–Blockchain Synergy, Privacy, Security, Data Integrity, Decentralized Systems, Differential Privacy, Smart Contracts, Transparency, Scalability, Regulatory Compliance, Ethical AI, Trustworthy Systems, Digital Ecosystems.

I. INTRODUCTION

Digital technology is growing rapidly, but it has also created challenges in maintaining our data's privacy and security. Challenges in data-driven systems can be handled by two key technologies: Artificial Intelligence and Blockchain. Artificial Intelligence offers smart analysis, decision-making, and automation, while Blockchain ensures decentralization, transparency, and resistance to tampering. Combining these technologies also raises new problems related to safeguarding sensitive information. Protecting privacy and security in AI and blockchain systems is essential for supporting individual rights. It also helps to maintain trust, reliability, and scalability. Recent studies show that methods like data encryption, de-identification, multi-tier distributed ledgers, and K-anonymity play a vital role in keeping information safe in these ecosystems. Authorization management, access control, data protection, and network security still face challenges in the current framework. Moreover, blockchain can support decentralized decisionmaking and consensus methods. This could foster trust and cooperation among different stakeholders. This paper examines how AI and blockchain can collaborate to safeguard privacy. It discusses various ways they can be utilized, the challenges they encounter, and how they could be enhanced. This study aims to create more secure, efficient, and scalable AI and blockchain systems by summarizing privacy protection methods. The goal is to build AI and blockchain systems that are not only more secure but also more efficient and scalable. This combination can create a strong, efficient, and privacy-friendly digital system. It explains different uses of these technologies and points out the main challenges currently encountered. In conclusion, the integration of AI and Blockchain can play a vital role in enhancing the security and privacy of IoT (Internet of Things) systems. It also reviews recent methods and approaches that researchers have used to study their applications. This paper discusses how these two technologies can work together and support each other. In addition, the cooperative use of AI and blockchain opens new opportunities for smart cities, healthcare, supply chains, and other IoT-based applications. Overall, the findings show that AI and blockchain integration can help protect systems from security risks while opening new areas for research and development.

II. LITERATURE REVIEW

The mixture of artificial intelligence and blockchain is attracting the world. Smart data analysis, automate tasks, and decision-making is allowed by AI. Immutable digital ledger and clear record-keeping is enabled by Blockchain. Problem relating to data integrity can be solved by these technologies. This also secures the sensitive information and reliable auditing. This blending of AI and blockchain brings new provocation especially around the privacy of individuals and system scalability. This research has investigated various technical tactics to balance the advantages and risk of blending AI and blockchain. combining federated learning with blockchain is an important approach. The collaborative training of machine learning models that improves privacy is enabled by federated learning. Blockchain has been used in healthcare and IoT, but it faces issues caused by blockchain protocols. In addition to this many cryptographic techniques like Differential Privacy, Homomorphic encryption has been used to reduce the leakage of information.

The study emphasizes the importance of utilizing AI technologies to overcome the cybersecurity issues and suggests AI to provide security and privacy. The study demonstrates that, there is a need for more research in this area. The study emphasizes the importance of utilizing AI technologies to overcome the cybersecurity issues and suggests AI to provide security and privacy. The study demonstrates that, there is a need for more research in this area. Blockchain technology always relies on decentralized network. There is no centralized authority for this. Vehicular network and Smart city systems also supports blockchain’s security and AI’s predictive capability. Despite this development, problems like scalability and regular compliance may arise.

III. OBJECTIVES FOR PRIVACY AND SECURITY IN BLOCKCHAIN–AI SYNERGY

1. Protect Personal Data One main aim is to keep personal details safe, like hospital records or bank information. When a large amount of data is stored and used, there is a chance it may get misused or stolen. To avoid this, the system should keep the data safe by allowing the right people to see it. 2. Stop Unauthorized Access Security means the system should be used only by the right people. Outsiders or hackers must not get a chance to enter or misuse the data. Just like we lock our home for safety, a secure system gives access only with permission and keeps watch for anything unusual. 3. Build Trust Trust is important when people share information. Blockchain keeps the records safe and makes sure no one can secretly change them. Once something is stored, it stays the same. 4. Keep Identities Safe

When data is used for research; personal details should stay hidden. Things like name, address, phone number, or account should not be shown. Data can be made anonymous, so nobody knows whose it is. This lets us use the information without revealing the person. 5. Detect and Prevent Fraud Fraud is a big problem in banks, hospitals, and online businesses. The system can look at the data and find fake transactions or unusual activities quickly. Blockchain keeps records safe so they cannot be changed. Together, this helps to easily stop and control fraud.

IV. TRUST, THREAT, AND SECURITY IN AI-BLOCKCHAIN SYNERGY

The combination of Blockchain and Artificial Intelligence (AI) is used together; they become a strong way to improve privacy and security in digital systems.

Blockchain provides a secure foundation which is essential for effective operation of AI models. Combining Blockchain with AI can supply verified data to AI, mitigating manipulation risks, while AI can bolster Blockchain by enhancing its scalability and detecting suspicious activities instantaneously. Together, they form a system that safeguards sensitive information and adjusts to emerging security threats. This collaboration is also crucial for fostering trust and accountability. The convergence of these technologies creates a transparent and auditable trail for AI actions and decisions, which is crucial for building public confidence, especially in sensitive applications like healthcare or finance. Threat Identification on Blockchain: Blockchain technology protects records securely, but it does not possess the capability to autonomously detect new threats. On the other hand, AI can observe transactions in real-time and spot unusual activities, such as fraud or hacking attempts. The integration of AI enhances the safety of blockchain by finding and fixing vulnerabilities. Moreover, AI can address issues by forecasting future attacks based on past data. This means AI also prevents them, making blockchain systems more secure and trustworthy.

V. CHALLENGES IN PRIVACY AND SECURITY IN BLOCKCHAIN–AI SYNERGY

i. Privacy issue: Public blockchain ledgers offer a secure method for data storage, and the gathered information may be accessed by anybody. This accessibility may result in worries about the possible privacy abuses. The IOT sensing devices are continuously gathering sensitive and private data from people; posting this data on public ledgers can create privacy issues.

ii. Security in Blockchain: The decentralized nature of blockchain makes it helpless to exploitation and abuse. Even though blockchain take over robust security features for predictive analysis and the Internet of Things, 51 percentages of cyberattacks may target blockchain networks. To tackle this problem, newly introduced blockchain platforms like Intel SGX are equipped with hardware that enables execution within Trusted Execution Environments(TEEs).

iii. Governance: Successfully launching, developing, and managing a blockchain platform involving multiple users and stakeholders requires substantial effort. There are major challenges in deciding which type of blockchain to implement (e.g., Hyperledger or Ethereum), even when considering a private or consortium blockchain. This involves research focused on establishing robust governance structures. Developing governance in blockchain systems also demands careful coordination between technical protocols and organizational policies to ensure long-term sustainability.

iv. Interoperability: An analysis indicates that many industries have an interest in blockchain technology. However, there is no standard protocol to facilitate collaborate and integrate among them. This situation is called a lack of interoperability, which could reduce the growth of the blockchain industry. The absence of interoperability allows blockchain developers to code in different programming languages. Therefore, standardization is essential for companies to work together on application development, share blockchain-based solutions, and integrate with existing systems.

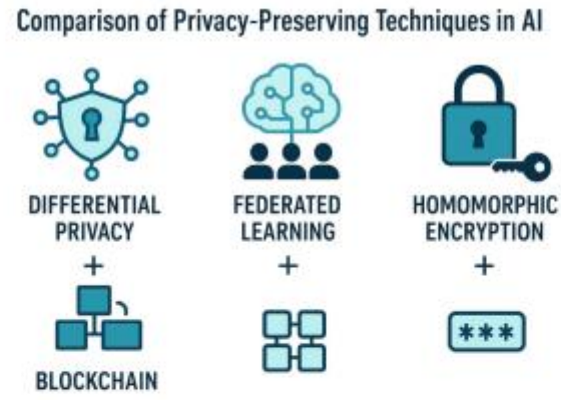

Fig. 2. Comparison of Privacy-Preserving Techniques In AI

Differential privacy is a technique which is used to safeguard personal data while still allowing data to be used. It works by incorporating a small amount of random ”noise” or changes into the data. These modifications make it difficult to associate specific data with any individual, yet they are minimal enough not to significantly affect the overall outcomes. Consider a health survey; a small piece of extra information is appended by the system when you submit your answer. Consequently, it becomes impossible to determine which response is linked to you, thereby safeguarding your privacy while enabling researchers and healthcare experts to analyze broader trends, such as the rates of heart disease and diabetes. By utilizing the aggregated data, public health initiatives can be enhanced through the insights gained. In online shopping platforms, AI analyses the buying behaviours of numerous users to showcase the most sought after products, without disclosing individual transactions. This approach protects personal details while providing companies with enhanced insights. When integrated with blockchain technology, differential privacy strengthens the security of our information. Blockchain ensures that records are secure and unchangeable. Differential privacy allows AI to make conclusions based on data and protect personal information to restrict potential misuse.

VII. APPLICATIONS

1. Healthcare Systems Health records can be stored safely on blockchain so that no one can change or misuse them. AI can then use hidden or anonymous patient data to help doctors find diseases and predict what care is needed in the future. Privacy is protected because new methods allow AI to learn without showing personal details. This keeps patient information safe while also helping doctors make better decisions. Real-world example: Companies like MediLedger and BurstIQ use blockchain with AI to secure the sharing of health data.

2. Financial Services In banking and insurance, blockchain is used to keep records safe. AI helps by catching fraud and checking risks early. Privacy is also protected because banks

can follow rules without showing personal information. This makes the system safer from money laundering and online attacks.

3. Identity Management and Authentication Blockchain allows the creation of decentralised digital identities. It helps users control over their personal data. AI proves identity claims without showing sensitive certification. It ensures safe access to digital services like e-governance. Real-world example: Sovrin Network and Uport provide Blockchain based digital identities with AI-powered confirmation.

4. Supply Chain Management AI enhances the supply chain management through sales forecasting. Blockchain ensures the clarity of goods from origin to destination. Privacy is declared by safeguarding sensitive business information, and safety is improved by verifying the importance of food and other critical products. Real-world example: Walmart uses IBM Food Trust to ensure the food is secure and safe.

5. Autonomous Vehicles and Robotics AI plays an important role in self-driving cars and drones. Blockchain secures communication between devices to prevent tampering. Privacy-preserving methods protect the passenger and user data. Real-world example: Bosch and Volkswagen experiment the Blockchain and AI in secure driving systems.

6. Regulatory Compliance and Governance Always, AI driven decision-making has difficulties with transparency and accountability. Many Companies can meet regulatory requirements by recording AI processes on Blockchain. Real-world example: AI with Blockchain technology is used by the Estonia's e-governance system to secure public services.

VIII. CONCLUSION AND FUTURE DIRECTIONS

This work looks into the privacy protection technologies merged with artificial intelligence and blockchain, introducing their related methodologies. Furthermore, it explores the problems within current systems and recommendations for improvement. Finally, these technologies are classified based on application scenarios and technical solutions. This study holds benchmark for the development of AI and blockchain fusion and provides deep, clear realization and directions for future research. AI and blockchain form a complementary structure that not only improves privacy but also strengthens trust, accountability across fields such as healthcare, finance, supply chain and smart governance. Overall, the use of AI and blockchain enhances secure, efficient, and privacy-preserving systems. By ensuring trust and preventing frauds, it develops the safe digital infrastructures. It also opens new chances in IoT, smart cities and healthcare. To overcome the restrictions and to realise the power of these technologies, continued research is required.

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A Survey on Smart Contracts in Blockchain

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ABSTRACT - Smart contracts are among the most revolutionary advancements in blockchain technology, facilitating the automated, secure, and transparent execution of agreements without requiring intermediaries. As self-executing digital protocols, they aim to bolster trust, lower costs, and reduce human errors in transactions across various sectors. This review paper delves into the core principles of smart contracts, their architecture, and operational mechanisms within blockchain networks. It underscores their applications in finance, supply chain management, healthcare, and governance while critically assessing the challenges, including scalability concerns, security risks, and legal uncertainties. Additionally, the paper surveys current research trends and pinpoints future avenues for enhancing interoperability, regulatory frameworks, and integration with emerging technologies such as the Internet of Things (IoT) and Artificial Intelligence (AI). The review concludes that although smart contracts possess significant potential to transform digital transactions, it is crucial to address technical and regulatory challenges to facilitate their widespread implementation.

Keywords: Blockchain, Smart Contracts, Decentralization, Automation, Security, Scalability, Interoperability, Digital Transactions, Artificial Intelligence (AI), Internet of Things (IoT).

A Review on Role of Cryptography in Block Chain

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Abstract: Blockchain technology employs cryptographic methods to guarantee secure, transparent, and immutable data storage and transactions. It utilizes various techniques, including hashing, digital signatures, public-key infrastructure, and Merkle trees, to uphold confidentiality, authenticity, and integrity. As blockchain underpins cryptocurrencies, smart contracts, decentralized finance (DeFi), and numerous other applications, comprehending the significance of cryptography is essential for tackling security issues and fostering future innovations. Blockchain technology relies fundamentally on cryptography to ensure secure, transparent, and tamper-resistant digital transactions. Cryptographic techniques such as hash functions, digital signatures, and public–private key encryption provide the backbone for data integrity, authentication, and confidentiality within distributed ledgers. Hashing secures block linkage and prevents data modification, while asymmetric cryptography enables secure identity management and transaction validation. Additionally, advanced methods like zero-knowledge proofs and homomorphic encryption are emerging to enhance privacy and scalability in blockchain applications.

Keywords: Blockchain Technology, Cryptography, Data Security, Distributed Ledger, Hashing Algorithms, Digital Signatures, Smart Contracts.

Smart High Beam Assist Using EmbeddedVision and Sensor Logic (SHBA–EVSL)

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Abstract—The Smart High Beam Assist System enhances road safety by automatically controlling a vehicle's headlight beams. Using embedded vision (camera or LDR sensors) and sensor logic, the system detects oncoming vehicle headlights and dims the high beam to avoid glare for other drivers. The core logic is powered by a microcontroller (Arduino) that processes light intensity or image data to determine when to switch between high and low beam. Is an advanced yet cost-effective automotive safety system aimed at enhancing night-time driving conditions. The system uses real-time image processing through a vision-based camera and light sensors to detect oncoming or preceding vehicles. Based on the detection, it automatically switches the vehicle’s headlights from high beam to low beam to prevent glare and reduce the risk of accidents. Unlike the expensive automated lighting systems found in luxury vehicles, SHBA-EVSL is designed to be affordable and compatible with all types of vehicles, including passenger cars, commercial trucks, buses, and two-wheelers. This solution promotes safer roads by combining embedded hardware, sensor logic, and intelligent lighting control.

Keywords: Automatic Headlight Dimming--Embedded Vision--High Beam Assist--Road Safety—Arduino--Raspberry Pi—IoT--Smart Vehicles--Adaptive Lighting System--Light Sensor.

MOBILE APP: DETECT WARD OF WEEDS WITH DEEP LEARNING

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Abstract— Weed infestation is a major constraint in achieving maximum crop yields, as unwanted plants compete with cultivated crops for essential resources. Traditional weed detection methods such as manual identification are labor-intensive, while chemical control approaches increase costs and environmental risks. This paper presents a mobile application that employs deep learning to detect weeds in real time through smartphone-based image capture. A lightweight pre-trained convolutional neural network (CNN), optimized for mobile deployment, is utilized to analyze images and accurately identify weed-affected regions. The proposed framework involves the construction of an annotated weed–crop image dataset, customized model training, mobile optimization, and the design of an intuitive user interface for field usability. Image inference is performed locally on the device or via cloud processing, with results providing visual highlights of weed regions and actionable insights for precision crop management. Field evaluations demonstrate reliable performance across diverse weed and crop scenarios. The proposed system reduces reliance on manual labor and excessive herbicide usage, thereby contributing to sustainable and cost-effective agricultural practices.

Index Terms— Deep learning, weed detection, convolutional neural networks (CNN), precision agriculture, mobile application.

Nano Banana: Diffusion-Based Image Editing, Ethical Challenges, and Responsible AI Innovation

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Abstract

Lately, diffusion-based generative mechanisms have provided the best landscape for image editing, allowing for alterations that are highly realistic with semantically relevant and user-oriented controls. Gemini 2.5 Flash Image by Google, better known as Nano Banana, marks a great milestone in this development process. Nano Banana is built atop diffusion models and text-driven editing frameworks like Imagic, UniTune, and a few others, bringing along the ability to perform multi-image fusion, sanitize character consistency across edits, and provide fine-grained prompt-level control; all of which pack tremendous prospects towards making image editing fast, intuitive, and accessible to the layperson. These capabilities, though powerful, carry certain ethical considerations. Scholarly review surveys denote risks such as the spread of deepfakes, controversies surrounding intellectual property, biased representation, confined cynical representation, to name a few. Google's choice to implement watermarking technologies such as SynthID and preference-alignment solutions signals an industry-aware presence, though striking a balance between freedom for the creator and prevention against misuse remains unresolved. The paper theorizes Nano Banana within the broad.

Keywords: Diffusion Models, Generative AI, Image Editing, Multi-Image Fusion, Semantic Control, Ethical AI, Deepfake Risks, Intellectual Property, Watermarking (SynthID), Responsible Innovation.

Artificial Intelligence and Public Safety: Balancing Innovation with Regulation

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Abstract:

In future, AI is becoming popular now a days as it enhancing human life in many areas. intelligent machines will replace or enhance human capabilities in many areas. Artificial intelligence is the intelligence exhibited by machines and software .In last two decades AI has improved performance of manufacturing, service sector, education, medical, etc. The areas employing the technology of artificial intelligence have seen an increase in the quality and efficiency. This study shows overview of technology and scope of AI in field of education and what steps government have taken to stop AI from destructing ourself.

Keywords: Artificial intelligence, software, technology, machines, scope of AI.

Study on Architecture of Block Chain

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Abstract

Blockchain technology utilizes a decentralized and distributed ledger system peer network) to securely record transactions across a network of computers. Using this technology, participants can transfer value across the Internet without the need for a central third party. Blockchain is the cornerstone of cryptocurrencies, offering a secure and transparent framework for recording transactions along with its unknowns and benefits. Blockchain is a decentralized and distributed ledger technology designed to securely record transactions across a peer-to-peer network. Its architecture is composed of several key layers: the data layer, which structures information into cryptographically linked blocks; the network layer, which manages communication among nodes; the consensus layer, which ensures agreement on the validity of transactions through mechanisms such as Proof of Work or Proof of Stake; and the application layer, which supports smart contracts and decentralized applications. Together, these layers provide immutability, transparency, and trust without the need for central authorities. While the architecture strengthens security and reliability, it also faces challenges related to scalability, energy efficiency, and interoperability. Understanding the architecture of blockchain is fundamental to exploring its applications in domains such as finance, supply chain, healthcare, and governance.

AI-Driven Consultancy for Mental Health: A Step Towards Accessible and Scalable Psychological Support

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Abstract— Mental health disorders such as depression, anxiety, and stress are increasing worldwide, while access to timely consultancy and treatment remains limited. Artificial Intelligence (AI) offers a new approach by providing scalable, affordable, and continuous mental health support. Using Natural Language Processing (NLP) and Machine Learning (ML), AI systems can analyze user input, detect emotional patterns, and deliver evidence-based guidance inspired by therapeutic methods like Cognitive Behavioral Therapy (CBT). This research explores how AI-driven consultancy can complement human professionals, improve accessibility, and ensure early intervention, while also addressing ethical concerns such as data privacy, accuracy, and the need for human oversight.

Smart Drone for Delivery Package

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Abstract
The rapid growth of e-commerce and logistics has accelerated the need for innovative, efficient, and cost-effective delivery solutions. Smart drones for package delivery represent a transformative approach that leverages unmanned aerial vehicles (UAVs) integrated with advanced navigation, real-time tracking, and autonomous decision-making systems. These drones utilize GPS, sensors, and artificial intelligence to ensure safe route planning, obstacle avoidance, and timely delivery. Compared to conventional methods, smart drone delivery reduces transportation time, minimizes human intervention, and offers significant advantages in remote or congested urban areas. However, challenges remain in terms of payload capacity, flight endurance, regulatory compliance, and data security. This study examines the architecture, benefits, and limitations of smart drone delivery systems, highlighting their potential to revolutionize logistics and supply chain operations while addressing future research directions.

Improving Quality Education by Increasing Student Retention Ratio in Higher Education Institutes through Machine Learning

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Abstract
Student retention is a major problem in higher education institutions (HEIs) across the world. Low retention rates negatively impact institutional reputation, resource allocation, and, most importantly, students' life outcomes. This paper proposes a comprehensive solution that uses Machine Learning (ML) to predict student attrition and to inform targeted interventions. We present a framework that covers data collection, feature engineering, model selection, evaluation, and deployment strategies for HEIs. The framework emphasizes interpretability, ethical considerations, and integration into institutional workflows. Works on publicly available and simulated datasets demonstrate that ensemble models and gradient boosting machines achieve high predictive performance while feature importance and SHAP analyses enable actionable insights for counseling and resource prioritization. Recommendations for policy and future work are also provided.

Keywords: Student Retention, Machine Learning, Higher Education, Predictive Analytics, Early Warning Systems

AI POWERED-PHYSICAL STRESS DETECTION USING FASTER R-CNN

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Abstract

Now a day, psychological problems are becoming a major problem to peoples life. It is pre-dominant to detect manage stress before it turns into a severe health issues. Also, traditional stress detection methods like(ECG), EMG, EEG are time consuming and costly. In this project, we are going to detect the stress states of user by monitoring the facial expression using faster R-CNN which comes under deep learning technology. This study presents a novel approach for detecting physical stress using Faster R-CNN, a state-of-the-art object detection framework. The proposed method leverages visual data, such as facial expressions, body posture, and physiological signals, to identify stress-related patterns. The Faster R-CNN model is trained on a curated dataset containing labeled instances of stress indicators. The system demonstrates high accuracy in detecting stress levels under diverse environmental conditions, making it a robust solution for applications in healthcare, workplace monitoring, and fitness tracking. The proposed framework outperforms traditional machine learning methods in terms of speed and precision, paving the way for advanced stress management systems.

Keywords: Region based convolution neural network(r-cnn), Electrocardiography(ecg) Electromyography(emg), Electroencephalography(eeg).

AI-Based NEET College Predictor: An Integrated Platform for Medical College Admission Prediction Using Deep Learning and Mock Test Analysis

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Abstract

The increasing competition and limited availability of seats in the National Eligibility cum Entrance Test (NEET) have highlighted the necessity of intelligent and data-driven approaches for medical college admission prediction. This study presents an AI-based NEET College Predictor, an integrated platform designed using deep learning models, mock test analysis, and historical cutoff data to provide accurate and personalized guidance for medical aspirants. The system incorporates machine learning algorithms, neural networks, and predictive analytics to estimate a candidate's admission probability based on examination scores, category-based reservations, institutional seat matrices, and previous year admission trends. Unlike traditional prediction tools, the proposed framework goes beyond static cutoff comparison by incorporating dynamic score projections derived from continuous mock test performance analysis, enabling the system to simulate real-time exam scenarios and assess potential improvements in student readiness. By processing large-scale datasets and applying student performance profiling, the model enhances prediction accuracy while adapting to evolving exam patterns and policy changes.

The outcome of this research is an intelligent platform that not only recommends the most probable colleges but also provides strategic insights for exam preparation and admission planning. The proposed system contributes to transparency, fairness, and efficiency in the admission process by offering students data-driven recommendations, thus reducing uncertainty and anxiety. With its integration of artificial intelligence, educational data mining, predictive modeling, and admission analytics, this platform has the potential to serve as a valuable decision-support tool for students, parents, and academic counselors in the highly competitive medical education landscape.

Keywords: NEET, Deep Learning, Mock Test Analysis, College Predictor, Machine Learning, Predictive Analytics, Student Performance Profiling, Admission Probability Estimation, Educational Data Mining, Medical Education

Blockchain in Education: Certificates & Academic Records

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Abstract

Certificates and academic records are the backbone of education and career opportunities. However, traditional systems often suffer from issues like forgery, paper based delays, and difficulty in verification. Blockchain technology provides a modern solution by creating tamper proof, transparent, and easily shareable academic credentials. This paper discusses how blockchain can transform the way certificates and academic records are stored, verified, and accessed.

Brain–Computer Interface Powered by AI for Next-Generation Healthcare

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Abstract

Brain–Computer Interfaces (BCIs) are redefining the future of healthcare by enabling direct communication between neural activity and external systems. The integration of Artificial Intelligence (AI) significantly enhances their capabilities, improving signal decoding, adaptability, and clinical applicability. AI-powered BCIs offer promising solutions for neurorehabilitation, assistive communication, mental health monitoring, and personalized treatment strategies. By learning from individual neural patterns, these systems achieve higher accuracy and responsiveness, paving the way for patient- centered care in conditions such as paralysis, stroke, and neurodegenerative disorders. Beyond treatment, AI- augmented BCIs may also support preventive and predictive healthcare by detecting early signs of cognitive decline or neurological dysfunction. However, critical challenges remain, including signal variability, data privacy, ethical considerations, and the need for robust clinical validation. This work highlights the convergence of AI and BCI technologies and their transformative potential to drive the next generation of intelligent and inclusive healthcare solutions.

An Intelligent and Transparent Framework for Tamper-Proof Cricket Scoring

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Abstract - Cricket scoring has long faced challenges of human error, bias and disputes over the verification of match records, making reliability and transparency a pressing issue in the sport. This work introduces a novel framework that combines automated event detection with a secure validation system to address these challenges in a simple therefore, also impactful way. Using computer vision techniques the system automatically identifies ball-by-ball events-such as runs, wickets and extras so on. While predictive analytics help minimize scoring errors and improve the accuracy. To make sure that when the recorded match data cannot be manipulated or altered, an immutable and transparent record-keeping mechanism is employed governed by pre-defined rules that validate events consistently. Beyond safety of integrity, the framework also increase the new opportunities for players, officials, members and broadcasters by enabling access to trustworthy match data in real time encouraging fair play and fostering greater engagement with the sport. Ultimately, the proposed solution not only revamps the way cricket scores are recorded but also handles the real-world problem of trust in sports scoring systems, paving the way for reliable and inclusive data-driven experiences.

Keywords – *Computer Vision, Predict Analysis, Event Detection, Data Integrity*

AI Driven Deepfake Detection and Prevention System to safeguard women's digital identity in online platforms

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Abstract

The sudden development of deepfake technology has posed serious issues concerning online security, particularly of women whose online identities are exceedingly susceptible to abuse. Deepfake images and videos created by advanced AI methods can be used for harassment, defamation, and invasion of privacy, causing critical psychological and social impacts. This paper introduces an AI-based Deepfake Detection System to protect women's online identity on the internet. The framework being proposed utilizes a hybrid approach that integrates handcrafted image artifact analysis with deep learning models, i.e., convolutional neural networks (CNNs). Using compression artifacts, color inconsistencies, and texture-level irregularities, the system improves its robustness in distinguishing between real and manipulated media. Experimental testing is set up to show increased robustness under different levels of compression and under real-world scenarios. The research contributes to the emerging debate on online security through the provision of a credible mechanism of detection that emphasizes the safety of women's online presence against threats enabled by artificial intelligence.

Keywords: Deepfake Detection, Convolutional Neural Networks (CNNs), Image Artifacts, Compression Artifacts, Artificial Intelligence.

Leftover Food Sharing Website

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Abstract

Food destruction is a critical issue affecting both developed and developing countries. While tons of edible food are discarded daily by households, restaurants, supermarkets, and event organizers, millions of people continue to suffer from hunger and malnutrition. In response to this growing problem, this project proposes the development of a Leftover Food Sharing Website, a digital platform that bridges the gap between food donors and recipients. The core objective of the platform is to minimize food waste by facilitating the sharing and redistribution of surplus or leftover food in a structured, safe, and efficient manner. The website enables individual users, restaurants, caterers, and grocery stores to register and list leftover food items that are still safe for consumption. Each listing includes important details such as food type, quantity, expiry time, pickup address, and contact information. On the recipient side, NGOs, food banks, orphanages, shelters, and volunteers can browse or search for available food donations in their vicinity using location-based filters. The platform provides features like user authentication, real-time notifications, request management, and feedback ratings to ensure transparency, security, and trust among users. Additionally, the system encourages responsible behavior through a points or rewards system for frequent donors and active volunteers. It also incorporates basic AI/ML logic to suggest donation matches based on previous activity, location proximity, and food types. This initiative not only contributes to solving the problem of food wastage and hunger but also promotes community involvement, sustainable living, and social responsibility. By leveraging modern web technologies, the project provides a scalable, user-friendly, and impactful solution that can be implemented in local communities and expanded to a broader network over time

Keywords: Food Waste Management, Leftover Food Redistribution, Community, Food Sharing, Sustainable Living, User Authentication, Real-time Notifications, Loc

Online Blood Bank Web Application

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Abstract-The Online Blood Bank Web Application is a digital platform designed to enhance the efficiency and accessibility of blood donation and distribution. This system addresses the critical need for timely blood availability during emergencies by connecting donors, recipients, hospitals, and blood banks in a seamless and efficient manner. The application provides a user-friendly interface for donors to register, maintain health records, and schedule donation appointments. Recipients can search for available blood groups, place requests, and track the status of their requirements. Blood banks and hospitals can update their inventory in real-time, ensuring accurate information is always available. Geolocation services allow users to locate nearby blood banks, while automated notifications and alerts inform donors about eligibility and notify recipients of urgent requirements. The system is designed to ensure data security, protecting sensitive information of donors and recipients.

Keywords: Geolocation Services, Appointment Scheduling, Hospital Integration, Healthcare System, Database Management, Scalability, Voluntary Donation, Search and Filter, Blood Type Compatibility.

Integrate – AI-Powered Unified Notification Hub

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Abstract— Modern users are constantly overwhelmed by the sheer volume of notifications and messages spread across multiple applications such as email, chat platforms, social media, and productivity tools, often leading to missed updates, distraction, and decreased efficiency. To solve this challenge, we propose Integrate, an AI-powered unified notification and workflow assistant that consolidates all incoming messages and alerts into a single platform. Leveraging automation frameworks such as n8n and Microsoft Power Automate, Integrate fetches and synchronizes data from diverse applications, processes it intelligently, and employs AI models to summarize, prioritize, and transform raw information into clear, actionable insights. Users receive streamlined updates, task reminders, and contextual highlights without the noise of fragmented notifications, ensuring a more focused and productive digital experience. In addition, Integrate offers advanced features like automated meeting summaries, missed call follow-ups, and personalized daily digests, all accessible through a unified dashboard. By blending multi-app automation with AI-driven transformation, Integrate eliminates redundancy, reduces notification fatigue, and provides a scalable, user-friendly solution for managing digital information overload.

Keywords*:* Unified Notification Hub, AI-Powered Message Summarization, Workflow Automation, Multi-App Integration, Productivity Enhancement

Decoding Sign Language: An EEG-Based Motor Imagery Brain-Computer Interface Approach

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Abstract— Communication barriers faced by individuals with hearing and speech impairments highlight the urgent need for innovative assistive technologies. This research explores the use of an Electroencephalogram (EEG)-based Motor Imagery Brain-Computer Interface (MI-BCI) for sign language recognition. By capturing and analyzing brainwave patterns generated during motor imagery tasks, the system aims to decode intended sign language gestures without requiring physical movement. The proposed framework integrates EEG signal acquisition, feature extraction, and machine learning algorithms to achieve accurate classification of motor imagery signals corresponding to sign gestures. This approach has the potential to provide a direct communication channel between the brain and external devices, reducing dependency on conventional sign interpretation methods. While promising, challenges such as signal noise, inter-subject variability, and real-time processing efficiency must be addressed. The study underscores the potential of EEG-based MI-BCI systems in creating inclusive communication platforms and advancing human-computer interaction for assistive applications.

Keywords— EEG, Motor Imagery, Brain-Computer Interface (BCI), Sign Language Recognition, Assistive Technology, Signal Processing, Machine Learning, Human-Computer Interaction

AI-Based Plastic Hunter and Blue Sweeper

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Abstract - Water pollution caused by floating waste is one of the most pressing environmental challenges, affecting aquatic ecosystems, human health, and overall water quality. Conventional cleaning methods are labor-intensive, hazardous, and often inefficient. To address this, an AI-based robotic boat is proposed for the autonomous detection and collection of floating debris in water bodies such as lakes, rivers, and ponds. The system integrates artificial intelligence algorithms with computer vision to detect and classify garbage in real time. A camera module, combined with AI-based image recognition, enables the boat to differentiate between waste and natural objects such as leaves or aquatic plants. The navigation system is supported by GPS, ultrasonic sensors, and obstacle avoidance algorithms, ensuring smooth movement and efficient coverage of the cleaning area. The collected waste is stored in an onboard compartment, which can be periodically emptied. The robotic boat operates on solar or battery power, making it eco-friendly and sustainable. Additionally, IoT-based connectivity can be integrated for real-time monitoring, remote control, and data analysis. This innovation reduces human intervention, increases efficiency, and contributes to cleaner waterways. The proposed solution has potential applications in municipal waste management, tourism sectors, and industrial water bodies.

Index Terms—Artificial Intelligence, Computer Vision, Autonomous Robot, Waste Management, IoT, Water Pollution.

Speak AR: An AI and AR-Powered Gamified Language Learning System

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Abstract - Speak AR is an innovative mobile-based language learning system that integrates Artificial Intelligence (AI), Augmented Reality (AR), and gamification to deliver an immersive and personalized experience. Unlike traditional methods that rely on rote memorization, SpeakAR adapts to each learner’s progress through AI-generated flashcards, intelligent recommendations, and an AI-powered chatbot for realistic conversations. The AR module allows users to scan real-world objects and instantly learn vocabulary in the target language, building strong contextual associations and improving retention. Real-time speech recognition ensures learners receive instant pronunciation feedback, enhancing fluency and speaking confidence. To sustain motivation, the system incorporates levels, badges, streaks, and leaderboards, while location-based suggestions provide vocabulary relevant to the learner’s surroundings, ensuring practical application. Built as a scalable cross-platform app using Flutter and cloud services, SpeakAR bridges the gap between theory and practice, transforming language learning into an engaging, interactive, and future-ready process.

Smart AI-Powered Port Terminal and Logistics Transportation Management System

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Abstract--In an era defined by global trade acceleration, supply chain disruption, and rising customer expectations, the logistics and maritime industry is undergoing a digital revolution. Traditional port and transport operations—once driven by manual coordination and siloed systems—are evolving into smart ecosystems powered by Artificial Intelligence (AI), Internet of Things (IoT), and cloud technologies. The Smart AI-Powered Port, Terminal, and Logistics Transportation Management System (SPTL-TMS) offers a unified platform integrating port operations, terminal handling, gate automation, transport dispatch, and warehouse synchronization. This paper presents its concept, objectives, uniqueness, potential applications, market impact, and implementation roadmap.This intelligent system is not just an upgrade, it is a strategic enabler for future-ready, resilient, and sustainable logistics infrastructure, essential for global competitiveness in the dynamic landscape of the 21st century.

Keywords: Intelligent Transportation System (ITS) - Supply Chain Optimization - Internet of Things (IoT) - Real-time Tracking - Automation in Logistics - Smart Container Management - Artificial Intelligence (AI) in Ports - Blockchain for Logistics

Prognosis of Hepatitis Variants

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Abstract

Hepatitis is a serious liver disease caused by different viral infections, toxins, and autoimmune factors, which leads to high global morbidity and mortality. The major variants of viral hepatitis include types A, B, C, D, and E, each differing in mode of transmission, clinical outcome, and risk of chronic progression. Accurate prognosis of these variants is essential for early diagnosis, timely intervention, and effective treatment planning. This study focuses on the prognosis of hepatitis variants by analyzing clinical, biochemical, and epidemiological features that influence disease progression. Prognostic assessment incorporates liver function tests, serological markers, and molecular diagnostics to distinguish between acute, chronic, and severe outcomes. Additionally, modern computational and machine learning approaches are being increasingly applied to predict the disease course and categorize patient risk levels. A systematic understanding of hepatitis variants not only enhances patient management but also supports the development of preventive strategies, ultimately reducing the burden of liver-related complications worldwide.

ECommerce Data Analysis System

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Abstract

The rapid growth of e-commerce has led to the accumulation of massive volumes of structured and unstructured data This project proposes the development of an AI-powered e-commerce assistance platform designed to benefit both consumers and businesses. On the consumer side, the platform enables users to easily discover desired products, view truthful ratings, and analyze sales percentages of specific items across different shops. A unique feature of the system is the AI call integration, where by simply tapping the "Contact Shop" button, the consumer is automatically connected to the respective shop owner without leaving the platform. This ensures seamless communication, trust, and convenience in the shopping process. On the business side, the platform provides a powerful tool for sales growth and data-driven decision making. Businesses can promote their products, track sales data, analyze frequently sold items, and receive AI-based recommendations to optimize stock management and enhance sales performance. The dual focus on consumer convenience and business efficiency makes the system a sustainable solution that bridges the gap between customers and retailers. By ensuring transparency, reliability, and advanced AI-driven support, the project aims to create a next-generation e-commerce ecosystem that fosters trust, efficiency, and mutual growth

Keywords - E Commerce, AI CallIntegration, AI Recommendations, Suistanable solutions, Datadriven decision making, Transparency

AI Rules the World

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Abstract
Artificial Intelligence (AI) is no longer a futuristic concept confined to research labs or science fiction—it has emerged as a global driving force that is reshaping industries, governance, culture, and daily life. Its applications extend from healthcare and education to defense, finance, and climate change mitigation, influencing decisions at both individual and societal levels. The statement "AI rules the world" reflects the unprecedented influence of intelligent systems on human progress, innovation, and global power structures. This paper examines AI's transformative role across multiple sectors, highlighting both opportunities and challenges. It explores how AI can fuel economic growth, enhance problem-solving, and democratize knowledge, while also raising critical concerns about ethical dilemmas, job displacement, privacy, and security risks. The analysis concludes that AI has the potential to act as humanity's greatest ally—but only if governed responsibly, ethically, and inclusively.

Closing the Employability Gap: An AI Approach to Curriculum Updates

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Abstract—The rapid evolution of technology and industry demands has created a persistent gap between academic curricula and the skills required in the job market. Traditional curriculum design often fails to keep pace with dynamic employment trends, leaving graduates underprepared for real-world challenges. This paper proposes an AI-driven framework for curriculum up- dates that leverages labor market data from platforms such as LinkedIn, Naukri, and Indeed. The model continuously scans and analyzes job postings to identify emerging skills, technologies, and competencies in high demand. Using natural language processing (NLP) and machine learning (ML) techniques, the system categorizes and prioritizes these skills, recommending corresponding updates to academic syllabi. By aligning education with industry needs, the proposed framework aims to reduce the employability gap, foster adaptive learning environments, and ensure that students are equipped with relevant, future-ready skills. This research highlights the potential of AI as a bridge between academia and industry, offering an innovative solution for dynamic and evidence-based curriculum development.

Keywords—Artificial Intelligence, Curriculum Development, Job Market Trends, Skill Gap, Employability, Natural Language Processing, Machine Learning, Education-Industry Alignment, Adaptive Learning, Future Skills

Kidolens – AR Based Teaching Software

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Abstract: Our application leverages Augmented Reality (AR) technology to bring educational content to life, enabling children to interact with 3D animated objects—such as animals, letters, numbers, and shapes—directly in their real-world environment. Instead of passively watching or reading, kids actively engage with the learning material, which promotes better understanding and enjoyment. The app is designed with a user-friendly interface, colorful visuals, and child-safe interactions suitable for early learners. The central idea is to use an AR-enabled device with a 360-degree camera to transform a child's real-world environment—like their living room, a classroom, or a library—into a dynamic learning space. Instead of simply placing 3D models on a flat surface, the app captures the child and their entire surroundings, allowing for a seamless integration of virtual content into the real world. This creates a deeply personal and interactive experience where the child is not just a viewer, but an active participant in their own learning adventure. To create an engaging and interactive learning experience using AR for children aged 3–8. To improve conceptual understanding of basic topics like alphabets, numbers, animals, and shapes through visual and tactile interaction. To increase retention and interest in learning by turning abstract ideas into tangible 3D experiences. To offer an accessible digital learning tool that complements classroom and home education.

Keywords: Augmented reality, Smart Education, AR based Education, Gamification in Education, Interactive Education, STEM education

AI- Driven Predictive Analytics Powered Mental Health Monitoring Using Internet of Things

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Abstract: Mental health has become a global concern, with conditions such as stress, anxiety, and depression increasing rapidly in today's society, making early detection and continuous monitoring essential for improved treatment outcomes. Recent advancements in Artificial Intelligence (AI) have enabled innovative approaches to mental health monitoring, including chatbots, virtual assistants, social media analysis, speech recognition, and wearable devices, with existing literature highlighting their ability to provide timely assessments, personalized interventions, and stigma-free private support. The methodology involves data collection from text, speech, and physiological signals, followed by preprocessing for accuracy and the application of machine learning and natural language processing models for prediction. Results from recent studies indicate that AI systems can detect mental health disorders with significant accuracy, provide immediate digital support, and assist healthcare professionals in decision-making; however, challenges such as privacy concerns, ethical issues, and the absence of human empathy remain. The study concludes that while AI cannot replace human care, it has great potential to improve accessibility, effectiveness, and personalization of mental health support when integrated responsibly.

AI Glass with Voice-Assisted Navigation and Object Detection for the Visually Impaired

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Abstract— This research proposes an innovative AI- enhanced smart glasses system designed to empower visually impaired individuals through real-time scene understanding, object detection, and multimodal navigation assistance. The system integrates thermal imaging sensors, high-resolution cameras, GPS modules, and embedded AI processing units to deliver comprehensive environmental awareness in diverse conditions. Key innovations include offline AI inference using optimized deep learning models, regional language voice guidance through bone-conduction speakers, and a modular MSME-friendly design that ensures affordability and scalability. Our experimental analysis demonstrates significant improvements in navigation safety, object detection accuracy, and user independence compared to traditional assistive technologies. The proposed system achieves real-time processing latency under 200ms, supports 24/7 operation through thermal-visual sensor fusion, and provides seamless hands-free interaction. This research establishes a foundation for next-generation wearable assistive technologies, with potential applications extending to elderly care, industrial safety, and emergency response scenarios.

Keywords—Visually impaired, smart glasses, thermal imaging, AI navigation, scene understanding, assistive technology.

The Integration of AI and Blockchain in Healthcare: Ensuring

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Abstract: AI and blockchain in the healthcare sector are slowly emerging as a modern and innovative approach towards the achievement of secure and accurate data management, as well as effective process optimization. AI has the ability to work with big data to help get insights into large sets of medical data; on the other hand, blockchain provides a decentralized place to store this data securely. While progression in healthcare management has been led by an emphasis on digitization, risks arising from privacy violations, endeavor to manipulate contents of the records for malevolent purposes and gaining access to patients' information also escalate. AI, when combined with blockchain, does this by ensuring that data across the different facilities is shared securely and that patients can be monitored in real-time without the risk of information manipulation or forgery. In this paper, the research considers the advantages of applying Artificial Intelligence and Blockchain technology in the healthcare sector; data security, patient privacy, and healthcare systems. It also discusses AI and Blockchain's current use cases, theoretical frameworks, and future in this area of application. Furthermore, the integration of AI and Blockchain through various pictorial representations of the origin process also sets the workflow diagram micrograph structure. Some of the technical, ethical, and regulatory issues involved in this integration will also be explained, and suggestions will be provided. Finally, this paper makes the following policy implications for the various stakeholders namely the healthcare providers, policy makers and the technology developers.

AI-Based Sign Language to Speech Converter

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Abstract— This research presents an Artificial Intelligence (AI)-based framework for translating sign language into text and speech, aimed at bridging communication barriers between the hearing-impaired community and non-sign language users. The proposed system integrates Convolutional Neural Networks (CNNs) for spatial feature extraction of hand gestures and Recurrent Neural Networks (RNNs) for temporal sequence modeling, enabling robust recognition of dynamic sign patterns. A large dataset of sign language gestures, including static alphabets and continuous word sequences, is utilized to train the model. Experimental results demonstrate that the system achieves an accuracy of 93.4%, with a precision of 91.2% and recall of 90.6%, surpassing traditional image processing-based approaches. To enhance interpretability and reliability, SHAP (Shapley Additive Explanations) values are incorporated, allowing users and developers to identify critical gesture features influencing the predictions. The findings emphasize the potential of deep learning in fostering inclusive communication, improving accessibility, and supporting real-time applications such as education, healthcare, and public services. Future enhancements include multilingual sign language support, deployment on mobile and wearable devices, and integration with real-time speech synthesis for broader societal impact.

Keywords— Sign Language Recognition, Artificial Intelligence, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Gesture Recognition, Human–Computer Interaction, Deep Learning, Explainable AI, SHAP Values, Assistive Technology, Accessibility, Speech Conversion

AI Powered Panic Detection and Support System Using Machine Learning

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Abstract-Panic disorder is a severe mental health condition wherein people abruptly go into episodes of great worry, chest pain, dizziness, or even fainting. Particularly when immediate medical or mental help is not accessible, these abrupt assaults can generate a sense of helplessness. This project offers an AI-Powered Panic Attack Detection and Support System to solve this problem. The system is meant to serve as a virtual emergency coach that senses panic or anxiety symptoms from user input and offers quick solutions. It cleans and assesses text using Natural Language Processing (NLP) techniques and uses Machine Learning (ML) algorithms including Naive Bayes and Support Vector Machine (SVM) will sort the input into two categories: People are vulnerable during these crucial events since immediate help is frequently not present. Early indicators of panic from text input are recognized by this AI-Powered Panic Attack Detection and Support System, which also offers immediate assistance. By Using Natural Language Processing (NLP) and Machine Learning (ML) methods including logistic regression, naive Bayes, and support vector machines (SVM), the system properly categorizes. Normal or panic detected user inputs.Acting as a virtual emergency coach, the system offers practical solutions including motivational reminders, guided breathing exercises, and relaxing messages in addition to detection. This method guarantees that people may control their worry right now, therefore lowering the possibility of major physical or mental consequences. Emphasizing the application of artificial intelligence for preventative mental health care, the project shows how technology may help for real-time intervention and emotional well-being. With coming improvements Like mobile app deployment and integration with wearable sensors, the system can become a thorough mental health assistant, providing continuous support and monitoring for individuals vulnerable to panic attacks

Keywords: machine learning-natural language processing-anxiety management- panic detection- mental health assistance

NLP based Domain-Specific Auto Spelling Check and Correction: A Survey

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Abstract

Spelling error detection and correction is an important area in Natural Language Processing (NLP). This paper reviews 25 recent works (2023–2025) focusing on domain-specific auto spell checking and correction.The approaches range from traditional rule-based and hybrid techniques to deep learning models such as BiLSTM, CNN-LSTM, Transformer-based architectures like BERT and BART, and large language models (LLMs).Results are evaluated using accuracy, precision, recall, F1-score, BLEU, and robustness.The findings show that deep learning and hybrid methods provide better results than traditional approaches, while LLMs and adversarial training models improve robustness and adaptability.Low-resource languages such as Tibetan, Wolof, Assamese, and Persian also benefited from these models.Overall, the survey highlights that contextual embeddings, bi-directional correction, phonetic features, and multilingual zero-shot models are effective, and future work can focus on real- time, multilingual, and domain-adaptive applications.

Keywords: Spell checking, Error correction, NLP, Deep learning, Transformers, LLM, OCR, Low- resource languages

Smart Community Health Monitoring and Early Warning System for Water Borne Diseases

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Abstract-

Water-borne diseases such as cholera, typhoid, and dysentery continue to pose significant threats to public health, particularly in developing communities with limited access to safe water and timely medical intervention. This project proposes a Smart Community Health Monitoring and Early Warning System that integrates IoTenabled sensors, data analytics, and machine learning techniques to detect, predict, and mitigate outbreaks of water-borne diseases. The system continuously monitors water quality parameters such as pH, turbidity, temperature, and contamination levels through distributed sensor networks. The collected data is transmitted to a cloud-based platform, where advanced algorithms analyze trends and detect anomalies that may indicate potential health risks. A predictive model is employed to assess the likelihood of disease outbreaks based on environmental data, historical health records, and community reports. The system further provides realtime alerts and early warnings to public health officials, local authorities, and community members via mobile applications and dashboards, enabling proactive intervention and resource allocation. By combining smart sensing, predictive analytics, and community engagement, this system enhances public health surveillance, reduces response time, and contributes to the prevention and control of water-borne diseases in vulnerable populations.

Immersive Learning: AR/VR in Smart Education

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Abstract

Immersive learning, driven by Augmented Reality (AR) and Virtual Reality (VR), is revolutionizing modern education by creating interactive, engaging, and experiential learning environments. Unlike traditional classroom methods, AR and VR technologies enable learners to visualize complex concepts, simulate real-world scenarios, and actively participate in the learning process. This transformative approach enhances understanding, retention, and collaboration, while bridging the gap between theoretical knowledge and practical application. Smart education systems that integrate AR/VR offer personalized and adaptive learning experiences, catering to diverse learning styles and improving accessibility. As educational institutions embrace digital transformation, immersive technologies are emerging as vital tools for shaping the future of education through innovation, interactivity, and inclusivity.

Keywords: Immersive Learning, Augmented Reality (AR), Virtual Reality (VR), Smart Education, Interactive Learning, Digital Transformation, Experiential Learning, EdTech, Simulation, Personalized Learning

Data Security and Privacy on Blockchain

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Abstract

Blockchain technology has emerged as a transformative paradigm for secure and decentralized data management. Its inherent features—immutability, transparency, and distributed consensus—offer strong protections against data tampering and unauthorized access. However, while blockchain ensures data integrity and availability, it also raises significant challenges concerning privacy, scalability, and compliance with data protection regulations. Public blockchains, for instance, expose transaction details that may inadvertently compromise user confidentiality, whereas private and permissioned blockchains attempt to balance transparency with controlled access. Recent advancements, including cryptographic techniques such as zero-knowledge proofs, homomorphic encryption, and differential privacy, aim to strengthen privacy preservation without undermining the trustless nature of the system. This paper explores the dual aspects of security and privacy in blockchain, reviews emerging solutions, and highlights open research challenges in designing blockchain-based systems that achieve both robust data protection and user confidentiality.

Keywords: Blockchain, Data Security, Privacy, Cryptography, ZKPs, Decentralizati

A Firewall Rule Simulation Approach for Network Security

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Abstract— Network security is crucial in today's computing world because of the rise in cyberattacks, harmful traffic, and unauthorized access attempts. These threats endanger the confidentiality, integrity, and availability of information systems. Among various security tools, firewalls are widely used and essential for network protection. A firewall acts as a barrier between trusted internal networks and untrusted external ones. It filters traffic according to a set of defined rules. However, beginners and students often find the internal logic and operation of firewall rules hard to grasp since they usually work at the system or enterprise level. This project addresses that issue by introducing a firewall rule simulation approach for network security. This aims to provide an educational and experimental platform for understanding packet filtering mechanisms. The project focuses on designing and implementing a lightweight simulation system that imitates how a firewall operates by applying user-defined rules to simulated network packets. The simulator allows users to create filtering rules based on criteria like source IP address, destination IP address, port number, and protocol type (e.g., TCP, UDP, ICMP). After users set the rules, the system checks incoming or outgoing packets against these conditions to decide whether to allow or block the traffic. This gives users hands-on experience in creating, managing, and testing firewall rules in a controlled setting without needing access to a real network firewall.

Keywords— *Firewall Simulation, Network Security, Traffic Filtering, Cybersecurity Education, Network Protection*

Organ Donation Management System

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Abstract— The Organ Donation Management System is a computerized solution designed to streamline and manage the process of organ donation, allocation, and transplantation. The system facilitates effective communication between donors, recipients, and healthcare organizations by maintaining a centralized database. It ensures transparency and efficiency in matching donors with recipients based on medical compatibility and urgency. By automating record-keeping, registration, and tracking, the system minimizes manual errors and reduces delays in critical cases. Furthermore, it enhances awareness about organ donation by providing accurate information and easy access for potential donors. Overall, the system aims to save lives by improving the speed, reliability, and accessibility of the organ donation process.

Keywords: Organ Donation - Management System – Healthcare – Database – Donor – Recipient – Transplantation – Automation – Transparency - Life-Saving

Detection of Phishing Urls Through the Use of Machine Learning

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Abstract: Phishing represents a significant threat in the realm of cybersecurity, where attackers deceive individuals into disclosing sensitive information like passwords, banking details, or personal data by masquerading malicious websites as genuine. Traditional methods for identifying phishing URLs, including blacklists and rule-based systems, often fall short due to the fast-paced evolution and sophisticated obfuscation tactics employed by cybercriminals. To address these issues, this project introduces a machine learning-based strategy for detecting phishing URL.The proposed system extracts various features from URLs, focusing on both lexical and host-based characteristics, such as the length of the URL, the presence of special characters, domain specifics, and patterns of redirection. These features are subsequently utilized to train machine learning models, including Decision Trees, Random Forest, Logistic Regression, and Support Vector Machines.

Keywords: cyber security, fraud detection, URL classification, Machine Learning, Malicious URL detection, Fraudlent websites, Online fraud detection, Character distribution, URLlength, Phishing

AI-Powered Crop Yeild Prediction and Optimization

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Abstract— Food security depends heavily on agriculture, yet estimating crop yields is difficult due to uncontrollable factors including climate change, soil quality, pest infestations, and erratic rainfall. Conventional forecasting techniques frequently lack precision and flexibility, which results in ineffective farming methods. By utilizing past agricultural data, weather patterns, and soil properties, artificial intelligence (AI) provides sophisticated data-driven strategies to address these issues. This study uses machine learning algorithms to accurately forecast crop production depending on a number of variables, including temperature, rainfall, soil nutrients, and seed type. More accurate forecasting is made possible by the integration of deep learning models, which are used to identify intricate patterns in big datasets. To recommend the best crop selections, appropriate fertilizers, and irrigation schedules for optimal productivity, optimization techniques are used

Keywords: Machine Learning (ML), Predictive Analytics, Internet of Things, Rasspberry Pi, Optimization, Prediction, Sensors, Dhasboard, Blockchain Technology